

CHAPTER 10

FUEL SYSTEM

CHAPTER OVERVIEW

SECTION	TITLE
10-1	Equipment Description and Data
10-2	Troubleshooting
10-3	On Aircraft Inspections
10-4	Maintenance Procedures
10-5	Maintenance Procedures (BENCH)

10-1/(10-2 Blank)

WARNING – This document contains technical data subject to the ITAR; and the EAR. Authorization is required prior to providing this technical data to any company, entity, person, or destination. ITAR/USML Category: VIII(i), EAR Export Classification: ECCN 9E610.a

SECTION 10-1

EQUIPMENT DESCRIPTION AND DATA

SECTION OVERVIEW

PARAGRAPH	TITLE
10-1-1	Fuel Low-Level Warning System
10-1-2	Submerged Fuel Boost Pump System
10-1-3	Fuel Prime Boost System
10-1-4	Pressure Refuel/Defuel System
10-1-5	Fuel Quantity System

10-1-1/(10-1-2 Blank)

10-1-1 FUEL LOW-LEVEL WARNING SYSTEM.

10-1-1.1 System Description.

The fuel low-level warning system operates in conjunction with the No. 1 and No. 2 main fuel tanks to alert the pilots when fuel in the main fuel tanks has dropped to the 20-minute reserve level. Major components of the system include a fuel low-level sensor on each fuel quantity probe, a low-level warning conditioner, a fuel low indicator test circuit, and two caution capsules labeled #1 FUEL LOW and #2 FUEL LOW on the caution/advisory panel (Figure 10-1-1-1).

10-1-1.2 Principles Of Operation.

10-1-1.2.1 Power Distribution.

The fuel low-level warning system is supplied 28 vdc from the No. 1 dc primary bus through the FUEL LOW WARN circuit breaker on the copilot's circuit breaker panel. The 28 vdc is routed to the low-level warning conditioner and the FUEL IND TEST switch on the miscellaneous switch panel.

10-1-1.2.2 System Operation.

The fuel low-level warning circuits for the No. 1 and No. 2 main fuel tanks are identical in operation; therefore, only the No. 1 main fuel tank low-level warning circuit will be described (Figure 10-1-1-2).

The No. 1 main fuel tank low-level warning circuit is supplied dc current from the low-level warning conditioner. The sensor at the lower portion of the fuel quantity probe contains a fuel low-level sensor (thermistor) which senses the changes in fuel level. When the fuel level in the fuel tank is greater than 170 to 180 pounds, it covers the thermistor, keeps it cool, and increases its electrical resistance. This increase in thermistor resistance causes the low-level warning conditioner to open the supply path to the #1 FUEL LOW caution capsule causing the capsule to stay off. When the fuel level in the fuel tank drops below 170 pounds, the thermistor is uncovered, its temperature rises, and its resistance is lowered. This decrease in thermistor resistance causes the low-level warning conditioner to close

the supply path to the #1 FUEL LOW caution capsule, causing the capsule to flash. The operation of the low-level warning circuit can be checked by pressing the FUEL IND TEST button on the miscellaneous switch panel.

10-1-1.3 Component Description.

10-1-1.3.1 Fuel Low-Level Sensor.

The fuel low-level sensor (thermistor) has a self-contained heating element which will dissipate heat at two distinct rates: immersed in fuel or exposed to free air/fuel vapor. When dry, the heating element heats the thermistor. Since the thermistor has a negative temperature coefficient of resistance, the net result is a decrease in resistance. This lower resistance is sensed by the fuel low-level warning conditioner and, after approximately 7 seconds, causes either the #1 or #2 FUEL LOW capsule to go on.

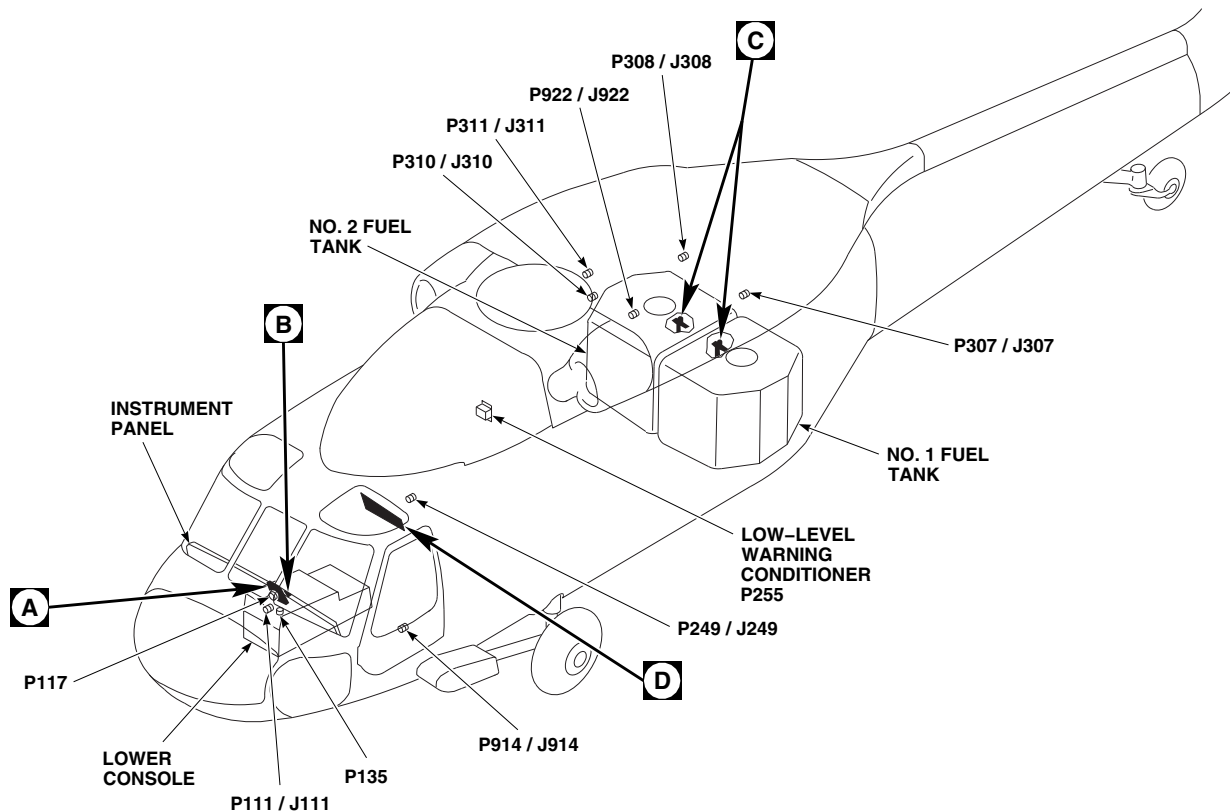
10-1-1.3.2 Fuel Quantity Probe.

The fuel quantity probe is a tube-type probe that has two concentric metal tubes which form the plates of a capacitor. The tubes are connected to the fuel quantity signal conditioner. The total fuel probe capacitance varies with the height of fuel (which relates to volume), and with the dielectric constant of the fuel (which relates to density).

The probe includes the fuel low-level sensor assembly which contains a sensor thermistor, a reference thermistor, two 0.25 amp fuses, and two 2.2 microfarad capacitors. The fuses and capacitors provide a safety circuit protection against a misapplied voltage. Such a voltage might otherwise present a fire hazard.

10-1-1.3.3 Low-Level Warning Conditioner.

The low-level warning conditioner receives signals from the fuel low-level sensors in the main fuel tanks. Each set of signals energizes or de-energizes a relay in the low-level warning conditioner to turn the #1 or #2 FUEL LOW capsule on the caution/advisory panel off or on when the fuel level is above or below the 20-minute reserve level.



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P117	CAUTION / ADVISORY PANEL
P135	MISCELLANEOUS SWITCH PANEL
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P255	LOW-LEVEL WARNING CONDITIONER
P307 / J307	CABIN CEILING, BL 15 LH, STA 430
P308 / J308	CABIN CEILING, BL 15 RH, STA 430
P310 / J310	CABIN CEILING, BL 17 RH, STA 387
P311 / J311	CABIN CEILING, BL 16.5 LH, STA 383
P914 / J914	COCKPIT, BL 24 LH, STA 247
P922 / J922	CABIN CEILING, BL 10.7 LH, STA 380

FB1563_1
SA

**FIGURE 10-1-1-1. FUEL LOW-LEVEL WARNING SYSTEM LOCATION DIAGRAM
(SHEET 1 OF 3)**

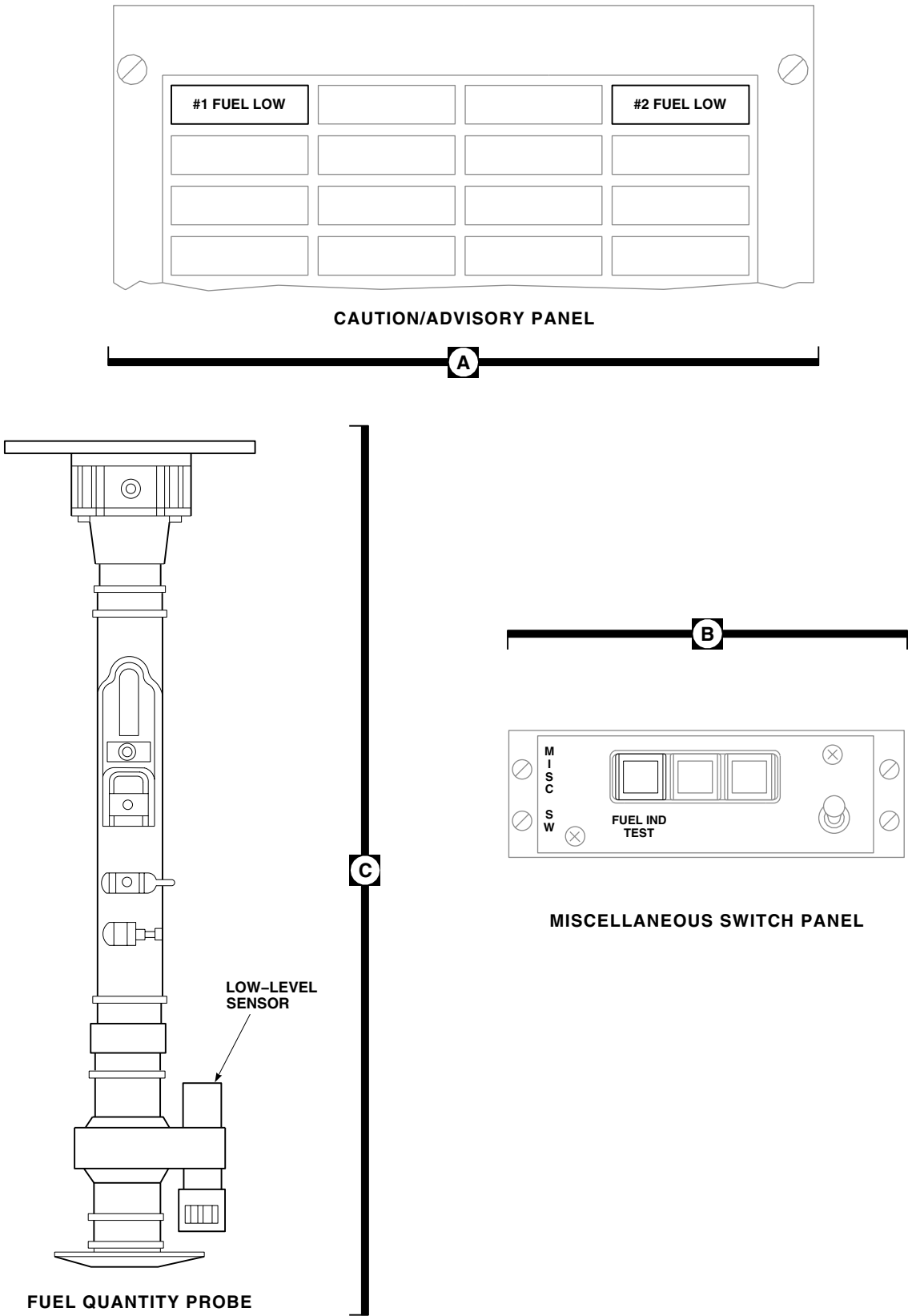
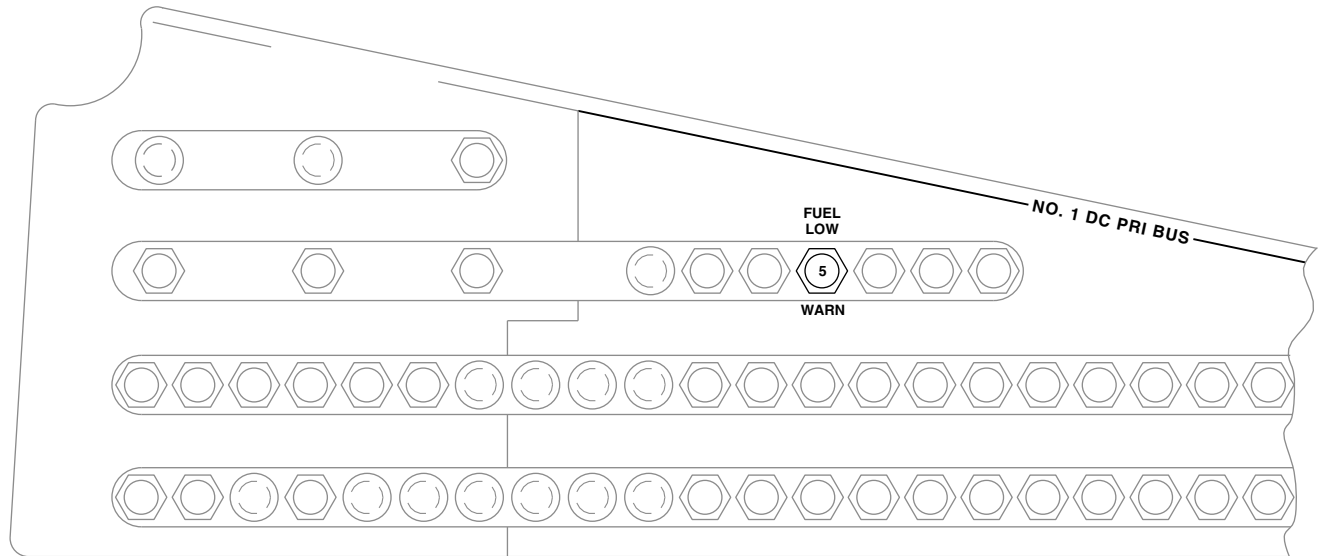


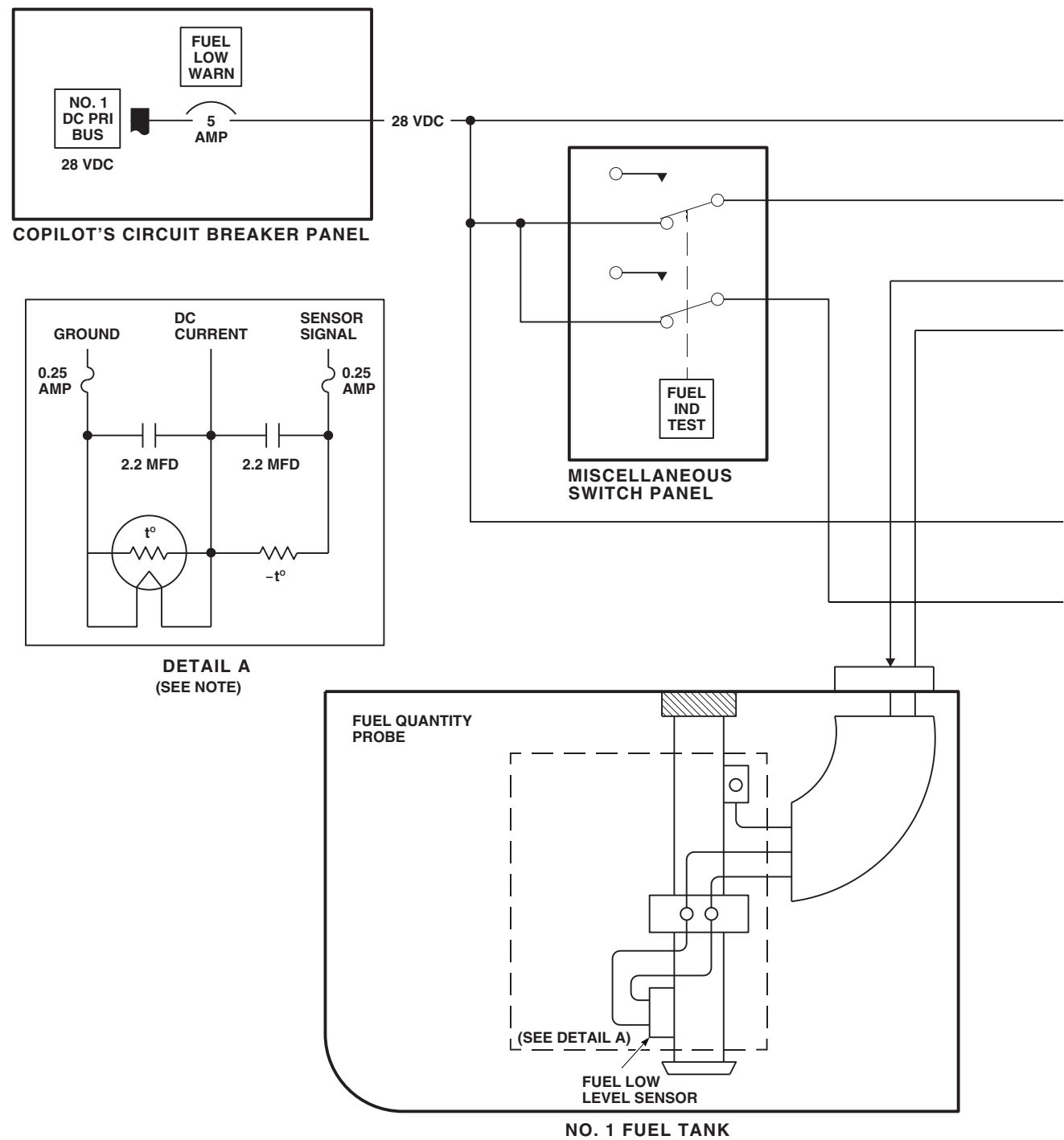
FIGURE 10-1-1-1. FUEL LOW-LEVEL WARNING SYSTEM LOCATION DIAGRAM
(SHEET 2 OF 3)

D



COPLOT'S CIRCUIT BREAKER PANEL

FB1563_3
SA**FIGURE 10-1-1-1. FUEL LOW-LEVEL WARNING SYSTEM LOCATION DIAGRAM
(SHEET 3 OF 3)**



NOTE
THERMISTORS HAVE NEGATIVE TEMPERATURE COEFFICIENT.

AA7814_1
SA

FIGURE 10-1-1-2. FUEL LOW-LEVEL WARNING SYSTEM BLOCK DIAGRAM (SHEET 1 OF 2)

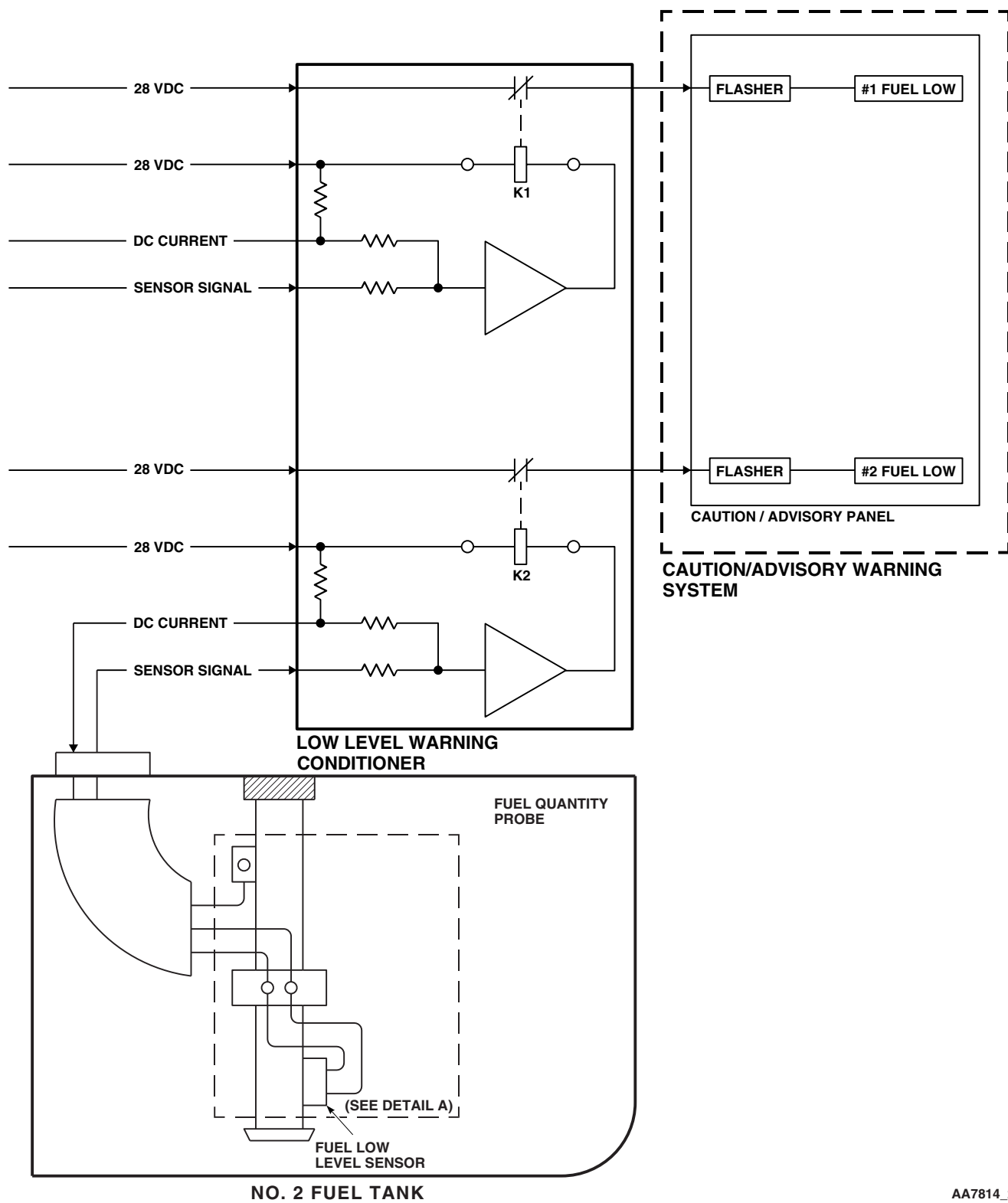
AA7814_2A
SA

FIGURE 10-1-1-2. FUEL LOW-LEVEL WARNING SYSTEM BLOCK DIAGRAM (SHEET 2 OF 2)

10-1-2 SUBMERGED FUEL BOOST PUMP SYSTEM.

10-1-2.1 System Description.

The submerged fuel boost pump system provides fuel under pressure to the engine-driven fuel pumps that feed the No. 1 and No. 2 engines. Major components of the system include a centrifugal boost pump in the bottom of each main fuel tank, two pressure switches, two fuel tank inlet check valves, and a fuel boost pump control panel (Figure 10-1-2-1).

10-1-2.2 Principles Of Operation.

10-1-2.2.1. Power Distribution.

Electrical power for the submerged fuel boost pump system is supplied from the No. 1 ac primary bus, No. 2 ac primary bus, No. 1 dc primary bus, and No. 2 dc primary bus. The No. 1 boost pump is supplied three-phase 115 vac from the No. 1 ac primary bus through the NO. 1 FUEL BOOST PUMP circuit breaker on the mission readiness circuit breaker panel. The No. 2 boost pump is supplied three-phase 115 vac from the No. 2 ac primary bus through the NO. 2 FUEL BOOST PUMP circuit breaker on the mission readiness circuit breaker panel (Figure 10-1-2-2).

The NO. 1 and NO. 2 PUMP indicator lights on the fuel boost pump control panel receive 28 vdc from the No. 1 dc primary bus through the LIGHTS ADVSY circuit breaker on the copilot's circuit breaker panel. The 28 vdc is then routed through the de-energized contacts of dimming relay K40 in the left relay panel and out to the indicator lights. Both the No. 1 and the No. 2 dc primary buses also provide 28 vdc through the NO. 1 and NO. 2 ENG WARN LTS circuit breakers to light the #1 and #2 FUEL PRESS caution lights on the caution/advisory panel when there is inadequate fuel pressure from one or both of the engine-driven fuel pumps.

10-1-2.2.2. System Operation.

The submerged fuel boost pump system consists of two independent circuits for the No. 1 and No. 2

fuel tanks that are identical in operation, therefore only the No. 1 submerged fuel boost pump circuit will be described.

Setting the NO. 1 PUMP ON/OFF switch on the fuel boost pump control panel to ON routes 115 vac from the No. 1 ac primary bus to the No. 1 boost pump to energize the pump. The energized boost pump then pumps pressurized fuel from the No. 1 fuel tank to the boost pump pressure switch, causing the NO. 1 PUMP indicator light on the fuel boost pump control panel to illuminate when the pump output pressure reaches approximately 7.5 to 8.5 psi. The pressurized fuel is also routed through the fuel tank inlet check valve to either the No. 1 or No. 2 engine-driven fuel pump, depending upon the selected position of the system fuel selector handle located on the engine control quadrant. When there is inadequate pressure from the No. 1 or No. 2 engine-driven fuel pump, 28 vdc from the No. 1 or No. 2 dc primary bus is routed through the closed contacts of the applicable engine fuel pressure sensor to light the #1 or #2 FUEL PRESS caution capsule on the caution/advisory panel.

10-1-2.3 Component Description.

10-1-2.3.1. Boost Pump.

Each fuel boost pump consists of a three-phase AC driven motor and a centrifugal pump. The electric motor is a continuous duty, 200-volt AC type, that contains a non-resettable thermal protection circuit designed to limit boost pump temperatures. The mechanical pump is internally mounted in each main fuel tank.

10-1-2.3.2. Fuel Tank Inlet Check Valve.

The fuel tank inlet check valve has an integral thermal pressure relief valve that operates in the opposite direction of normal fuel flow to maintain fuel line prime. The valve opens at 45 psi and resets (closes) at 10 psi.

10-1-2.3.3. Boost Pump Pressure Switch.

The boost pump pressure switch is a normally-open switch that will close when fuel pump output pressure reaches approximately 7.5 to 8.5 psi.

10-1-2.3.4. Fuel Boost Pump Control Panel.

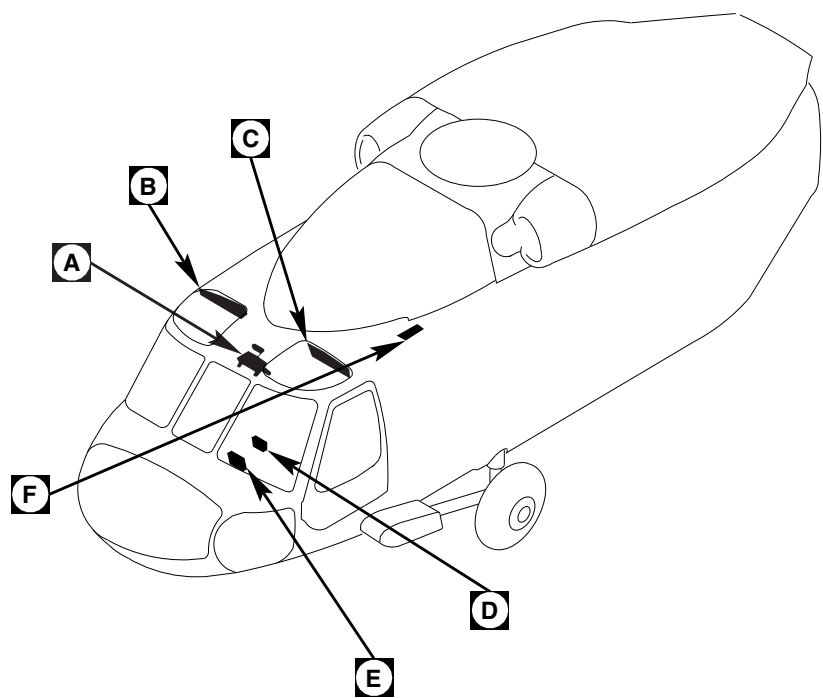
The fuel boost pump control panel is located on the lower console. It contains two green PUMP ON indicator lights, and two, two-position PUMP ON/OFF switches used to route 115 volt, three-phase ac power to the No. 1 and No. 2 boost pumps.

10-1-2.3.5. System Fuel Selector Handle.

The system fuel selector handle, one for each engine, provides manual control movement of the

fuel selector valves. Each system fuel selector handle has three positions: OFF, DIR (direct), and XFD (crossfeed). With the No. 1 engine system fuel selector handle in the OFF position, the fuel selector valves are closed to prevent fuel from entering the engine-driven fuel pumps. In the DIR position, fuel is routed from the No. 1 fuel tank to the No. 1 engine-driven fuel pump and on to the No. 1 engine. With the system fuel selector handle in the XFD position, fuel is routed from the No. 1 fuel tank to the No. 2 engine-driven fuel pump and on to the No. 2 engine.

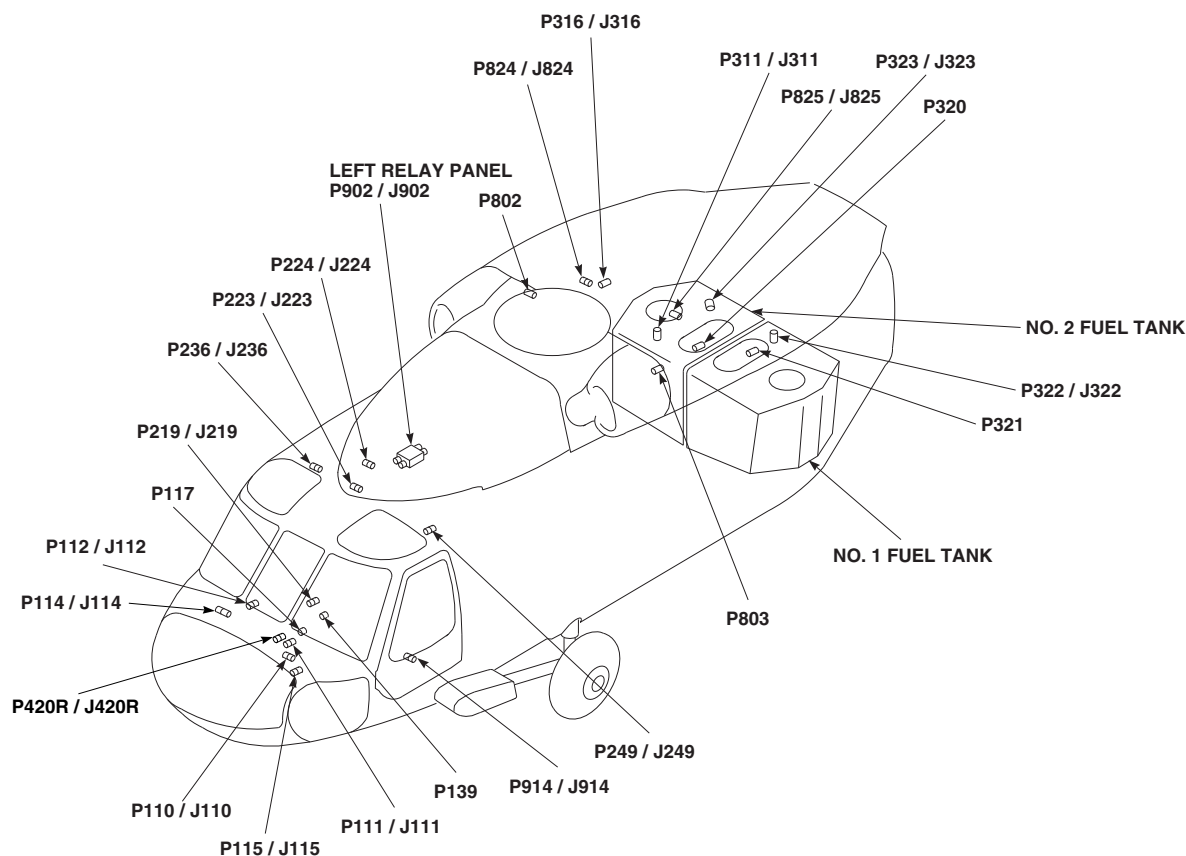
The No. 2 engine system fuel selector handle operates in the same manner as the No. 1 engine fuel selector handle.



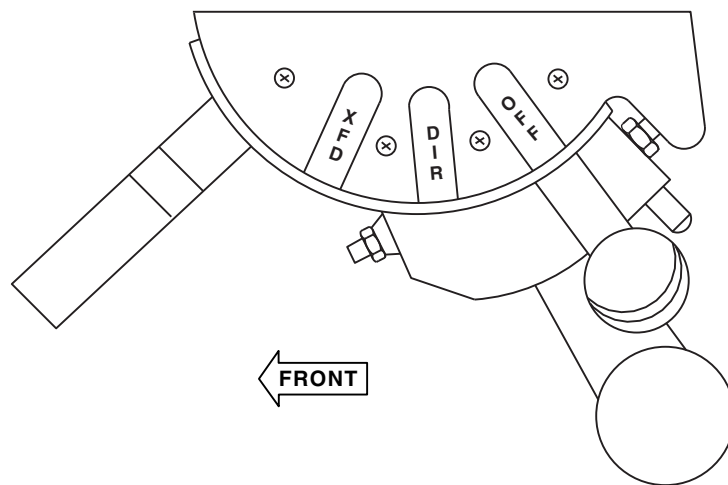
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P110 / J110	COCKPIT BL 7.5 RH, STA 197
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P112 / J112	COCKPIT BL 6 RH, STA 197
P114 / J114	COCKPIT BL 8 RH, STA 200
P115 / J115	COCKPIT BL 6 RH, STA 210
P117	CAUTION / ADVISORY PANEL
P139 / J1	FUEL BOOST PUMP CONTROL PANEL
P219 / J219	CABIN FLOOR BL 25 RH, STA 247
P223 / J223	JUNCTION BOX MAIN ROTOR PYLON DECK
P224 / J224	JUNCTION BOX MAIN ROTOR PYLON DECK
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 23 RH
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P311 / J311	CABIN CEILING BL 16.5 RH, STA 383
P316 / J316	CABIN CEILING BL 17 RH, STA 387
P320	TOP OF PLATE NO. 2 FUEL TANK
P321	TOP OF PLATE NO. 1 FUEL TANK

FB1571_1B
SA

FIGURE 10-1-2-1. SUBMERGED FUEL BOOST PUMP SYSTEM LOCATION DIAGRAM
(SHEET 1 OF 4)



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P322 / J322	ABOVE NO. 1 FUEL TANK BL 7, WL 269
P323 / J323	ABOVE NO. 2 FUEL TANK BL 7, WL 269
P420R / J420R	COCKPIT BL 0, STA 191.5
P802	NO. 2 ENGINE INTERFACE DISCONNECT
P803	NO. 1 ENGINE INTERFACE DISCONNECT
P824 / J824	FIREWALL DISCONNECT BL 14 RH, STA 356.25
P825 / J825	FIREWALL DISCONNECT BL 14 LH, STA 358.25
P902 / J902	LEFT RELAY PANEL
P914 / J914	COCKPIT BL 24 LH, STA 247



ENGINE CONTROLS QUADRANT

FB1571_2B
SAFIGURE 10-1-2.1. SUBMERGED FUEL BOOST PUMP SYSTEM LOCATION DIAGRAM
(SHEET 2 OF 4)

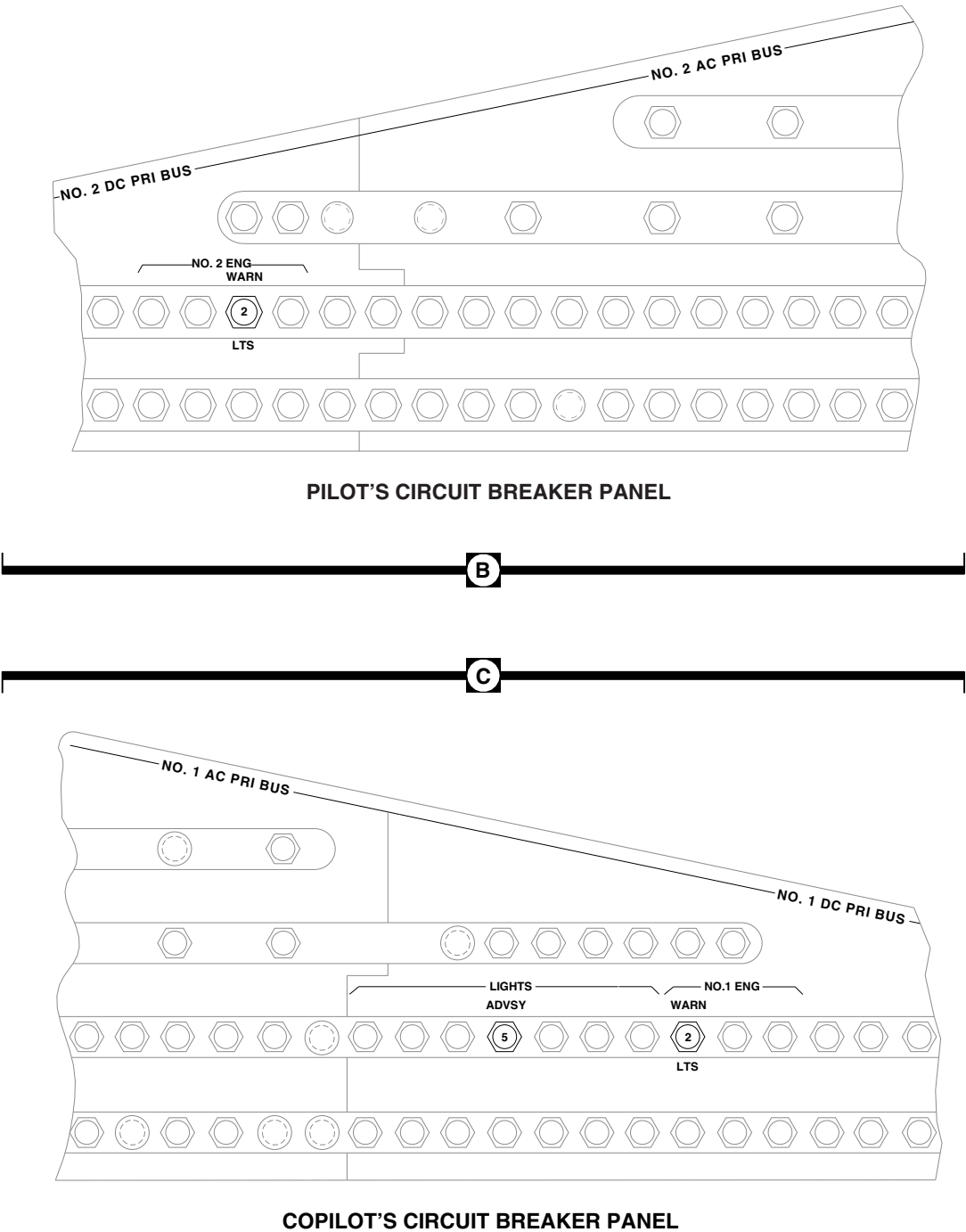
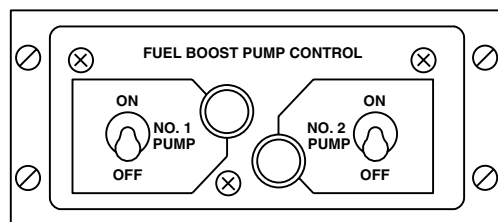


FIGURE 10-1-2-1. SUBMERGED FUEL BOOST PUMP SYSTEM LOCATION DIAGRAM
 (SHEET 3 OF 4)

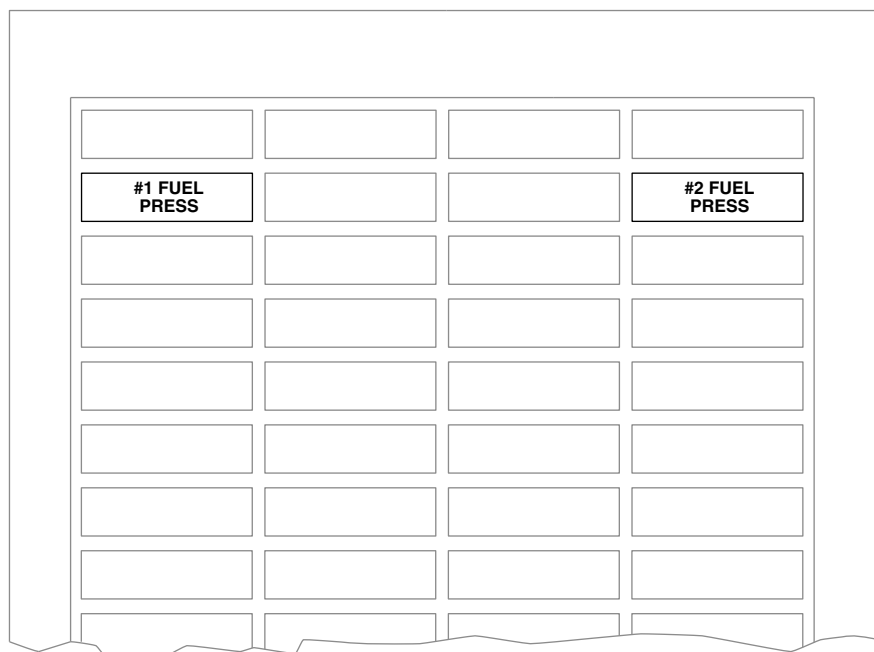
FB1571_3A
 SA



FUEL BOOST PUMP CONTROL PANEL

D

E



CAUTION/ADVISORY PANEL

FB1571_4
SA

**FIGURE 10-1-2-1. SUBMERGED FUEL BOOST PUMP SYSTEM LOCATION DIAGRAM
(SHEET 4 OF 4)**

10-1-2-6

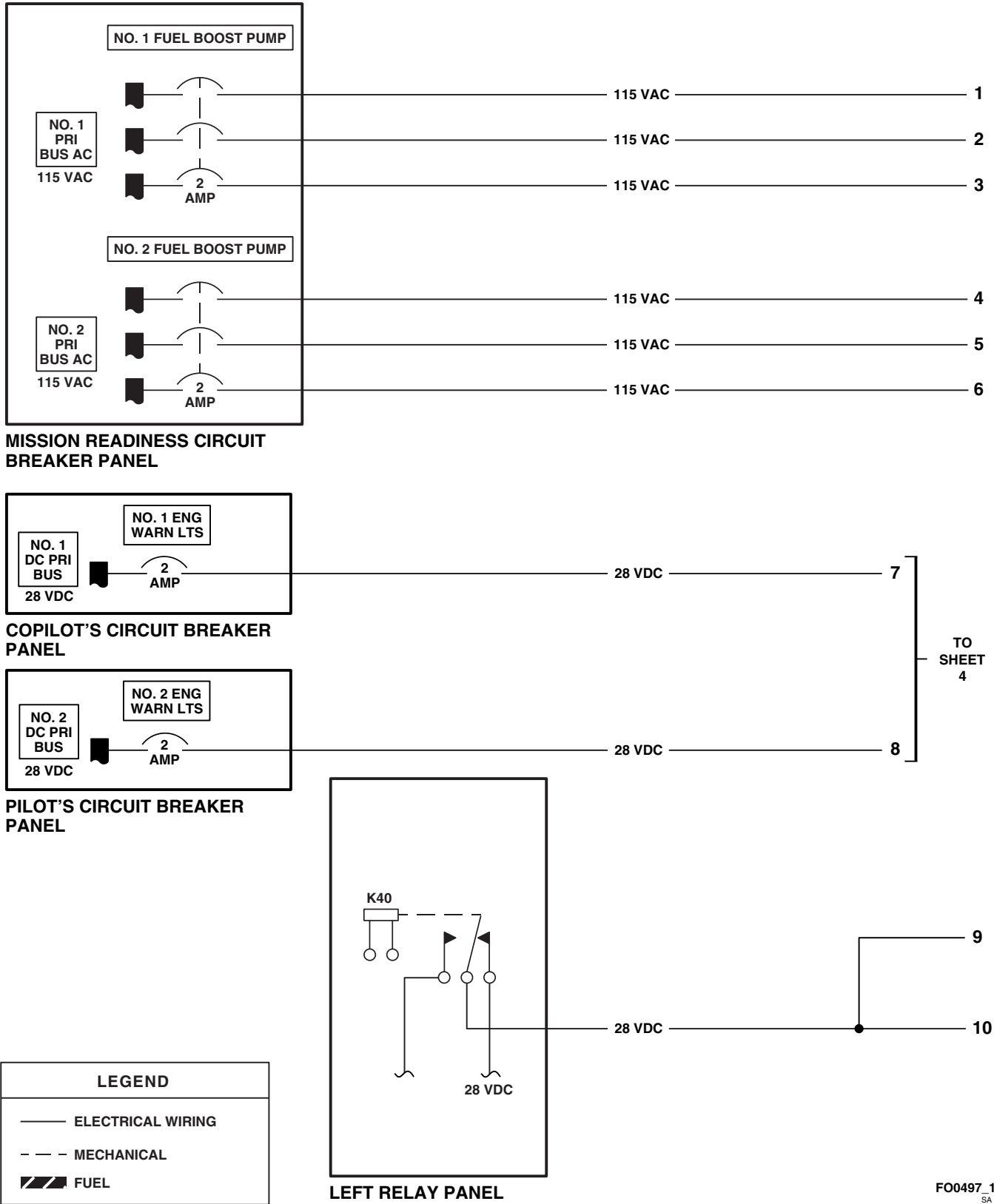
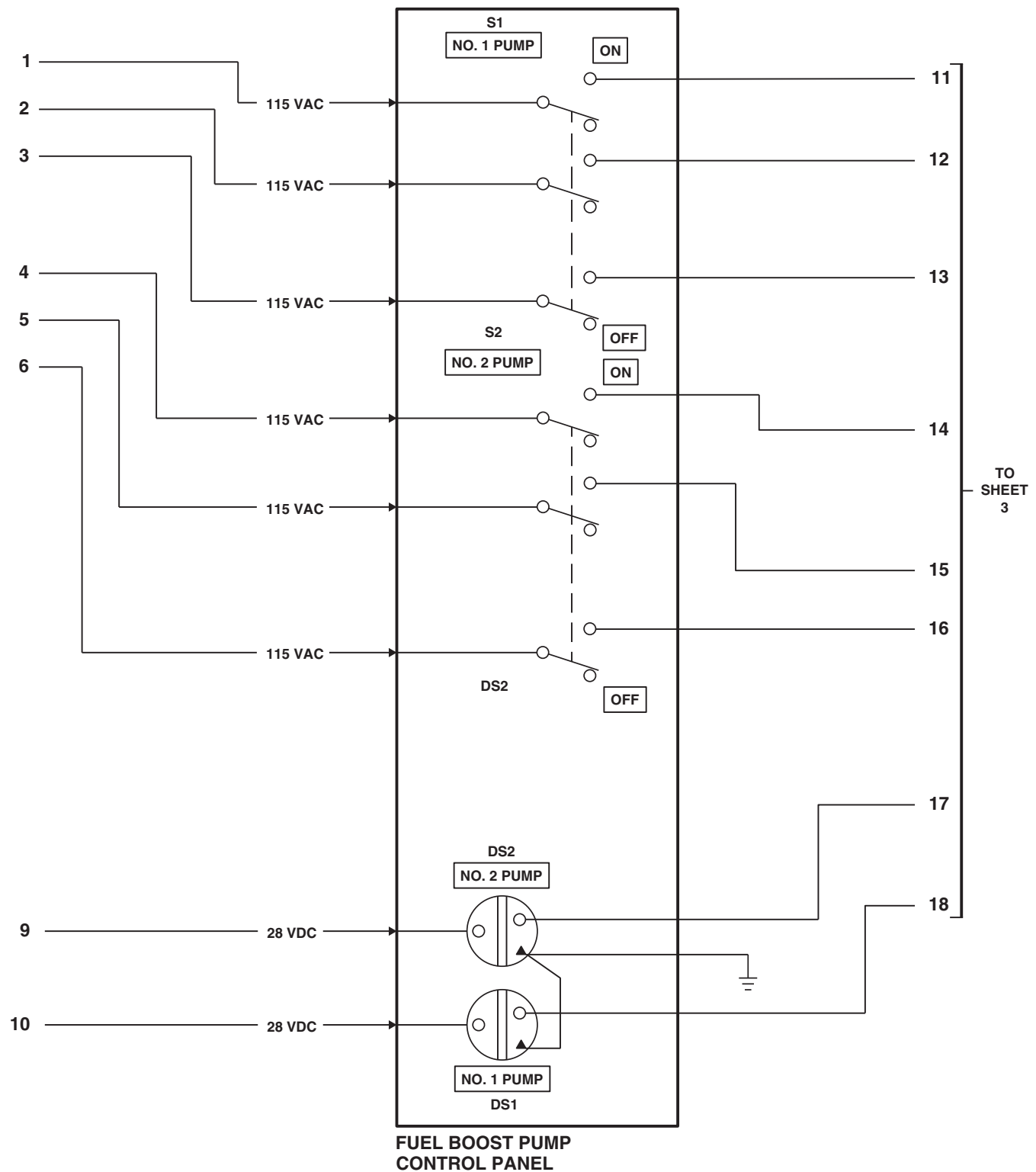


FIGURE 10-1-2-2. SUBMERGED FUEL BOOST PUMP SYSTEM BLOCK DIAGRAM (SHEET 1 OF 4)

FO0497_1
SA

FC3180_2
SA

**FIGURE 10-1-2-2. SUBMERGED FUEL BOOST PUMP SYSTEM BLOCK DIAGRAM
(SHEET 2 OF 4)**

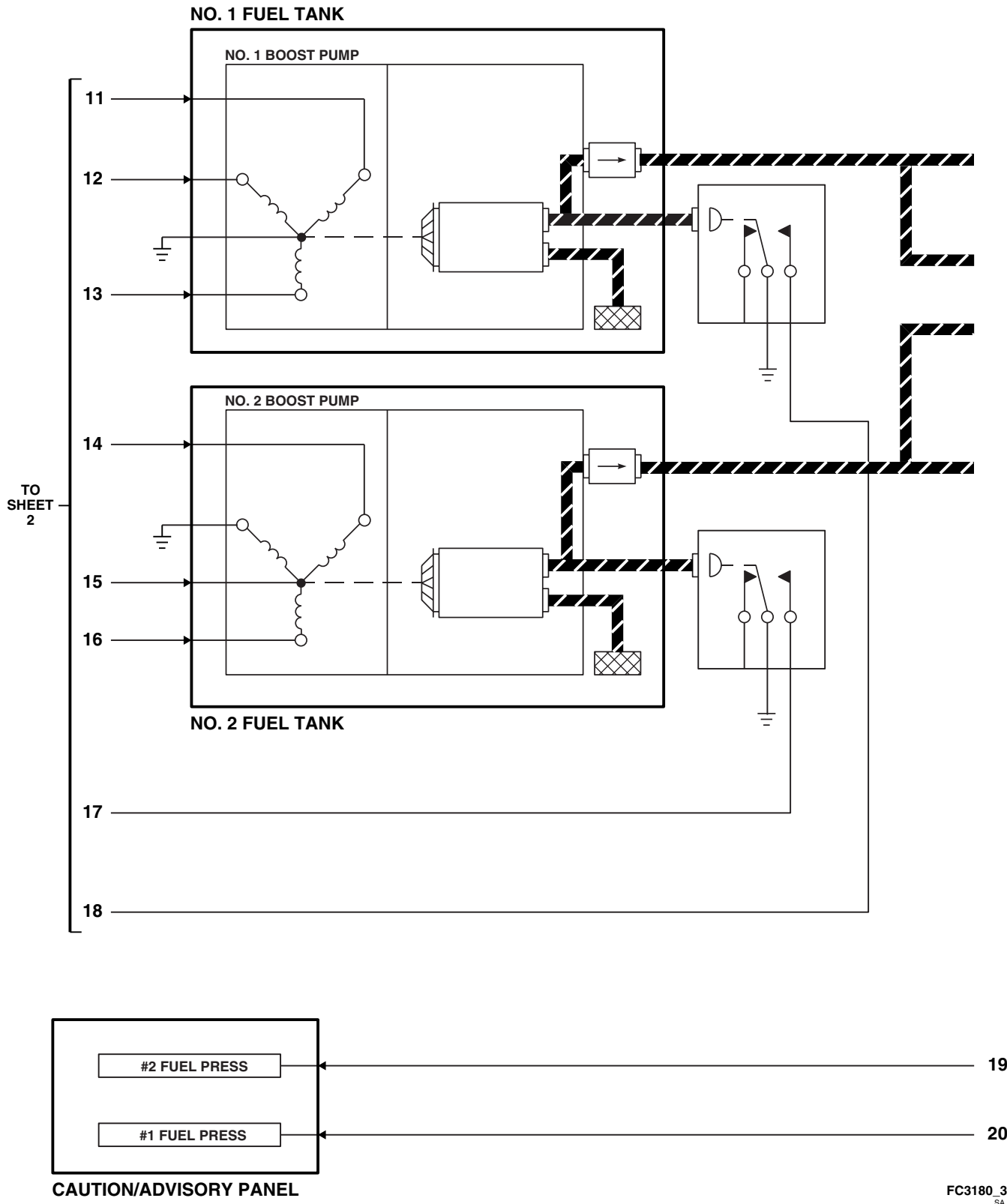
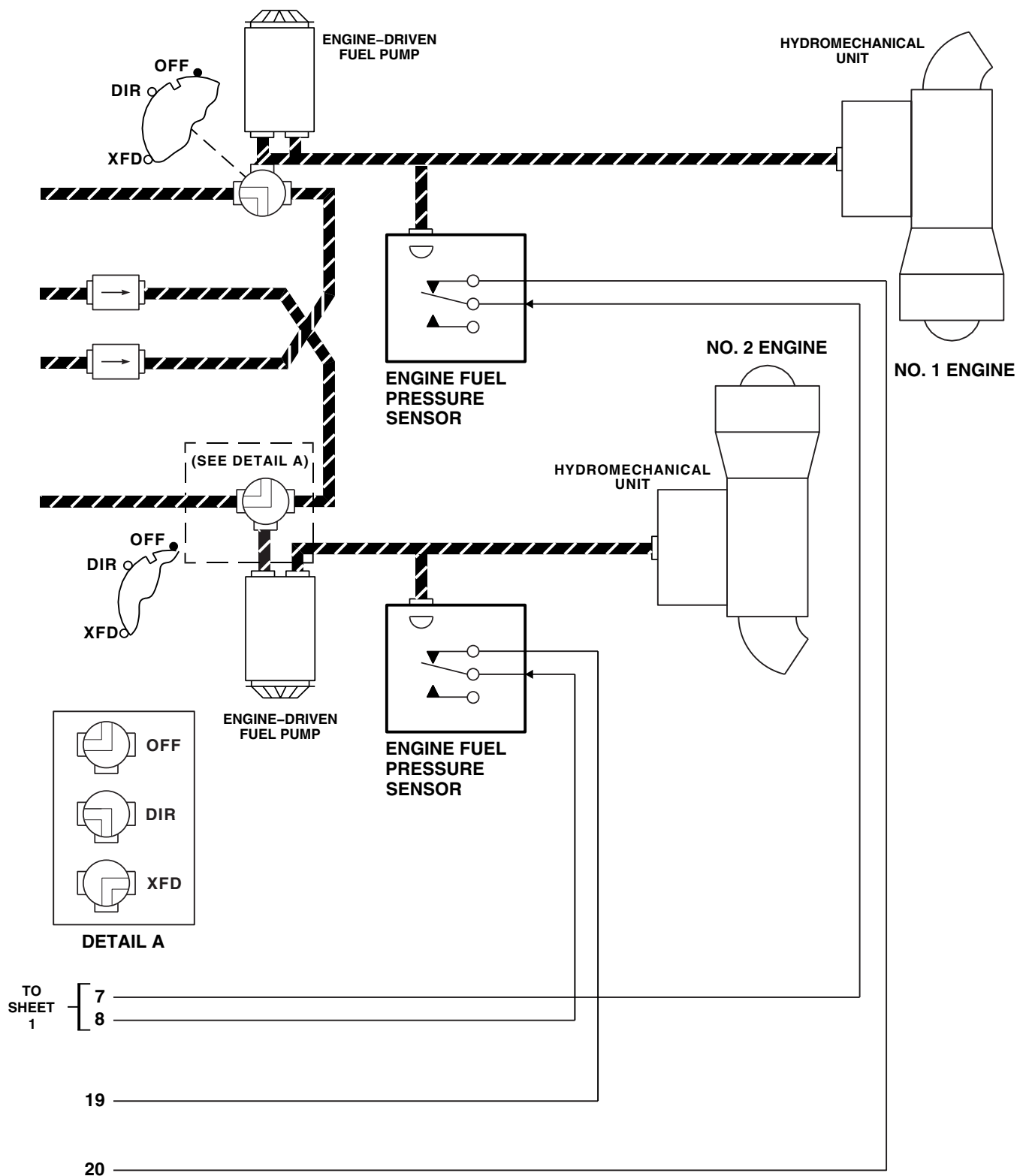


FIGURE 10-1-2-2. SUBMERGED FUEL BOOST PUMP SYSTEM BLOCK DIAGRAM
(SHEET 3 OF 4)



FC3180_4
SA

FIGURE 10-1-2-2. SUBMERGED FUEL BOOST PUMP SYSTEM BLOCK DIAGRAM
(SHEET 4 OF 4)

10-1-3 FUEL PRIME BOOST SYSTEM.

10-1-3.1 System Description.

The fuel prime boost system provides fuel under pressure to prime all fuel lines and supply fuel boost for proper APU operation. Major components of the system include an APU fuel shutoff valve, system fuel selector handles, a prime boost pump, and No. 1 and No. 2 engine prime shutoff valves (Figure 10-1-3-1).

10-1-3.2 Principles Of Operation.

10-1-3.2.1 Power Distribution.

Electrical power of 28 vdc for the prime boost system is supplied from the battery bus through the FUEL PRIME BOOST circuit breaker on the lower console battery bus circuit breaker panel (Figure 10-1-3-2).

10-1-3.2.2 System Operation.

Setting the FUEL PUMP switch on the upper console to APU BOOST routes 28 vdc from the battery bus through the closed contacts of APU T-handle switch S101. This voltage is then simultaneously routed to the caution/advisory panel to turn on the PRIME BOOST PUMP ON capsule, to the prime boost pump to energize the pump, and through the right relay panel to open the APU fuel prime shutoff valve. With the prime boost pump energized and the APU fuel prime shutoff valve open, pressurized fuel from the No. 1 fuel tank is routed directly to the APU for operation.

Setting the FUEL PUMP switch on the upper console to FUEL PRIME routes 28 vdc from the battery bus through the closed contacts of FUEL PUMP switch S19 and APU T-handle S101. This voltage is then simultaneously routed to the caution/advisory panel to turn on the PRIME BOOST PUMP ON capsule, to the prime boost pump to energize the pump, to the No. 1 and No. 2 engine prime shutoff valves to open the valves, and through the right relay panel to open the APU fuel prime shutoff valve. With the prime boost pump energized and all three shutoff valves open, pressurized fuel from the No. 1 fuel tank is routed to

the APU and the No. 1 and No. 2 engine feed lines to prime the lines for APU and engine operation.

10-1-3.3 Component Description.

10-1-3.3.1 APU Fuel Prime Shutoff Valve.

The APU fuel prime shutoff valve is a normally-closed, solenoid-operated valve connecting the prime pump outlet to the APU. The APU fuel prime shutoff valve is used to control the flow of fuel to the APU when the FUEL PUMP switch is placed to APU BOOST.

10-1-3.3.2 System Fuel Selector Handle.

The system fuel selector handle, one for each engine, provides manual control movement of the fuel selector valves. Each system fuel selector handle has three positions: OFF, DIR (direct), and XFD (crossfeed). With the No. 1 engine system fuel selector handle in the OFF position, the fuel selector valves are closed to prevent fuel from entering the engine-driven fuel pumps. In the DIR position, fuel is routed from the No. 1 fuel tank to the No. 1 engine-driven fuel pump and on to the No. 1 engine. With the system fuel selector handle in the XFD position, fuel is routed from the No. 1 fuel tank to the No. 2 engine-driven fuel pump and on to the No. 2 engine.

The No. 2 engine system fuel selector handle operates in the same manner as the No. 1 engine fuel selector handle.

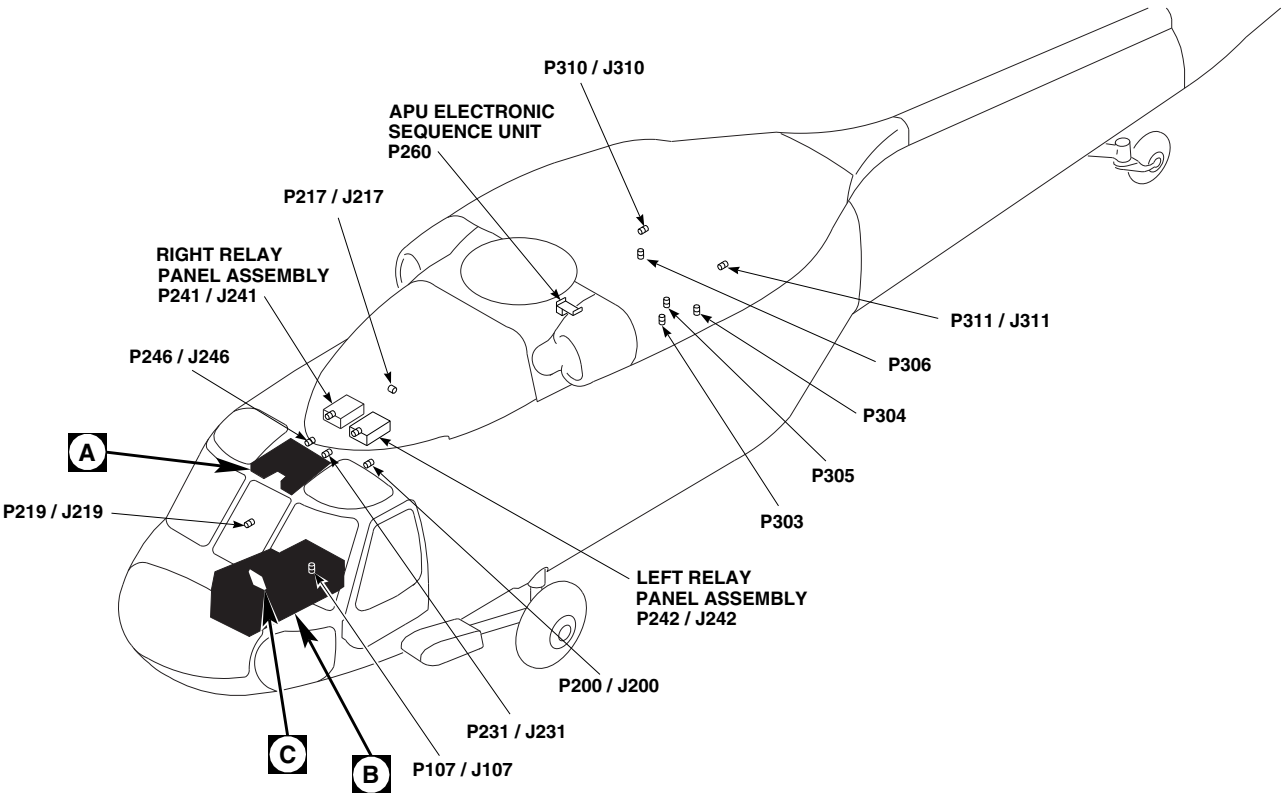
10-1-3.3.3 Prime Boost Pump.

The prime boost pump consists of an electric motor and a mechanical pump with pressure relief and suction bypass. The electric motor is a continuous duty, permanent magnet type. The pump design is a vane-type which delivers 40 gph at 5 psi discharge pressure. The relief valve is set for 15 gph discharge flow at 12 psid and will bypass at 30 psid maximum. The prime boost pump primes all fuel lines and acts as an APU boost for APU start and operation.

10-1-3.3.4 Prime Shutoff Valves.

The prime shutoff valves are normally-closed, solenoid-operated hydraulic valves used to control

the flow of fuel to the No. 1 and No. 2 engines when the FUEL PUMP switch on the upper console is placed to FUEL PRIME.

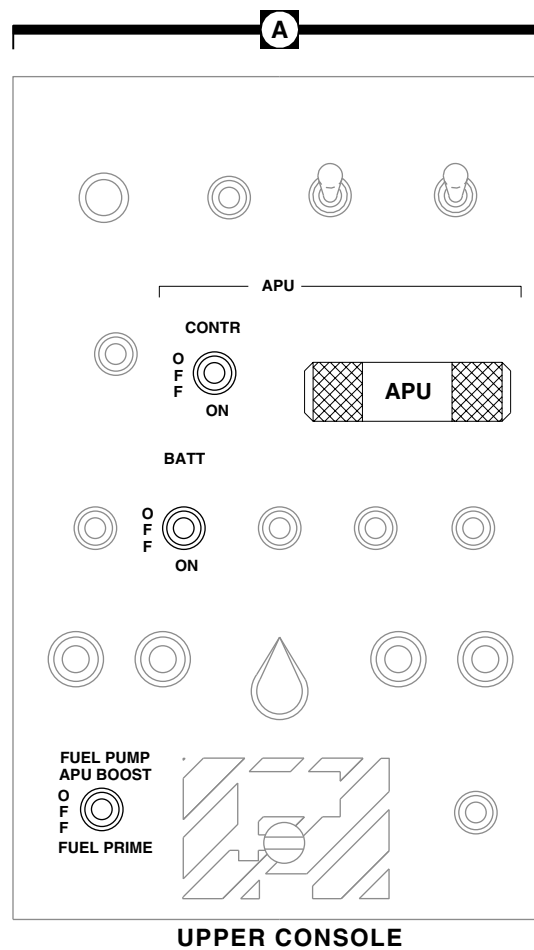
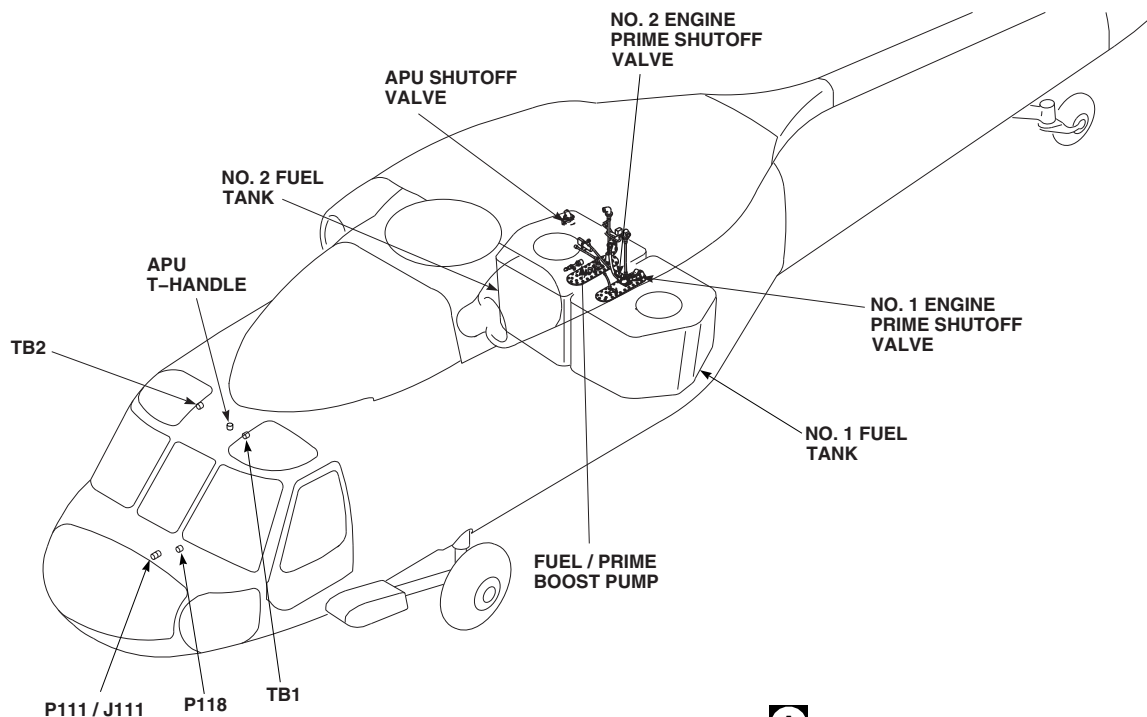


TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P107 / J107	COCKPIT BL 9 LH, STA 236
P111 / J111	COCKPIT B. 7.5 LH, STA 197
P118	CAUTION / ADVISORY PANEL
P200 / J200	CABIN CEILING BL 13 LH, STA 246
P217 / J217	CABIN CEILING BL 0, STA 284
P219 / J219	CABIN FLOOR BL 25 RH, STA 247
P231 / J231	CABIN CEILING BL 9 LH, STA 247
P241 / J241	RIGHT RELAY PANEL ASSEMBLY
P242	LEFT RELAY PANEL ASSEMBLY
P246 / J246	CABIN CEILING BL 5 RH, STA 247

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P260	APU ELECTRONIC SEQUENCE UNIT
P303	PRIME BOOST PUMP
P304	NO. 1 ENGINE PRIME SHUTOFF VALVE
P305	NO. 2 ENGINE PRIME SHUTOFF VALVE
P306	APU FUEL SHUTOFF VALVE
P310 / J310	CABIN CEILING BR 17 LH, STA 387
P311 / J311	CABIN CEILING BL 165 LH, STA 343
TB1	COCKPIT CEILING BL 9 LH, STA 236
TB2	COCKPIT CEILING BL 9 RH, STA 236

AA2763_1
 SA

FIGURE 10-1-3-1. FUEL PRIME BOOST SYSTEM LOCATION DIAGRAM (SHEET 1 OF 3)



UPPER CONSOLE

FN1507_2B
SA

FIGURE 10-1-3-1. FUEL PRIME BOOST SYSTEM LOCATION DIAGRAM (SHEET 2 OF 3)

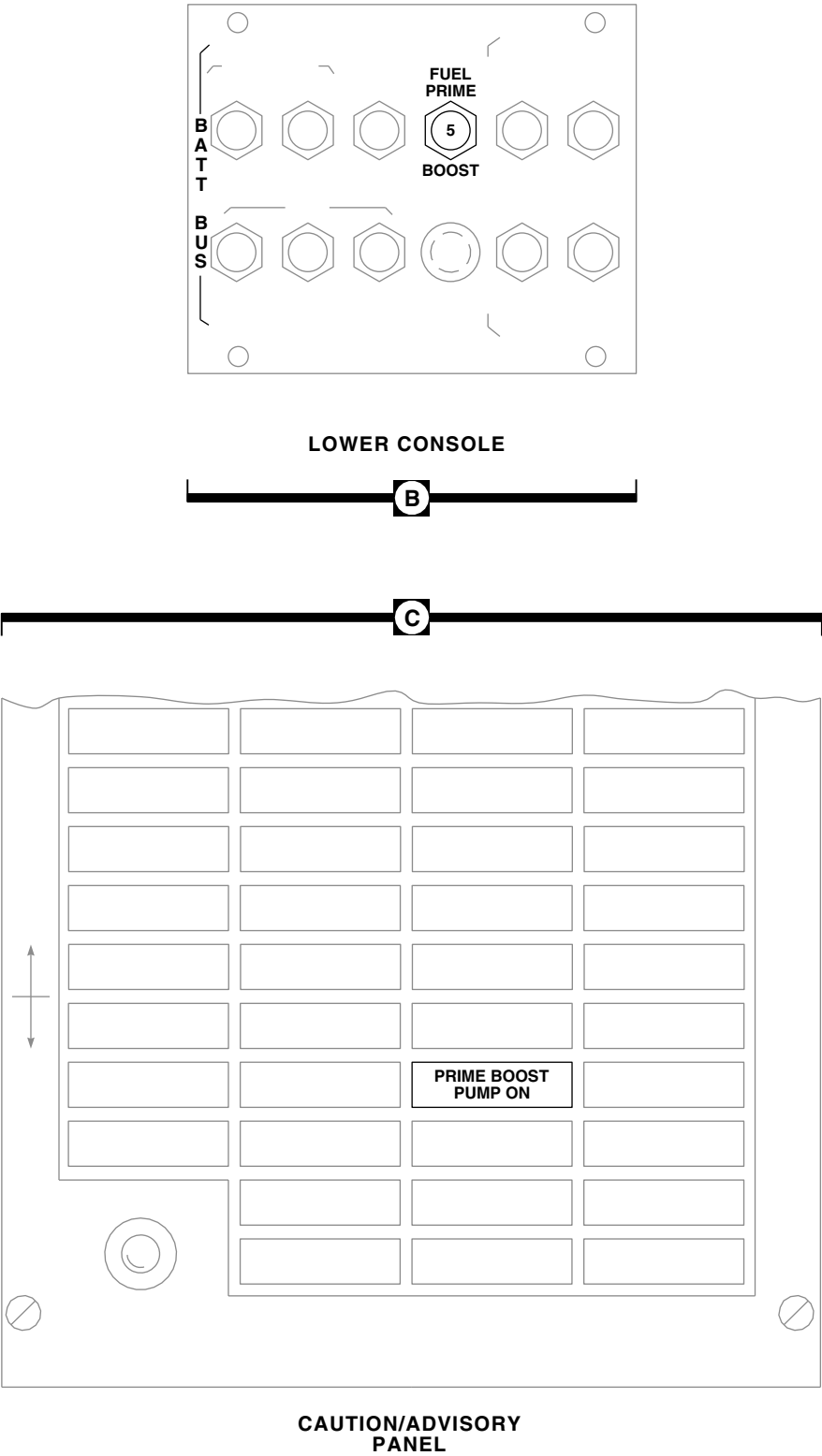


FIGURE 10-1-3-1. FUEL PRIME BOOST SYSTEM LOCATION DIAGRAM (SHEET 3 OF 3)

AA2763_3
SA

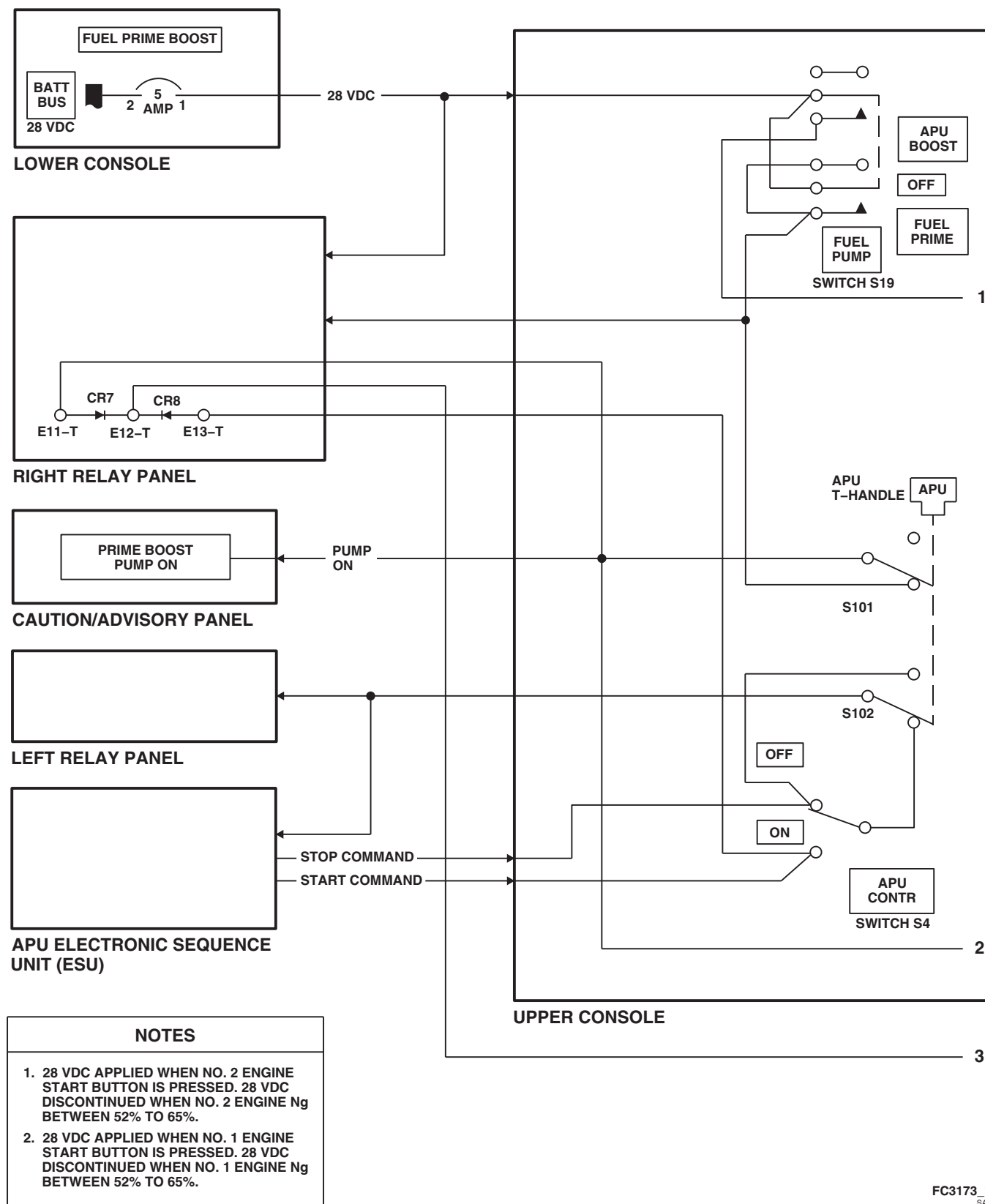


FIGURE 10-1-3-2. FUEL PRIME BOOST SYSTEM BLOCK DIAGRAM (SHEET 1 OF 3)

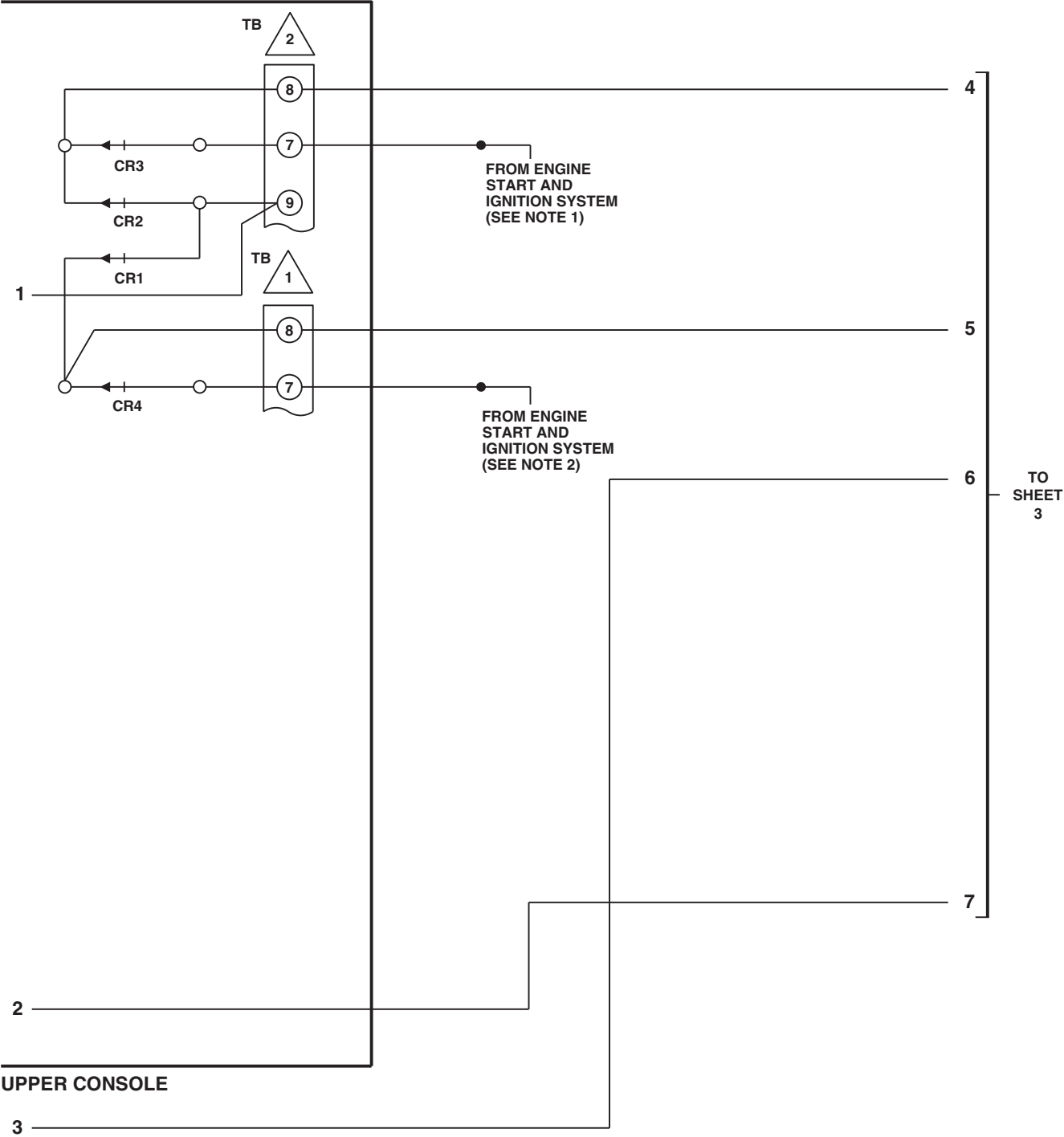


FIGURE 10-1-3-2. FUEL PRIME BOOST SYSTEM BLOCK DIAGRAM (SHEET 2 OF 3)

FC3173_2
SA

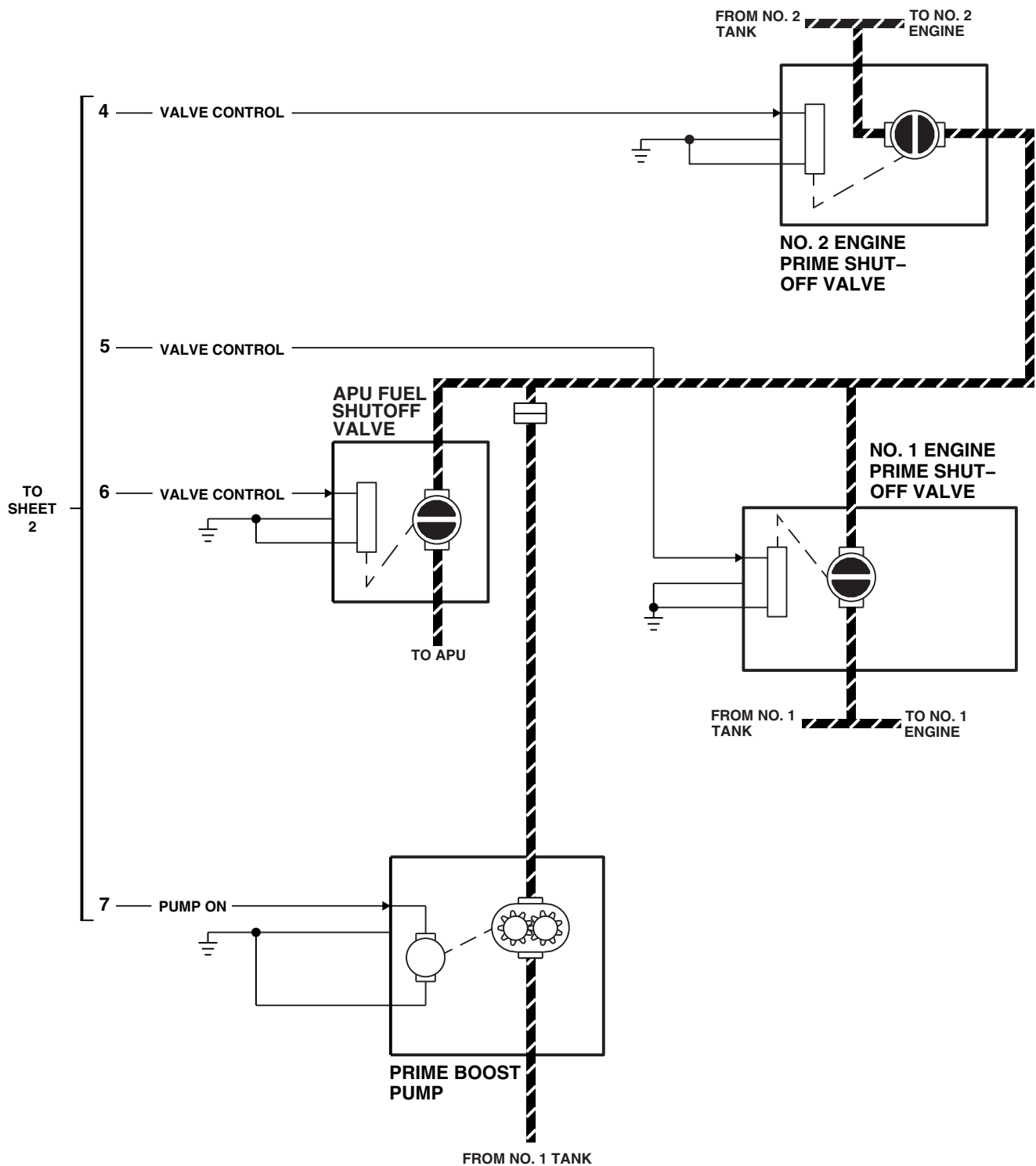
FC3173_3
SA

FIGURE 10-1-3-2. FUEL PRIME BOOST SYSTEM BLOCK DIAGRAM (SHEET 3 OF 3)

10-1-4 PRESSURE REFUEL/DEFUEL SYSTEM.

10-1-4.1 System Description.

The pressure refuel/defuel system provides several ways of refueling and defueling the helicopter. Refueling each fuel tank can be done by using gravity, pressure, or closed-circuit refueling methods. Defueling each fuel tank can be done using suction or gravity defueling. Major components of the system include the fuel tanks, single-point pressure refuel receptacle, gravity refuel ports, a closed-circuit refuel receptacle, hi-level shutoff valves, low-level shutoff valves, and pressure refueling valves (Figure 10-1-4-1).

10-1-4.2 Principles Of Operation.

10-1-4.2.1 Pressure Refueling.

Pressure refueling of the No. 1 and No. 2 fuel tanks is performed through the single-point pressure refuel receptacle. Fuel pressure from the refueling unit opens the pressure refueling valve to allow fuel to flow into the tank. When the fuel reaches the predetermined high level, the hi-level shutoff valve closes, causing back pressure to shut the pressure refueling valve and stop the flow of fuel into the tank (Figure 10-1-4-2).

10-1-4.2.2 Suction Defueling.

Suction defueling of the No. 1 and No. 2 fuel tanks is performed through the single-point pressure refuel receptacle. Connection of the defueling unit opens the pressure refueling valve to allow fuel to flow from the fuel tank. When the fuel reaches the predetermined low level, the low-level shutoff valve closes, stopping the suction of fuel from the tank.

10-1-4.2.3 Gravity Refueling.

Gravity refueling of the No. 1 and No. 2 fuel tanks can be performed through the gravity refuel ports on the left and right sides of the helicopter.

10-1-4.2.4 Gravity Defueling.

Gravity defueling of the No. 1 and No. 2 fuel tanks can be performed by opening the sump drain

valves, located on the bottom of each fuel tank, or by pumping fuel through the gravity refuel ports by means of a special adapter.

10-1-4.3 Component Description.

10-1-4.3.1 Fuel Tanks.

The No. 1 and No. 2 fuel tanks, located in the transition section, incorporate crashworthy and self-sealing protection. Each tank consists of a rubber compound with a nylon filament added for strength and an outer coating of vithane used for scuff resistance. Each tank's usable capacity varies with the method of refueling used as follows:

- (1) Single-point pressure refueling - 180.5 gallons (approximately 1263.5 lbs of fuel).
- (2) Closed circuit refueling - 179 gallons (approximately 1253 lbs of fuel).
- (3) Gravity refueling - 181 gallons (approximately 1267 lbs of fuel).

10-1-4.3.2 Single Point Pressure Refuel Receptacle.

The single-point pressure refueling receptacle (receptacle) located on the left side of the helicopter, can be used for pressure refueling and suction defueling the No. 1 and No. 2 fuel tanks. The receptacle consists of an integral, spring-loaded valve which opens when the fuel nozzle is attached to prevent fuel spillage. A protective cap covers the receptacle when it is not in use.

10-1-4.3.3 Gravity Refuel Ports.

Gravity refuel ports, located on the left and right sides of the helicopter, provide a means of gravity refueling and defueling the main fuel tanks individually. Each gravity refuel port consists of a spring-loaded valve which opens when the gravity refuel nozzle is inserted. A protective cap covers the receptacle when it is not in use.

10-1-4.3.4 Closed Circuit Refuel Receptacle.

The closed-circuit refuel receptacle, located on the left side of the helicopter, can be used to low pres-

sure refuel the No. 1 and No. 2 fuel tanks when normal refueling equipment is not available.

10-1-4.3.5 Hi-Level Shutoff Valves.

Hi-level shutoff valves, located on the top of the No. 1 and No. 2 fuel tanks, are float type shutoff valves used to provide tank full indications to the pressure refueling valves.

10-1-4.3.6 Low-Level Shutoff Valves.

Low-level shutoff valves, located on the bottom of the No. 1 and No. 2 fuel tanks, are float type shut-

off valves used to provide tank empty indications to the pressure refueling valves.

10-1-4.3.7 Pressure Refueling Valves.

Pressure fueling valves, located in the No. 1 and No. 2 fuel tanks, are pressure-operated valves used to control the flow of fuel into the fuel tanks. During pressure refueling, the pressure fueling valves will be open until a tank full condition is sensed by the hi-level shutoff valves. During suction defueling, the pressure refueling valves will be open until a tank low condition is sensed by the low-level shutoff valves.

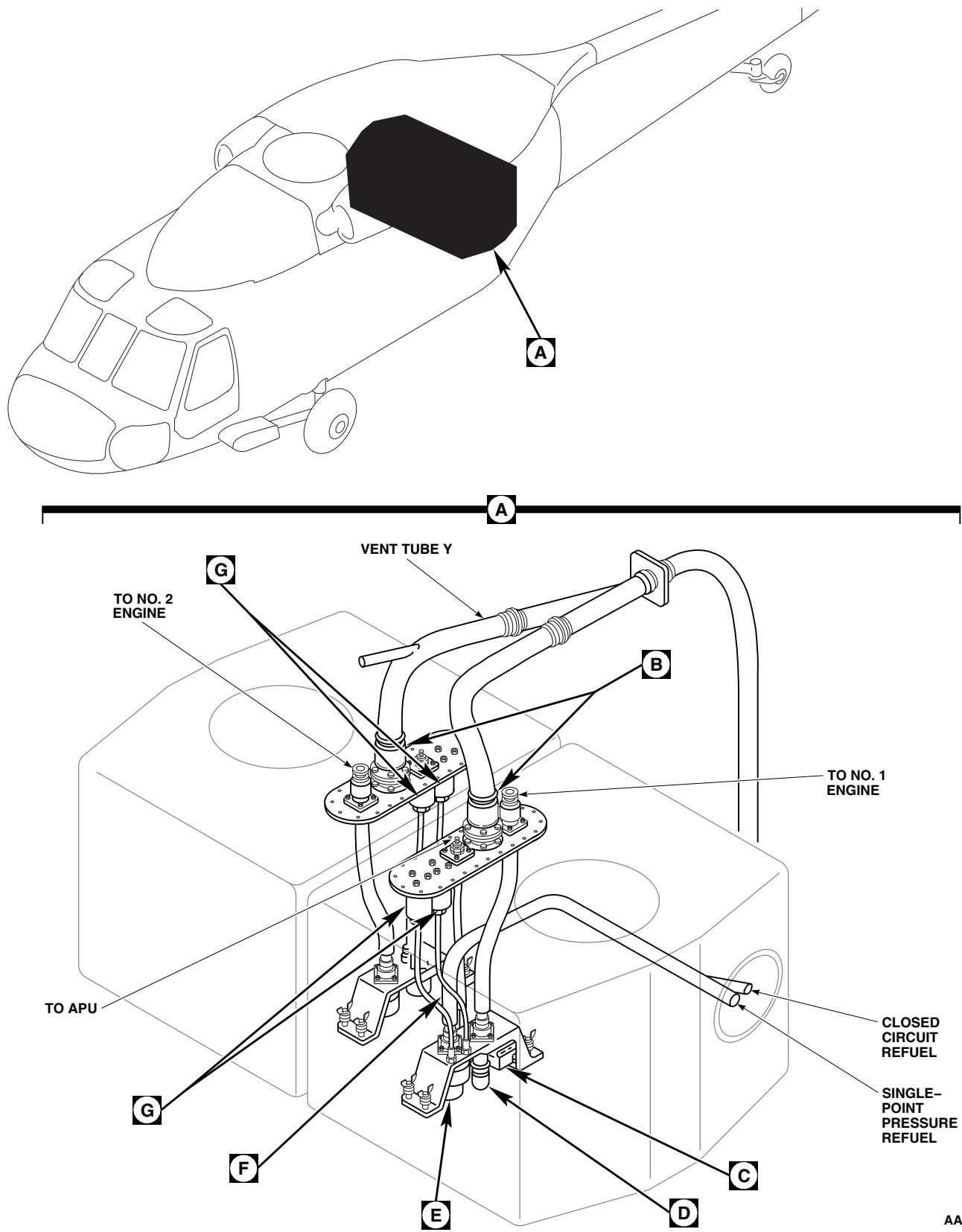
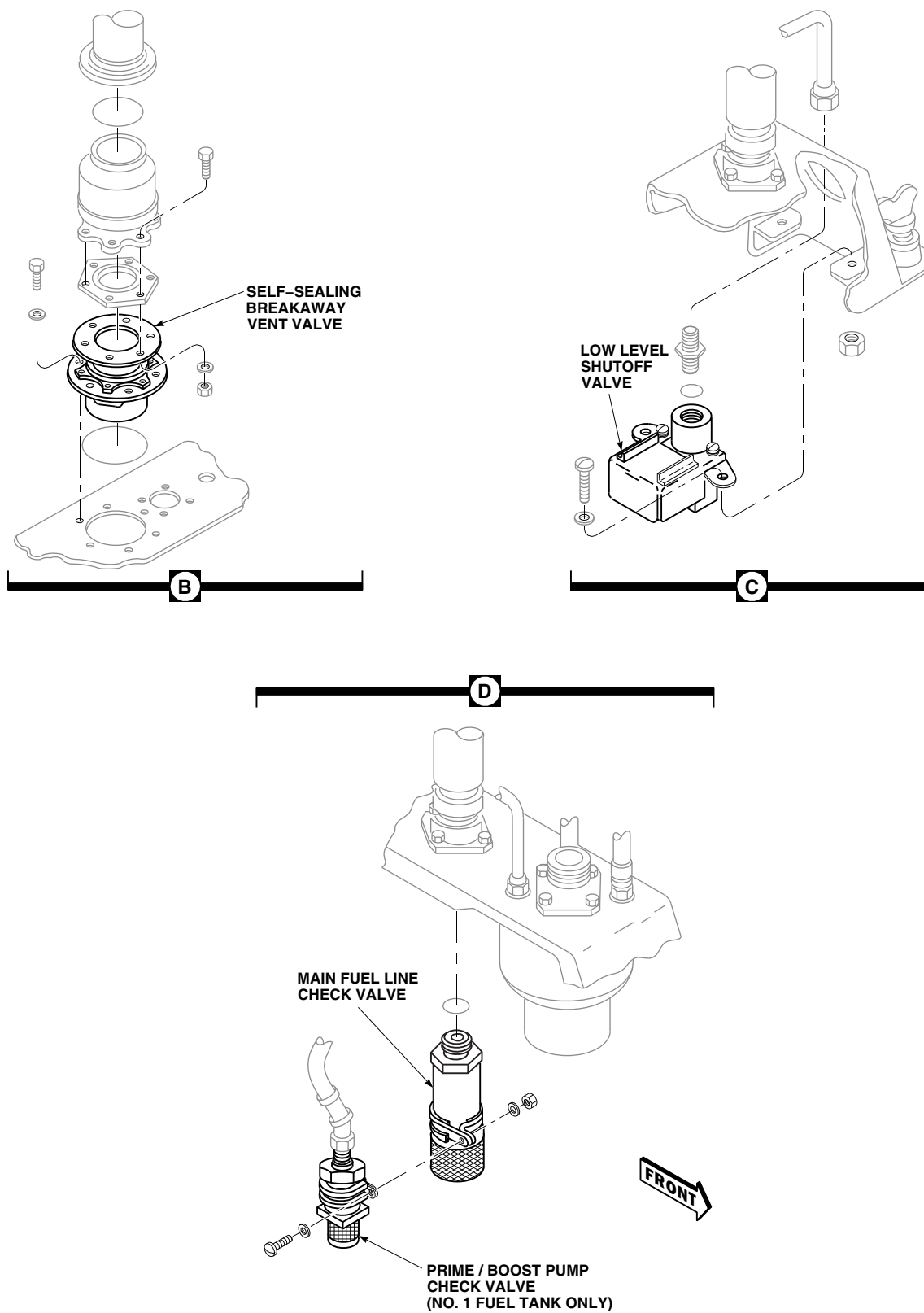
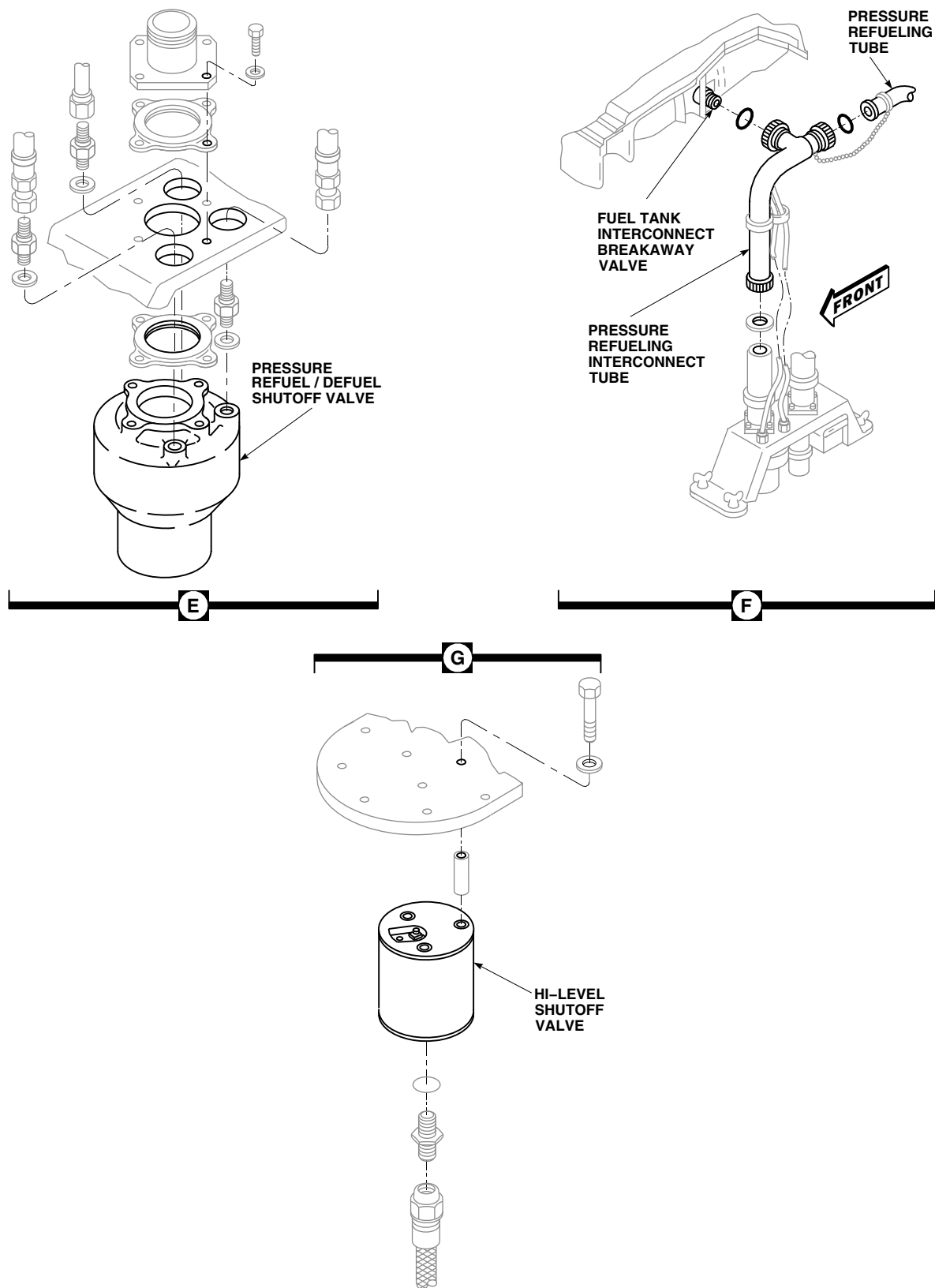


FIGURE 10-1-4-1. PRESSURE REFUEL/DEFUEL SYSTEM LOCATION DIAGRAM
(SHEET 1 OF 3)

AA2797_1
SA

FO0423_2
SA

**FIGURE 10-1-4-1. PRESSURE REFUEL/DEFUEL SYSTEM LOCATION DIAGRAM
(SHEET 2 OF 3)**



FO0423_3
SA

FIGURE 10-1-4-1. PRESSURE REFUEL/DEFUEL SYSTEM LOCATION DIAGRAM (SHEET 3 OF 3)

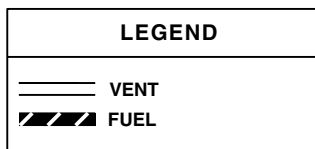
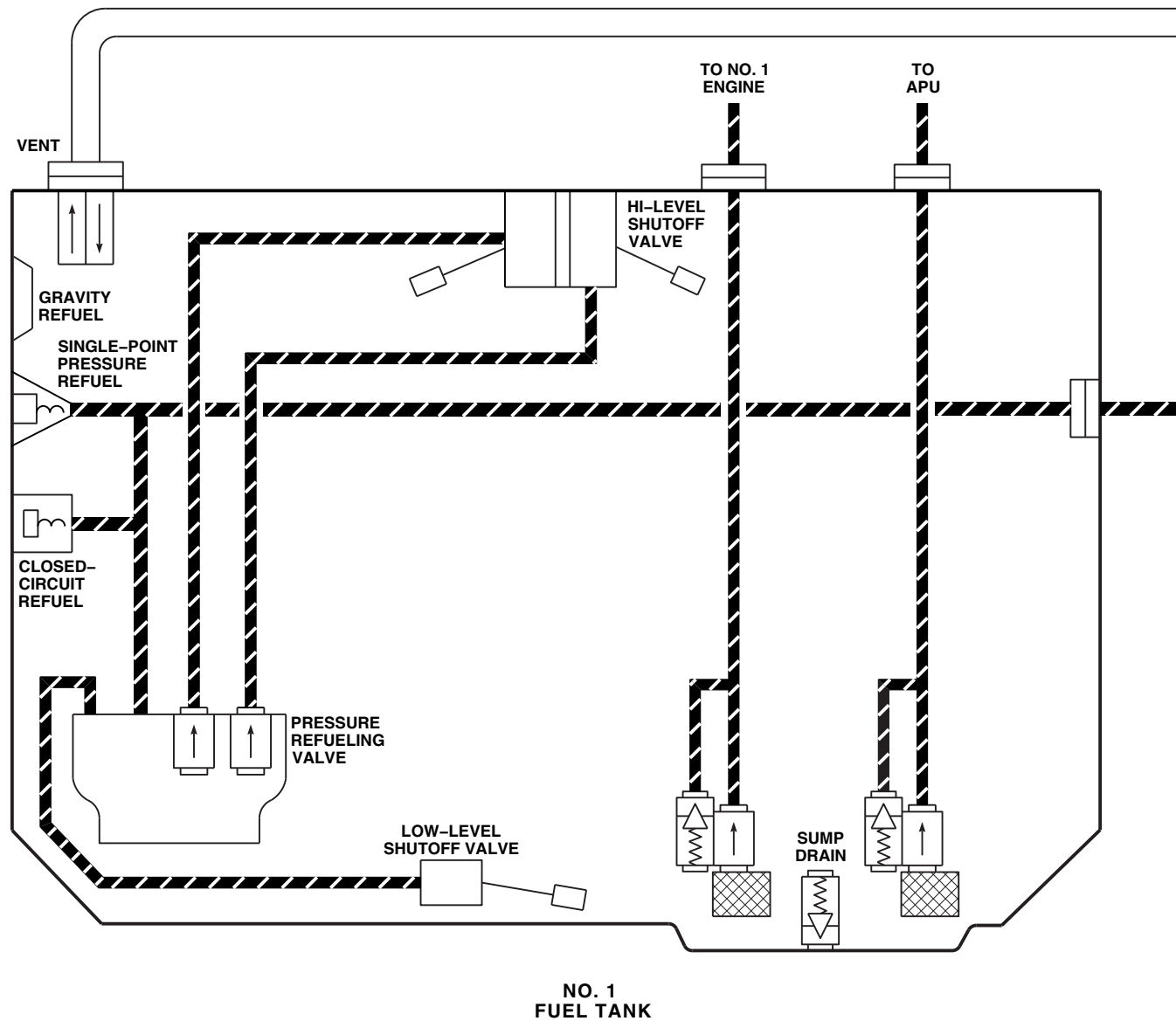
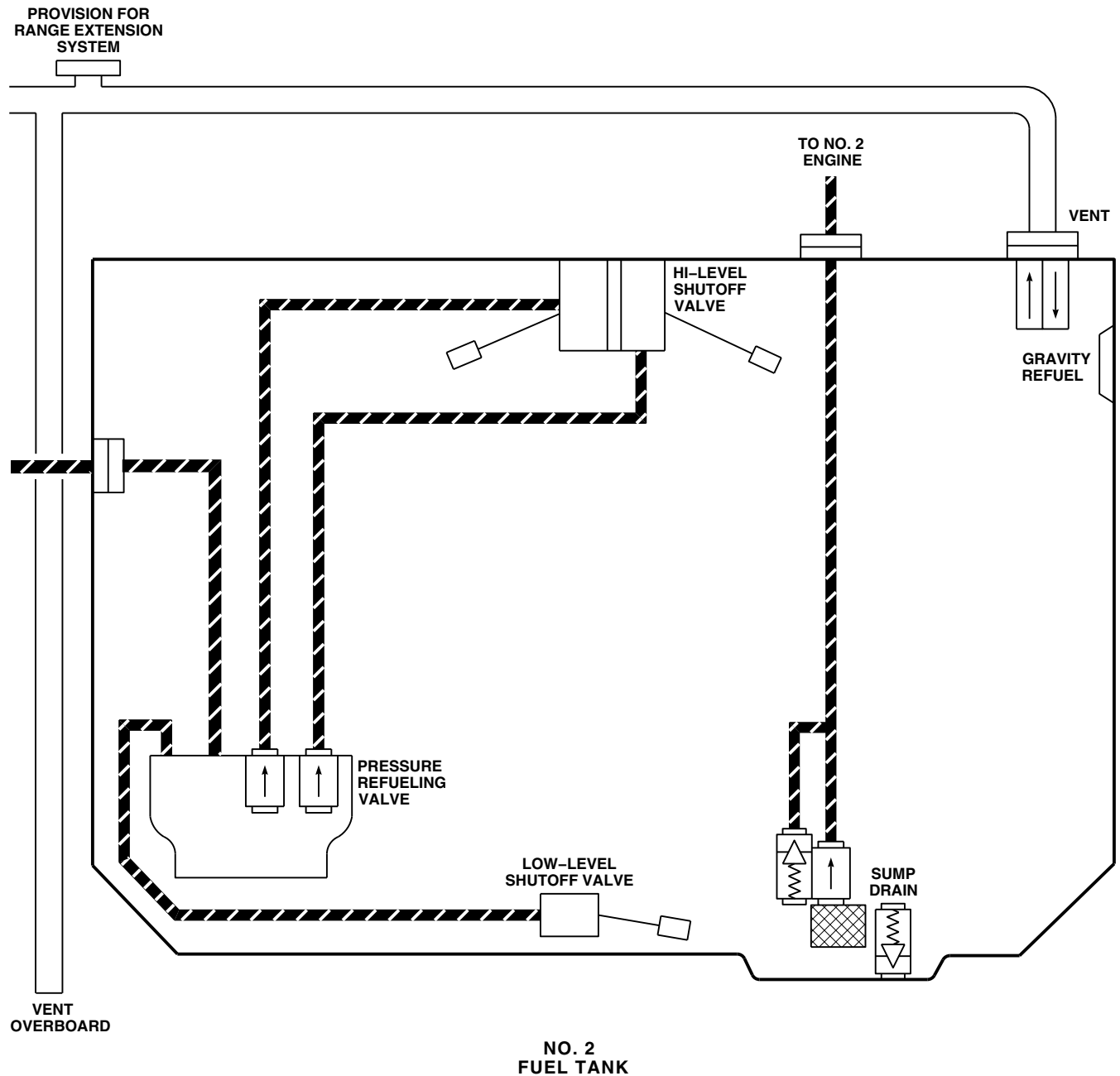
AA2798_1
SA

FIGURE 10-1-4-2. PRESSURE REFUEL/DEFUEL SYSTEM BLOCK DIAGRAM (SHEET 1 OF 2)



AA2798_2
SA

FIGURE 10-1-4-2. PRESSURE REFUEL/DEFUEL SYSTEM BLOCK DIAGRAM (SHEET 2 OF 2)

10-1-4-7/(10-1-4-8 Blank)

10-1-5 FUEL QUANTITY SYSTEM.

10-1-5.1 System Description.

The fuel quantity system consists of two independent systems that provide a visual indication in pounds, of the total amount of fuel in the No. 1 and No. 2 fuel tanks. The major components of the system include a fuel quantity probe located in each tank, fuel quantity signal conditioner, FUEL IND TEST switch located on the miscellaneous switch panel, and No. 1 and No. 2 fuel vertical fuel quantity scales located on the central display unit (Figure 10-1-5-1).

10-1-5.2 Principles Of Operation.

10-1-5.2.1 Power Distribution.

Electrical power of 115 vac for the fuel quantity system is supplied from the No. 1 ac primary bus through the NO. 1 AC INST circuit breaker on the copilot's circuit breaker panel (Figure 10-1-5-2).

10-1-5.2.2 System Operation.

The fuel quantity system consists of two independent circuits which are identical in operation; therefore only the No. 1 fuel quantity system is described.

The fuel quantity circuit is based on the principles that total capacitance in an ac circuit can be affected by the kind of insulating material (dielectric) between the capacitor plates. The fuel quantity probe acts as the capacitor and the fuel acts as the dielectric. As the total amount of fuel in the tank is changed, the total capacitance of the fuel quantity circuit will vary in direct proportion to the fuel level causing the fuel quantity probe output signal to change. This changing output signal is then fed back to the fuel quantity signal conditioner where it is amplified, demodulated and amplified again. The amplified signal from the fuel quantity signal conditioner is then routed to the No. 1 signal data converter (SDC) where it is sent to central display unit to generate the fuel quantity vertical scales.

10-1-5.3 Component Description.

10-1-5.3.1 Fuel Quantity Probe.

The fuel quantity probe, located in each fuel tank, has two concentric metal tubes that form the plates of a capacitor. The probe is a capacitive-type sensor, employing fuel as the dielectric. The value of capacitance varies with the height of the fuel and the dielectric constant of the fuel. The greater the amount of the fuel in the tank, the more capacitance each probe produces.

10-1-5.3.2 Fuel Quantity Signal Conditioner.

The fuel quantity signal conditioner applies an ac signal to each fuel quantity probe. The output signal from the probe varies with its capacitance. This signal is applied back to the fuel quantity signal conditioner and is proportional to the amount of fuel in the tank. The fuel quantity signal conditioner then converts the varying ac signal to a dc output voltage ranging from 0 to 7 vdc which is applied to both SDCs. The fuel quantity signal conditioner includes tank empty and tank full adjustments for the No. 1 and No. 2 fuel tanks.

10-1-5.3.3 Signal Data Converter (SDC).

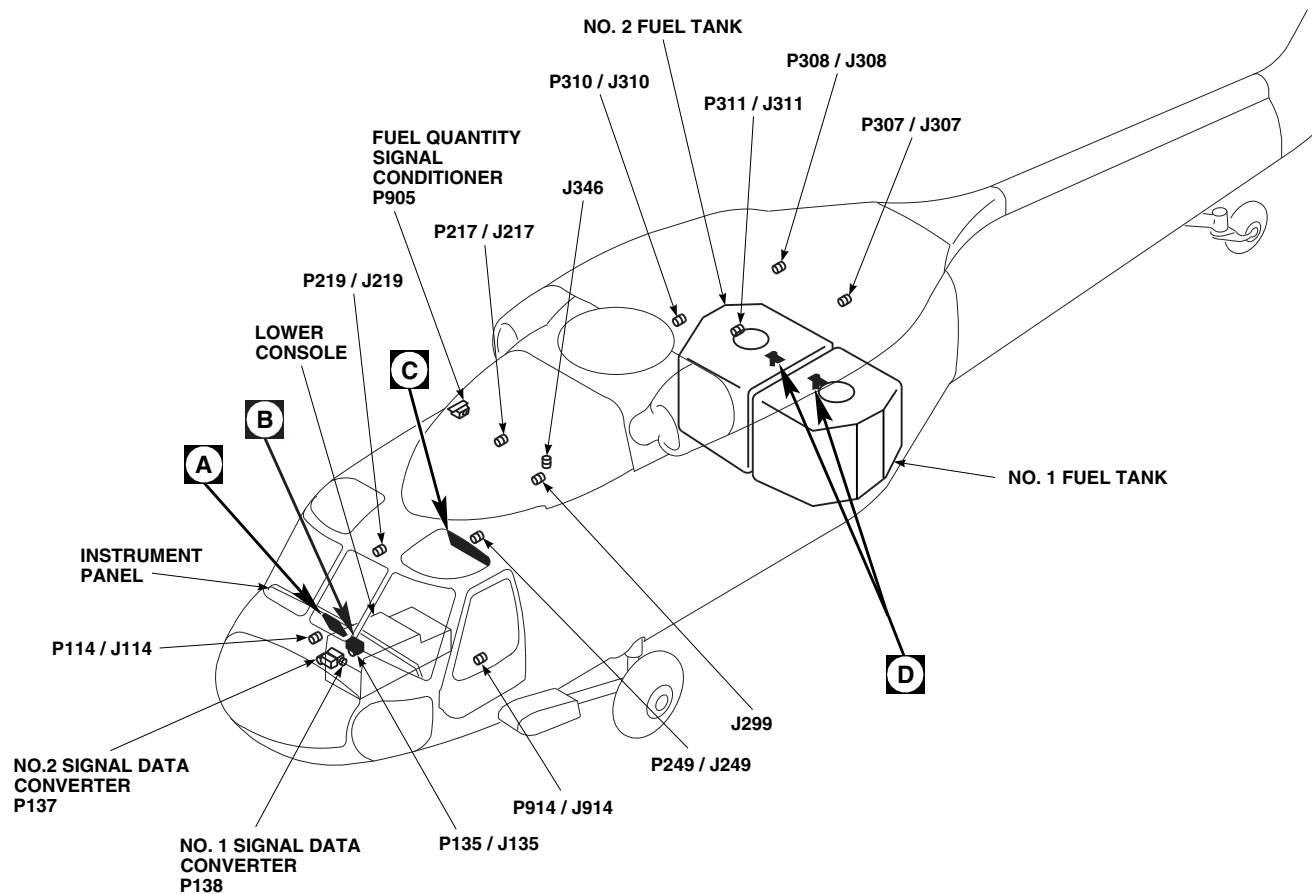
The SDCs are components of the vertical instrument display system (VIDS). Each unit is identical and is physically mounted on a shelf in the nose avionics bay. The SDCs apply signals from the fuel quantity signal conditioners to the central display unit. For a discussion of SDC operation, refer to Instrument Display System (PARA 8-1-3).

10-1-5.3.4 Central Display Unit.

The central display unit located on the center instrument panel consists of two vertical FUEL QTY LB X 100 indicator scales. Each vertical scale consists of two multicolored segments which indicate as follows: amber for 0 - 200 pounds of fuel and green for 200 - 1500 pounds of fuel. A DIGITS ON-OFF switch controls power to the

digital displays. The digital readout of total fuel in both tanks is displayed directly under the vertical indicator. For a discussion of central display unit

operation, refer to Instrument Display System (PARA 8-1-3).

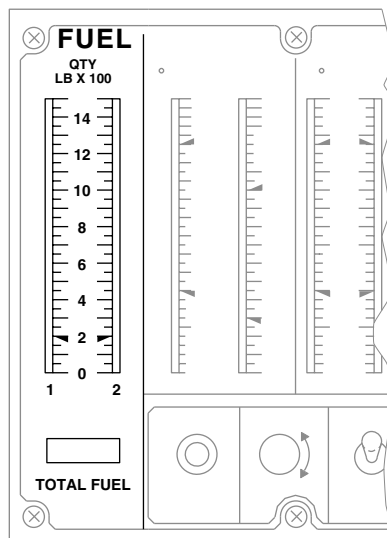


TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
J299	CABIN CEILING BL 10 LH, STA 302
J346	CABIN CEILING BL 18 LH, STA 301
P114 / J114	COCKPIT BL 8 RH, STA 200
P135	MISCELLANEOUS SWITCH PANEL
P137	INSTRUMENT DISPLAY SYSTEM NO. 2 SIGNAL DATA CONVERTER

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P138	INSTRUMENT DISPLAY SYSTEM NO. 1 SIGNAL DATA CONVERTER
P217 / J217	MAIN ROTOR PYLON DECK BL 0, STA 284
P219 / J219	CABIN FLOOR BL 25 RH, STA 247
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P307 / J307	TRANSITION DECK BL 15 LH, STA 430
P308 / J308	TRANSITION DECK BL 15 RH, STA 430
P310 / J310	CABIN CEILING BL 17 RH, STA 387
P311 / J311	CABIN CEILING BL 16 LH, STA 383
P905	FUEL QUANTITY SIGNAL CONDITIONER
P914 / J914	COCKPIT BL 24 LH, STA 247

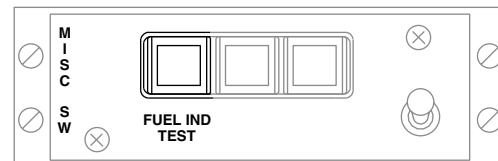
FB1597_1
SA

FIGURE 10-1-5-1. FUEL QUANTITY SYSTEM LOCATION DIAGRAM (SHEET 1 OF 2)



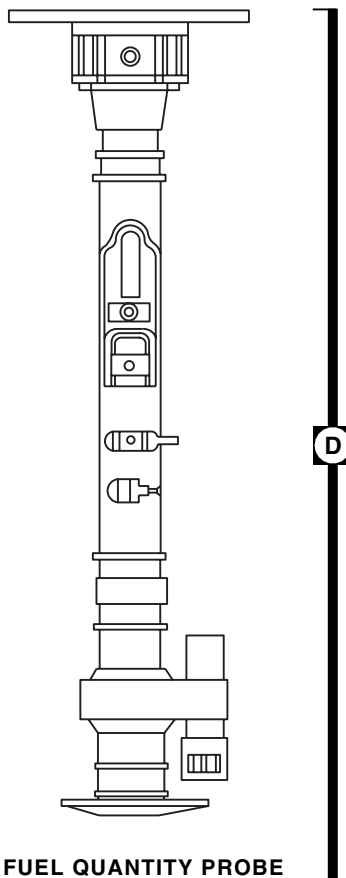
CENTRAL DISPLAY UNIT

A



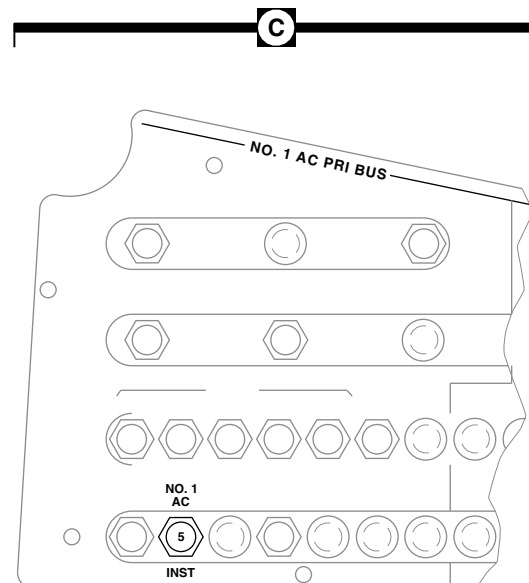
MISCELLANEOUS SWITCH PANEL

B



FUEL QUANTITY PROBE

D

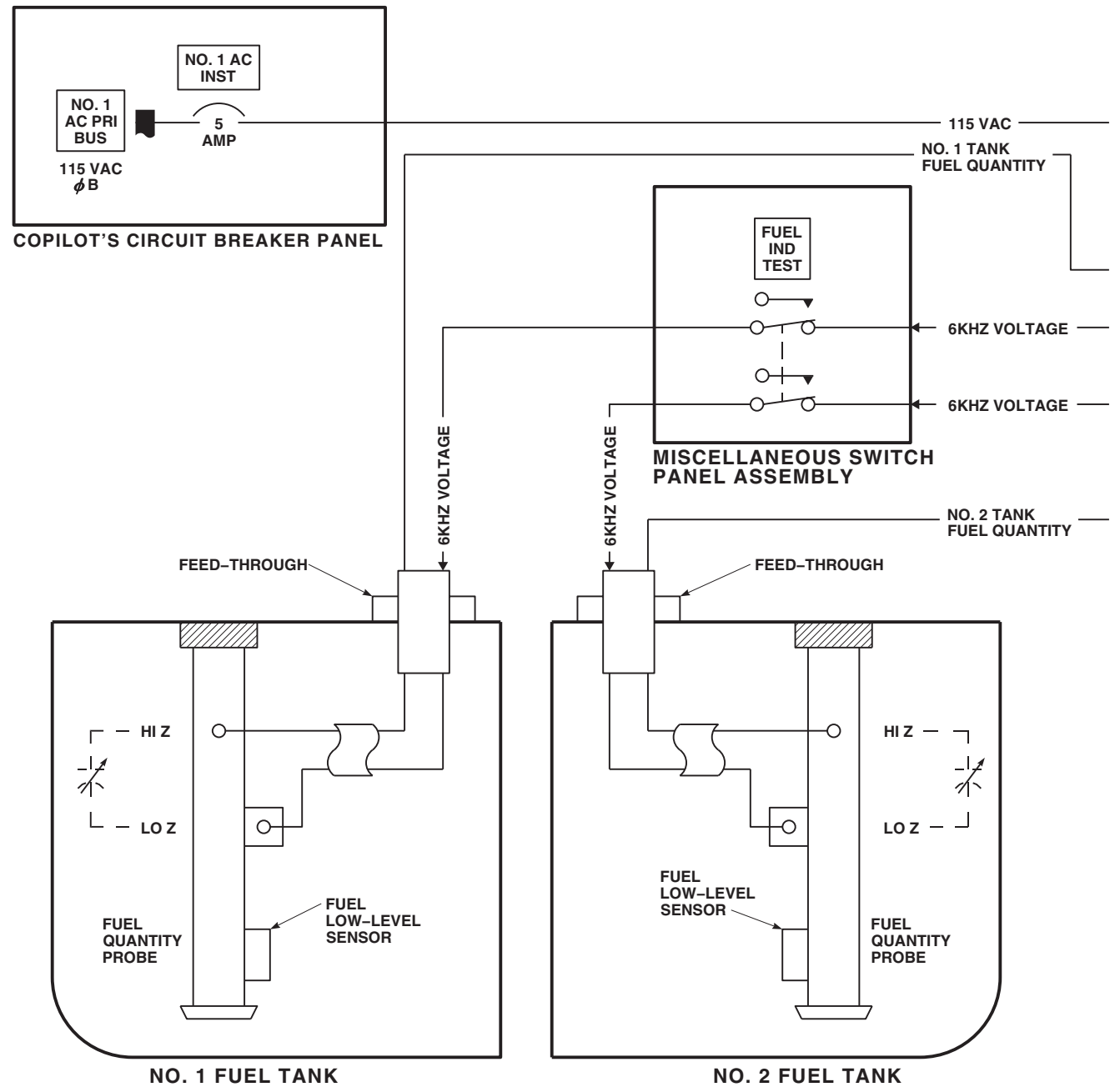


COPLOT'S CIRCUIT BREAKER PANEL

C

AA2789_2
SA

FIGURE 10-1-5-1. FUEL QUANTITY SYSTEM LOCATION DIAGRAM (SHEET 2 OF 2)



NOTE
ESSS

FN2251_1
 SA

FIGURE 10-1-5-2. FUEL QUANTITY SYSTEM BLOCK DIAGRAM (SHEET 1 OF 2)

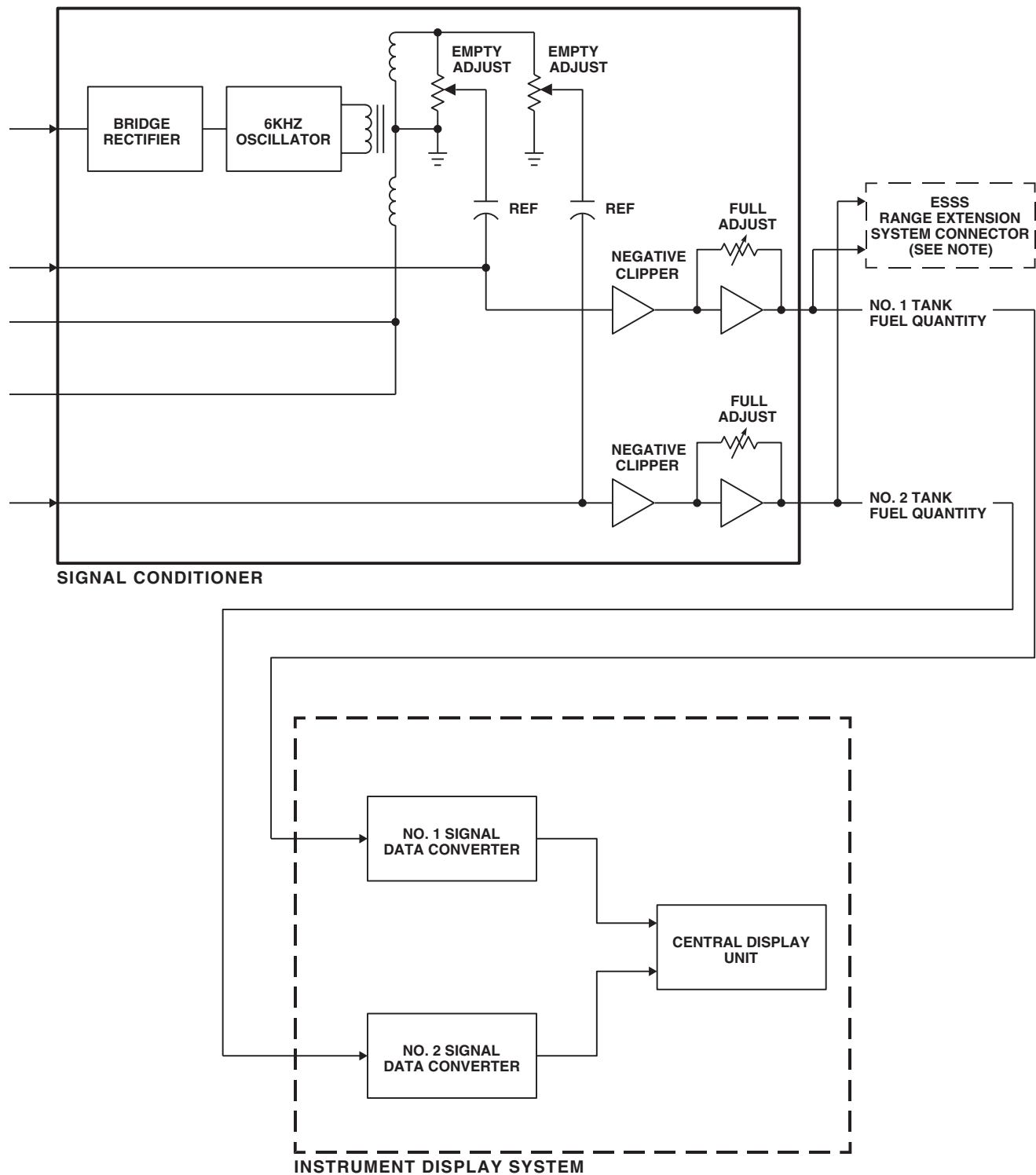
FN2251_2
SA

FIGURE 10-1-5-2. FUEL QUANTITY SYSTEM BLOCK DIAGRAM (SHEET 2 OF 2)

SECTION 10-2

TROUBLESHOOTING

SECTION OVERVIEW

PARAGRAPH	TITLE
10-2-1	Fuel Low-Level Warning System
10-2-2	Submerged Fuel Boost Pump System
10-2-3	Fuel Prime Boost System
10-2-4	Pressure Refuel/Defuel System
10-2-5	Fuel Quantity System (Using Test Set PSD-40)
10-2-6	Fuel Quantity System (Using Test Set PSD-60)
10-2-7	Engine Feed Fuel Lines

10-2-1 FUEL LOW-LEVEL WARNING SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 27/AN

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Power Available or APU Operational

- No. 1 and No. 2 Fuel Tanks Each Filled With **200 pounds** of Fuel

NOTE

If #1 or #2 FUEL LOW capsule on caution/advisory panel is ever on steadily (not flashing), replace caution/advisory panel (PARA 8-4-19), and then continue with operational/troubleshooting procedure.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

10-2-1.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in except those noted as being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).

RESULT:

- No. 1 and No. 2 fuel quantity indications on central display unit shall indicate approximately 200 pounds.
 - #1 FUEL LOW capsule on caution/advisory panel shall be off.
 - #2 FUEL LOW capsule on caution/advisory panel shall be off.
- If results (1), (2), and (3) are not as specified, pressure fuel No. 1 and No. 2 fuel tanks until 200 pounds of fuel is indicated on central display unit (PARA 1-3-17).
 - If result (2) is not as specified, go to TABLE NO. 10-2-1-1.

- If result (3) is not as specified, go to TABLE NO. 10-2-1-2.
- If results (2) and (3) are not as specified, go to TABLE NO. 10-2-1-3.
- c. Place and hold caution/advisory panel BRT DIM/TEST switch to TEST.

RESULT: #1 and #2 FUEL LOW capsules on caution/advisory panel shall flash.

- If result is not as specified, troubleshoot caution/advisory warning system (PARA 8-2-4).
- d. Release caution/advisory panel BRT DIM/TEST switch.

RESULT: #1 and #2 FUEL LOW capsules on caution/advisory panel shall go off.

- If result is not as specified, troubleshoot caution/advisory warning system (PARA 8-2-4).
- e. Press and hold miscellaneous switch panel FUEL IND TEST button.

RESULT: #1 and #2 FUEL LOW capsules on caution/advisory panel shall flash.

- If result is not as specified, go to TABLE NO. 10-2-1-4.

f. Release miscellaneous switch panel FUEL IND TEST button.

RESULT: #1 and #2 FUEL LOW capsules on caution/advisory panel shall go off.

- If result is not as specified, replace miscellaneous switch panel (PARA 9-4-7).
- g. While defueling No. 1 fuel tank to approximately **150 pounds**, observe #1 FUEL LOW capsule on caution/advisory panel. **Record poundage when #1 FUEL LOW capsule begins to flash.**

RESULT: #1 FUEL LOW capsule shall flash with less than **170 to 180 pounds** of fuel in No. 1 fuel tank.

- If result is not as specified, go to TABLE NO. 10-2-1-5.

h. While defueling No. 2 fuel tank to approximately **150 pounds**, observe #2 FUEL LOW capsule on caution/advisory panel. **Record poundage when #2 FUEL LOW capsule begins to flash.**

RESULT: #2 FUEL LOW capsule shall flash with less than **170 to 180 pounds** of fuel in No. 2 tank.

- If result is not as specified, go to TABLE NO. 10-2-1-6.
- i. Turn off electrical power (PARA 1-6-16).

TABLE NO. 10-2-1-1. #1 FUEL LOW Capsule Flashing When No. 1 Fuel Tank Has 200 Pounds Of Fuel.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between P255-K and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.
2. Check continuity between J307-B and P255-B.	Step 1. If continuity is present, replace low-level warning conditioner (PARA 10-4-35). Go to 6. Step 2. If trouble remains, replace malfunctioning fuel quantity probe (PARA 10-4-33). Go to 6. Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 6.
3. Check continuity between P135-P and P255-K.	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 6.
4. Check for 28 vdc between P135-R and ground.	

**TABLE NO. 10-2-1-1. #1 FUEL LOW Capsule Flashing When No. 1 Fuel Tank Has 200 Pounds Of Fuel.
(Cont)**

TEST OR INSPECTION	CORRECTIVE ACTION
Step 1. If voltage is as specified, replace miscellaneous switch panel (PARA 9-4-7). Go to 6. Step 2. If voltage is not as specified, go to 5.	5. Check continuity between P135-R and FUEL LOW WARN circuit breaker terminal 1. Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to 6. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 6. 6. Procedure completed.

TABLE NO. 10-2-1-2. #2 FUEL LOW Capsule Flashing When No. 2 Fuel Tank Has 200 Pounds Of Fuel.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between P255-J and ground. Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.	2. Check continuity between J308-B and P255-F. Step 1. If continuity is present, replace low-level warning conditioner (PARA 10-4-35). Go to 6. Step 2. If trouble remains, replace malfunctioning fuel quantity probe (PARA 10-4-33). Go to 6. Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 6. 3. Check continuity between P135-M and P255-J. Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 6. 4. Check for 28 vdc between P135-N and ground.

**TABLE NO. 10-2-1-2. #2 FUEL LOW Capsule Flashing When No. 2 Fuel Tank Has 200 Pounds Of Fuel.
(Cont)**

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If voltage is as specified, replace miscellaneous switch panel (PARA 9-4-7). Go to 6. Step 2. If voltage is not as specified, go to 5.
5. Check continuity between P135-N and FUEL LOW WARN circuit breaker terminal 1.	Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to 6. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 6.
6. Procedure completed.	

TABLE NO. 10-2-1-3. #1 And #2 FUEL LOW Capsules Flashing When No. 1 And No. 2 Fuel Tanks Have 200 Pounds Of Fuel.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between: P255-J and ground P255-K and ground	Step 1. If voltage is as specified, replace low-level warning conditioner (PARA 10-4-35). Go to 2. Step 2. If voltage is not as specified, replace miscellaneous switch panel (PARA 9-4-7). Go to 2.
2. Procedure completed.	

TABLE NO. 10-2-1-4. #1 And #2 FUEL LOW Capsules Do Not Flash When FUEL IND TEST Button Is Pressed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check helicopter configuration.	

TABLE NO. 10-2-1-4. #1 And #2 FUEL LOW Capsules Do Not Flash When FUEL IND TEST Button Is Pressed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. EMEP , go to 2. Step 2. W/O EMEP , go to 4.
EMEP	2. Remove and check pin filtered adapter P117 (PARA 9-4-98). Step 1. If pin filtered adapter is good, go to 3. Step 2. If pin filtered adapter is not good, replace pin filtered adapter (PARA 9-4-98). Go to 8.
EMEP	3. Install pin filtered adapter (PARA 9-4-98). Go to 4. 4. Check for 28 vdc between: P117-1 and ground P117-49 and ground Step 1. If voltage is as specified, replace caution/advisory panel (PARA 8-4-19). Go to 8. Step 2. If voltage is not as specified, go to 5. 5. Check for 28 vdc between: P255-N (No. 1 tank) and ground P255-S (No. 2 tank) and ground Step 1. If voltage is as specified, go to 6. Step 2. If voltage is not as specified, go to 7. 6. Check continuity between: P117-1 and P255-P P117-49 and P255-T Step 1. If continuity is present, replace low-level warning conditioner (PARA 10-4-35). Go to 8. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8. 7. Check continuity between: Terminal 1 of FUEL LOW WARN circuit breaker and P255-S Terminal 1 of FUEL LOW WARN circuit breaker and P255-N Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to 8. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8.

TABLE NO. 10-2-1-4. #1 And #2 FUEL LOW Capsules Do Not Flash When FUEL IND TEST Button Is Pressed. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

8. Procedure completed.

TABLE NO. 10-2-1-5. #1 FUEL LOW Capsule Does Not Flash When Tank Has Less Than 170 To 180 Pounds Of Fuel.

TEST OR INSPECTION
CORRECTIVE ACTION

- 1. Check thermistor resistance between:**
 P255-A and P255-B (650-1500 Ohms)
 P255-A and P255-C (360-440 Ohms)

Step 1. If resistance is as specified, replace low-level warning conditioner (PARA 10-4-35). Go to **3**.

Step 2. If resistance is not as specified, go to **2**.

- 2. Check continuity between:**
 P255-A and J307-C
 P255-C and J307-A
 P255-B and J307-B

Step 1. If continuity is present, replace malfunctioning fuel quantity probe or wiring harness (PARA 10-4-33). Go to **3**.

Step 2. If continuity is not present, repair/replace wiring (Appendix F) as required. Go to **3**.

3. Procedure completed.

TABLE NO. 10-2-1-6. #2 FUEL LOW Capsule Does Not Flash When Tank Has Less Than 170 To 180 Pounds Of Fuel.

TEST OR INSPECTION
CORRECTIVE ACTION

- 1. Check thermistor resistance between:**
 P255-E and P255-F (650-1500 Ohms)
 P255-E and P255-G (360-440 Ohms)

TABLE NO. 10-2-1-6. #2 FUEL LOW Capsule Does Not Flash When Tank Has Less Than 170 To 180 Pounds Of Fuel. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
Step 1. If resistance is as specified, replace low-level warning conditioner (PARA 10-4-35). Go to 3. Step 2. If resistance is not as specified, go to 2.	2. Check continuity between: P255-E and J308-C P255-G and J308-A P255-F and J308-B Step 1. If continuity is present, replace malfunctioning fuel quantity probe or wiring harness (PARA 10-4-33). Go to 3. Step 2. If continuity is not present, repair/replace wiring (Appendix F) as required. Go to 3.
3. Procedure completed.	

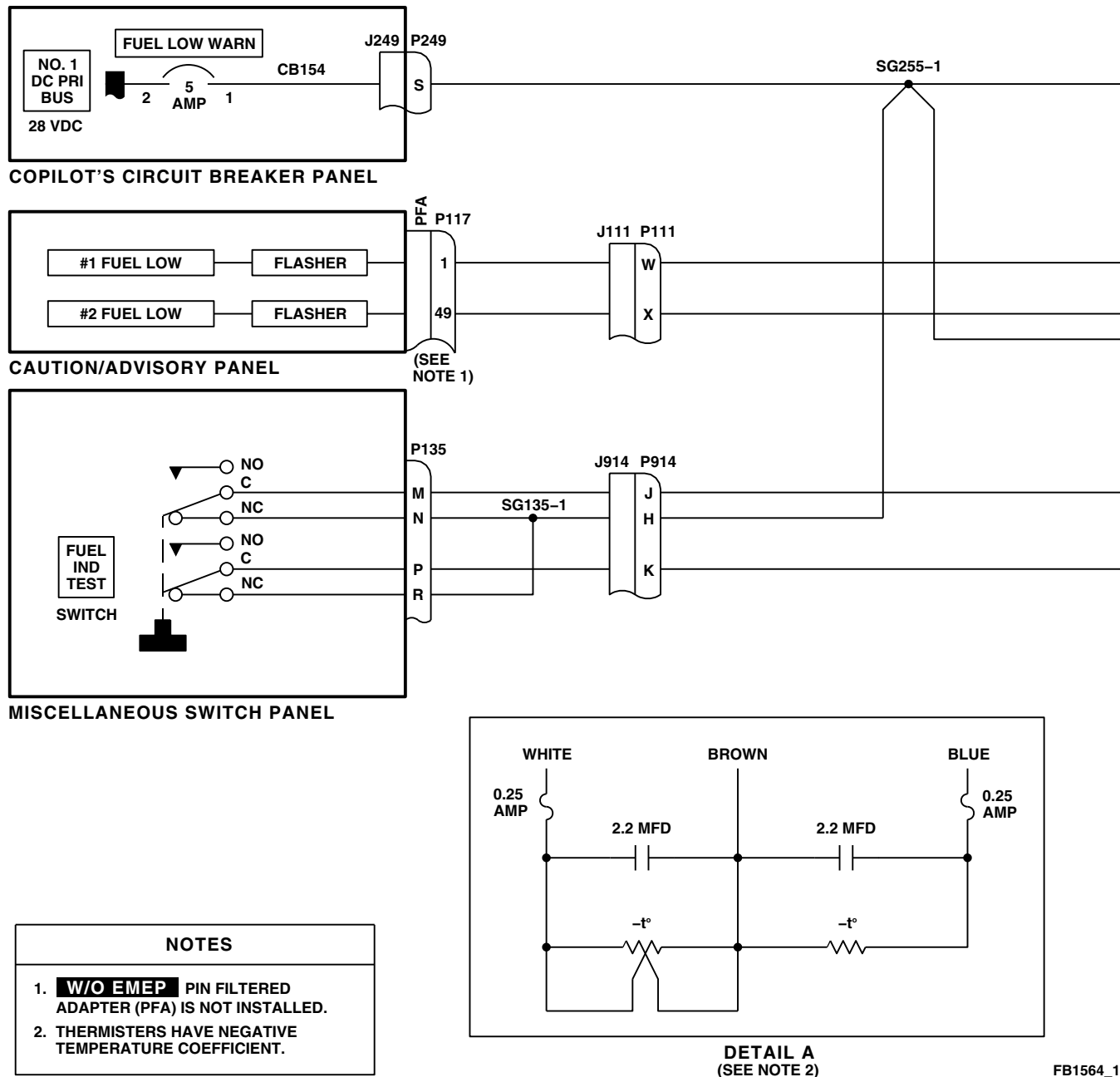
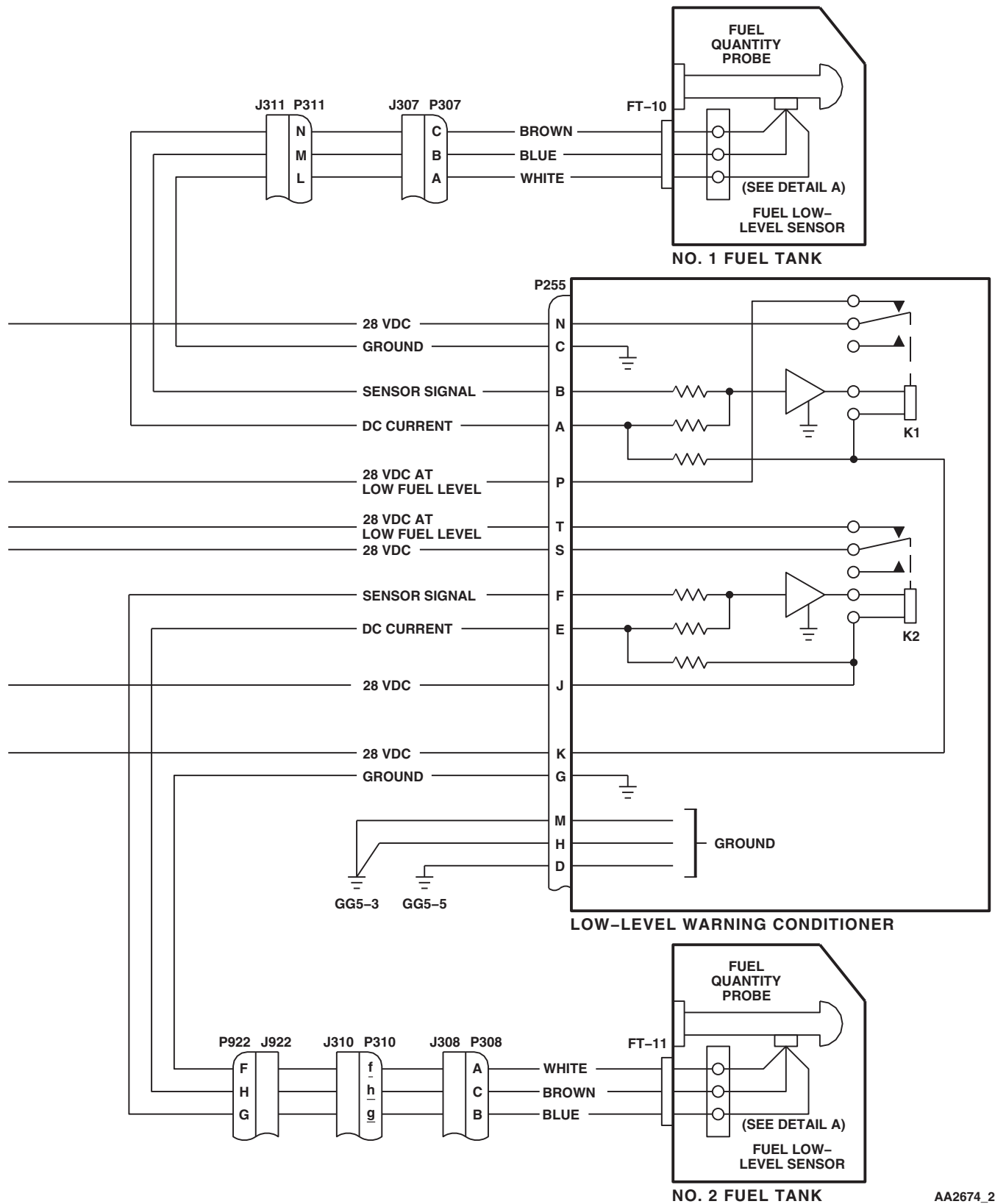


FIGURE 10-2-1-1. FUEL LOW-LEVEL WARNING SYSTEM SCHEMATIC DIAGRAM
(SHEET 1 OF 2)

10-2-1-8



AA2674_2
SA

FIGURE 10-2-1-1. FUEL LOW-LEVEL WARNING SYSTEM SCHEMATIC DIAGRAM
(SHEET 2 OF 2)

10-2-1-9/(10-2-1-10 Blank)

10-2-2 SUBMERGED FUEL BOOST PUMP SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer Toolkit

Materials/Parts

- Suitable Container

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational
- No. 1 and No. 2 Engines Primed
- No. 1 and No. 2 Fuel Tanks Serviced

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

10-2-2.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).

RESULT:

- #1 FUEL PRESS capsule on caution/advisory panel shall be on.
- #2 FUEL PRESS capsule on caution/advisory panel shall be on.
 - If result (1) is not as specified, go to TABLE NO. 10-2-2-1.
 - If result (2) is not as specified, go to TABLE NO. 10-2-2-2.
- Place engine controls quadrant NO. 1 ENG FUEL SYS selector handle to DIR.
- Place NO. 1 PUMP switch on fuel boost pump control panel to ON.

RESULT:

- NO. 1 PUMP pressure light shall go on.
- #1 FUEL PRESS (caution/advisory panel) capsule shall go off.
 - If results (1) and (2) are not as specified, go to TABLE NO. 10-2-2-3.
 - If result (1) is not as specified, go to TABLE NO. 10-2-2-4.
 - If result (2) is not as specified, go to TABLE NO. 10-2-2-5.
- Move engine controls quadrant NO. 1 ENG FUEL SYS selector handle to OFF.
- Place NO. 1 PUMP switch on fuel boost pump control panel to OFF.

RESULT: NO. 1 PUMP pressure light shall go off.

- If result is not as specified, go to TABLE NO. 10-2-2-6.

- g. Place engine controls quadrant NO. 2 ENG FUEL SYS selector handle to DIR.
- h. Place NO. 2 PUMP switch on fuel boost pump control panel to ON.

RESULT:

- (1) NO. 2 PUMP pressure light shall go on.
- (2) #2 FUEL PRESS (caution/advisory panel) capsule shall go off.
- If results (1) and (2) are not as specified, go to TABLE NO. 10-2-2-7.
- If result (1) is not as specified, go to TABLE NO. 10-2-2-8.
- If result (2) is not as specified, go to TABLE NO. 10-2-2-9.
- i. Place engine controls quadrant NO. 2 ENG FUEL SYS selector handle to OFF.
- j. Place NO. 2 PUMP switch on fuel boost pump control panel to OFF.

RESULT: NO. 2 PUMP pressure light shall go off.

- If result is not as specified, go to TABLE NO. 10-2-2-10.
- k. Place engine controls quadrant NO. 1 ENG FUEL SYS selector handle to XFD.
- l. Place NO. 2 PUMP switch on fuel boost pump control panel to ON.

RESULT:

- (1) NO. 2 PUMP pressure light shall go on.
- (2) #1 FUEL PRESS (caution/advisory panel) capsule shall go off.

- If results (1) and (2) are not as specified, go to TABLE NO. 10-2-2-7.

- If result (1) is not as specified, go to TABLE NO. 10-2-2-8.

- If result (2) is not as specified, replace fuel selector valve (PARA 10-4-25).

- m. Move engine controls quadrant NO. 1 ENG FUEL SYS selector handle to OFF.
- n. Place NO. 2 PUMP switch on fuel boost pump control panel to OFF.
- o. Place engine controls quadrant NO. 2 ENG FUEL SYS selector handle to XFD.
- p. Place NO. 1 PUMP switch on fuel boost pump control panel to ON.

RESULT:

- (1) NO. 1 PUMP pressure light shall go on.
- (2) #2 FUEL PRESS (caution/advisory panel) capsule shall go off.

- If results (1) and (2) are not as specified, go to TABLE NO. 10-2-2-3.

- If result (1) is not as specified, go to TABLE NO. 10-2-2-4.

- If result (2) is not as specified, replace fuel selector valve (PARA 10-4-25).

- q. Move engine controls quadrant NO. 2 ENG FUEL SYS selector handle to OFF.
- r. Place NO. 1 PUMP switch on fuel boost pump control panel to OFF.
- s. Turn off electrical power (PARA 1-6-16).

TABLE NO. 10-2-2-1. #1 FUEL PRESS Caution/Advisory Capsule Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between P803-23 and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.
2. Check continuity between P803-23 and P117-2.	Step 1. If continuity is present, replace engine fuel pressure sensor (TM 1-2840-248-23). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
3. Check continuity between NO. 1 ENG WARN LTS circuit breaker terminal 1 and P803-23.	Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
4. Procedure completed.	

TABLE NO. 10-2-2-2. #2 FUEL PRESS Caution/Advisory Capsule Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between P802-23 and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.
2. Check continuity between P802-24 and P117-50.	Step 1. If continuity is present, replace engine fuel pressure sensor (TM 1-2840-248-23). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
3. Check continuity between NO. 2 ENG WARN LTS circuit breaker terminal 1 and P802-23.	Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.

TABLE NO. 10-2-2-2. #2 FUEL PRESS Caution/Advisory Capsule Does Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

4. Procedure completed.

TABLE NO. 10-2-2-3. NO. 1 PUMP Pressure Light Does Not Go On And #1 Or #2 FUEL PRESS Capsule Does Not Go Off.

TEST OR INSPECTION
CORRECTIVE ACTION

- 1. Check for 115 vac between:**
 J322-A and J322-D
 J322-B and J322-D
 J322-C and J322-D

Step 1. If voltage is as specified, replace No. 1 boost pump (PARA 10-4-12).
 Go to **6.**

Step 2. If voltage is not as specified, go to **2.**

- 2. Check continuity between J322-D and ground.**

Step 1. If continuity is present, go to **3.**

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **6.**

- 3. Check continuity between:**
 P139-G and J322-A
 P139-H and J322-B
 P139-J and J322-C

Step 1. If continuity is present, go to **4.**

Step 2. If continuity is not present, repair/replace wiring (Appendix F) as required.
 Go to **6.**

- 4. Check for 115 vac between:**
 P139-V and ground
 P139-U and ground
 P139-T and ground

Step 1. If voltage is as specified, replace fuel boost pump control panel (PARA 9-4-40). Go to **6.**

Step 2. If voltage is not as specified, go to **5.**

TABLE NO. 10-2-2-3. NO. 1 PUMP Pressure Light Does Not Go On And #1 Or #2 FUEL PRESS Capsule Does Not Go Off. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
5. Check for 115 vac between NO. 1 FUEL BOOST PUMP circuit breaker: Terminal A1 and ground Terminal B1 and ground Terminal C1 and ground Step 1. If voltage is as specified, repair/replace wiring (Appendix F) as required, then go to 6: Terminal A1 and P139-V Terminal B1 and P139-U Terminal C1 and P139-T Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 6. 6. Procedure completed.

TABLE NO. 10-2-2-4. #1 Or #2 FUEL PRESS Capsule Goes Off But NO. 1 PUMP Pressure Light Does Not Go On.

TEST OR INSPECTION CORRECTIVE ACTION
1. Press face of NO. 1 PUMP light. Check that light goes on. Step 1. If light goes on, go to 2. Step 2. If light does not go on, go to 5. 2. With NO. 1 PUMP switch at ON, check continuity between P139-M and ground. Step 1. If continuity is present, replace fuel boost pump control panel (PARA 9-4-40). Go to 8. Step 2. If continuity is not present, go to 3. 3. Check continuity between P139-M and P321-1. Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8. 4. Check continuity between P321-2 and ground.

TABLE NO. 10-2-2-4. #1 Or #2 FUEL PRESS Capsule Goes Off But NO. 1 PUMP Pressure Light Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If continuity is present, replace No. 1 boost pump pressure switch (PARA 10-4-14). Go to 8. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8.
5. Check NO. 1 PUMP switch lamp.	Step 1. If lamp is good, go to 6. Step 2. If lamp is not good, replace lamp (PARA 9-4-42). Go to 8.
6. Check for 28 vdc between P139-R and ground.	Step 1. If voltage is as specified, replace fuel boost pump control panel (PARA 9-4-40). Go to 8. Step 2. If voltage is not as specified, go to 7.
7. Check continuity between P139-R and P902-X.	Step 1. If continuity is present, troubleshoot instrument panel and consoles indicator lights dimming (PARA 9-2-10). Go to 8. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8.
8. Procedure completed.	

TABLE NO. 10-2-2-5. NO. 1 PUMP Pressure Light Goes On But #1 FUEL PRESS Capsule Does Not Go Off.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Place suitable container beneath left side of helicopter fuselage at No. 1 engine overboard drain. Go to 2.	
2. Make sure NO. 1 ENG WARN LTS circuit breaker is pushed in. Go to 3.	
3. Place engine controls quadrant NO. 1 ENG FUEL SYS selector handle in OFF and DIR positions and observe that push/pull cable actuates fuel selector valve.	
	Step 1. If fuel selector valve is actuated, go to 4.

TABLE NO. 10-2-2-5. NO. 1 PUMP Pressure Light Goes On But #1 FUEL PRESS Capsule Does Not Go Off. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
Step 2. If fuel selector valve is not actuated, replace push/pull cable (PARA 4-4-30). Go to 10 .	Step 3. If trouble remains, replace control box (PARA 4-4-31). Go to 10 .
4. Place ENG POWER CONT lever to LOCKOUT. Go to 5.	
5. Check for fuel flow overboard.	
Step 1. If fuel flow is present, go to 6 .	Step 2. If fuel flow is not present, go to 8 .
6. Return engine controls quadrant NO. 1 ENG FUEL SYS selector handle and ENG POWER CONT lever to OFF. Go to 7.	
7. Replace engine fuel pressure sensor (TM 1-2840-248-23). Go to 10.	
8. Return engine controls quadrant NO. 1 ENG FUEL SYS selector handle and ENG POWER CONT lever to OFF. Go to 9.	
9. Replace fuel tank inlet check valve (PARA 10-4-13). Go to 10.	
10. Procedure completed.	

TABLE NO. 10-2-2-6. NO. 1 PUMP Pressure Light Does Not Go Off When NO. 1 PUMP Switch Is Placed To OFF.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Place suitable container beneath left side of helicopter fuselage at No. 1 engine overboard drain. Go to 2.	
2. Make sure NO. 1 ENG WARN LTS circuit breaker is pushed in. Go to 3.	
3. Move engine controls quadrant NO. 1 ENG FUEL SYS selector handle to DIR. Go to 4.	
4. Place ENG POWER CONT lever to LOCKOUT. Fuel should not flow overboard.	

TABLE NO. 10-2-2-6. NO. 1 PUMP Pressure Light Does Not Go Off When NO. 1 PUMP Switch Is Placed To OFF. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
Step 1. If fuel does not flow overboard, go to 5 . Step 2. If fuel does flow overboard, go to 10 .
5. Return engine controls quadrant NO. 1 ENG FUEL SYS selector handle to OFF. Go to 6.
6. Disconnect P321 from No. 1 boost pump pressure switch. Go to 7.
7. Check continuity between pressure switch terminals 2 and 3.
Step 1. If continuity is present, go to 8 . Step 2. If continuity is not present, replace pressure switch (PARA 10-4-14). Go to 12 .
8. Connect P321 to No. 1 boost pump pressure switch. Go to 9.
9. Check continuity between P139-M and ground.
Step 1. If continuity is present, repair/replace wiring (Appendix F). Go to 12 . Step 2. If continuity is not present, replace fuel boost pump control panel (PARA 9-4-40). Go to 12 .
10. Return engine controls quadrant NO. 1 ENG FUEL SYS selector handle to OFF. Go to 11.
11. Replace fuel boost pump control panel (PARA 9-4-40). Go to 12.
12. Procedure completed.

TABLE NO. 10-2-2-7. NO. 2 PUMP Pressure Light Does Not Go On And #2 Or #1 FUEL PRESS Capsule Does Not Go Off.

TEST OR INSPECTION CORRECTIVE ACTION
1. Check for 115 vac between: J323-A and J323-D J323-B and J323-D J323-C and J323-D

TABLE NO. 10-2-2-7. NO. 2 PUMP Pressure Light Does Not Go On And #2 Or #1 FUEL PRESS Capsule Does Not Go Off. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
Step 1. If voltage is as specified, replace No. 2 boost pump (PARA 10-4-12). Go to 6. Step 2. If voltage is not as specified, go to 2.	2. Check continuity between J323-D and ground.
Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring (Appendix F) as required. Go to 6.	3. Check continuity between: J323-A and P139-D J323-B and P139-E J323-C and P139-F
Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F) as required. Go to 6.	4. Check for 115 vac between: P139-A and ground P139-B and ground P139-C and ground
Step 1. If voltage is as specified, replace fuel boost pump control panel (PARA 9-4-40). Go to 6. Step 2. If voltage is not as specified, go to 5.	5. Check for 115 vac between NO. 2 FUEL BOOST PUMP circuit breaker: Terminal A2 and ground Terminal B2 and ground Terminal C2 and ground
Step 1. If voltage is as specified, repair/replace wiring (Appendix F), as required, then go to 6: Terminal A2 and P139-C Terminal B2 and P139-B Terminal C2 and P139-A Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 6.	

TABLE NO. 10-2-2-7. NO. 2 PUMP Pressure Light Does Not Go On And #2 Or #1 FUEL PRESS Capsule Does Not Go Off. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
6. Procedure completed.	

TABLE NO. 10-2-2-8. #2 Or #1 FUEL PRESS Capsule Goes Off But NO. 2 PUMP Pressure Light Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Press face of NO. 2 PUMP light.	Step 1. If light goes on, go to 2. Step 2. If light does not go on, go to 5.
2. With NO. 2 PUMP switch at ON, check continuity between P139-N and ground.	Step 1. If continuity is present, replace fuel boost pump control panel (PARA 9-4-40). Go to 8. Step 2. If continuity is not present, go to 3.
3. Check continuity between P139-N and P320-1.	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8.
4. Check continuity between P320-2 and ground.	Step 1. If continuity is present, replace No. 2 boost pump pressure switch (PARA 10-4-14). Go to 8. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8.
5. Check NO. 2 PUMP switch lamp.	Step 1. If lamp is good, go to 6. Step 2. If lamp is not good, replace lamp (PARA 9-4-42). Go to 8.
6. Check for 28 vdc between P139-P and ground.	Step 1. If voltage is as specified, replace fuel boost pump control panel (PARA 9-4-40). Go to 8.

TABLE NO. 10-2-2-8. #2 Or #1 FUEL PRESS Capsule Goes Off But NO. 2 PUMP Pressure Light Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If voltage is not as specified, go to 7.
7. Check continuity between P139-P and P902-X.	Step 1. If continuity is present, troubleshoot instrument panel and consoles indicator lights dimming (PARA 9-2-10). Go to 8. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8.
8. Procedure completed.	

TABLE NO. 10-2-2-9. NO. 2 PUMP Pressure Light Goes On But #2 FUEL PRESS Capsule Does Not Go Off.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Place suitable container beneath right side of helicopter fuselage at No. 2 engine overboard drain. Go to 2.	
2. Make sure NO. 2 ENG WARN LTS circuit breaker is pushed in. Go to 3.	
3. Place engine controls quadrant NO. 2 ENG FUEL SYS selector handle in OFF and DIR positions and observe that push/pull cable actuates fuel selector valve.	Step 1. If fuel selector valve is actuated, go to 4. Step 2. If fuel selector valve is not actuated, replace push/pull cable (PARA 4-4-30). Go to 10. Step 3. If trouble remains, replace control box (PARA 4-4-31). Go to 10.
4. Place ENG POWER CONT lever to LOCKOUT. Go to 5.	
5. Check for fuel flow overboard.	Step 1. If fuel flows overboard, go to 6. Step 2. If fuel does not flow overboard, go to 8.
6. Return engine controls quadrant NO. 2 ENG FUEL SYS selector handle and ENG POWER CONT lever to OFF. Go to 7.	

TABLE NO. 10-2-2-9. NO. 2 PUMP Pressure Light Goes On But #2 FUEL PRESS Capsule Does Not Go Off. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	7. Replace engine fuel pressure sensor (TM 1-2840-248-23). Go to 10. 8. Return engine controls quadrant NO. 2 ENG FUEL SYS selector handle and ENG POWER CONT lever to OFF. Go to 9. 9. Replace fuel tank inlet check valve (PARA 10-4-13). Go to 10. 10. Procedure completed.

TABLE NO. 10-2-2-10. NO. 2 PUMP Pressure Light Does Not Go Off When NO. 2 PUMP Switch Is Placed To OFF.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. With NO. 2 PUMP switch at OFF, place suitable container beneath right side of helicopter fuselage at No. 2 engine overboard drain. Go to 2. 2. Make sure NO. 2 ENG WARN LTS circuit breaker is pushed in. Go to 3. 3. Move engine controls quadrant NO. 2 ENG FUEL SYS selector handle to DIR. Go to 4. 4. Place ENG POWER CONT lever to LOCKOUT and check that fuel does not flow overboard. - Step 1. If fuel does not flow overboard, go to 5. - Step 2. If fuel flows overboard, go to 9. 5. Return engine controls quadrant NO. 2 ENG FUEL SYS selector handle to OFF. Go to 6. 6. Disconnect P320 and check continuity between No. 2 boost pump pressure switch terminals 2 and 3. - Step 1. If continuity is present, go to 7. - Step 2. If continuity is not present, replace pressure switch (PARA 10-4-14). Go to 11.

TABLE NO. 10-2-2-10. NO. 2 PUMP Pressure Light Does Not Go Off When NO. 2 PUMP Switch Is Placed To OFF. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
7. Connect P320 to pressure switch. Go to 8.	
8. Check continuity between P139-N and ground.	
	Step 1. If continuity is present, repair/replace wiring (Appendix F). Go to 11.
	Step 2. If continuity is not present, replace fuel boost pump control panel (PARA 9-4-40). Go to 11.
9. Return engine controls quadrant NO. 2 ENG FUEL SYS selector handle to OFF. Go to 10.	
10. Replace fuel boost pump control panel (PARA 9-4-40). Go to 11.	
11. Procedure completed.	

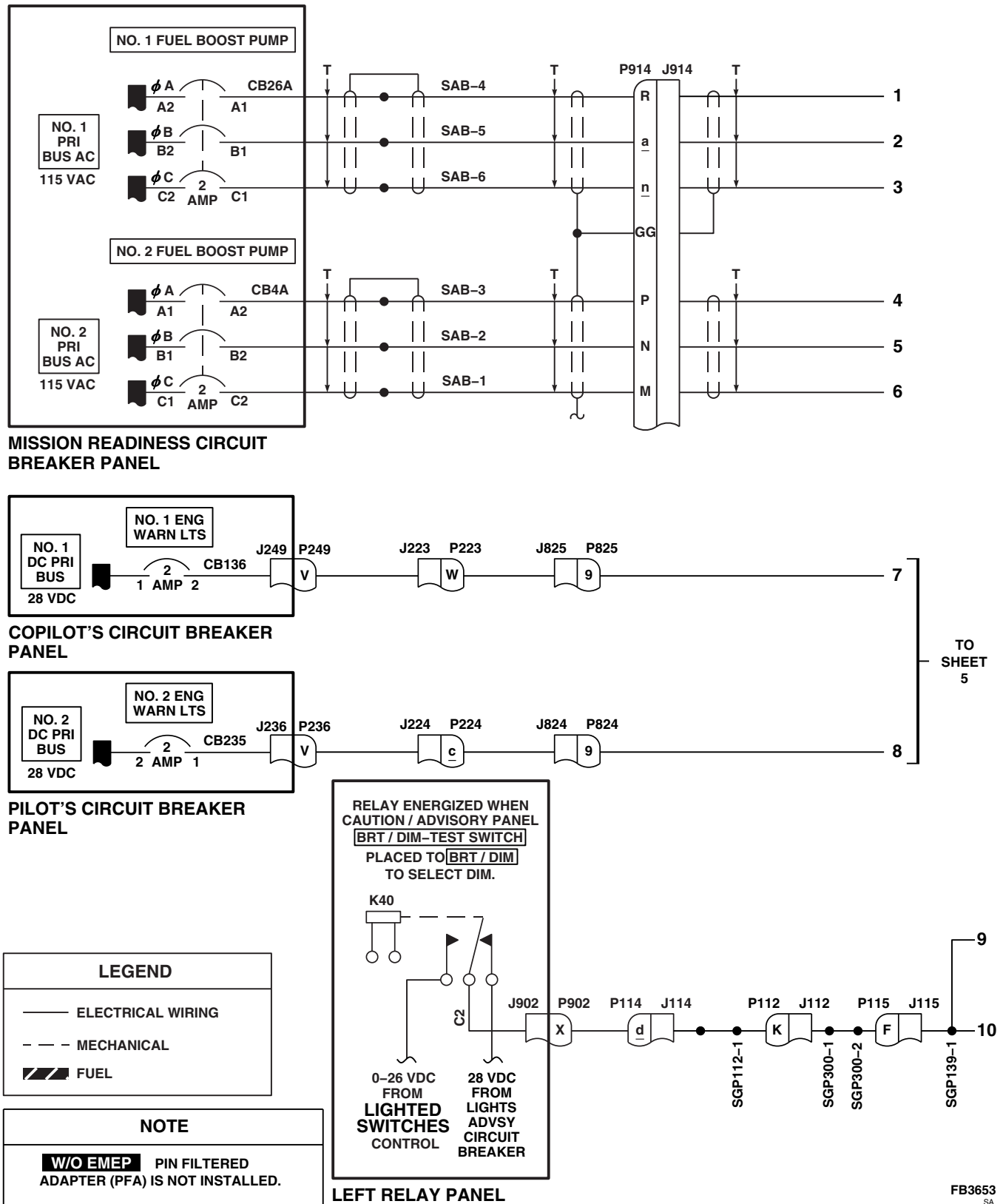


FIGURE 10-2-2-1. SUBMERGED FUEL BOOST PUMP SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 5)

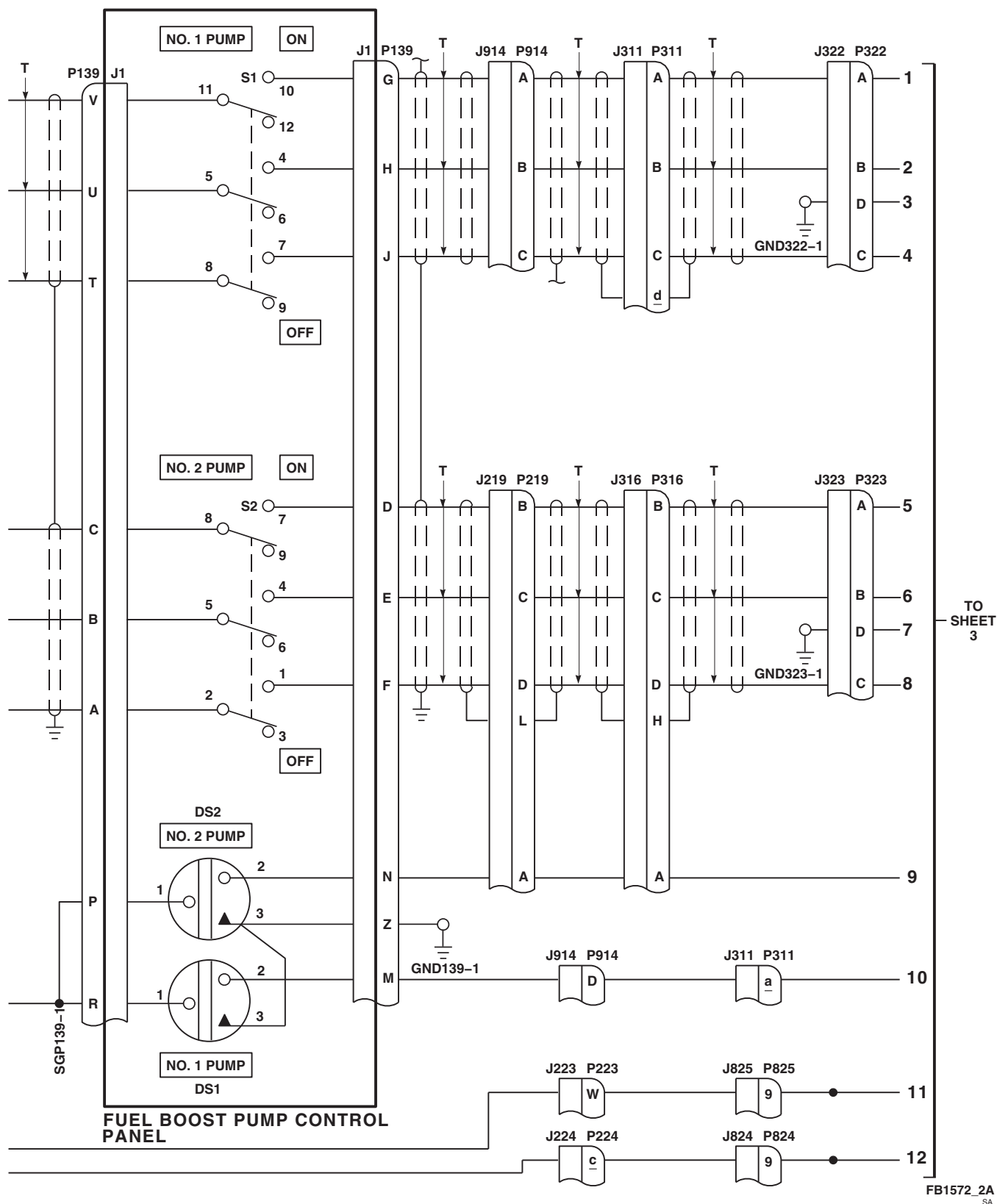


FIGURE 10-2-2-1. SUBMERGED FUEL BOOST PUMP SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 5)

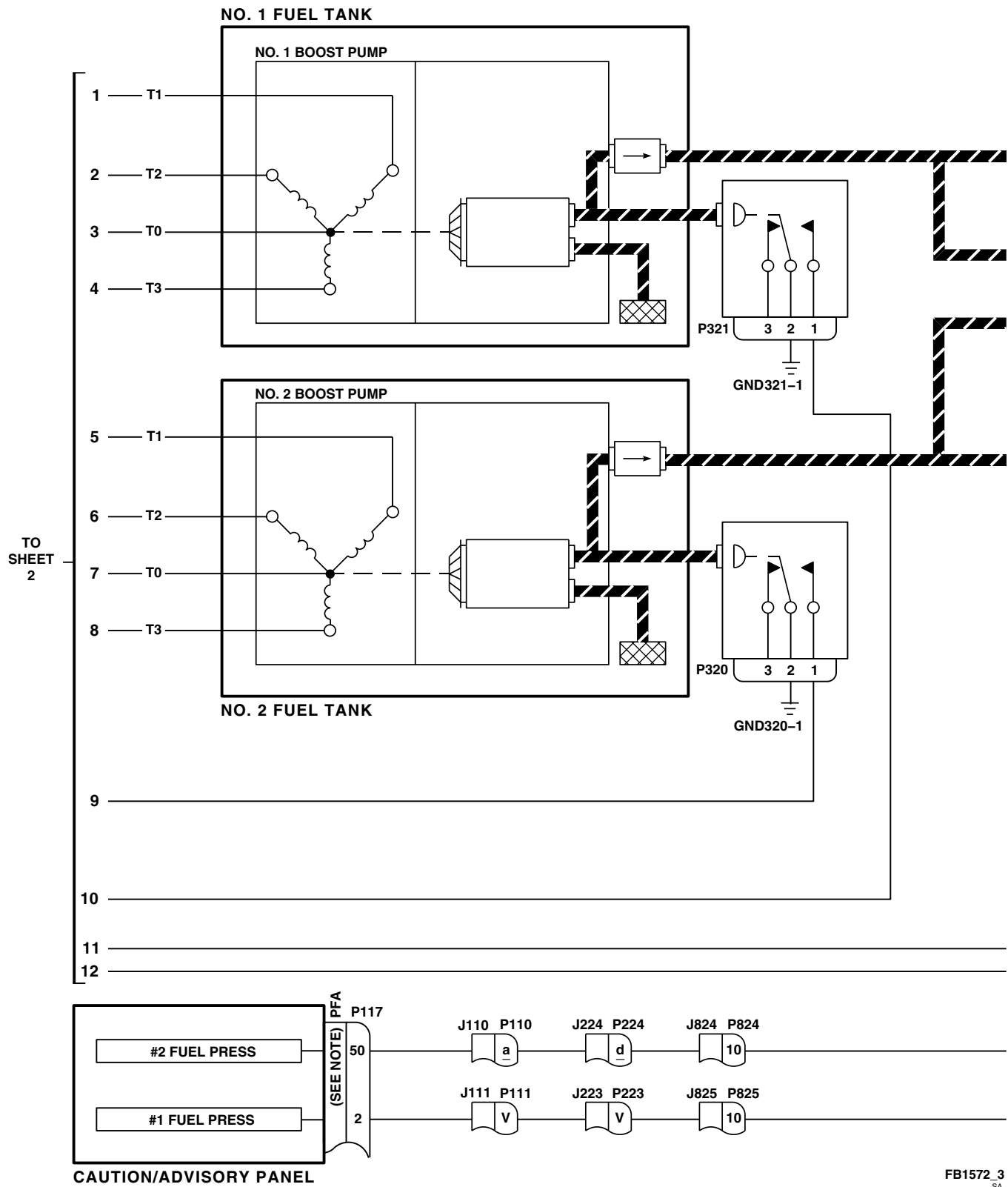


FIGURE 10-2-2-1. SUBMERGED FUEL BOOST PUMP SYSTEM SCHEMATIC DIAGRAM
(SHEET 3 OF 5)

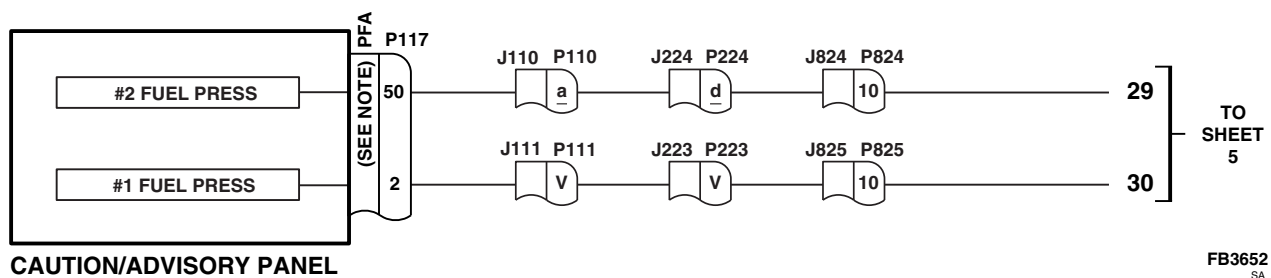
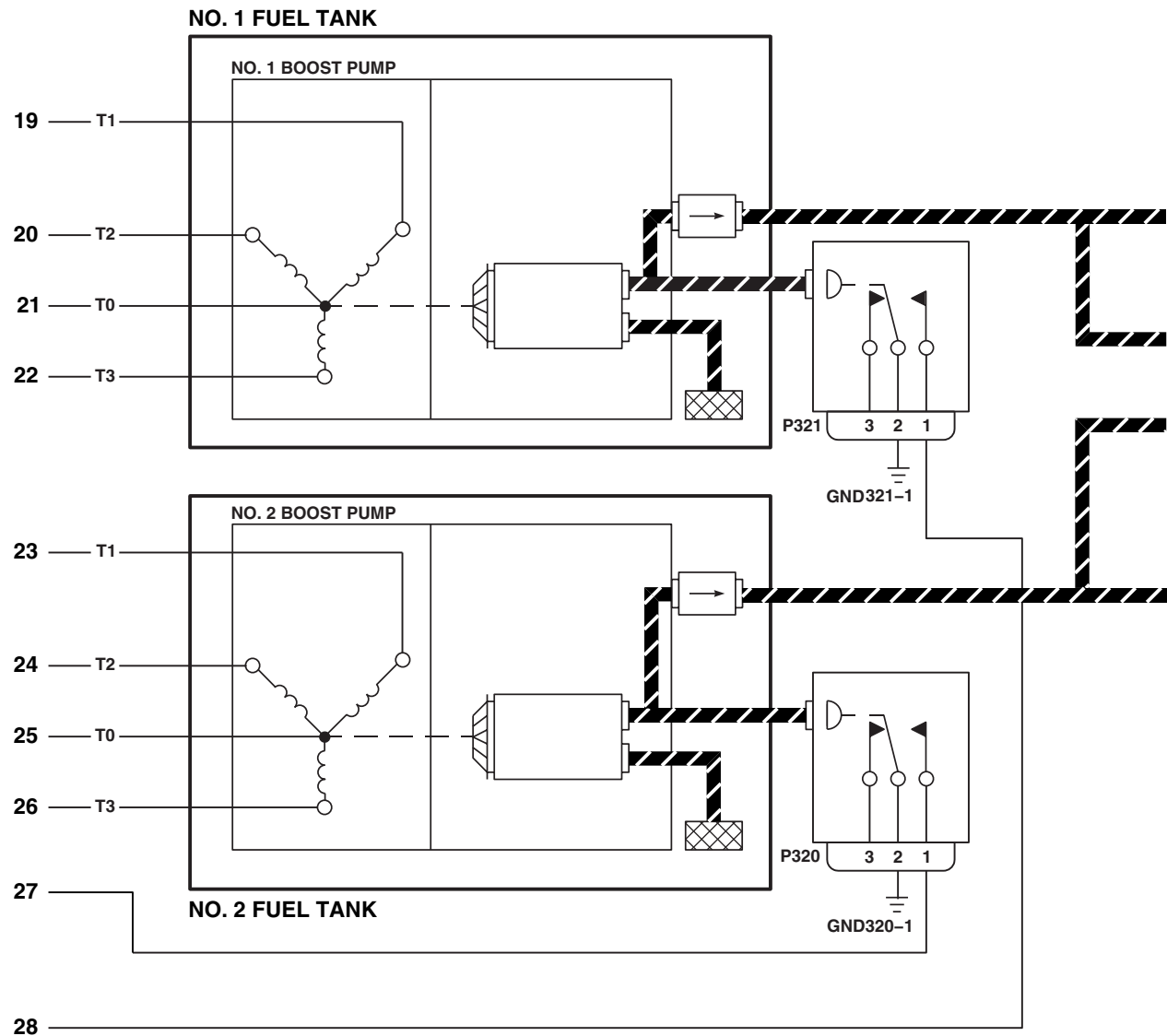


FIGURE 10-2-2-1. SUBMERGED FUEL BOOST PUMP SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 5)

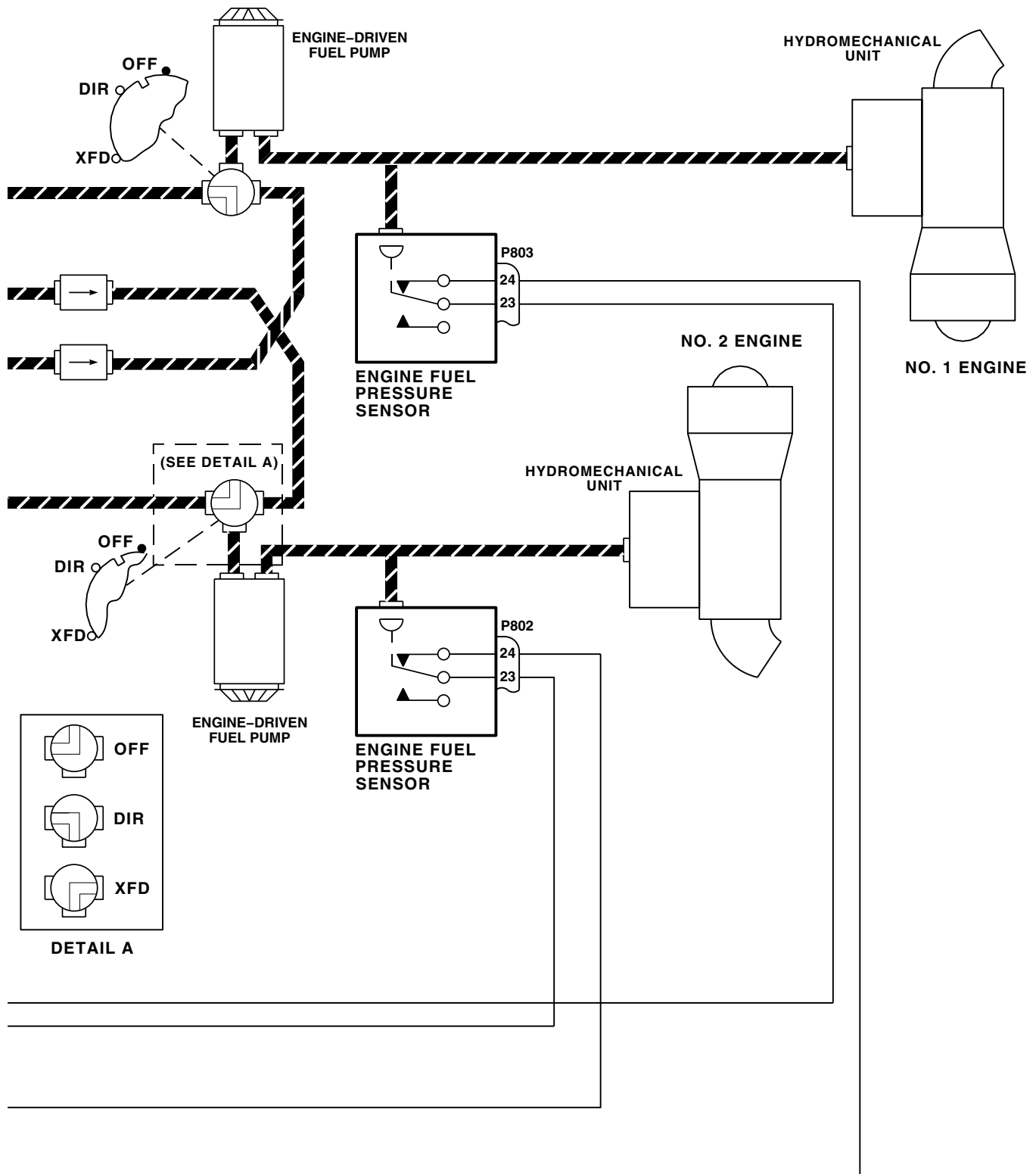
FB1572_4
SA

FIGURE 10-2-2-1. SUBMERGED FUEL BOOST PUMP SYSTEM SCHEMATIC DIAGRAM
(SHEET 5 OF 5)

10-2-3 FUEL PRIME BOOST SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer Toolkit

Materials/Parts

- Suitable Container

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available
- No. 1 Fuel Tank Serviced

WARNING

To prevent damage to equipment or injury to personnel, use normal fire precautions while doing procedures.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

10-2-3.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).

NOTE

Make sure APU T-handle (on upper console panel) is pushed in or fuel prime/boost pump will not operate.

- Place upper console BATT switch to ON.

RESULT: PRIME BOOST PUMP ON capsule on caution/advisory panel shall not go on.

- If result is not as specified, troubleshoot engine start and ignition system (PARA 4-2-8).

- Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST.

RESULT:

- All capsules on caution/advisory, pilot's and copilot's master warning panels, and all switch legends on pilot's and copilot's HSI/VSI mode select panels shall go on.
 - #1 FUEL LOW, #2 FUEL LOW, and LOW ROTOR RPM capsules shall flash.
- If results are not as specified, troubleshoot caution/advisory warning system (PARA 8-2-4).

- Release BRT/DIM-TEST switch.
- Place a rag under B-nut on APU fuel control enclosure. Loosen nut enough to allow fuel to leak.
- Place upper console FUEL PUMP switch to APU BOOST.

RESULT:

- (1) PRIME BOOST PUMP ON capsule on caution/advisory panel shall go on.
 - (2) Boost pump operation shall be heard.
 - (3) Fuel shall leak from B-nut.
- If results (1) and (2) are not as specified, go to TABLE NO. 10-2-3-1.

- If result (1) is not as specified, check helicopter configuration. **EMEP**, go to (a), **W/O EMEP**, go to (b).

- (a) **EMEP** Remove pin filtered adapter P118 (PARA 9-4-98), and check.

- 1 If pin filtered adapter is good, go to 3.

- 2 If pin filtered adapter is not good, replace it (PARA 9-4-98).

- 3 Install pin filtered adapter (PARA 9-4-98). Go to (b).

- (b) **W/O EMEP** Check for 28 vdc between P118-K and ground.

- 1 If voltage is as specified, replace caution/advisory panel (PARA 8-4-19).

- 2 If voltage is not as specified, repair/replace wiring between P118-K and S101-C (Appendix F).

- If result (2) is not as specified, go to TABLE NO. 10-2-3-2.
- If result (3) is not as specified, go to TABLE NO. 10-2-3-3.

- h. Tighten nut on APU fuel control enclosure and wipe fuel from B-nut.

- i. Place upper console FUEL PUMP switch to OFF.

- j. Move and hold engine controls quadrant NO. 1 ENG POWER CONT lever in cockpit to LOCKOUT position.

- k. Move engine controls quadrant NO. 1 ENG FUEL SYS selector lever to DIR detent.

RESULT: NO. 1 ENG FUEL SYS selector lever shall move without difficulty.

- If result is not as specified, go to TABLE NO. 10-2-3-4.

- l. Place suitable container beneath left side of helicopter fuselage at No. 1 engine overboard drain.

- m. Place upper console FUEL PUMP switch to FUEL PRIME until fuel is coming from overboard drain for 3 minutes, then place switch to OFF.

RESULT:

- (1) PRIME BOOST PUMP ON capsule shall go on.

- (2) Fuel from No. 1 engine overboard drain line shall drain into container.

- (3) Operation of boost pump shall be heard.

- If results (1), (2), and (3) are not as specified, go to TABLE NO. 10-2-3-5.

- n. Place engine controls quadrant NO. 1 ENG FUEL SYS selector lever to OFF.

- o. Pull engine controls quadrant NO. 1 ENG POWER CONT lever back to OFF.

- p. Move and hold engine controls quadrant NO. 2 ENG POWER CONT lever in cockpit to LOCKOUT position.

- q. Move engine controls quadrant NO. 2 ENG FUEL SYS selector lever in DIR detent.

RESULT: NO. 2 ENG FUEL SYS selector lever shall move without difficulty.

- If result is not as specified, go to TABLE NO. 10-2-3-4.

10-2-3-2

- r. Place suitable container beneath right side of helicopter fuselage at No. 2 engine overboard drain.
- s. Place upper console FUEL PUMP switch to FUEL PRIME until fuel is coming from overboard drain for 3 minutes, then place switch to OFF.

RESULT:

- (1) PRIME BOOST PUMP ON capsule shall go on.
- (2) Fuel from No. 2 engine overboard drain line shall drain into container.
- (3) Operation of boost pump shall be heard.
- If results (1), (2), and (3) are not as specified, go to TABLE NO. 10-2-3-6.

- t. Place engine controls quadrant NO. 2 ENG FUEL SYS selector lever to OFF.
- u. Pull engine controls quadrant NO. 2 ENG POWER CONT lever back to OFF.
- v. Press engine controls quadrant No. 1 or No. 2 engine start button.

RESULT: PRIME BOOST PUMP ON capsule on caution/advisory panel shall go on.

- If result is not as specified, troubleshoot engine start and ignition system (PARA 4-2-8).

- w. Turn off electrical power (PARA 1-6-16).
- x. Remove container from under helicopter.

TABLE NO. 10-2-3-1. PRIME BOOST PUMP ON Capsule Not On And Boost Pump Not Operating.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between APU T-handle switch S101-C and ground.	Step 1. If voltage is as specified, repair/replace wiring between switch and J231-Y (Appendix F). Go to 7. Step 2. If voltage is not as specified, go to 2.
2. Check for 28 vdc between APU T-handle switch terminal NC and ground.	Step 1. If voltage is as specified, replace switch (PARA 12-4-9). Go to 7. Step 2. If voltage is not as specified, go to 3.
3. Check for 28 vdc between FUEL PUMP switch terminal 6 and ground.	Step 1. If voltage is as specified, repair/replace wiring between FUEL PUMP switch and APU T-handle switch S101 (Appendix F). Go to 7. Step 2. If voltage is not as specified, go to 4.
4. Check for 28 vdc between FUEL PUMP switch terminal 5 and ground.	Step 1. If voltage is as specified, replace switch (PARA 9-4-5). Go to 7.

**TABLE NO. 10-2-3-1. PRIME BOOST PUMP ON Capsule Not On And Boost Pump Not Operating.
(Cont)**

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If voltage is not as specified, go to 5 .
5. Check for 28 vdc between FUEL PUMP switch terminal 2 and ground.	
	Step 1. If voltage is as specified, repair/replace wiring between switch terminal 2 and 5 (Appendix F). Go to 7 .
	Step 2. If voltage is not as specified, go to 6 .
6. Check for 28 vdc between FUEL PRIME BOOST circuit breaker terminal 1 and ground.	
	Step 1. If voltage is as specified, repair/replace wiring between circuit breaker and switch (Appendix F). Go to 7 .
	Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 7 .
7. Procedure completed.	

TABLE NO. 10-2-3-2. PRIME BOOST PUMP ON Capsule On But Boost Pump Not Operating.

TEST OR INSPECTION	CORRECTIVE ACTION
1. With FUEL PUMP switch placed to APU BOOST, check for 28 vdc between P303-A and P303-B.	
	Step 1. If voltage is as specified, replace prime boost pump (PARA 10-4-20). Go to 3 .
	Step 2. If voltage is not as specified, go to 2 .
2. Check for 28 vdc between P303-A and ground.	
	Step 1. If voltage is as specified, repair/replace wiring between P303-B and ground (Appendix F). Go to 3 .
	Step 2. If voltage is not as specified, repair/replace wiring between P303-A and APU T-handle switch S101-C (Appendix F). Go to 3 .
3. Procedure completed.	

TABLE NO. 10-2-3-3. PRIME BOOST PUMP ON Capsule On, Pump Operating, But No Fuel Coming Out Of B-Nut.

TEST OR INSPECTION CORRECTIVE ACTION
1. Check breakaway valve for visible yellow line in center of valve. Step 1. If yellow line is visible, replace breakaway valve (PARA 10-4-22). Go to 6. Step 2. If yellow line is not visible, go to 2. 2. With FUEL PUMP switch placed to APU BOOST, check for 28 vdc between P306-A and P306-C. Step 1. If voltage is as specified, replace APU fuel prime shutoff valve (PARA 15-4-10). Go to 6. Step 2. If voltage is not as specified, go to 3. 3. Check for 28 vdc between P306-A and ground. Step 1. If voltage is as specified, repair/replace wiring between P306-C and ground (Appendix F). Go to 6. Step 2. If voltage is not as specified, go to 4. 4. Check continuity between P306-A and P241-S. Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 6. 5. Check continuity between P241-T and APU T-handle switch S101-C. Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to 6. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 6. 6. Procedure completed.

TABLE NO. 10-2-3-4. No. 1 Or No. 2 ENG FUEL SYS Selector Lever Does Not Move Or Is Difficult To Move.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Inspect inside of engine controls quadrant for loose items that may cause interference. Go to 2.	
2. Disconnect push/pull cable from control box. Go to 3.	
3. Cycle ENG FUEL SYS selector lever back and forth. Movement of lever shall be smooth.	Step 1. If movement is smooth, replace control box or fuel selector valve (PARA 4-4-31 or PARA 10-4-25). Go to 4. Step 2. If movement is not smooth, replace fuel selector push/pull cable (PARA 4-4-30). Go to 4.
4. Procedure completed.	

TABLE NO. 10-2-3-5. PRIME BOOST PUMP ON Capsule Not On, No Fuel Coming Out Of No. 1 Engine Fuel Drain, And Pump Not Operating.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Visually check drain tube for obvious obstructions.	Step 1. If obstructions are visible, remove obstructions. Go to 10. Step 2. If obstructions are not visible, go to 2.
2. Position ENG POWER CONT lever to LOCKOUT. As lever is advanced to LOCKOUT, check to hear a click in hydromechanical control unit (HMU).	Step 1. If click is heard in HMU, go to 3. Step 2. If click is not heard in HMU, go to 9.
3. Check push/pull cable to fuel selector valve for proper rigging (PARA 4-4-33).	Step 1. If push/pull cable is properly rigged, go to 4. Step 2. If push/pull cable is not properly rigged, correctly rig push/pull cable (PARA 4-4-33). Go to 10.

TABLE NO. 10-2-3-5. PRIME BOOST PUMP ON Capsule Not On, No Fuel Coming Out Of No. 1 Engine Fuel Drain, And Pump Not Operating. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
4. With FUEL PUMP switch pressed to FUEL PRIME, check for 28 vdc between P304-A and P304-C.	Step 1. If voltage is as specified, replace No. 1 engine prime shutoff valve (PARA 10-4-15). Go to 10. Step 2. If voltage is not as specified, go to 5.
5. With FUEL PUMP switch pressed to FUEL PRIME, check for 28 vdc between P304-A and ground.	Step 1. If voltage is as specified, repair/replace wiring (Appendix F) between, then go to 10: P304-B and ground P304-C and ground Step 2. If voltage is not as specified, go to 6.
6. With FUEL PUMP switch pressed to FUEL PRIME, check for 28 vdc between TB1-8 and ground.	Step 1. If voltage is as specified, repair/replace wiring between TB1-8 and P304-A (Appendix F). Go to 10. Step 2. If voltage is not as specified, go to 7.
7. With FUEL PUMP switch pressed to FUEL PRIME, check for 28 vdc between TB2-9 and ground.	Step 1. If voltage is as specified, replace diode board (PARA 9-4-43). Go to 10. Step 2. If voltage is not as specified, go to 8.
8. Check continuity between FUEL PUMP switch terminal 1 and TB2-9.	Step 1. If continuity is present, replace switch S19 (PARA 9-4-5). Go to 10. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 10.
9. Check engine control for restrictions/proper rigging at LOCKOUT (PARA 4-4-33).	Step 1. If engine controls are properly rigged, replace HMU (SEI 671 and TM 55-2840-248-23). Go to 10. Step 2. If engine controls are not properly rigged, rig engine controls (PARA 4-4-33). Go to 10.

TABLE NO. 10-2-3-5. PRIME BOOST PUMP ON Capsule Not On, No Fuel Coming Out Of No. 1 Engine Fuel Drain, And Pump Not Operating. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

10. Procedure completed.

TABLE NO. 10-2-3-6. PRIME BOOST PUMP ON Capsule Not On, No Fuel Coming Out Of No. 2 Engine Fuel Drain, And Pump Not Operating.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Visually check drain tube for obvious obstructions.

Step 1. If obstructions are visible, remove obstructions. Go to **8**.

Step 2. If obstructions are not visible, go to **2**.

2. Position ENG POWER CONT lever to LOCKOUT. As lever is advanced to LOCKOUT, check to hear click in HMU.

Step 1. If click is heard in HMU, go to **3**.

Step 2. If click is not heard in HMU, go to **7**.

3. Check push/pull cable to fuel selector valve for proper rigging (PARA 4-4-33).

Step 1. If push/pull cable is properly rigged, go to **4**.

Step 2. If push/pull cable is not properly rigged, correctly rig push/pull cable (PARA 4-4-33). Go to **8**.

4. With FUEL PUMP switch pressed to FUEL PRIME, check for 28 vdc between P305-A and P305-C.

Step 1. If voltage is as specified, replace No. 2 engine prime shutoff valve (PARA 10-4-15). Go to **8**.

Step 2. If voltage is not as specified, go to **5**.

5. With FUEL PUMP switch pressed to FUEL PRIME, check for 28 vdc between P305-A and ground.

Step 1. If voltage is as specified, repair/replace wiring (Appendix F) between, then go to **8**:

TABLE NO. 10-2-3-6. PRIME BOOST PUMP ON Capsule Not On, No Fuel Coming Out Of No. 2 Engine Fuel Drain, And Pump Not Operating. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	P305-B and ground
	P305-C and ground
	Step 2. If voltage is not as specified, go to 6 .
6.	With FUEL PUMP switch pressed to FUEL PRIME, check for 28 vdc between TB2-8 and ground.
	Step 1. If voltage is as specified, repair/replace wiring between TB2-8 and P305-A (Appendix F). Go to 8 .
	Step 2. If voltage is not as specified, replace diode board (PARA 9-4-43). Go to 8 .
7.	Check engine control for restrictions/proper rigging at LOCKOUT (PARA 4-4-33).
	Step 1. If engine controls are properly rigged, replace HMU (SEI 671 and TM 55-2840-248-23). Go to 8 .
	Step 2. If engine controls are not properly rigged, rig engine controls (PARA 4-4-33). Go to 8 .
8.	Procedure completed.

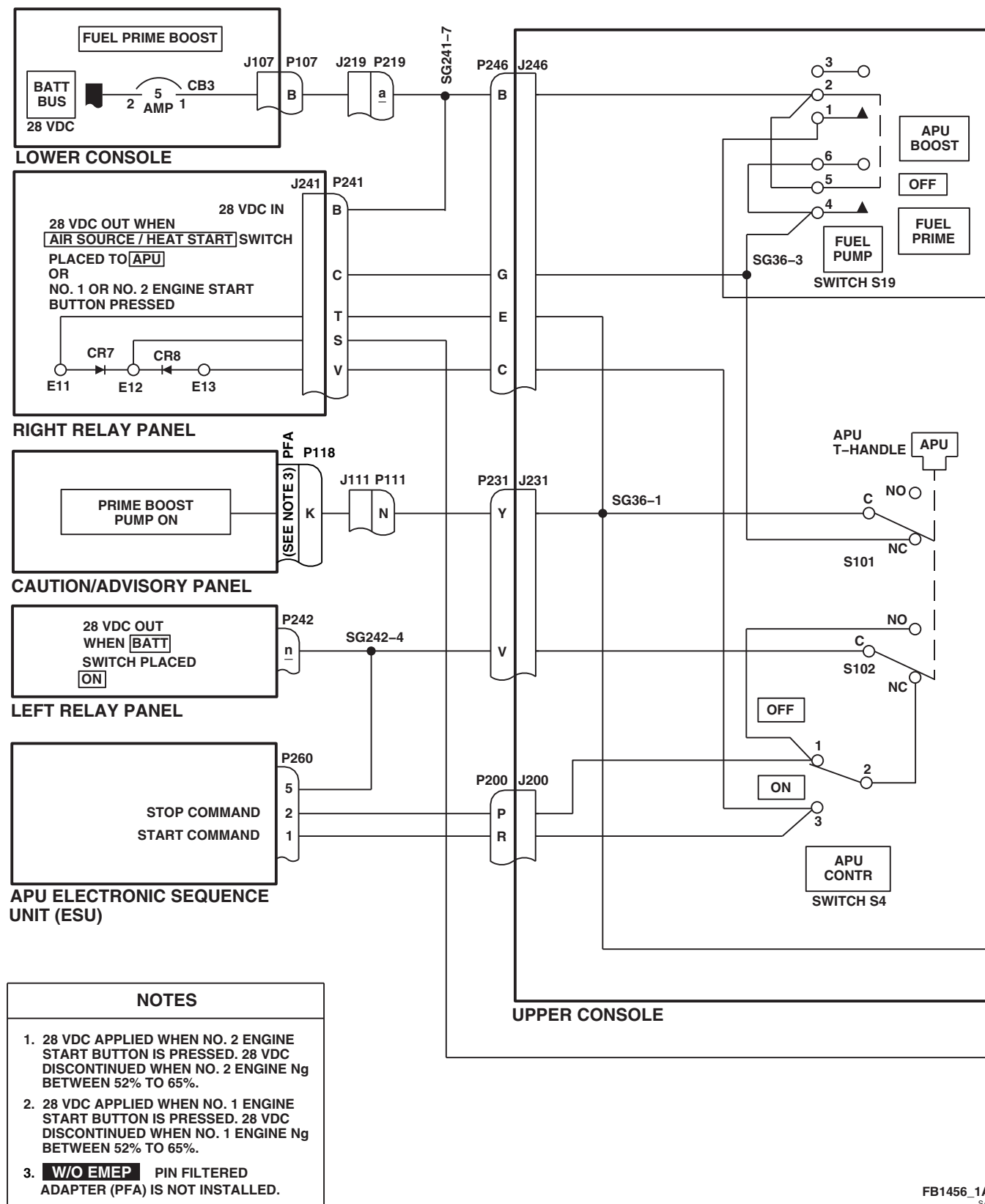


FIGURE 10-2-3-1. FUEL PRIME BOOST SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 3)

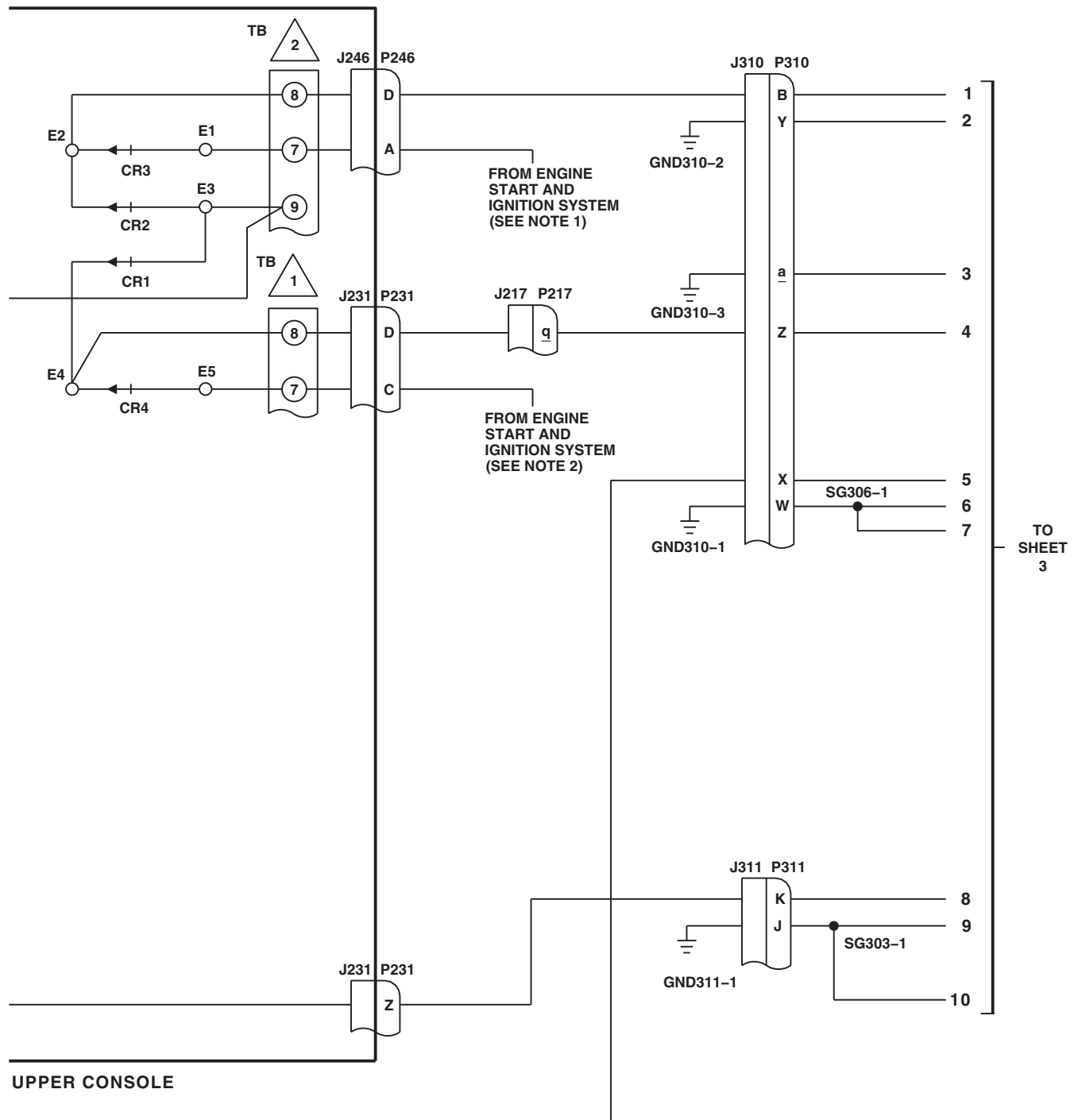


FIGURE 10-2-3-1. FUEL PRIME BOOST SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 3)

FB1456_2
SA

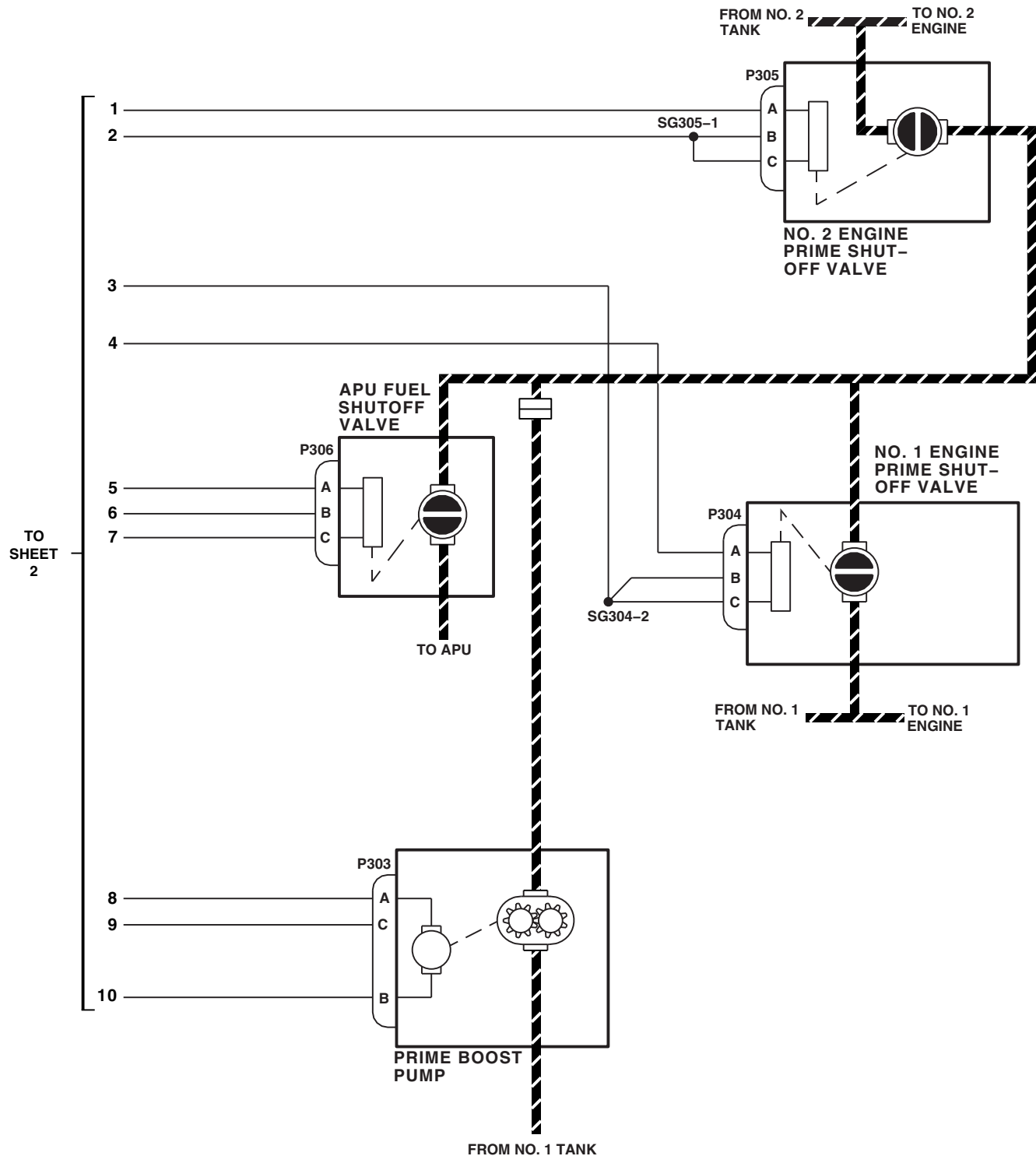
FB1456.3
SA

FIGURE 10-2-3-1. FUEL PRIME BOOST SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 3)

10-2-4 PRESSURE REFUEL/DEFUEL SYSTEM.

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 1-3-17
- PARA 1-3-20
- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available



Use extreme care when removing gravity fill port cap; fuel may come out with pressure.

Special Instructions

The pressure refuel/defuel system operational/troubleshooting procedure is subdivided as follows:

- (1) Pressure Fueling
- (2) Pressure Defueling

10-2-4.1 Pressure Fueling.

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).

NOTE

- Monitor fuel quantity indication (on main instrument panel) during pressure fueling to verify that fuel is flowing into tank.
- When pressure fueling is completed, remove gravity fill port cap to verify proper high-level shutoff valve operation.
- c. Pressure fuel both tanks (PARA 1-3-17).

RESULT:

- (1) When the tanks are full, the high-level shutoff valve shall cause the pressure-fueling valve to close.

- (2) Fuel flow from the refueling truck shall stop. At this time there shall be a noticeable jump from the fueling hose.
 - (3) Fuel shall not come from the vent overboard tube at the bottom of the helicopter.
 - (4) When the gravity fill port cap is removed, fuel shall not spill out.
- For minor spills from gravity fill cap, first check if helicopter is in ground level attitude, then check installation of high level sensor.
 - If result (3) or (4) is not as specified, replace both high-level shutoff valves in overfilled fuel tank (PARA 10-4-4).
 - (a) If trouble remains, replace refuel/defuel valve in bottom of tank (PARA 10-4-5).
 - If only tank No. 2 fills, check for hanging floats in high level sensor of tank No. 1. Replace pressure fueling valve if floats are not hung up (PARA 10-4-5).

- If only tank No. 1 fills, check for hanging float in high level sensor of tank No. 2, then visually inspect the breakaway interconnect valve for activating and sealing off tank No. 2. If breakaway interconnect valve is broken, replace valve (PARA 10-4-18).

d. Turn off electrical power (PARA 1-6-16).

10-2-4.2 Pressure Defueling.

a. Turn on electrical power (PARA 1-6-16).

NOTE

Monitor fuel quantity indication (on main instrument panel) during defueling to verify that fuel is leaving tank.

b. Pressure defuel both tanks (PARA 1-3-20).

RESULT: Both tanks shall empty at about the same time.

- If result is not as specified (as shown on fuel quantity indicator), do the following checks on tank with fuel remaining:

(a) Visually check breakaway vent valve at top of tank. If a yellow paint stripe can be seen, replace vent valve (PARA 10-4-17).

(b) If yellow paint stripe is not seen on vent valve, replace low-level shutoff valve (PARA 10-4-9).

(c) If tank still cannot be properly pressure-defueled, replace pressure fuel/defuel valve (PARA 10-4-5).

(d) If tank No. 2 does not empty, visually check breakaway interconnect valve to verify if valve has activated and sealed off tank No. 2. If breakaway interconnect valve is broken, replace valve (PARA 10-4-18).

c. Turn off electrical power (PARA 1-6-16).

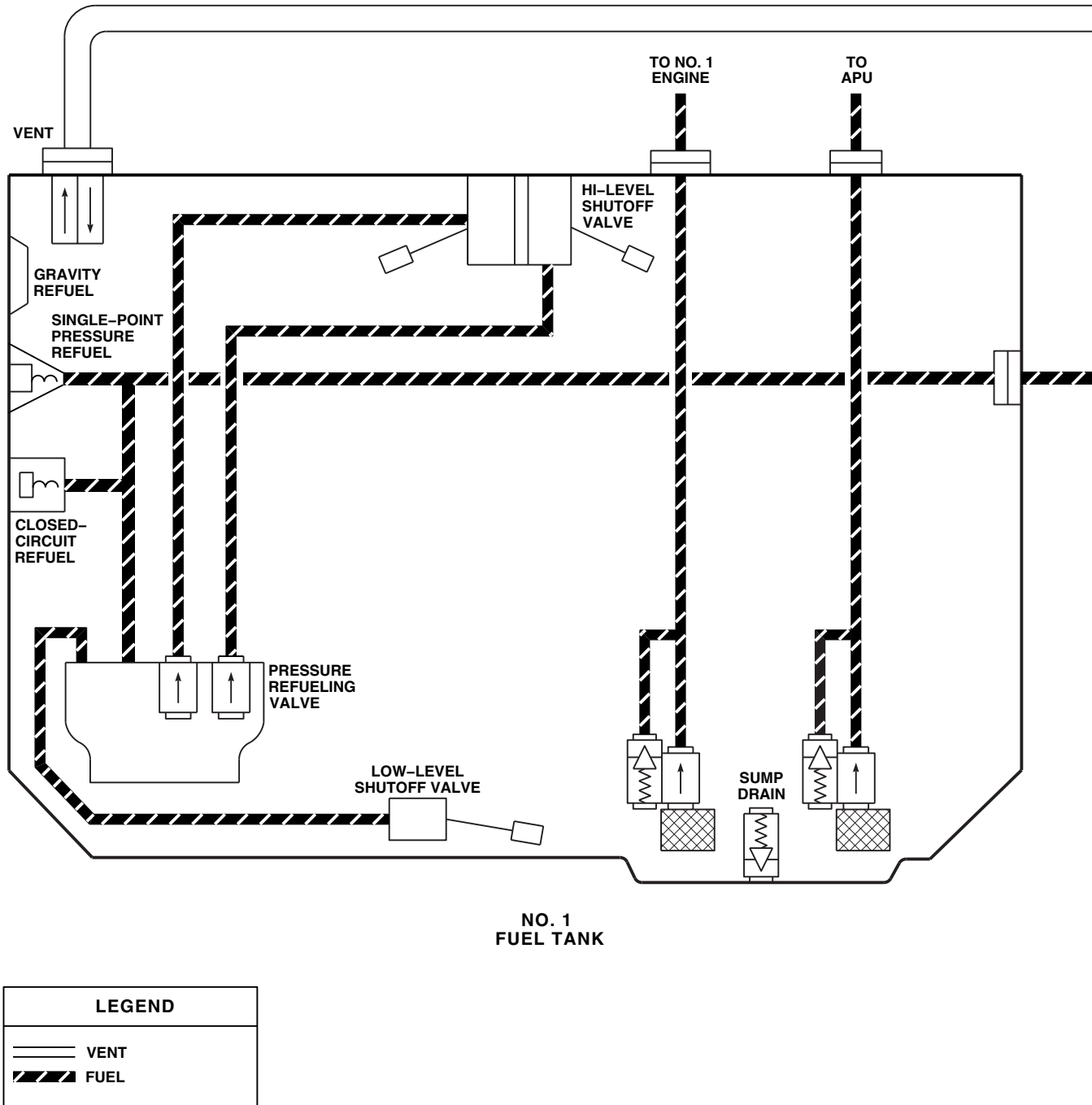


FIGURE 10-2-4-1. PRESSURE REFUEL/DEFUEL SYSTEM SCHEMATIC DIAGRAM
 (SHEET 1 OF 2)

AA2798_1
 SA

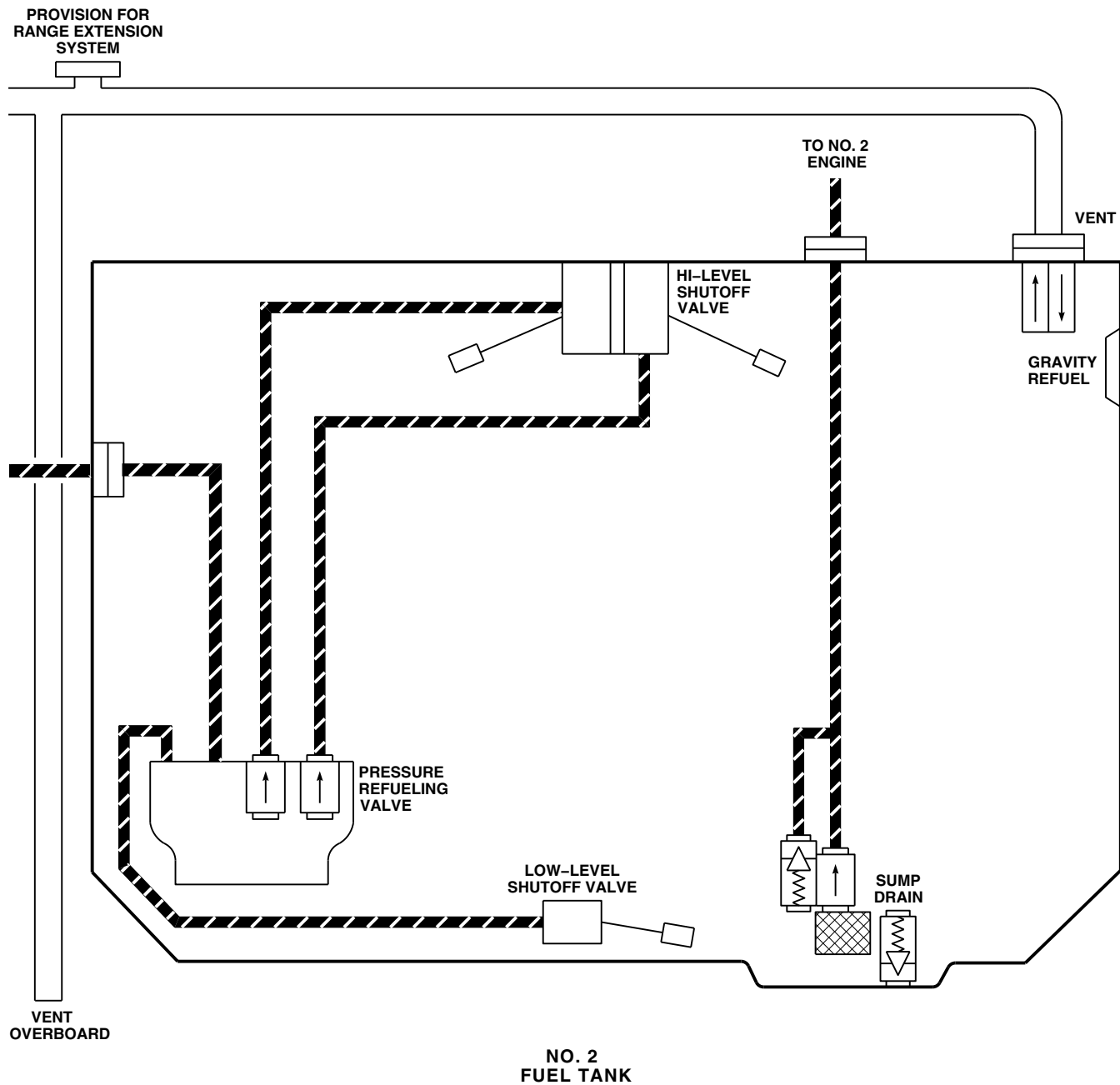
AA2798_2
SA

FIGURE 10-2-4-1. PRESSURE REFUEL/DEFUEL SYSTEM SCHEMATIC DIAGRAM
(SHEET 2 OF 2)

10-2-4-4

10-2-5 FUEL QUANTITY SYSTEM (USING TEST SET PSD-40).

INITIAL SETUP

Test Equipment

- AC/DC Capacitance Gauging System Test Set, PSD 40-1
- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit
- Fuel Quantity Gauging System Test Set Adapter Harness, PSD 40-542-1

References

- Operating Instructions, PSD 40-1
- PARA 1-3-18 or PARA 1-3-17
- PARA 1-6-16
- PARA 2-4-69

Equipment Conditions

- External Electrical Power Available
- Fuel Tanks Empty And Purged

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

Special Instructions

The fuel quantity system operational/troubleshooting procedure is subdivided as follows:

- (1) Setup
- (2) Test Harness Capacitance
- (3) Tank Probe Capacitance
- (4) Tank Probe Insulation Resistance
- (5) Tank Probe Empty Adjustment
- (6) Tank Probe Full Adjustment
- (7) System Check
- (8) Shutdown

The operational/troubleshooting procedure must be performed in sequence from 1 to 8.

10-2-5.1 Setup.

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).
- Connect test set CHASSIS (J9) to any convenient ground.

- Place test set power switch (S1) to ON and allow 3 minutes for warmup.

NOTE

If colons and decimal points flash on test set digital display, replace test set battery.

- Perform capacitance functional self-test using procedures contained in test set operating instructions.

10-2-5.2 Test Harness Capacitance.**NOTE**

- Capacitance check is the same for TANK 1 and TANK 2 harness.
 - Do not connect adapter harness J1 and P1 to aircraft when checking harness capacitance. Do not allow J1 and P1 to touch any conductive surface or each other.
- Connect adapter harness TANK 1 H1 lead to test set HI Z (J5) and T lead to test set LO Z (J4).
 - Place test set control switches as follows:

CONTROL	POSITION
FUNCTION SEL	EXT CAP
FUNCTION MEASURE	TU LOZ-HIZ

RESULT: Test set digital display shall indicate between 0 and 3 pF. Record indication.

- If result is not as specified, replace adapter harness and restart operational/troubleshooting procedure.
- Disconnect TANK 1 leads from test set.
 - Connect adapter harness TANK 2 H2 lead to test set HI Z (J5) and T lead to test set LO Z (J4).
 - Place test set control switches as follows:

CONTROL	POSITION
FUNCTION SEL	EXT CAP
FUNCTION MEASURE	TU LOZ-HIZ

RESULT: Test set digital display shall indicate between 0 and 3 pF. Record indication.

- If result is not as specified, replace adapter harness and restart operational/troubleshooting procedure.

10-2-5.3 Tank Probe Capacitance.

- Remove soundproofing to gain access to fuel quantity signal conditioner (PARA 2-4-69).
- With adapter harness TANK 2 leads connected to test set, disconnect P905 from helicopter and connect it to adapter harness J1.
- Make sure test set control switches are as follows:

CONTROL	POSITION
FUNCTION SEL	EXT CAP
FUNCTION MEASURE	TU LOZ-HIZ

- Record No. 2 tank probe capacitance.
- Subtract TANK 2 harness capacitance recorded in Test Harness Capacitance, step e.

RESULT: Calculation shall be between 69.1 and 72.1 pF.

- If result is not as specified, go to TABLE NO. 10-2-5-1.
- Disconnect TANK 2 leads from test set and connect TANK 1 H1 lead to test set HIZ (J5) and TANK 1 T lead to test set LOZ (J4).
 - Record No. 1 tank probe capacitance.
 - Subtract TANK 1 harness capacitance recorded in Test Harness Capacitance, step b.

RESULT: Calculation shall be between 69.1 and 72.1 pF.

- If result is not as specified, go to TABLE NO. 10-2-5-1.

10-2-5-2

10-2-5.4 Tank Probe Insulation Resistance.

- a. Place test set control switches as follows:

CONTROL	POSITION
FUNCTION SEL	MEG
FUNCTION MEASURE	HIZ-GND

RESULT: Test set digital display shall indicate greater than 10 megohms.

- If result is not as specified, go to TABLE NO. 10-2-5-2.
- b. Place test set FUNCTION MEASURE control switch to LOZ-GND.
- RESULT: Test set digital display shall indicate greater than 10 megohms.
- If result is not as specified, go to TABLE NO. 10-2-5-2.
- c. Disconnect TANK 1 leads from test set and connect TANK 2 H2 lead to test set HIZ (J5) and the T lead to test set LOZ (J4).

- d. Place test set control switches as follows:

CONTROL	POSITION
FUNCTION SEL	MEG
FUNCTION MEASURE	HIZ-GND

RESULT: Test set digital display shall indicate greater than 10 megohms.

- If result is not as specified, go to TABLE NO. 10-2-5-2.
- e. Place test set FUNCTION MEASURE control switch to LOZ-GND.

RESULT: Test set digital display shall indicate greater than 10 megohms.

- If result is not as specified, go to TABLE NO. 10-2-5-2.

- f. Place test set control switches as follows:

CONTROL	POSITION
FUNCTION SEL	INT CAP
FUNCTION MEASURE	TU LOZ-HIZ

- g. Disconnect TANK 2 leads from test set.

10-2-5.5 Tank Probe Empty Adjustment.

- a. Connect adapter harness P1 to J905 of signal conditioner.
- b. Connect multimeter to TANK 1 HI and LO test points on adapter harness.
- c. Loosen retaining screws on signal conditioner adjustment cover plate and slide plate back to expose adjustment potentiometers.
- d. Adjust TK 1 E potentiometer until multimeter indicates between -0.005 and 0.005 vdc.

RESULT:

- (1) Multimeter shall indicate between -0.005 and 0.005 vdc.
- (2) FUEL QTY 1 vertical indicator on CDU shall indicate 0 pounds of fuel (disregard digital fuel quantity readout on CDU at this time).

- If result (1) is not as specified, replace signal conditioner (PARA 10-4-36).
- If result (2) is not as specified, troubleshoot instrument display system (PARA 8-2-3).

- e. Connect multimeter to TANK 2 HI and LO test points on adapter harness.

- f. Adjust TK 2 E potentiometer until multimeter indicates between -0.005 and 0.005 vdc.

RESULT:

- (1) Multimeter shall indicate between -0.005 and 0.005 vdc.
 - (2) FUEL QTY 2 vertical indicator on CDU shall indicate 0 pounds of fuel.
 - (3) TOTAL FUEL or MAIN FUEL digital display on CDU shall indicate 0 pounds of fuel.
- If result (1) is not as specified, replace signal conditioner (PARA 10-4-36).
 - If results (2) and (3) are not as specified, troubleshoot instrument display system (PARA 8-2-3).

10-2-5.6 Tank Probe Full Adjustment.

- a. Place test set control switches as follows:

CONTROL	POSITION
FUNCTION SEL	INT CAP
FUNCTION MEASURE	TU LOZ-HIZ
SIMULATOR TU (x100 PF)	00

- b. Connect TANK 1 H1 lead to test set HIZ (J2) and T lead to test set LOZ (J1). Attach shorting cap to test set COMP (J3).
- c. Subtract TANK 1 harness capacitance recorded in Test Harness Capacitance, step b., from 39.9 to 41.3 pF to determine simulated HI full capacitance.

NOTE

It is important to adjust SIMULATOR TU 0-200 PF control knob as accurately as possible.

- d. Adjust test set SIMULATOR TU 0-200 PF control knob to show on test set digital display the calculation recorded in step c.
- e. Place test set FUNCTION SEL control switch to SIM.
- f. Connect multimeter to TANK 1 H1 and LO test points on adapter harness.
- g. Adjust TK 1 F potentiometer on signal conditioner until FUEL QTY 1 vertical indicator on CDU indicates 1150 pounds.
- h. Slowly adjust TK 1 F potentiometer until FUEL QTY 1 vertical indicator lights a steady 1200 pounds indication.

RESULT:

- (1) TOTAL FUEL or MAIN FUEL digital display on CDU shall indicate 1200 pounds.
 - (2) Multimeter shall indicate between 5.97 and 6.03 vdc.
- If result (1) is not as specified, troubleshoot instrument display system (PARA 8-2-3).
 - If result (2) is not as specified, replace signal conditioner (PARA 10-4-36).
- i. Press and hold FUEL IND TEST pushbutton on the miscellaneous switch panel.

RESULT: TOTAL FUEL or MAIN FUEL digital display and FUEL QTY 1 vertical indicators shall change.

- If result is not as specified, go to TABLE NO. 10-2-5-3.
- j. Release FUEL IND TEST pushbutton.

RESULT:

- (1) TOTAL FUEL or MAIN FUEL digital display and FUEL QTY 1 vertical indicators shall indicate between 1180 and 1220 pounds.

10-2-5-4

- (2) Multimeter shall indicate between 5.97 and 6.03 vdc.
- If result (1) is not as specified, replace miscellaneous switch panel (PARA 9-4-7).
 - (a) If trouble remains, replace signal conditioner (PARA 10-4-36).
- k. Disconnect TANK 1 leads from test set and connect TANK 2 H2 lead to test set HIZ (J2) and T lead to test set LOZ (J1).
- l. Place test set FUNCTION SEL control switch to INT CAP.
- m. Subtract TANK 2 harness capacitance recorded in Test Harness Capacitance, step e., from 39.9 to 41.3 pF to determine simulated TANK 2 full capacitance.

NOTE

It is important to adjust SIMULATOR TU 0-200 PF control knob as accurately as possible.

- n. Adjust test set SIMULATOR TU 0-200 PF control knob to show on test set digital display the calculation recorded in step m.
- o. Place test set FUNCTION SEL control switch to SIM.
- p. Connect multimeter to TANK 2 HI and LO test points on adapter harness.
- q. Adjust TK 2 F potentiometer on signal conditioner until FUEL QTY 2 vertical indicator on CDU indicates 1150 pounds.
- r. Slowly adjust TK 2 F potentiometer until FUEL QTY 2 vertical indicator lights a steady 1200 pounds indication.

RESULT:

- (1) TOTAL FUEL or MAIN FUEL digital display on CDU shall indicate 1200 pounds.

- (2) Multimeter shall indicate between 5.97 and 6.03 vdc.
- If result (1) is not as specified, troubleshoot instrument display system (PARA 8-2-3).
- If result (2) is not as specified, replace signal conditioner (PARA 10-4-36).
- s. Press and hold FUEL IND TEST pushbutton on the miscellaneous switch panel.

RESULT: TOTAL FUEL or MAIN FUEL digital display and FUEL QTY 2 vertical indicators shall change.

- If result is not as specified, go to TABLE NO. 10-2-5-3.
- t. Release FUEL IND TEST pushbutton.

RESULT:

- (1) TOTAL FUEL or MAIN FUEL digital display and FUEL QTY 2 vertical indicators shall indicate between 1180 and 1220 pounds.
- (2) Multimeter shall indicate 5.97 to 6.03 vdc.
- If result (1) is not as specified, replace miscellaneous switch panel (PARA 9-4-7).
 - (a) If trouble remains, replace signal conditioner (PARA 10-4-36).
- u. Disconnect TANK 2 leads from test set.

10-2-5.7 System Check.

- a. Turn off test set.
- b. Turn off electrical power (PARA 1-6-16).
- c. Disconnect adapter harness from signal conditioner and reconnect P905 to signal conditioner.
- d. Turn on electrical power (PARA 1-6-16).

RESULT: FUEL QTY 1 and 2 vertical indicators shall indicate 0 pounds.

- If result is not as specified, perform Tank Probe Empty Adjustment.

10-2-5.8 Shutdown.

- Slide signal conditioner cover plate back into place and tighten retaining screws.

- Install soundproofing (PARA 2-4-69).

- Turn off electrical power (PARA 1-6-16).

- Refuel main tanks (PARA 1-3-18 or PARA 1-3-17).

TABLE NO. 10-2-5-1. Probe Capacitance Is Not As Specified.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between: P905-M and J307-D P905-N and J308-D P905-B and J307-G P905-D and J308-G	Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, go to 3.
2. Check continuity of fuel tank wiring harness.	Step 1. If continuity is present, replace fuel quantity probe (PARA 10-4-33). Go to 4. Step 2. If continuity is not present, replace fuel tank wiring harness (PARA 10-4-34). Go to 4.
3. Check continuity on miscellaneous switch panel between: J135-L and J135-K J135-H and J135-J	Step 1. If continuity is present, repair/replace wiring (Appendix F) as required, between, then go to 4: P135-J and J307-D P135-H and P905-M P135-K and J308-D P135-L and P905-N P905-B and J307-G P905-D and J308-G Step 2. If continuity is not present, replace miscellaneous switch panel (PARA 9-4-7). Go to 4.
4. Procedure completed.	

TABLE NO. 10-2-5-2. Multimeter Does Not Indicate Greater Than 10 Megohms.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for greater than 10 megohms between:	P905-B and P905-A P905-B and P905-P P905-D and P905-E P905-D and P905-R P307-G and P307-F P307-G and P307-H P307-G and P307-K P307-G and P307-E P308-G and P308-F P308-G and P308-H P308-G and P308-K P308-G and P308-E
	Step 1. If resistance is as specified, replace fuel quantity probe (PARA 10-4-33). Go to 2.
	Step 2. If resistance is not as specified, repair/replace wiring (Appendix F) as required. Go to 2.
2. Procedure completed.	

TABLE NO. 10-2-5-3. No Change When FUEL IND TEST Pushbutton Is Pressed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between:	P905-N and P135-L P905-M and P135-H P135-K and J308-D P135-J and J307-D
	Step 1. If continuity is present, replace miscellaneous switch panel (PARA 9-4-7). Go to 2.
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 2.

TABLE NO. 10-2-5-3. No Change When FUEL IND TEST Pushbutton Is Pressed. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

2. Procedure completed.

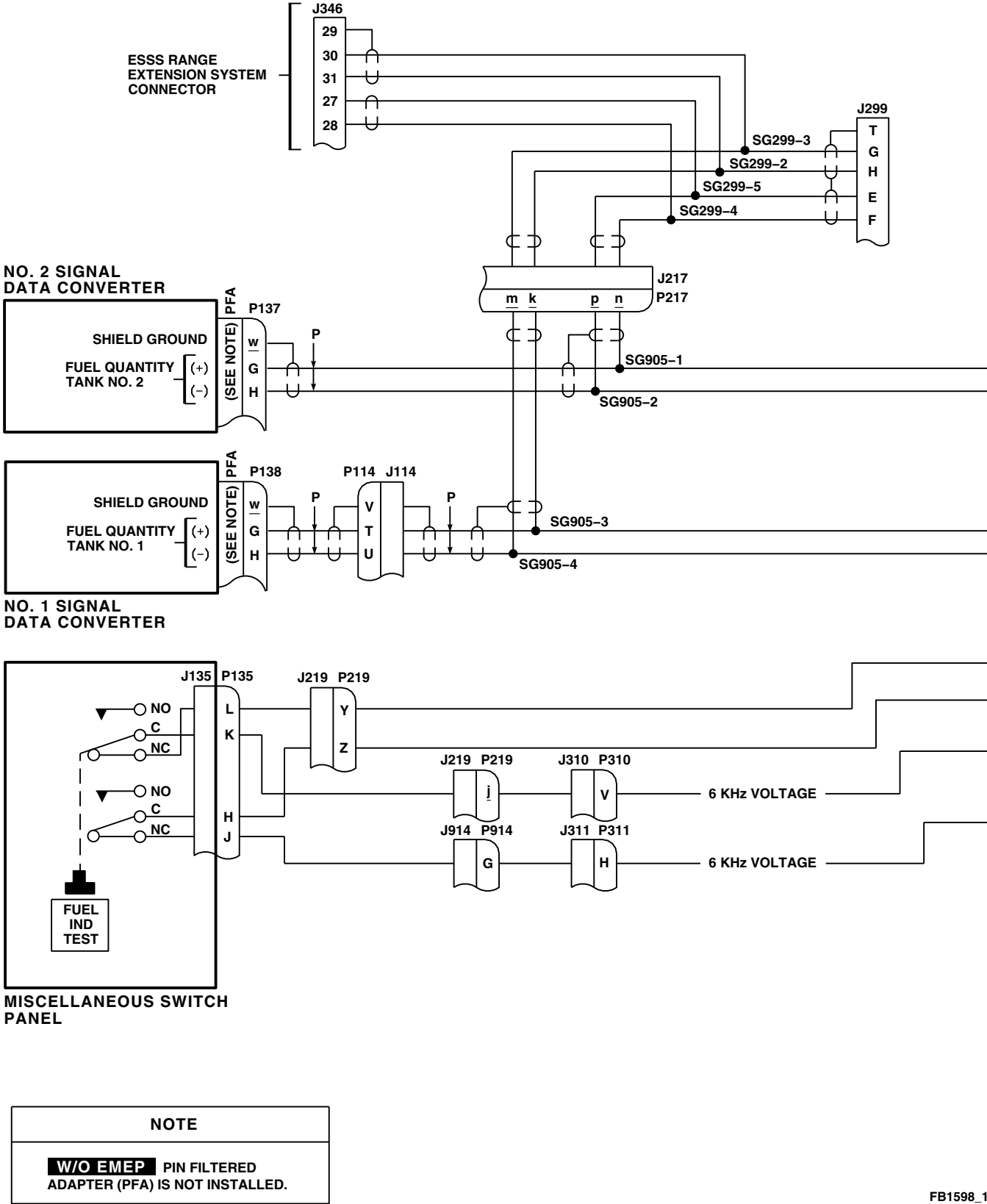


FIGURE 10-2-5-1. FUEL QUANTITY SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 2)

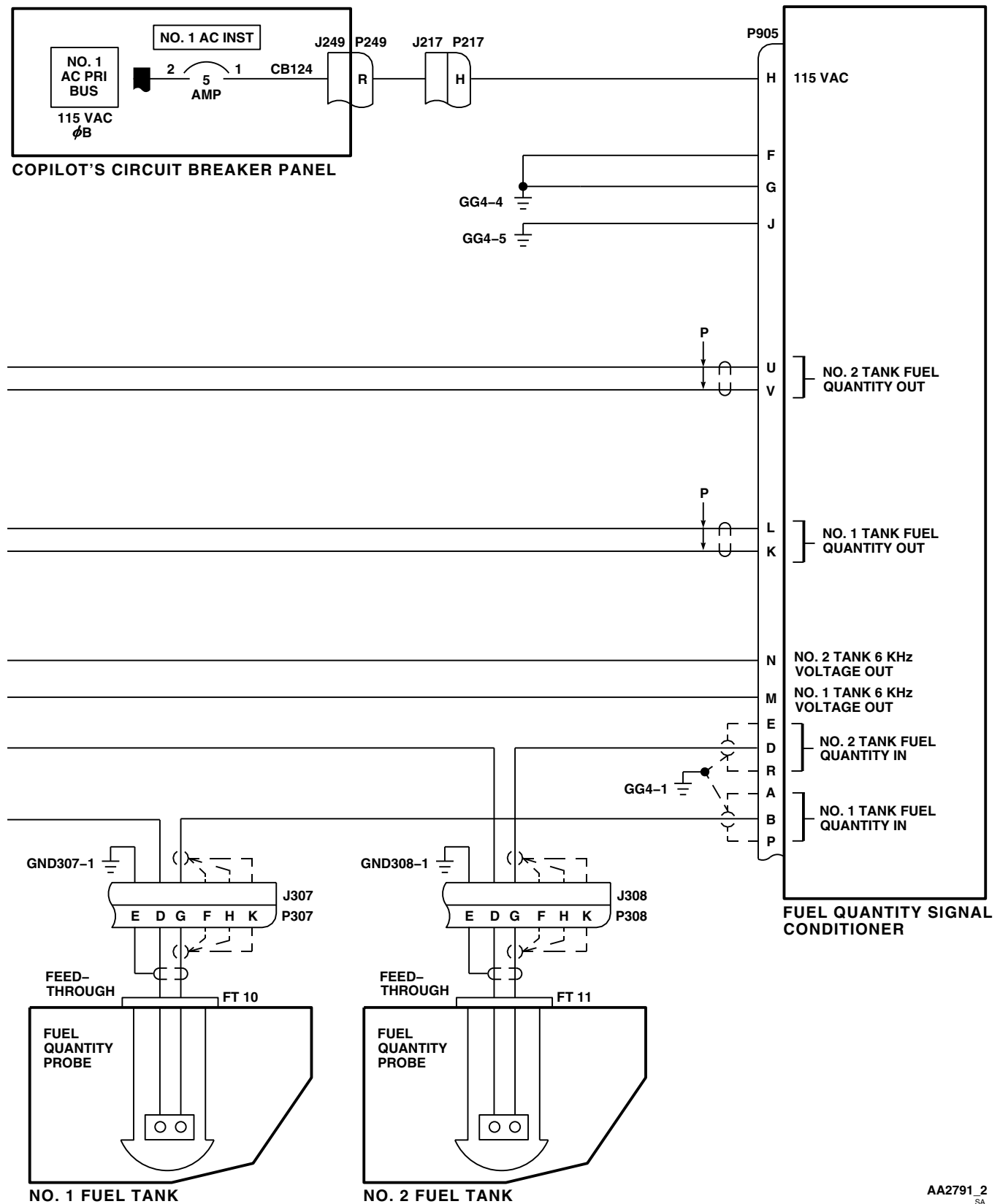


FIGURE 10-2-5-1. FUEL QUANTITY SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 2)

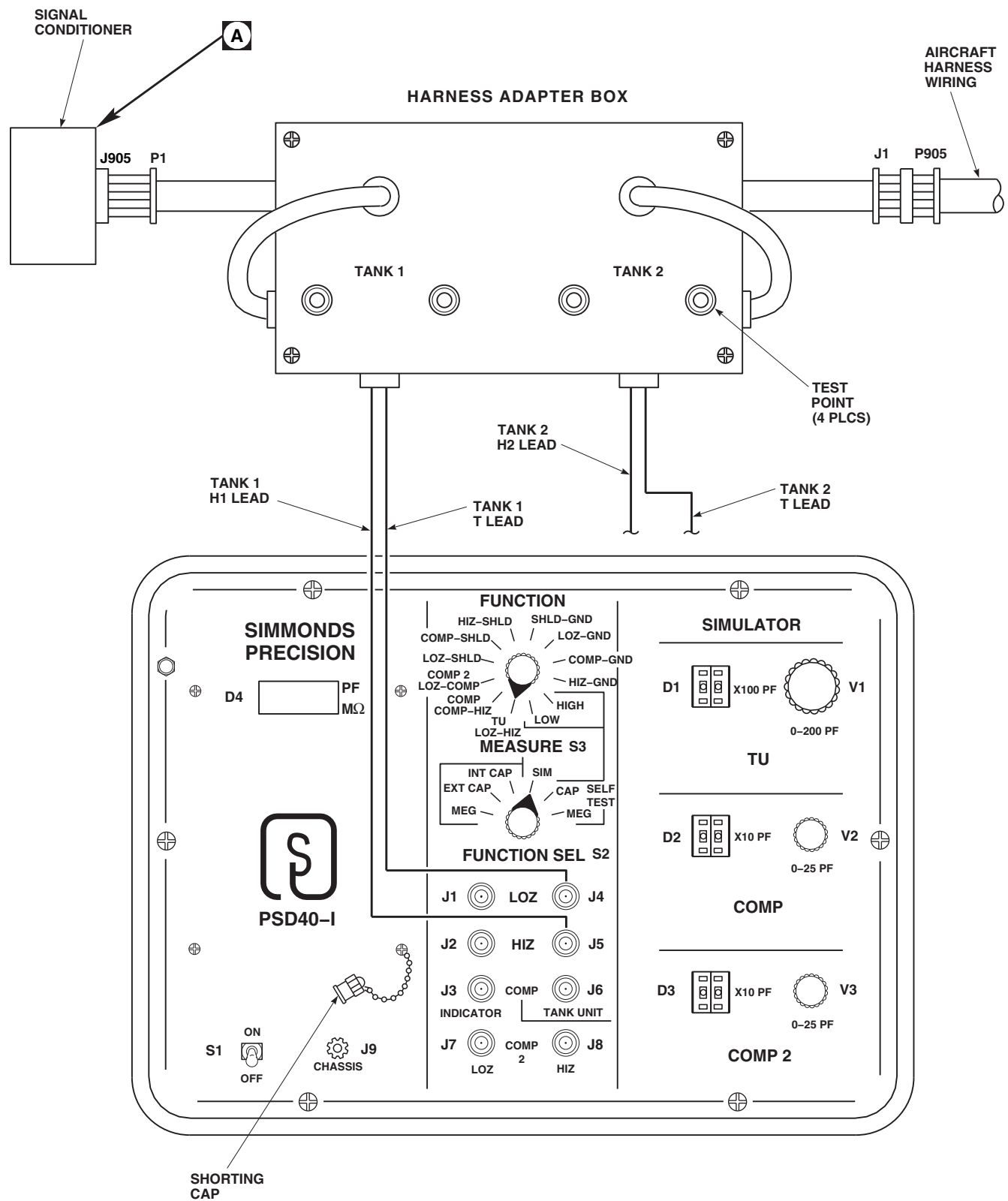
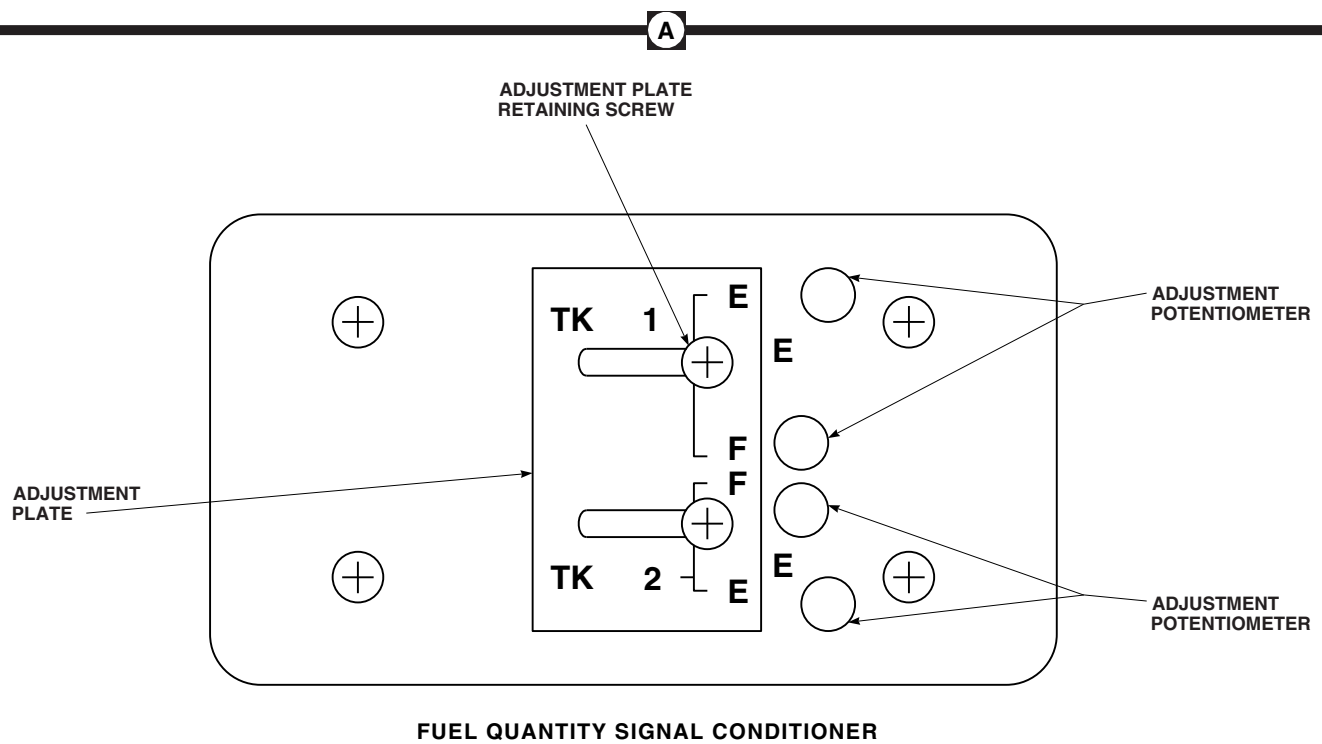


FIGURE 10-2-5-2. FUEL QUANTITY SYSTEM TEST SETUP DIAGRAM (SHEET 1 OF 2)



FB1601_2
SA

FIGURE 10-2-5-2. FUEL QUANTITY SYSTEM TEST SETUP DIAGRAM (SHEET 2 OF 2)

10-2-5-12

10-2-6 FUEL QUANTITY SYSTEM (USING TEST SET PSD-60).

INITIAL SETUP

Test Equipment

- AC/DC Capacitance Gauging System Test Set, PSD 60-2
- Multimeter, Fluke 27/AN

Tools

- Electrical Repairer Toolkit
- Fuel Quantity Gauging System Harness Adapter Test Box, PSD 40-542-1 or T7030-01014-041 with Test Leads T7030-01014-042 and -043

References

- Operating Instructions, PSD 60-2
- PARA 1-3-18 or PARA 1-3-17
- PARA 1-6-16
- PARA 2-4-69

Equipment Conditions

- External Electrical Power Available
- Fuel Tanks Empty and Purged (Dry method only)

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

NOTE

The fuel quantity system operational/troubleshooting procedure can be per-

formed in either of two ways: dry method (preferred) or wet method (alternate). The dry method should be used whenever possible for a complete checkout. The wet method can be used only when main fuel tanks cannot be defueled. The wet method will not check out the fuel probe or its harness.

NOTE

If test set digital display indicates all zeros and flashing colons, the measurement is out of range of the test set.

Special Instructions

The fuel quantity system operational/troubleshooting procedure is subdivided as follows:

Using Harness Adapter Test Box, PSD 40-542-1

- (1) Setup
- (2) Dry Method
- (3) Wet Method

Using Harness Adapter Test Box, T7030-01014-041, with Test Leads T7030-01014-042 and -043

- (4) Setup
- (5) Dry Method
- (6) Wet Method

Using Either Harness Adapter Test Box

- (7) System Check
- (8) Shutdown

10-2-6.1 Setup.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- b. Turn on electrical power (PARA 1-6-16).
- c. On harness adapter test box, connect jumper harness P2 through P5 to J2 through J5.

NOTE

- If LO BAT is indicated on test set digital display, turn in test set for battery replacement.
 - Do not connect harness adapter J1 and P1 to helicopter when checking harness capacitance. Do not allow J1 and P1 to touch any conductive surface.
- d. Place test set power switch (S1) to ON and allow 3 minutes for test set to stabilize.
 - e. Connect harness adapter TANK 2 HIZ lead to test set TANK UNITS HIZ (J5) and TANK 2 LOZ lead to test set TANK UNITS LOZ (J4).
 - f. Connect harness adapter P1 to J1.
 - g. Set test set control switches as follows:

CONTROL	POSITION
FUNCT (S2)	MEASURE EXT
SELECT (S3)	TU

- h. Record digital display on test set; this is the TANK 2 harness adapter test box capacitance.
- i. Disconnect TANK 2 HIZ lead and TANK 2 LOZ lead from test set, and connect TANK 1 HIZ lead to test set TANK UNITS HIZ (J5) and TANK 1 LOZ lead to test set TANK UNITS LOZ (J4).
- j. Record digital display on test set; this is the TANK 1 harness adapter test box capacitance.

10-2-6.2 Dry Method.**10-2-6.2.1 No. 1 Main Fuel Tank.****NOTE**

Make sure main fuel tanks are empty and purged (PARA 1-3-1).

- a. Do Setup, if not already done.
- b. Pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
- c. Remove interior soundproofing (PARA 2-4-69) as necessary to gain access to fuel quantity signal conditioner.
- d. With harness adapter TANK 1 leads connected to test set, disconnect helicopter harness P905 from signal conditioner and connect it to harness adapter test box J1.
- e. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
- f. Make sure test set SELECT (S3) control switch is at TU.
- g. Calculate left main tank probe capacitance by subtracting TANK 1 harness adapter test box capacitance (recorded in Setup, step j.) from test set digital display.

RESULT: Calculation shall be between 69.1 and 72.1 pF.

- If result is not as specified, go to TABLE NO. 10-2-6-1.
- h. Set test set SELECT (S3) control switch to HIZ-GND position.

RESULT: Test set digital display shall indicate greater than 10 megohms.

- If result is not as specified, go to TABLE NO. 10-2-6-2.
- i. Set test set SELECT (S3) control switch to LOZ-GND position.

10-2-6-2

RESULT: Test set digital display shall indicate greater than 10 megohms.

- If result is not as specified, go to TABLE NO. 10-2-6-2.
- j. Set test set FUNCT (S2) control switch to AIRCRAFT ONLY position.
- k. Connect TANK 1 HIZ lead to test set INDICATOR HIZ (J2) and TANK 1 LOZ lead to test set INDICATOR LOZ (J1).
- l. Connect multimeter to TANK 1 test points on the harness adapter test box.
- m. Connect harness adapter test box P1 to signal conditioner J905.
- n. Loosen retaining screws on signal conditioner adjustment cover plate and slide plate back to expose adjustment potentiometers.

NOTE

- Failure to adjust TK 1 E potentiometer as described can result in a miscalculation of the fuel quantity system.
- The following procedure requires observing three possible results while performing an adjustment. Read and understand the following procedure before starting.
- o. While observing multimeter voltage and polarity, use a jeweler's screwdriver to adjust TK 1 E potentiometer on signal conditioner counterclockwise.
 - (1) If polarity on multimeter is positive (+) and adjustment decreases the voltage, continue counterclockwise adjustment until multimeter indicates between -0.005 and 0.005 vdc.
 - (2) If polarity on multimeter is positive (+) and adjustment increases the voltage, turn TK 1 E potentiometer clockwise until multimeter indicates negative (-) polarity.

Continue clockwise adjustment until multimeter indicates between -0.005 and 0.005 vdc.

- (3) If polarity on multimeter is negative (-), turn TK 1 E potentiometer clockwise until multimeter indicates between -0.005 and 0.005 vdc.

RESULT:

- (1) Multimeter shall indicate between -0.005 and 0.005 vdc.
- (2) FUEL QTY 1 vertical indicator on central display unit (CDU) shall indicate 0 pounds of fuel (disregard digital fuel quantity display on CDU at this time).

- If result (1) is not as specified, check for 115 vac between P905-H and ground.

(a) If voltage is as specified, replace fuel quantity signal conditioner (PARA 10-4-36).

(b) If voltage is not as specified, check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.

1 If voltage is as specified, repair/replace wiring between P905-H and NO. 1 AC INST circuit breaker terminal 1 (Appendix F).

2 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

- If result (2) is not as specified, check continuity between the following:

P905-L	and	P138-G
P905-K	and	P138-H

(a) If continuity is present, troubleshoot instrument display system (PARA 8-2-3).

- (b) If continuity is not present, repair/replace wiring (Appendix F) as required.

p. Connect test set CHASSIS (J7) to ground.

q. Set test set control switches as follows:

CONTROL	POSITION
FUNCT (S2)	MEASURE INT
SELECT (S3)	TU
SIMULATOR	00
COMP (V2)	
SIMULATOR	002
TU (V1)	

NOTE

It is important to adjust SIMULATOR TU 10-8000 PF vernier control knob as accurately as possible.

- r. Adjust test set SIMULATOR TU 10-8000 PF decade switches and vernier control knob until test set digital display indicates between 40.9 and 41.3 pF to simulate a full left main tank.
- s. Set test set FUNCT (S2) control switch to SIM TU ONLY.
- t. Using a jeweler's screwdriver, adjust TK 1 F potentiometer on signal conditioner until FUEL QTY 1 vertical indicator on CDU indicates 1150 pounds.
- u. Slowly adjust TK 1 F potentiometer until FUEL QTY 1 vertical indicator lights a steady 1200 pounds indication.

RESULT:

- (1) FUEL QTY 1 vertical indicator on CDU shall indicate a steady 1200 pounds of fuel.
- (2) Multimeter shall indicate between 5.97 and 6.03 vdc.

- If result (1) is not as specified, check continuity between the following:

P905-L	and	P138-G
P905-K	and	P138-H

- (a) If continuity is present, troubleshoot instrument display system (PARA 8-2-3).
- (b) If continuity is not present, repair/replace wiring (Appendix F) as required.

- If result (2) is not as specified, check for 115 vac between P905-H and ground.

- (a) If voltage is as specified, replace fuel quantity signal conditioner (PARA 10-4-36).
- (b) If voltage is not as specified, check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.

1 If voltage is as specified, repair/replace wiring between P905-H and NO. 1 AC INST circuit breaker terminal 1 (Appendix F).

2 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

10-2-6.2.2 No. 2 Main Fuel Tank.

- a. Pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
- b. Disconnect test set CHASSIS (J7) from ground.
- c. Disconnect harness adapter TANK 1 leads from test set, and connect TANK 2 HIZ lead to test set TANK UNITS HIZ (J5) and TANK 2 LOZ lead to test set TANK UNITS LOZ (J4).

10-2-6-4

- d. Disconnect harness adapter test box P1 from signal conditioner.
- e. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
- f. Set test set control switches as follows:

CONTROL	POSITION
FUNCT (S2)	MEASURE EXT
SELECT (S3)	TU

- g. Calculate right main tank probe capacitance by subtracting TANK 2 harness adapter test box capacitance (recorded in Setup, step h.) from test set digital display.

RESULT: Calculation shall be between 69.1 and 72.1 pF.

- If result is not as specified, go to TABLE NO. 10-2-6-1.

- h. Set test set SELECT (S3) control switch to HIZ-GND position.

RESULT: Test set digital display shall indicate greater than 10 megohms.

- If result is not as specified, go to TABLE NO. 10-2-6-2.

- i. Set test set SELECT (S3) control switch to LOZ-GND position.

RESULT: Test set digital display shall indicate greater than 10 megohms.

- If result is not as specified, go to TABLE NO. 10-2-6-2.

- j. Set test set FUNCT (S2) control switch to AIRCRAFT ONLY position.

- k. Connect TANK 2 HIZ lead to test set INDICATOR HIZ (J2) and TANK 2 LOZ lead to test set INDICATOR LOZ (J1).

- l. Connect harness adapter test box P1 to signal conditioner.

- m. Connect multimeter to TANK 2 test points on the harness adapter test box.

NOTE

- Failure to adjust TK 2 E potentiometer as described can result in a miscalculation of the fuel quantity system.

- The following procedure requires observing three possible results while performing an adjustment. Read and understand the following procedure before starting.

- n. While observing multimeter voltage and polarity, use a jeweler's screwdriver to adjust TK 2 E potentiometer on signal conditioner counterclockwise.

- (1) If polarity on multimeter is positive (+) and adjustment decreases the voltage, continue counterclockwise adjustment until multimeter indicates between -0.005 and 0.005 vdc.

- (2) If polarity on multimeter is positive (+) and adjustment increases the voltage, turn TK 2 E potentiometer clockwise until multimeter indicates negative (-) polarity. Continue clockwise adjustment until multimeter indicates between -0.005 and 0.005 vdc.

- (3) If polarity on multimeter is negative (-), turn TK 2 E potentiometer clockwise until multimeter indicates between -0.005 and 0.005 vdc.

RESULT:

- (1) Multimeter shall indicate between -0.005 and 0.005 vdc.
- (2) FUEL QTY 2 vertical indicator and TOTAL FUEL or MAIN FUEL digital display on CDU shall indicate 0 pounds of fuel.

- If result (1) is not as specified, check for 115 vac between P905-H and ground.

(a) If voltage is as specified, replace fuel quantity signal conditioner (PARA 10-4-36).

(b) If voltage is not as specified, check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.

1 If voltage is as specified, repair/replace wiring P905-H and NO. 1 AC INST circuit breaker terminal 1 (Appendix F).

2 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

- If result (2) is not as specified, check continuity between the following:

P905-U	and	P137-G
P905-V	and	P137-H

(a) If continuity is present, troubleshoot instrument display system (PARA 8-2-3).

(b) If continuity is not present, repair/replace wiring (Appendix F) as required.

o. Connect test set CHASSIS (J7) to ground.

p. Set test set control switches as follows:

CONTROL	POSITION
FUNCT (S2)	MEASURE INT
SELECT (S3)	TU
SIMULATOR	00
COMP (V2)	
SIMULATOR	002
TU (V1)	

NOTE

It is important to adjust SIMULATOR TU 10-8000 PF vernier control knob as accurately as possible.

- q. Adjust test set SIMULATOR TU 10-8000 PF decade switches and vernier control knob until test set digital display indicates between 40.9 and 41.3 pF to simulate a full right main tank.
- r. Set test set FUNCT (S2) control switch to SIM TU ONLY.
- s. Using a jeweler's screwdriver, adjust TK 2 F potentiometer on signal conditioner until FUEL QTY 2 vertical indicator on CDU indicates 1150 pounds.
- t. Slowly adjust TK 2 F potentiometer until FUEL QTY 2 vertical indicator lights a steady 1200 pounds indication.

RESULT:

- (1) FUEL QTY 2 vertical indicator and TOTAL FUEL or MAIN FUEL digital display on CDU shall indicate a steady 1200 pounds of fuel.
- (2) Multimeter shall indicate between 5.97 and 6.03 vdc.

- If result (1) is not as specified, check continuity between the following:

P905-U	and	P137-G
P905-V	and	P137-H

(a) If continuity is present, troubleshoot instrument display system (PARA 8-2-3).

(b) If continuity is not present, repair/replace wiring (Appendix F) as required.

- If result (2) is not as specified, check for 115 vac between P905-H and ground.

- (a) If voltage is as specified, replace fuel quantity signal conditioner (PARA 10-4-36).
- (b) If voltage is not as specified, check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.
 - 1 If voltage is as specified, repair/replace wiring between P905-H and NO. 1 AC INST circuit breaker terminal 1 (Appendix F).
 - 2 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).
- u. Do System Check.

10-2-6.3 Wet Method.

10-2-6.3.1 No. 1 Main Fuel Tank.

- a. Do Setup, if not already done.
- b. Pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
- c. Remove interior soundproofing (PARA 2-4-69) as necessary to gain access to fuel quantity signal conditioner.
- d. Disconnect helicopter harness P905 from signal conditioner and connect it to harness adapter test box J1.
- e. Set test set SELECT (S3) control switch to HIZ-GND position.

RESULT: Test set digital display shall indicate greater than 10 megohms.

- If result is not as specified, go to TABLE NO. 10-2-6-2.
- f. Set test set SELECT (S3) control switch to LOZ-GND position.

RESULT: Test set digital display shall indicate greater than 10 megohms.

- If result is not as specified, go to TABLE NO. 10-2-6-2.

- g. Disconnect TANK 1 HIZ lead and TANK 1 LOZ lead from test set, and connect TANK 2 HIZ lead to test set TANK UNITS HIZ (J5) and TANK 2 LOZ lead to test set TANK UNITS LOZ (J4).
- h. Set test set SELECT (S3) control switch to HIZ-GND position.

RESULT: Test set digital display shall indicate greater than 10 megohms.

- If result is not as specified, go to TABLE NO. 10-2-6-2.
- i. Set test set SELECT (S3) control switch to LOZ-GND position.

RESULT: Test set digital display shall indicate greater than 10 megohms.

- If result is not as specified, go to TABLE NO. 10-2-6-2.
- j. Disconnect TANK 2 leads from test set connections.
- k. Connect harness adapter test box P1 to signal conditioner J905.
- l. On harness adapter test box, disconnect harness adapter jumper harness connectors P2 through P5 from J2 through J5.
- m. Connect test set CHASSIS (J7) to ground.
- n. Connect harness adapter TANK 1 HIZ lead to test set INDICATOR HIZ (J2) and TANK 1 LOZ lead to test set INDICATOR LOZ (J1).
- o. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
- p. Set test set control switches as follows:

CONTROL	POSITION
FUNCT (S2)	MEASURE INT
SELECT (S3)	TU

CONTROL	POSITION
---------	----------

SIMULATOR	00
COMP (V2)	
SIMULATOR	005
TU (V1)	

- q. Calculate simulated empty tank capacitance by subtracting TANK 1 harness adapter test box capacitance (recorded in Setup, step j.) from 70.6 pF.

NOTE

It is important to adjust SIMULATOR TU 10-8000 PF vernier control knob as accurately as possible.

- r. Adjust test set SIMULATOR TU 10-8000 PF vernier control knob (V1) until test set digital display indicates simulated empty left main tank capacitance, calculated in step q.
- s. Set test set FUNCT (S2) control switch to SIM TU ONLY.
- t. Connect multimeter to TANK 1 test points on the harness adapter test box.
- u. Loosen retaining screws on signal conditioner adjustment cover plate and slide plate back to expose adjustment potentiometers.

NOTE

- Failure to adjust TK 1 E potentiometer as described can result in a miscalculation of the fuel quantity system.
 - The following procedure requires observing three possible results while performing an adjustment. Read and understand the following procedure before starting.
- v. While observing multimeter voltage and polarity, use a jeweler's screwdriver to adjust TK 1 E potentiometer on signal conditioner counterclockwise.

- (1) If polarity on multimeter is positive (+) and adjustment decreases the voltage, continue counterclockwise adjustment until multimeter indicates between -0.005 and 0.005 vdc.
- (2) If polarity on multimeter is positive (+) and adjustment increases the voltage, turn TK 1 E potentiometer clockwise until multimeter indicates negative (-) polarity. Continue clockwise adjustment until multimeter indicates between -0.005 and 0.005 vdc.
- (3) If polarity on multimeter is negative (-), turn TK 1 E potentiometer clockwise until multimeter indicates between -0.005 and 0.005 vdc.

RESULT:

- (1) Multimeter shall indicate between -0.005 and 0.005 vdc.
 - (2) FUEL QTY 1 vertical indicator on CDU shall indicate 0 pounds of fuel (disregard digital fuel quantity display on CDU).
- If result (1) is not as specified, check for 115 vac between P905-H and ground.
 - (a) If voltage is as specified, replace fuel quantity signal conditioner (PARA 10-4-36).
 - (b) If voltage is not as specified, check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.
 - 1 If voltage is as specified, repair/replace wiring P905-H and NO. 1 AC INST circuit breaker terminal 1 (Appendix F).
 - 2 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).
 - If result (2) is not as specified, check continuity between the following:

P905-L and P138-G
P905-K and P138-H

- (a) If continuity is present, troubleshoot instrument display system (PARA 8-2-3).
- (b) If continuity is not present, repair/replace wiring (Appendix F) as required.

w. Set test set control switches as follows:

CONTROL	POSITION
FUNCT (S2)	MEASURE INT
SELECT (S3)	TU
SIMULATOR COMP (V2)	00
SIMULATOR TU (V1)	005

- x. Calculate simulated full tank capacitance by subtracting TANK 1 harness adapter test box capacitance (recorded in Setup, step j.) from 111.7 pF.

NOTE

It is important to adjust SIMULATOR TU 10-8000 PF vernier control knob as accurately as possible.

- y. Adjust test set SIMULATOR TU 10-8000 PF vernier control knob (V1) until test set digital display indicates simulated full left main tank capacitance, calculated in step x.
- z. Set test set FUNCT (S2) control switch to SIM TU ONLY.
- aa. Using a jeweler's screwdriver, adjust TK 1 F potentiometer on signal conditioner until FUEL QTY 1 vertical indicator on CDU indicates 1150 pounds.
- ab. Slowly adjust TK 1 F potentiometer until FUEL QTY 1 vertical indicator lights a steady 1200 pounds indication.

RESULT:

- (1) FUEL QTY 1 vertical indicator on CDU shall indicate a steady 1200 pounds of fuel.
- (2) Multimeter shall indicate between 5.97 and 6.03 vdc.
- If result (1) is not as specified, check continuity between the following:

P905-L and P138-G
P905-K and P138-H

- (a) If continuity is present, troubleshoot instrument display system (PARA 8-2-3).
- (b) If continuity is not present, repair/replace wiring (Appendix F) as required.
- If result (2) is not as specified, check for 115 vac between P905-H and ground.
 - (a) If voltage is as specified, replace fuel quantity signal conditioner (PARA 10-4-36).
 - (b) If voltage is not as specified, check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.
 - 1 If voltage is as specified, repair/replace wiring between P905-H and NO. 1 AC INST circuit breaker terminal 1 (Appendix F).
 - 2 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

10-2-6.3.2 No. 2 Main Fuel Tank.

- a. Pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.

- b. On harness adapter test box, disconnect harness adapter TANK 1 leads from test set, and connect TANK 2 HIZ lead to test set INDICATOR HIZ (J2) and TANK 2 LOZ lead to test set INDICATOR LOZ (J1).
- c. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
- d. Set test set control switches as follows:

CONTROL	POSITION
FUNCT (S2)	MEASURE INT
SELECT (S3)	TU
SIMULATOR COMP (V2)	00
SIMULATOR TU (V1)	005

- e. Calculate simulated empty tank capacitance by subtracting TANK 2 harness adapter test box capacitance (recorded in Setup, step h.) from 70.6 pF.

NOTE

It is important to adjust SIMULATOR TU 10-8000 PF vernier control knob as accurately as possible.

- f. Adjust test set SIMULATOR TU 10-8000 PF vernier control knob (V1) until test set digital display indicates simulated empty right main tank capacitance, calculated in step e.
- g. Set test set FUNCT (S2) control switch to SIM TU ONLY.
- h. Connect multimeter to TANK 2 test points on the harness adapter test box.

NOTE

- Failure to adjust TK 2 E potentiometer as described can result in a miscalculation of the fuel quantity system.
- The following procedure requires observing three possible results while

performing an adjustment. Read and understand the following procedure before starting.

- i. While observing multimeter voltage and polarity, use a jeweler's screwdriver to adjust TK 2 E potentiometer on signal conditioner counterclockwise.
 - (1) If polarity on multimeter is positive (+) and adjustment decreases the voltage, continue counterclockwise adjustment until multimeter indicates between -0.005 and 0.005 vdc.
 - (2) If polarity on multimeter is positive (+) and adjustment increases the voltage, turn TK 1 E potentiometer clockwise until multimeter indicates negative (-) polarity. Continue clockwise adjustment until multimeter indicates between -0.005 and 0.005 vdc.
 - (3) If polarity on multimeter is negative (-), turn TK 1 E potentiometer clockwise until multimeter indicates between -0.005 and 0.005 vdc.

RESULT:

- (1) Multimeter shall indicate between -0.005 and 0.005 vdc.
- (2) FUEL QTY 2 vertical indication on CDU shall indicate between 0 and 50 pounds of fuel.
- If result (1) is not as specified, check for 115 vac between P905-H and ground.
 - (a) If voltage is as specified, replace fuel quantity signal conditioner (PARA 10-4-36).
 - (b) If voltage is not as specified, check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.
 - 1 If voltage is as specified, repair/replace wiring between P905-H

and NO. 1 AC INST circuit breaker terminal 1 (Appendix F).

- 2 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

- If result (2) is not as specified, check continuity between the following:

P905-U	and	P137-G
P905-V	and	P137-H

- (a) If continuity is present, troubleshoot instrument display system (PARA 8-2-3).
- (b) If continuity is not present, repair/replace wiring (Appendix F) as required.

- j. Set test set control switches as follows:

CONTROL	POSITION
FUNCT (S2)	MEASURE INT
SELECT (S3)	TU
SIMULATOR	00
COMP (V2)	
SIMULATOR	005
TU (V1)	

- k. Calculate simulated full tank capacitance by subtracting TANK 2 harness adapter test box capacitance (recorded in Setup, step h.) from 111.7 pF.

NOTE

It is important to adjust SIMULATOR TU 10-8000 PF vernier control knob as accurately as possible.

- l. Adjust test set SIMULATOR TU 10-8000 PF vernier control knob (V1) until test set digital display indicates simulated full right main tank capacitance, calculated in step k.

- m. Set test set FUNCT (S2) control switch to SIM TU ONLY.

- n. Using a jeweler's screwdriver, adjust TK 2 F potentiometer on signal conditioner until FUEL QTY 2 vertical indicator on CDU indicates 1150 pounds.

- o. Slowly adjust TK 2 F potentiometer until FUEL QTY 2 vertical indicator lights a steady 1200 pounds indication.

RESULT:

- (1) FUEL QTY 2 vertical indicator on CDU shall indicate a steady 1200 pounds of fuel.
- (2) Multimeter shall indicate between 5.97 and 6.03 vdc.

- If result (1) is not as specified, check continuity between the following:

P905-U	and	P137-G
P905-V	and	P137-H

- (a) If continuity is present, troubleshoot instrument display system (PARA 8-2-3).
- (b) If continuity is not present, repair/replace wiring (Appendix F) as required.

- If result (2) is not as specified, check for 115 vac between P905-H and ground.

- (a) If voltage is as specified, replace fuel quantity signal conditioner (PARA 10-4-36).
- (b) If voltage is not as specified, check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.

- 1 If voltage is as specified, repair/replace wiring P905-H and NO. 1 AC INST circuit breaker terminal 1 (Appendix F).

- 2 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

p. Do System Check.

10-2-6.4 Setup.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- b. Turn on electrical power (PARA 1-6-16).
- c. Place test set power switch (S1) to ON and allow 3 minutes for test set to stabilize.

NOTE

- If LO BAT is indicated on test set digital display, replace battery.
 - Do not connect harness adapter J1 and P1 to helicopter when checking harness capacitance. Do not allow J1 and P1 to touch any conductive surface.
- d. Connect one end of HI-Z (-042) test lead to test set TANK UNITS HIZ (J5) and the other end of HI-Z test lead to adapter harness adapter (-041) No. 1 tank unit HI-Z center tee connection.
 - e. Connect one end of LO-Z (-043) test lead to test set TANK UNITS LOZ (J4) and the other end of LO-Z test lead to adapter harness adapter (-041) No. 1 tank unit LO-Z center tee connection.
 - f. Connect harness adapter P1 to J1.
 - g. Set test set control switches as follows:

CONTROL	POSITION
FUNCT (S2)	MEASURE EXT
SELECT (S3)	COMP

RESULT: Test set digital display shall indicate less than 0.2 pF.

- If result is not as specified, replace adapter harness (-041) and repeat procedure.

- h. Disconnect HI-Z (-042) and LO-Z (-043) test leads from adapter harness No. 1 HI-Z and LO-Z tank unit center tee connections, and connect to adapter harness No. 2 HI-Z and LO-Z tank unit center tee connections, respectively.

- i. Make sure test set control switches are as follows:

CONTROL	POSITION
FUNCT (S2)	MEASURE EXT
SELECT (S3)	COMP

RESULT: Test set digital display shall indicate less than 0.2 pF.

- If result is not as specified, replace adapter harness (-041) and repeat procedure.

- j. Disconnect adapter harness (-041) P1 from J1.

10-2-6.5 Dry Method.

10-2-6.5.1 No. 1 Main Fuel Tank.

NOTE

Make sure main fuel tanks are empty and purged (PARA 1-3-1).

- a. Do Setup, if not already done.
- b. Pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
- c. Remove interior soundproofing (PARA 2-4-69) as necessary to gain access to fuel quantity signal conditioner.
- d. Disconnect helicopter harness P905 from signal conditioner and connect it to harness adapter (-041) test box J1 and connect adapter harness P1 to signal conditioner.
- e. Disconnect adapter harness (-041) No. 1 tank unit HI-Z (2A) lead and LO-Z (4A) lead from tee connections.

10-2-6-12

- f. Connect one end of HI-Z (-042) test lead to test set TANK UNITS HIZ (J5) and the other end of HI-Z test lead to adapter harness (-041) No. 1 tank unit HI-Z center tee connection.
- g. Connect one end of LO-Z (-043) test lead to test set TANK UNITS LOZ (J4) and the other end of LO-Z test lead to adapter harness (-041) No. 1 tank unit LO-Z center tee connection.
- h. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.

NOTE

Bends, crimps, and excess slack in adapter harness wires can cause capacitance reading to be high. Do not allow adapter harness tee connectors to touch any conductive surface or each other.

- i. Set test set SELECT (S3) control switch to TU. Measure and record left main tank probe capacitance.

RESULT: Test set digital display shall be between 69.1 and 72.1 pF.

- If reading is high by 0.1 or 0.2 pF, pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel and straighten adapter harness wires.

- (a) If trouble remains, go to TABLE NO. 10-2-6-1.

- j. Set test set SELECT (S3) control switch to HIZ-GND position.

RESULT: Test set digital display shall indicate greater than 10 megohms.

- If result is not as specified, go to TABLE NO. 10-2-6-2.

- k. Set test set SELECT (S3) control switch to LOZ-GND position.

RESULT: Test set digital display shall indicate greater than 10 megohms.

- If result is not as specified, go to TABLE NO. 10-2-6-2.

- l. Disconnect HI-Z (-042) and LO-Z (-043) leads from adapter harness No. 1 HI-Z and LO-Z tank unit center tee connections.
- m. Connect adapter harness No. 1 tank unit HI-Z (2A) lead and LO-Z (4A) lead to tee connections.
- n. Connect multimeter to TANK 1 test points positive (+) and negative (-) on the adapter harness (-041) test box.
- o. Loosen retaining screws on signal conditioner adjustment cover plate and slide plate back to expose adjustment potentiometers.

NOTE

- Failure to adjust TK 1 E potentiometer as described can result in a miscalculation of the fuel quantity system.

- The following procedure requires observing three possible results while performing an adjustment. Read and understand the following procedure before starting.

- p. While observing multimeter voltage and polarity, use a jeweler's screwdriver to adjust TK 1 E potentiometer on signal conditioner counterclockwise.

- (1) If polarity on multimeter is positive (+) and adjustment decreases the voltage, continue counterclockwise adjustment until multimeter indicates between -0.005 and 0.005 vdc.
- (2) If polarity on multimeter is positive (+) and adjustment increases the voltage, turn TK 1 E potentiometer clockwise until multimeter indicates negative (-) polarity.

Continue clockwise adjustment until multimeter indicates between -0.005 and 0.005 vdc.

- (3) If polarity on multimeter is negative (-), turn TK 1 E potentiometer clockwise until multimeter indicates between -0.005 and 0.005 vdc.

RESULT:

- (1) Multimeter shall indicate between -0.005 and 0.005 vdc.
- (2) FUEL QTY 1 vertical indicator on CDU shall indicate 0 pounds of fuel (disregard digital fuel quantity display on CDU at this time).
- If result (1) is not as specified, check for 115 vac between P905-H and ground.

(a) If voltage is as specified, replace fuel quantity signal conditioner (PARA 10-4-36).

(b) If voltage is not as specified, check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.

1 If voltage is as specified, repair/replace wiring between P905-H and NO. 1 AC INST circuit breaker terminal 1 (Appendix F).

2 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

- If result (2) is not as specified, check continuity between the following:

P905-L	and	P138-G
P905-K	and	P138-H

(a) If continuity is present, troubleshoot instrument display system (PARA 8-2-3).

(b) If continuity is not present, repair/replace wiring (Appendix F) as required.

- q. Set test set FUNCT (S2) control switch to AIRCRAFT ONLY position.
- r. Connect one end of HI-Z (-042) test lead to test set INDICATOR HIZ (J2) and the other end of HI-Z test lead to adapter harness (-041) No. 1 tank unit HI-Z center tee connection.
- s. Connect one end of LO-Z (-043) test lead to test set INDICATOR LOZ (J1) and the other end of LO-Z test lead to adapter harness (-041) No. 1 tank unit LO-Z center tee connection.
- t. Connect test set CHASSIS (J7) to ground.
- u. Set test set control switches as follows:

CONTROL	POSITION
FUNCT (S2)	MEASURE INT
SELECT (S3)	TU
SIMULATOR	00
COMP (V2)	
SIMULATOR	002
TU (V1)	

NOTE

It is important to adjust SIMULATOR TU 10-8000 PF vernier control knob as accurately as possible.

- v. Adjust test set SIMULATOR TU 10-8000 PF decade switches and vernier control knob until test set digital display indicates between 40.9 and 41.3 pF to simulate a full left main tank.
- w. Set test set FUNCT (S2) control switch to SIM TU ONLY.
- x. Using a jeweler's screwdriver, adjust TK 1 F potentiometer on signal conditioner until FUEL QTY 1 vertical indicator on CDU indicates 1150 pounds.

- y. Slowly adjust TK 1 F potentiometer until FUEL QTY 1 vertical indicator lights a steady 1200 pounds indication.

RESULT:

- (1) FUEL QTY 1 vertical indicator on CDU shall indicate a steady 1200 pounds of fuel.
- (2) Multimeter shall indicate between 5.97 and 6.03 vdc.

- If result (1) is not as specified, check continuity between the following:

P905-L	and	P138-G
P905-K	and	P138-H

- (a) If continuity is present, troubleshoot instrument display system (PARA 8-2-3).
- (b) If continuity is not present, repair/replace wiring (Appendix F) as required.
- If result (2) is not as specified, check for 115 vac between P905-H and ground.
 - (a) If voltage is as specified, replace fuel quantity signal conditioner (PARA 10-4-36).
 - (b) If voltage is not as specified, check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.

1 If voltage is as specified, repair/replace wiring between P905-H and NO. 1 AC INST circuit breaker terminal 1 (Appendix F).

2 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

10-2-6.5.2 No. 2 Main Fuel Tank.

- a. Pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.

- b. Disconnect test set CHASSIS (J7) from ground.
- c. Disconnect adapter harness (-041) No. 2 tank unit HI-Z (2B) lead and LO-Z (4B) lead from tee connections.
- d. Connect one end of HI-Z (-042) test lead to test set TANK UNITS HIZ (J5) and the other end of HI-Z test lead to adapter harness (-041) No. 2 tank unit HI-Z center tee connection.
- e. Connect one end of LO-Z (-043) test lead to test set TANK UNITS LOZ (J4) and the other end of LO-Z test lead to adapter harness (-041) No. 2 tank unit LO-Z center tee connection.
- f. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.

NOTE

Bends, crimps, and excess slack in adapter harness wires can cause capacitance reading to be high. Do not allow adapter harness tee connectors to touch any conductive surface or each other.

- g. Set test set FUNCT (S2) control switch to MEASURE EXT and SELECT (S3) to TU. Measure right main tank probe capacitance.

RESULT: Test set digital display shall be between 69.1 and 72.1 pF.

- If reading is high by 0.1 or 0.2 pF, pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel, and straighten adapter harness wires.

(a) If trouble remains, go to TABLE NO. 10-2-6-1.

- h. Set test set SELECT (S3) control switch to HIZ-GND position.

RESULT: Test set digital display shall indicate greater than 10 megohms.

- If result is not as specified, go to TABLE NO. 10-2-6-2.

- i. Set test set SELECT (S3) control switch to LOZ-GND position.

RESULT: Test set digital display shall indicate greater than 10 megohms.

- If result is not as specified, go to TABLE NO. 10-2-6-2.
- j. Disconnect HI-Z (-042) and LO-Z (-043) leads from adapter harness No. 2 HI-Z and LO-Z tank unit center tee connections.
 - k. Reconnect adapter harness No. 1 tank unit HI-Z (2B) lead and LO-Z (4B) lead to tee connections.
 - l. Connect multimeter to TANK 2 test points positive (+) and negative (-) on the adapter harness (-041) test box.

NOTE

- Failure to adjust TK 2 E potentiometer as described can result in a miscalculation of the fuel quantity system.
 - The following procedure requires observing three possible results while performing an adjustment. Read and understand the following procedure before starting.
- m. While observing multimeter voltage and polarity, use a jeweler's screwdriver to adjust TK 2 E potentiometer on signal conditioner counterclockwise.
 - (1) If polarity on multimeter is positive (+) and adjustment decreases the voltage, continue counterclockwise adjustment until multimeter indicates between -0.005 and 0.005 vdc.
 - (2) If polarity on multimeter is positive (+) and adjustment increases the voltage, turn TK 2 E potentiometer clockwise until multimeter indicates negative (-) polarity. Continue clockwise adjustment until multimeter indicates between -0.005 and 0.005 vdc.

- (3) If polarity on multimeter is negative (-), turn TK 2 E potentiometer clockwise until multimeter indicates between -0.005 and 0.005 vdc.

RESULT:

- (1) Multimeter shall indicate between -0.005 and 0.005 vdc.
 - (2) FUEL QTY 2 vertical indicator and TOTAL FUEL or MAIN FUEL digital display on CDU shall indicate 0 pounds of fuel.
- If result (1) is not as specified, check for 115 vac between P905-H and ground.
 - (a) If voltage is as specified, replace fuel quantity signal conditioner (PARA 10-4-36).
 - (b) If voltage is not as specified, check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.
 - 1 If voltage is as specified, repair/replace wiring between P905-H and NO. 1 AC INST circuit breaker terminal 1 (Appendix F).
 - 2 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).
 - If result (2) is not as specified, check continuity between the following:

P905-U	and	P137-G
P905-V	and	P137-H

 - (a) If continuity is present, troubleshoot instrument display system (PARA 8-2-3).
 - (b) If continuity is not present, repair/replace wiring (Appendix F) as required.

- n. Set test set FUNCT (S2) control switch to AIRCRAFT ONLY position.
- o. Connect one end of HI-Z (-042) test lead to test set INDICATOR HIZ (J2) and the other end of HI-Z test lead to adapter harness (-041) No. 2 tank unit HI-Z center tee connection.
- p. Connect one end of LO-Z (-043) test lead to test set INDICATOR HIZ (J2) and the other end of LO-Z test lead to adapter harness (-041) No. 2 tank unit LO-Z center tee connection.
- q. Connect test set CHASSIS (J7) to any convenient ground.
- r. Set test set control switches as follows:

CONTROL	POSITION
FUNCT (S2)	MEASURE INT
SELECT (S3)	TU
SIMULATOR COMP (V2)	00
SIMULATOR TU (V1)	002

NOTE

It is important to adjust SIMULATOR TU 10-8000 PF vernier control knob as accurately as possible.

- s. Adjust test set SIMULATOR TU 10-8000 PF decade switches and vernier control knob until test set digital display indicates between 40.9 and 41.3 pF to simulate a full right main tank.
- t. Set test set FUNCT (S2) control switch to SIM TU ONLY.
- u. Using a jeweler's screwdriver, adjust TK 2 F potentiometer on signal conditioner until FUEL QTY 2 vertical indicator on CDU indicates 1150 pounds.
- v. Slowly adjust TK 2 F potentiometer until FUEL QTY 2 vertical indicator lights a steady 1200 pounds indication.

RESULT:

- (1) FUEL QTY 2 vertical indicator and TOTAL FUEL or MAIN FUEL digital display on CDU shall indicate a steady 1200 pounds of fuel.
- (2) Multimeter shall indicate between 5.97 and 6.03 vdc.

- If result (1) is not as specified, check continuity between the following:

P905-U	and	P137-G
P905-V	and	P137-H

- (a) If continuity is present, troubleshoot instrument display system (PARA 8-2-3).
- (b) If continuity is not present, repair/replace wiring (Appendix F) as required.
- If result (2) is not as specified, check for 115 vac between P905-H and ground.

- (a) If voltage is as specified, replace fuel quantity signal conditioner (PARA 10-4-36).
- (b) If voltage is not as specified, check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.

1 If voltage is as specified, repair/replace wiring between P905-H and NO. 1 AC INST circuit breaker terminal 1 (Appendix F).

2 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

- w. Do System Check.

10-2-6.6 Wet Method.**10-2-6.6.1 No. 1 Main Fuel Tank.**

- a. Do Setup, if not already done.
- b. Connect test set CHASSIS (J7) to ground.
- c. Pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
- d. Remove interior soundproofing (PARA 2-4-69) as necessary to gain access to fuel quantity signal conditioner.
- e. Disconnect helicopter harness P905 from signal conditioner and connect it to harness adapter test box J1.
- f. Connect harness adapter test box P1 to signal conditioner J905.
- g. Disconnect harness adapter No. 1 and No. 2 HI-Z and LO-Z (1A, 3A and 1B, 3B) test leads from tee connections.
- h. Connect one end of HI-Z (-042) test lead to test set INDICATOR HIZ (J2) and the other end of HI-Z test lead to adapter harness adapter (-041) No. 1 tank unit HI-Z center tee connection.
- i. Connect one end of LO-Z (-043) test lead to test set INDICATOR LOZ (J1) and the other end of LO-Z test lead to adapter harness (-041) No. 1 tank unit LO-Z center tee connection.
- j. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
- k. Set test set control switches as follows:

CONTROL	POSITION
FUNCT (S2)	MEASURE INT
SELECT (S3)	TU
SIMULATOR	00
COMP (V2)	

NOTE

It is important to adjust SIMULATOR TU 10-8000 PF vernier control knob as accurately as possible.

- l. Adjust test set SIMULATOR TU 10-8000 PF decade switches and vernier control knob until test set digital display indicates 70.6 pF to simulate an empty left main tank.
- m. Set test set FUNCT (S2) control switch to SIM TU ONLY.
- n. Loosen retaining screws on signal conditioner adjustment cover plate and slide plate back to expose adjustment potentiometers.

NOTE

- Failure to adjust TK 1 E potentiometer as described can result in a miscalculation of the fuel quantity system.
 - The following procedure requires observing three possible results while performing an adjustment. Read and understand the following procedure before starting.
- o. While observing multimeter voltage and polarity, use a jeweler's screwdriver to adjust TK 1 E potentiometer on signal conditioner counterclockwise.
 - (1) If polarity on multimeter is positive (+) and adjustment decreases the voltage, continue counterclockwise adjustment until multimeter indicates between -0.005 and 0.005 vdc.
 - (2) If polarity on multimeter is positive (+) and adjustment increases the voltage, turn TK 1 E potentiometer clockwise until multimeter indicates negative (-) polarity. Continue clockwise adjustment until multimeter indicates between -0.005 and 0.005 vdc.
 - (3) If polarity on multimeter is negative (-), turn TK 1 E potentiometer clockwise until multimeter indicates between -0.005 and 0.005 vdc.

RESULT:

- (1) Multimeter shall indicate between -0.005 and 0.005 vdc.
 - (2) FUEL QTY 1 vertical indicator on central display unit (CDU) shall indicate 0 pounds of fuel (disregard digital fuel quantity display on CDU at this time).
- If result (1) is not as specified, check for 115 vac between P905-H and ground.
 - (a) If voltage is as specified, replace fuel quantity signal conditioner (PARA 10-4-36).
 - (b) If voltage is not as specified, check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.
 - 1 If voltage is as specified, repair/replace wiring between P905-H and NO. 1 AC INST circuit breaker terminal 1 (Appendix F).
 - 2 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).
 - If result (2) is not as specified, check continuity between the following:

P905-L	and	P138-G
P905-K	and	P138-H

- (a) If continuity is present, troubleshoot instrument display system (PARA 8-2-3).
 - (b) If continuity is not present, repair/replace wiring (Appendix F) as required.
- p. Set test set control switches as follows:

CONTROL	POSITION
FUNCT (S2)	MEASURE INT

CONTROL	POSITION
SELECT (S3)	TU
SIMULATOR	00
COMP (V2)	

NOTE

It is important to adjust SIMULATOR TU 10-8000 PF vernier control knob as accurately as possible.

- q. Adjust test set SIMULATOR TU 10-8000 PF decade switches and vernier control knob until test set digital display indicates between 40.9 and 41.3 pF to simulate a full left main tank.
- r. Set test set FUNCT (S2) control switch to SIM TU ONLY.
- s. Using a jeweler's screwdriver, adjust TK 1 F potentiometer on signal conditioner until FUEL QTY 1 vertical indicator on CDU indicates 1150 pounds.
- t. Slowly adjust TK 1 F potentiometer until FUEL QTY 1 vertical indicator lights a steady 1200 pounds indication.

RESULT:

- (1) FUEL QTY 1 vertical indicator on CDU shall indicate a steady 1200 pounds of fuel.
 - (2) Multimeter shall indicate between 5.97 and 6.03 vdc.
- If result (1) is not as specified, check continuity between the following:

P905-L	and	P138-G
P905-K	and	P138-H

- (a) If continuity is present, troubleshoot instrument display system (PARA 8-2-3).

- (b) If continuity is not present, repair/replace wiring (Appendix F) as required.
- If result (2) is not as specified, check for 115 vac between P905-H and ground.
 - (a) If voltage is as specified, replace fuel quantity signal conditioner (PARA 10-4-36).
 - (b) If voltage is not as specified, check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.
 - 1 If voltage is as specified, repair/replace wiring between P905-H and NO. 1 AC INST circuit breaker terminal 1 (Appendix F).
 - 2 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

10-2-6.6.2 No. 2 Main Fuel Tank.

- a. Pull out NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
- b. Disconnect HI-Z (-042) and LO-Z (-043) test leads from adapter harness No. 1 HI-Z and LO-Z tank unit center tee connections and connect them to No. 2 tank unit center tee connections.
- c. Push in NO. 1 AC INST circuit breaker on copilot's circuit breaker panel.
- d. Set test set control switches as follows:

CONTROL	POSITION
FUNCT (S2)	MEASURE INT
SELECT (S3)	TU
SIMULATOR	00
COMP (V2)	

NOTE

It is important to adjust SIMULATOR TU 10-8000 PF vernier control knob as accurately as possible.

- e. Adjust test set SIMULATOR TU 10-8000 PF decade switches and vernier control knob until test set digital display indicates 70.6 pF to simulate an empty right main tank.
- f. Set test set FUNCT (S2) control switch to SIM TU ONLY.
- g. Connect multimeter to TANK 2 test points on the harness adapter test box.

NOTE

- Failure to adjust TK 2 E potentiometer as described can result in a miscalculation of the fuel quantity system.
 - The following procedure requires observing three possible results while performing an adjustment. Read and understand the following procedure before starting.
- h. While observing multimeter voltage and polarity, use a jeweler's screwdriver to adjust TK 2 E potentiometer on signal conditioner counterclockwise.
 - (1) If polarity on multimeter is positive (+) and adjustment decreases the voltage, continue counterclockwise adjustment until multimeter indicates between -0.005 and 0.005 vdc.
 - (2) If polarity on multimeter is positive (+) and adjustment increases the voltage, turn TK 2 E potentiometer clockwise until multimeter indicates negative (-) polarity. Continue clockwise adjustment until multimeter indicates between -0.005 and 0.005 vdc.
 - (3) If polarity on multimeter is negative (-), turn TK 2 E potentiometer clockwise until multimeter indicates between -0.005 and 0.005 vdc.

RESULT:

- (1) Multimeter shall indicate between -0.005 and 0.005 vdc.

- (2) FUEL QTY 2 vertical indicator and TOTAL FUEL or MAIN FUEL digital display on CDU shall indicate 0 pounds of fuel.

- If result (1) is not as specified, check for 115 vac between P905-H and ground.

- (a) If voltage is as specified, replace fuel quantity signal conditioner (PARA 10-4-36).

- (b) If voltage is not as specified, check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.

- 1 If voltage is as specified, repair/replace wiring between P905-H and NO. 1 AC INST circuit breaker terminal 1 (Appendix F).

- 2 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

- If result (2) is not as specified, check continuity between the following:

P905-U	and	P137-G
P905-V	and	P137-H

- (a) If continuity is present, troubleshoot instrument display system (PARA 8-2-3).

- (b) If continuity is not present, repair/replace wiring (Appendix F) as required.

- i. Set test set control switches as follows:

CONTROL	POSITION
FUNCT (S2)	MEASURE INT
SELECT (S3)	TU
SIMULATOR	00
COMP (V2)	

NOTE

It is important to adjust SIMULATOR TU 10-8000 PF vernier control knob as accurately as possible.

- j. Adjust test set SIMULATOR TU 10-8000 PF decade switches and vernier control knob until test set digital display indicates 111.7 pF to simulate a full right main tank.
- k. Set test set FUNCT (S2) control switch to SIM TU ONLY.
- l. Using a jeweler's screwdriver, adjust TK 2 F potentiometer on signal conditioner until FUEL QTY 2 vertical indicator on CDU indicates 1150 pounds.
- m. Slowly adjust TK 2 F potentiometer until FUEL QTY 2 vertical indicator lights a steady 1200 pounds indication.

RESULT:

- (1) FUEL QTY 2 vertical indicator and TOTAL FUEL or MAIN FUEL digital display on CDU shall indicate a steady 1200 pounds of fuel.

- (2) Multimeter shall indicate between 5.97 and 6.03 vdc.

- If result (1) is not as specified, check continuity between the following:

P905-U	and	P137-G
P905-V	and	P137-H

- (a) If continuity is present, troubleshoot instrument display system (PARA 8-2-3).

- (b) If continuity is not present, repair/replace wiring (Appendix F) as required.

- If result (2) is not as specified, check for 115 vac between P905-H and ground.

- (a) If voltage is as specified, replace fuel quantity signal conditioner (PARA 10-4-36).
- (b) If voltage is not as specified, check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.
 - 1 If voltage is as specified, repair/replace wiring between P905-H and NO. 1 AC INST circuit breaker terminal 1 (Appendix F).
 - 2 If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

n. Do System Check.

10-2-6.7 System Check.

- a. Turn off test set power switch (S1) and turn off electrical power to helicopter (PARA 1-6-16).
- b. Disconnect harness adapter test box from signal conditioner.
- c. Connect helicopter harness P905 to signal conditioner.
- d. If necessary, refuel main tanks (PARA 1-3-18 or PARA 1-3-17).
- e. Turn on electrical power (PARA 1-6-16).
- f. Press and hold FUEL IND TEST pushbutton on the miscellaneous switch panel.

RESULT:

- (1) FUEL QTY 1 and 2 vertical indicators, and TOTAL FUEL or MAIN FUEL digital display on CDU shall change indication.
- (2) #1 FUEL LOW caution capsule shall flash.
- (3) #2 FUEL LOW caution capsule shall flash.
- If result (1) is not as specified, go to TABLE NO. 10-2-6-3.
- If result (2) or (3) is not as specified, troubleshoot fuel low-level warning system (PARA 10-2-1).

g. Release FUEL IND TEST pushbutton.

RESULT: FUEL QTY 1 and 2 vertical indicators, and TOTAL FUEL or MAIN FUEL digital display shall return to original indications.

- If result is not as specified, replace miscellaneous switch panel (PARA 9-4-7).

h. Go to Shutdown.

10-2-6.8 Shutdown.

- a. Turn off electrical power (PARA 1-6-16).
- b. Slide signal conditioner cover plate back into place to cover adjustment potentiometers, and tighten retaining screws.
- c. Install interior soundproofing (PARA 2-4-69).

TABLE NO. 10-2-6-1. Probe Capacitance Is Not As Specified.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between:	

TABLE NO. 10-2-6-1. Probe Capacitance Is Not As Specified. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
P905-M and J307-D P905-N and J308-D P905-B and J307-G P905-D and J308-G	Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, go to 3.
2. Check continuity of wires on fuel tank wiring harness.	Step 1. If continuity is present, replace fuel quantity probe (PARA 10-4-33). Go to 4. Step 2. If continuity is not present, replace fuel tank wiring harness (PARA 10-4-34). Go to 4.
3. Check continuity on miscellaneous switch panel between: J135-L and J135-K J135-H and J135-J	Step 1. If continuity is present, repair/replace wiring (Appendix F) as required between, then go to 4: P135-J and J307-D P135-H and P905-M P135-K and J308-D P135-L and P905-N P905-B and J307-G P905-D and J308-G Step 2. If continuity is not present, replace miscellaneous switch panel (PARA 9-4-7). Go to 4.
4. Procedure completed.	

TABLE NO. 10-2-6-2. Multimeter Does Not Indicate Greater Than 10 Megohms.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for greater than 10 megohms between:	

TABLE NO. 10-2-6-2. Multimeter Does Not Indicate Greater Than 10 Megohms. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	P905-B and P905-A P905-B and P905-P P905-D and P905-E P905-D and P905-R P307-G and P307-F P307-G and P307-H P307-G and P307-K P307-G and P307-E P308-G and P308-F P308-G and P308-H P308-G and P308-K P308-G and P308-E
	Step 1. If resistance is as specified, replace fuel quantity probe (PARA 10-4-33). Go to 2 . Step 2. If resistance is not as specified, repair/replace wiring (Appendix F) as required. Go to 2 .
	2. Procedure completed.

TABLE NO. 10-2-6-3. TOTAL FUEL Or MAIN FUEL And/Or FUEL QTY 1 And/Or FUEL QTY 2 Indicators Do Not Change When FUEL IND TEST Pushbutton Is Pressed.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check continuity between: P905-N and P135-L P905-M and P135-H P135-K and J308-D P135-J and J307-D
	Step 1. If continuity is present, go to 2 . Step 2. If continuity is not present, repair/replace wiring (Appendix F) as required. Go to 3 .
	2. With FUEL IND TEST pushbutton pressed, check continuity between: J135- L and J135-K J135-H and J135-J

TABLE NO. 10-2-6-3. TOTAL FUEL Or MAIN FUEL And/Or FUEL QTY 1 And/Or FUEL QTY 2 Indicators Do Not Change When FUEL IND TEST Pushbutton Is Pressed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If continuity is present, replace miscellaneous switch panel (PARA 9-4-7). Go to 3 .
	Step 2. If continuity is not present, replace fuel quantity signal conditioner (PARA 10-4-36). Go to 3 .
3. Procedure completed.	

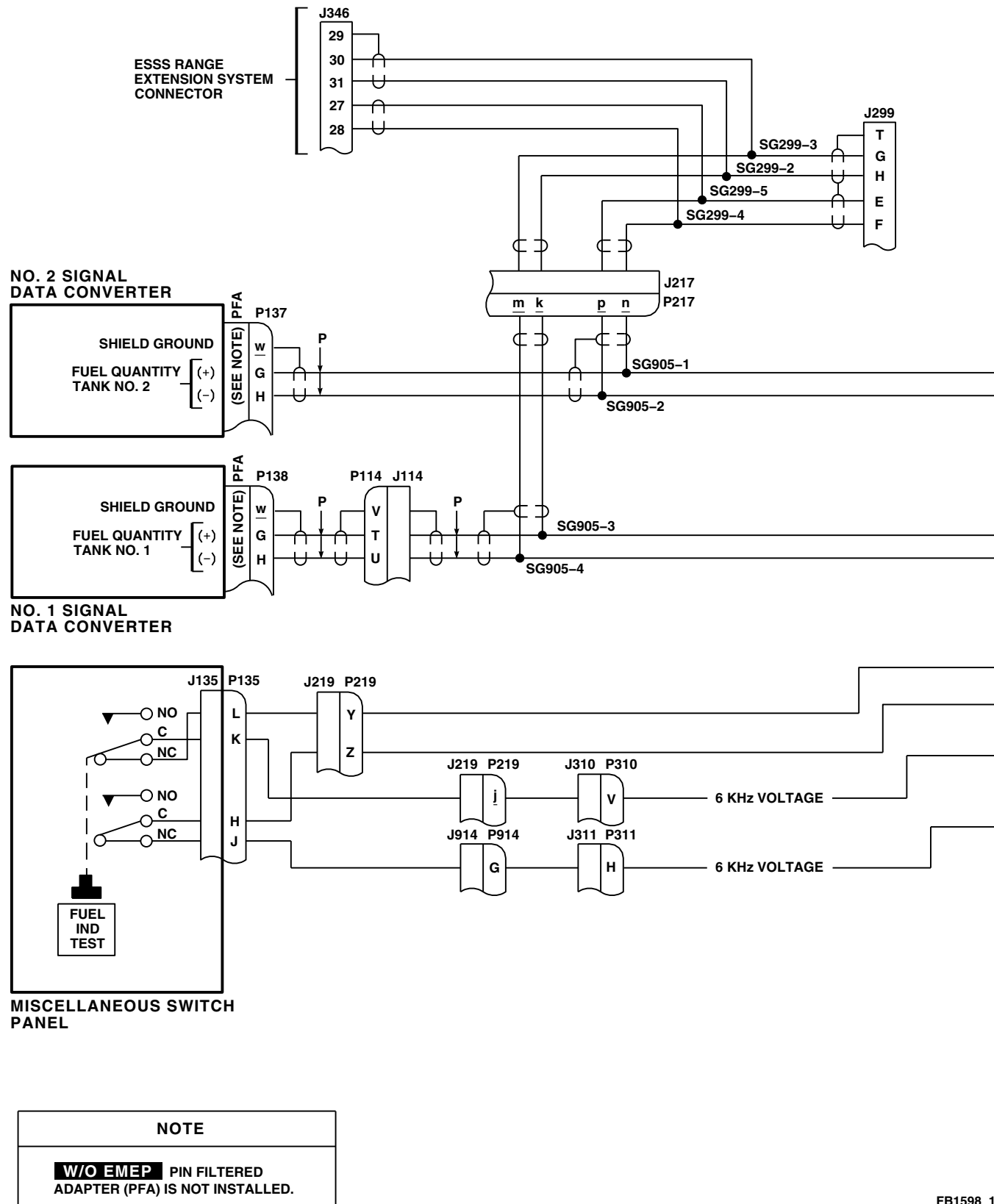


FIGURE 10-2-6-1. FUEL QUANTITY SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 2)

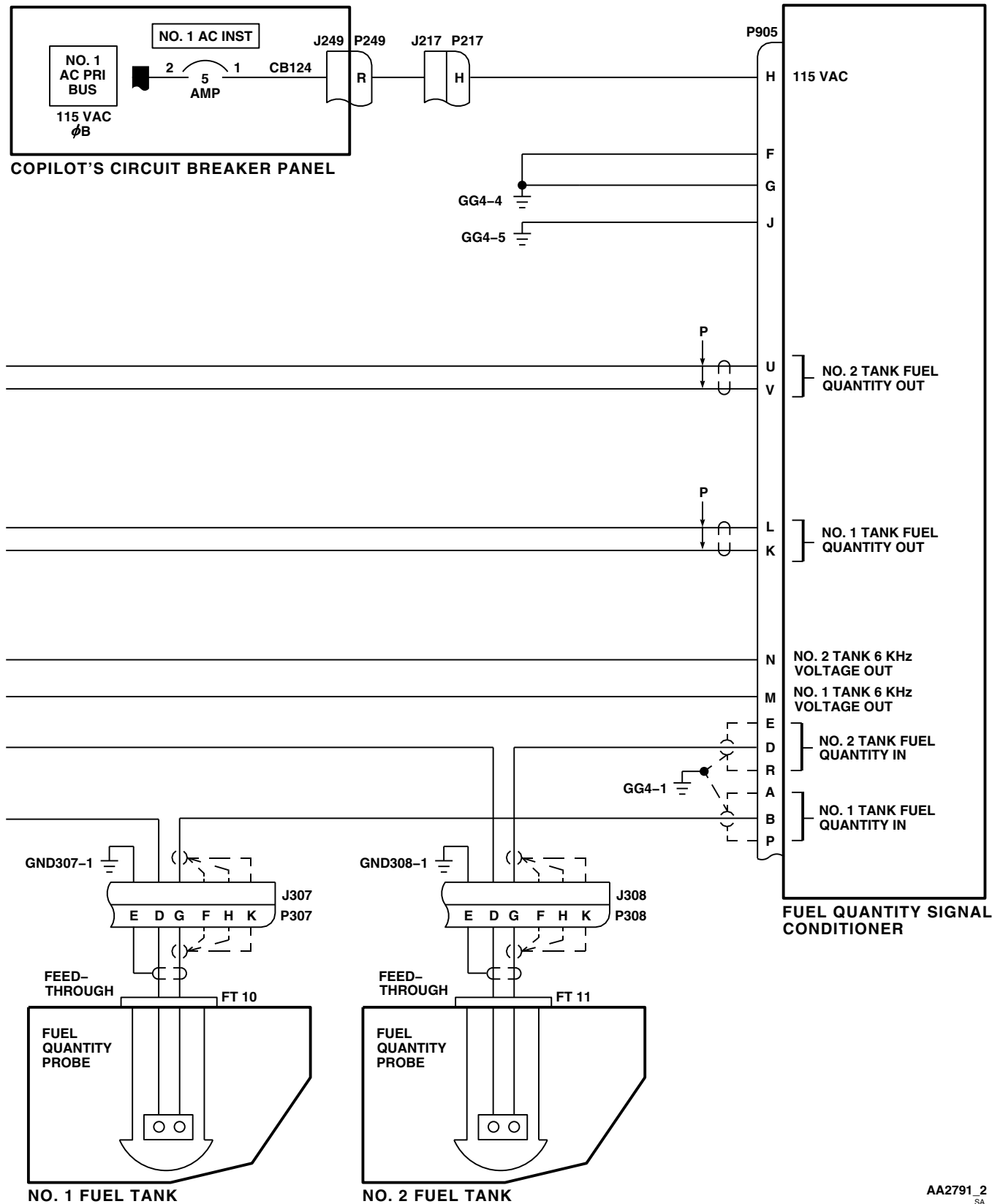


FIGURE 10-2-6-1. FUEL QUANTITY SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 2)

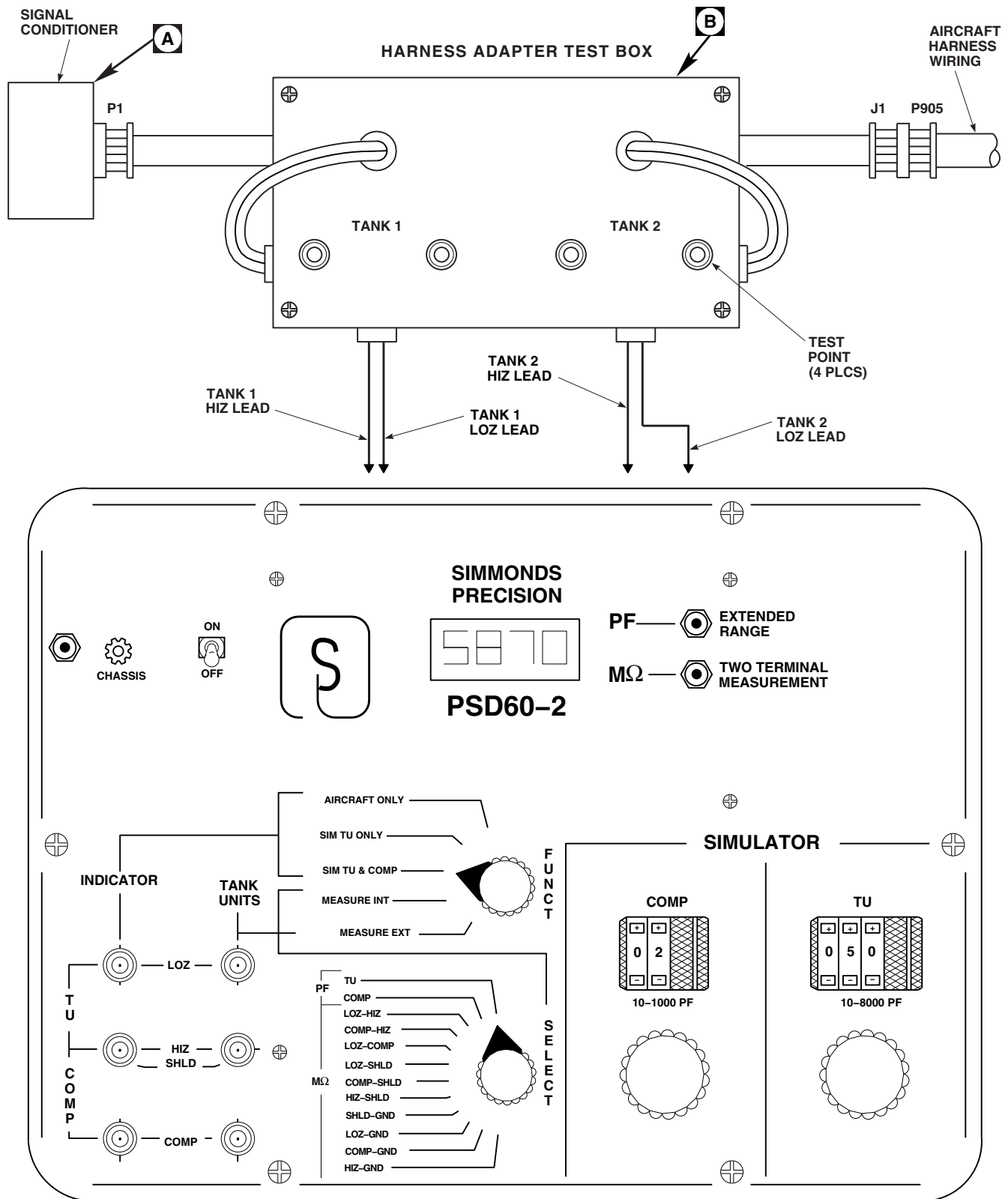


FIGURE 10-2-6-2. FUEL QUANTITY SYSTEM TEST SETUP DIAGRAM USING PSD 40-542-1 HARNESS ADAPTER (SHEET 1 OF 2)

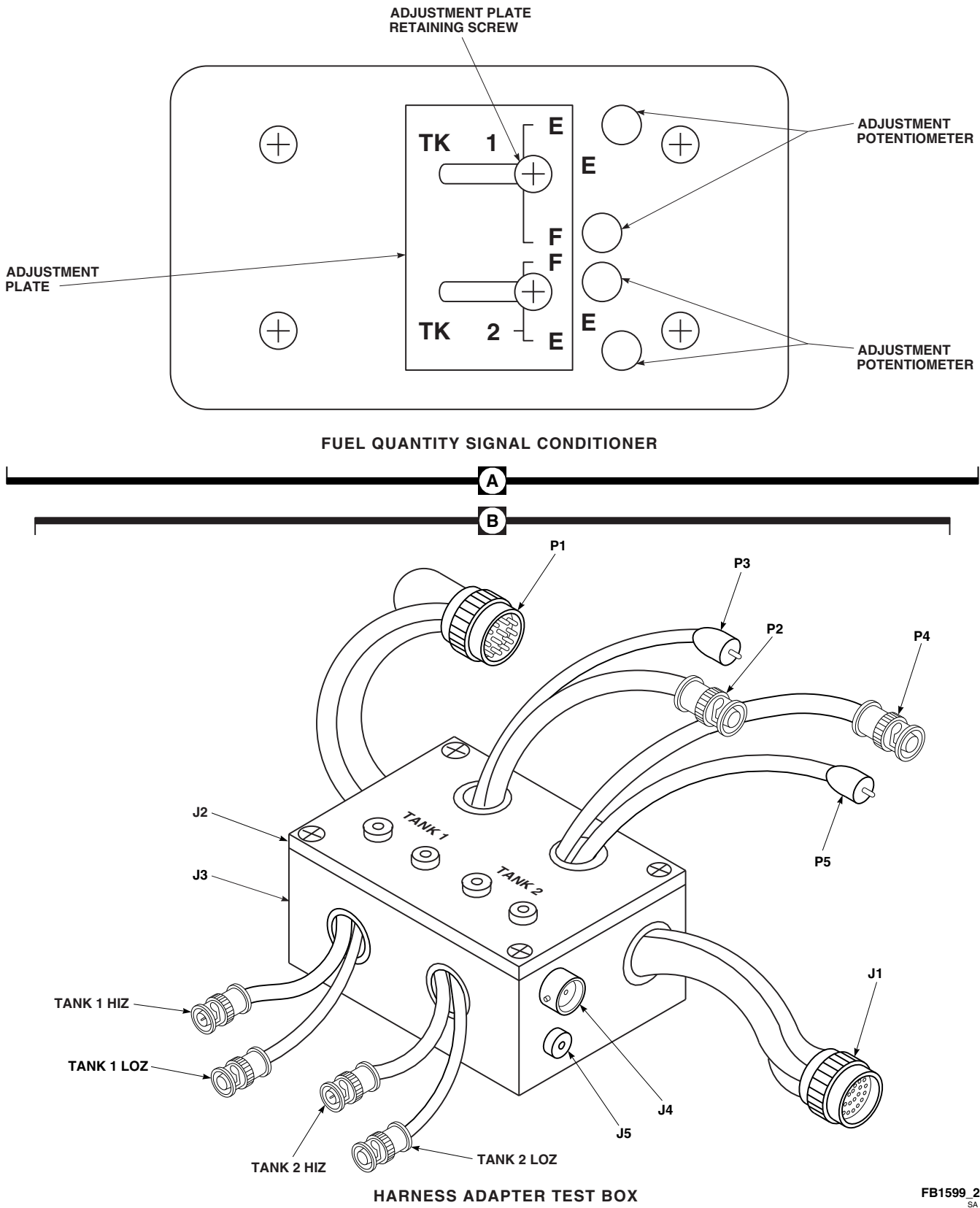


FIGURE 10-2-6-2. FUEL QUANTITY SYSTEM TEST SETUP DIAGRAM USING PSD 40-542-1 HARNESS ADAPTER (SHEET 2 OF 2)

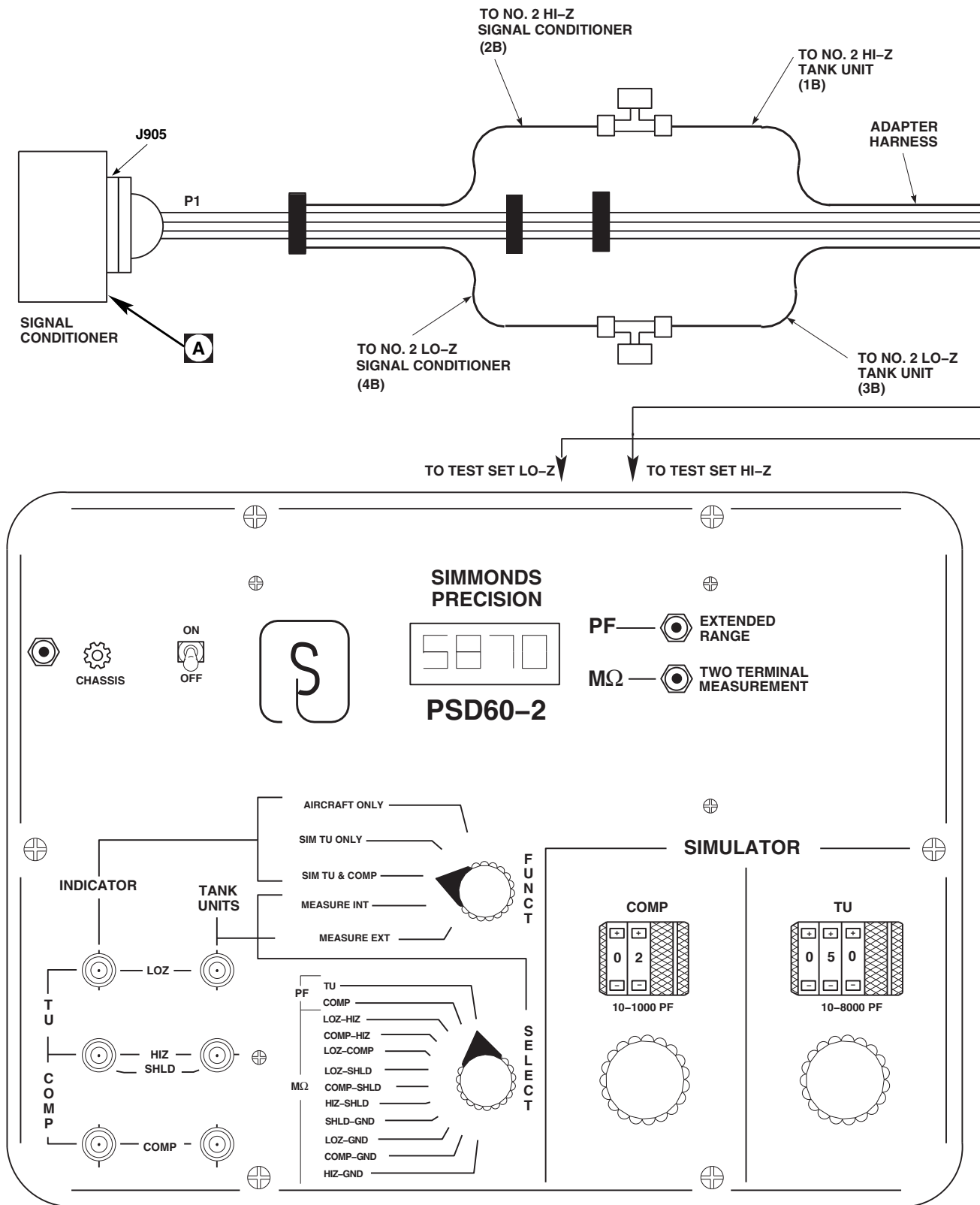
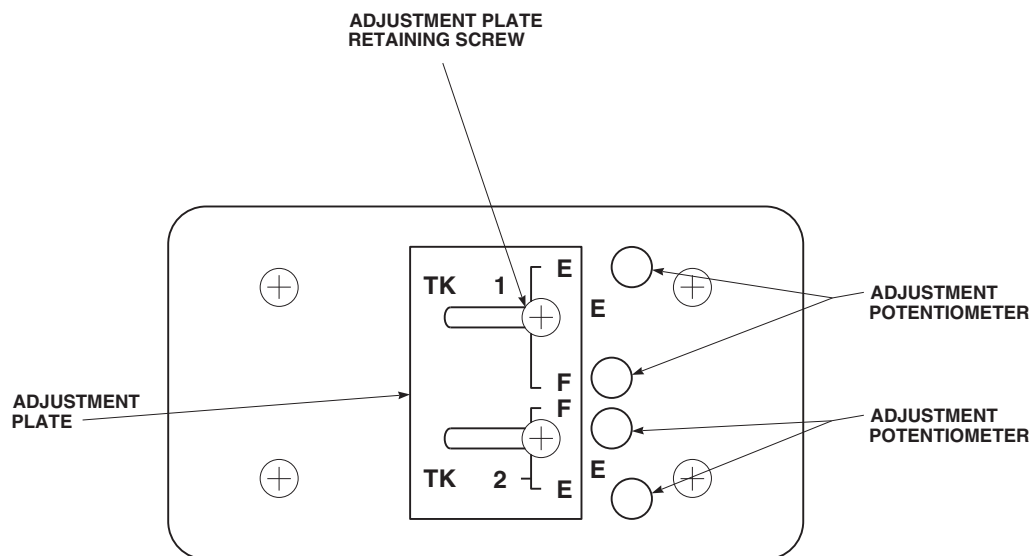
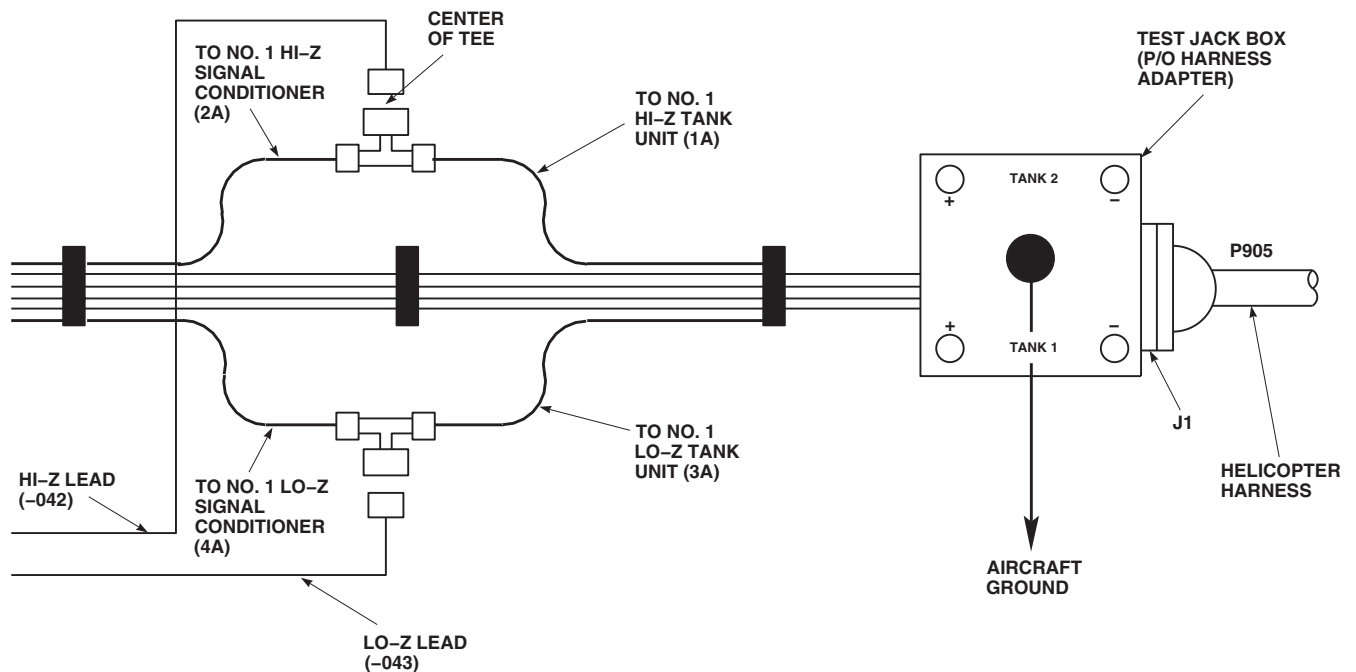
FB1600_1
SA

FIGURE 10-2-6-3. FUEL QUANTITY SYSTEM TEST SETUP DIAGRAM USING T7030-01014 HARNESS ADAPTER (SHEET 1 OF 2)

10-2-6-30



FUEL QUANTITY SIGNAL CONDITIONER

FF0187_2
SA

FIGURE 10-2-6-3. FUEL QUANTITY SYSTEM TEST SETUP DIAGRAM USING T7030-01014 HARNESS ADAPTER (SHEET 2 OF 2)

10-2-6-31/(10-2-6-32 Blank)

10-2-7 ENGINE FEED FUEL LINES.

INITIAL SETUP

Test Equipment

- Clear Line Assembly, 3/4 Inch diameter, 2 Feet (ft)
- Combination Vacuum/Pressure Gage, -16 Inches Of Mercury (in.Hg) To 50 Pounds Per Square Inch Gaged (psig)
- Discharge Hose, 1.5 in. Diameter, 40 ± 10 ft., JetA-JP-8 compatible
- Flow Meter, 1-5 Gallons Per Minute (gpm)
- Fuel Tanker Truck, with fueling capabilities as follows:
 Fueling: 175 gpm at 50 pounds psig nozzle pressure
 Defueling: 90 gpm at -3 psig nozzle pressure
- Spill Collection Pans, 5 gallons, 2 required
- Suction Hose, 1 in. diameter, 40 ± 10 ft., JetA/JP-8 compatible
- T-Fitting, fit 3/4 in. diameter hose
- Variable Speed Pump, JetA/JP-8 fuel, 0-5 gpm

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16
- PARA 10-2-2

Equipment Conditions

- External Electrical Power Available

NOTE

If a circuit breaker pops during the Engine Feed Fuel Lines Operational/Troubleshooting Procedure, a short circuit is indicated (PARA 9-2-3).

Special Instructions

The Engine Feed Fuel Lines Operational/Troubleshooting Procedure is subdivided as follows:

- 10-2-7.1 No. 1 Engine Setup
- 10-2-7.2 No. 1 Engine, Direct Feed, Restriction and Leak Test
- 10-2-7.3 No. 1 Engine, Cross Feed, Restriction and Leak Test
- 10-2-7.4 No. 2 Engine Setup
- 10-2-7.5 No. 2 Engine, Direct Feed, Restriction and Leak Test
- 10-2-7.6 No. 2 Engine, Cross Feed, Restriction and Leak Test
- 10-2-7.7 Engine Fuel Feed Lines Pressure Test

10-2-7.1 No. 1 Engine Setup.

WARNING

- Persons working at or near the helicopter will dissipate body static potential by gripping the static ground line. This procedure shall be done occasionally during servicing operations. Failure to do so could cause a fire should a static electricity discharge ignite the fuel vapors, resulting in injury or death to personnel.
- The nozzle must always be bonded to the helicopter before the fill cap is removed. This connection must remain in place until after the tank cap is replaced. Failure to follow this procedure could cause a fire should a static electricity discharge ignite the fuel vapors, resulting in injury or death to personnel.

CAUTION

Do not load the engine feed breakaway valve during the test. Attach the clear line assembly and pressure gage so that it is supported and does not cause the engine feed hose to load the breakaway valve attached at the firewall.

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).
- Open No. 1 tank gravity fill port.
- Fill No. 1 tank using fuel truck gravity fuel nozzle (over wing nozzle) to 200 pounds total as indicated on central display unit (CDU) FUEL QTY 1.
- Close No. 1 tank gravity fill port.

NOTE

Refer to Figure 10-2-7-1 for all test lines setup.

- Disconnect fuel feed line from No. 1 engine.
- Add T-fitting connector in 3/4-inch diameter from clear line assembly to No. 1 fuel feed line disconnected in step f.
- Connect combination vacuum/pressure gage to T-fitting.
- Connect variable speed pump using 1-inch diameter suction hose to 3/4-inch diameter clear line assembly.
- Connect flowmeter to discharge port of variable speed pump.
- Connect 1.5-inch discharge hose to flowmeter.
- Place open end of discharge hose in 55 gallon drum or other approved waste container.

10-2-7.2 No. 1 Engine, Direct Feed, Restriction and Leak Test.

CAUTION

To prevent damage to the fuel prime pump, do not run the fuel prime pump for more than 1 minute, if no fuel flow is visible in the clear line assembly.

- Do No. 1 Engine Setup, if not already done.
- Place No. 1 engine fuel selector to DIR and No. 2 engine fuel selector to OFF.
- Place and hold FUEL PUMP switch to FUEL PRIME and verify fuel is flowing through clear line assembly.
- Release FUEL PUMP SWITCH.
- Place NO. 1 PUMP switch (FUEL BOOST PUMP CONTROL - lower console) to ON.

RESULT: NO. 1 BOOST PUMP light shall go on.

- If light does not go on, check NO. 1 BOOST PUMP light by pressing press-to-test light.
- (a) If press-to-test light does not go on, replace lamp.
- f. Once fuel is seen flowing through clear line assembly, start variable speed pump and place NO. 1 PUMP switch to off.
- g. Adjust variable speed pump to flow rate of 2.5 ± 0.5 gpm on flowmeter.

RESULT: Vacuum gage shall not exceed -10 in. Hg.

- If result is not as specified, troubleshoot Boost Pump (PARA 10-2-2), then repeat steps e. through g.
- h. Run pump to clear any initial bubbles from Fuel System; continue to monitor clear line assembly for bubbles.
- i. Run leak-test pump until 70 ± 25 pounds of fuel remain in the fuel cell to make sure lines and fittings down to inlet check valve are tested for leakage.

RESULT: Bubbles less than 1/8-inch in diameter are acceptable, and shall not exceed 3 per 30 seconds flowing through clear line assembly.

- If result is not as specified, stop test and repair any leaks, then repeat step i.
- j. Shut down test pump.
- k. Empty 55 gallon drum or other approved waste fuel container.

10-2-7.3 No. 1 Engine, Cross Feed, Restriction and Leak Test.

- a. Do No. 1 Engine Setup, if not already done.
- b. Place No. 1 engine fuel selector to XFED.

- c. Place and hold FUEL PUMP switch to FUEL PRIME and verify fuel is flowing through clear line assembly.

- d. Release FUEL PUMP SWITCH.

- e. Place NO. 1 PUMP switch (FUEL BOOST PUMP CONTROL - lower console) to ON.

RESULT: NO. 1 BOOST PUMP light shall go on.

- If light does not go on, check NO. 1 BOOST PUMP light by pressing press-to-test light.

- (a) If press-to-test light does not go on, replace lamp.

- f. Once fuel is seen flowing through clear line assembly, start variable speed pump and place NO. 1 PUMP switch to off.

- g. Adjust variable speed pump to flow rate of 2.5 ± 0.5 gpm on flowmeter.

RESULT: Vacuum gage shall not exceed -10 in. Hg.

- If result is not as specified, troubleshoot Boost Pump (PARA 10-2-2), then repeat steps e. through g.

- h. Run pump to clear any initial bubbles from Fuel System; continue to monitor clear line assembly for bubbles.

- i. Run leak-test pump until 70 ± 25 pounds of fuel remain in the fuel cell to make sure lines and fittings down to inlet check valve are tested for leakage.

RESULT: Bubbles less than 1/8-inch in diameter are acceptable, and shall not exceed 3 per 30 seconds flowing through clear line assembly.

- If result is not as specified, stop test and repair any leaks, then repeat step i.

- j. Place No. 1 fuel selector to OFF.

NOTE

To minimize fuel spillage when removing clear line assembly, run test pump and crack hose fitting to evacuate line.

- k. Remove clear line assembly from No. 1 engine fuel feed line and reconnect feed line to No. 1 engine.
- l. Shut down test pump.
- m. Empty 55 gallon drum or other approved waste fuel container.

10-2-7.4 No. 2 Engine Setup.**WARNING**

- Persons working at or near the helicopter will dissipate body static potential by gripping the static ground line. This procedure shall be done occasionally during servicing operations. Failure to do so could cause a fire should a static electricity discharge ignite the fuel vapors, resulting in injury or death to personnel.

- The nozzle must always be bonded to the helicopter before the fill cap is removed. This connection must remain in place until after the tank cap is replaced. Failure to follow this procedure could cause a fire should a static electricity discharge ignite the fuel vapors, resulting in injury or death to personnel.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- b. Turn on electrical power (PARA 1-6-16).
- c. Open No. 2 tank gravity fill port.
- d. Fill No. 2 tank using fuel truck gravity fuel nozzle (over wing nozzle) to 200 pounds total as indicated on CDU FUEL QTY 2.

- e. Close No. 2 tank gravity fill port.

NOTE

Refer to Figure 10-2-7-1 for all test lines setup.

- f. Disconnect fuel feed line from No. 2 engine.
- g. Add T-fitting connector in 3/4-inch diameter from clear line assembly to No. 2 fuel feed line disconnected in step f.
- h. Connect combination vacuum/pressure gage to T-fitting.
- i. Connect variable speed pump using 1-inch diameter suction hose to 3/4-inch diameter clear line assembly.
- j. Connect flowmeter to discharge port of variable speed pump.
- k. Connect 1.5-inch discharge hose to flowmeter.
- l. Place open end of discharge hose in 55 gallon drum or other approved waste container.

10-2-7.5 No. 2 Engine, Direct Feed, Restriction and Leak Test.**CAUTION**

To prevent damage to the fuel prime pump, do not run the fuel prime pump for more than 1 minute, if no fuel flow is visible in the clear line assembly.

- a. Do No. 2 Engine Setup, if not already done.
- b. Place No. 2 engine fuel selector to DIR and No. 1 engine fuel selector to OFF.
- c. Place and hold FUEL PUMP switch to FUEL PRIME and verify fuel is flowing through clear line assembly.
- d. Release FUEL PUMP SWITCH.

- e. Place NO. 2 PUMP switch (FUEL BOOST PUMP CONTROL - lower console) to ON.

RESULT: NO. 2 BOOST PUMP light shall go on.

- If light does not go on, check NO. 2 BOOST PUMP light by pressing press-to-test light.
- (a) If press-to-test light does not go on, replace lamp.
- f. Once fuel is seen flowing through clear line assembly, start variable speed pump and place NO. 2 PUMP switch to off.
 - g. Adjust variable speed pump to flow rate of 2.5 ± 0.5 gpm on flowmeter.

RESULT: Vacuum gage shall not exceed -10 in. Hg.

- If result is not as specified, troubleshoot Boost Pump (PARA 10-2-2), then repeat steps e. through g.
- h. Run pump to clear any initial bubbles from Fuel System; continue to monitor clear line assembly for bubbles.
 - i. Run leak-test pump until 75 ± 25 pounds of fuel remain in the fuel cell to make sure lines and fittings down to inlet check valve are tested for leakage.

RESULT: Bubbles less than 1/8-inch in diameter are acceptable, and shall not exceed 3 per 30 seconds flowing through clear line assembly.

- If result is not as specified, stop test and repair any leaks, then repeat step i.
- j. Shut down test pump.
 - k. Empty 55 gallon drum or other approved waste fuel container.

10-2-7.6 No. 2 Engine, Cross Feed, Restriction and Leak Test.

- a. Do No. 2 Engine Setup, if not already done.
- b. Place No. 2 engine fuel selector to XFED.

- c. Place and hold FUEL PUMP switch to FUEL PRIME and verify fuel is flowing through clear line assembly.

- d. Release FUEL PUMP SWITCH.

- e. Place NO. 2 PUMP switch (FUEL BOOST PUMP CONTROL - lower console) to ON.

RESULT: NO. 2 BOOST PUMP light shall go on.

- If light does not go on, check NO. 2 BOOST PUMP light by pressing press-to-test light.
- (a) If press-to-test light does not go on, replace lamp.
- f. Once fuel is seen flowing through clear line assembly, start variable speed pump and place NO. 2 PUMP switch to off.
 - g. Adjust variable speed pump to flow rate of 2.5 ± 0.5 gpm on flowmeter.

RESULT: Vacuum gage shall not exceed -10 in. Hg.

- If result is not as specified, troubleshoot Boost Pump (PARA 10-2-2), then repeat steps e. through g.
- h. Run pump to clear any initial bubbles from Fuel System; continue to monitor clear line assembly for bubbles.
 - i. Run leak-test pump until 75 ± 25 pounds of fuel remain in the fuel cell to make sure lines and fittings down to inlet check valve are tested for leakage.

RESULT: Bubbles less than 1/8-inch in diameter are acceptable, and shall not exceed 3 per 30 seconds flowing through clear line assembly.

- If result is not as specified, stop test and repair any leaks, then repeat step i.
- j. Place No. 2 fuel selector to OFF.

NOTE

To minimize fuel spillage when removing clear line assembly, run test pump and crack hose fitting to evacuate line.

- k. Remove clear line assembly from No. 2 engine fuel feed line and reconnect feed line to No. 2 engine.
- l. Shut down test pump.
- m. Empty 55 gallon drum or other approved waste fuel container.

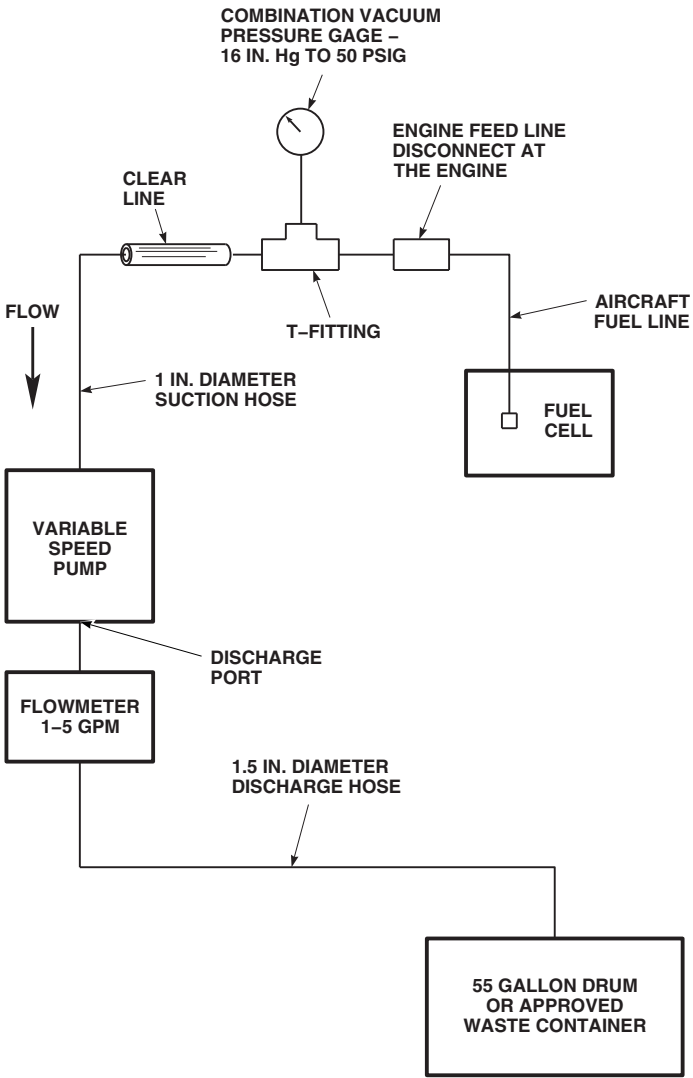
10-2-7.7 Engine Fuel Feed Lines Pressure Test.**NOTE**

Before beginning procedure, verify No. 1 and No. 2 tanks each have at least 70 pounds of fuel. Fuel tanks if required.

- a. Place No. 1 and No. 2 fuel selector valves to DIR.
- b. Place NO. 1 PUMP switch and NO. 2 PUMP switch (FUEL BOOST PUMP CONTROL - lower console) to ON.

RESULT: NO. 1 and NO. 2 BOOST PUMP lights shall go on.

- If lights do not go on, check NO. 1 and NO. 2 BOOST PUMP lights by pressing press-to-test light as required.
 - (a) If press-to-test light does not go on, replace lamp as required.
- c. Operate both boost pumps for 5 minutes.
- d. Place No. 1 and No. 2 fuel selector valves to XFED.
- e. Operate both boost pumps for additional 5 minutes.
- f. Place NO. 1 PUMP switch and NO. 2 PUMP switch (FUEL BOOST PUMP CONTROL - lower console) to OFF.
- g. Place No. 1 and No. 2 fuel selector valves to OFF.
- h. Turn off electrical power (PARA 1-6-16).
- i. Inspect all engine feed fuel lines (beginning from top of each tank) for signs of leakage.



FC1872
SA

FIGURE 10-2-7-1. TEST EQUIPMENT SETUP DIAGRAM

SECTION 10-3

ON AIRCRAFT INSPECTIONS

SECTION OVERVIEW

PARAGRAPH	TITLE
10-3-1	Main Fuel Cells Pressure Test (Primary Method)
10-3-2	Main Fuel Cells Pressure Test (Alternate Method)
10-3-3	Main Fuel System Inspections

10-3-1 MAIN FUEL CELLS PRESSURE TEST (PRIMARY METHOD).

PARAGRAPH BREAKDOWN

	PAGE
10-3-1.1 Main Fuel Cells Pressure Test (Primary Method)	10-3-1-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container
- Fuel Cell Cap Pressure Test Adapter, Locally-Made, Figure H-212, Appendix H (2)
- Manometer, A338
- Oil/Water Separator
- Radiator Plug, 2 ¼-inch

Materials/Parts

- Tee, AN824-6 (2)

References

- Appendix H
- PARA 1-3-17
- PARA 1-3-18
- PARA 1-3-19
- PARA 1-3-20
- PARA 1-3-21
- PARA 10-4-25
- SEI 671
- TM 1-2840-248-23

10-3-1.1 Main Fuel Cells Pressure Test (Primary Method).

- Defuel main fuel cells (PARA 1-3-20 or PARA 1-3-21).
- Drain fuel cell sumps (PARA 1-3-21).
- On engine control quadrant, make sure NO. 1 and NO. 2 ENG FUEL SYS levers are in OFF position.

- Locally make two fuel cell cap pressure test adapters (Figure H-212, Appendix H).
- Open both gravity fuel access covers (Figure 10-3-1-1, Detail A).
- Remove gravity fill caps as follows:
 - Release thumb lock on top of cap by pulling up on thumb lock.

CAUTION

Injury to personnel will result if main fuel cell pressure is not removed prior to removing filler cap. Before turning thumb lock, listen for escaping air and make sure no pressure remains in fuel cell.

- (2) Turn thumb lock counterclockwise until it stops.
- (3) Remove cap and let it hang by lanyard.
- g. Install main fuel cell cap pressure test adapters.
- h. Install 2 1/4-inch radiator plug into fuel vent at STA 443.0, left lower side of helicopter.
- i. Close inlet and regulator valves on oil/water separator.
- j. Connect air source to oil/water separator inlet valve.
- k. Connect hoses and tees to oil/water separator outlet valve, fuel cell cap pressure test adapters, and manometer.
- l. Open inlet and outlet valves on oil/water separator.
- m. Slowly open regulator valve on oil/water separator until gage on valve reads 0.5 psig. When manometer scale reads 13.8-inches H₂O, close outlet valve.
- n. After 15-minute period, look at manometer scale to see if dimension has decreased, indicating pressure leak. **GREATEST ALLOWABLE PRESSURE DECREASE IS 1/8-INCH ON MANOMETER SCALE FOR 15-MINUTE PERIOD.**
- o. Open outlet valve on oil/water separator.
- p. Slowly open regulator valve on oil/water separator until gage on valve reads 1 psig. When manometer scale reads 27.67-inches H₂O, close outlet valve.

- q. After 15-minute period, look at manometer scale to see if dimension has decreased, indicating pressure leak. **GREATEST ALLOWABLE PRESSURE DECREASE IS 1/8-INCH ON MANOMETER SCALE FOR 15-MINUTE PERIOD.**
- r. If leakage is beyond limits, check caps on gravity refueling ports for proper seating, and all line and hose connections. Retest main fuel cells for allowable pressure.
- s. Test No. 1 fuel selector valve as follows:
 - (1) Stabilize manometer pressure at 27.67-inches H₂O.
 - (2) On engine control quadrant, place NO. 1 ENG FUEL SYS lever to OFF position. Manometer pressure may or may not drop off.
 - (3) Allow manometer to stabilize to 27.67-inches H₂O.
 - (4) Using container to catch fuel, open fuel line nut at No. 1 engine boost pump elbow (SEI 671 and TM 1-2840-248-23). **NO DROP OFF ON MANOMETER PRESSURE ALLOWED.** If manometer pressure drops off, perform the following:
 - (a) Replace No. 1 fuel selector valve (PARA 10-4-25).
 - (b) Repeat step s. (1) through step s. (4).
 - (5) Connect fuel line at No. 1 engine boost pump elbow (SEI 671 and TM 1-2840-248-23).
 - (6) On engine control quadrant, place NO. 1 ENG FUEL SYS lever to DIR position.
 - (7) Allow manometer to stabilize to 27.67-inches H₂O.
 - (8) Using container to catch fuel, open fuel line nut at No. 1 engine boost pump elbow (SEI 671 and TM 1-2840-248-23). **PRESSURE ON MANOMETER**

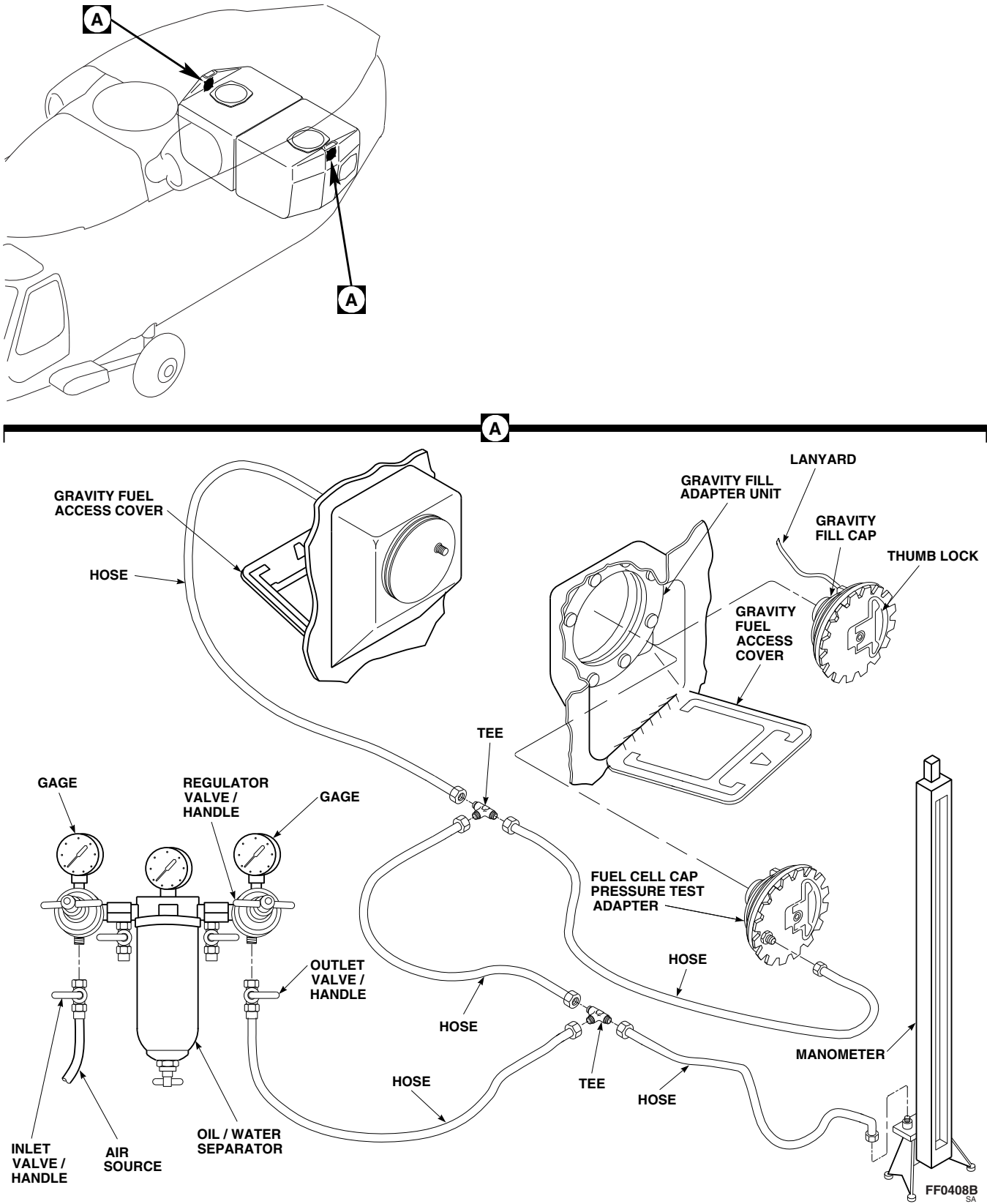


FIGURE 10-3-1-1. MAIN FUEL CELLS PRESSURE TEST (PRIMARY METHOD)

SHOULD DROP OFF. If manometer pressure does not drop off, perform the following:

- (a) Replace No. 1 fuel selector valve (PARA 10-4-25).
- (b) Repeat step s. (1) through step s. (8).
- (9) Connect fuel line nut at No. 1 engine boost pump elbow (SEI 671 and TM 1-2840-248-23).
- (10) On engine control quadrant, place NO. 1 ENG FUEL SYS lever to XFD position.
- (11) Allow manometer to stabilize to 27.67-inches H₂O.
- (12) Using container to catch fuel, open fuel line nut at No. 1 engine boost pump elbow (SEI 671 and TM 1-2840-248-23).
PRESSURE ON MANOMETER SHOULD DROP OFF. If manometer pressure does not drop off, perform the following:
 - (a) Replace No. 1 fuel selector valve (PARA 10-4-25).
 - (b) Repeat step s. (1) through step s. (12).
- (13) Connect fuel line at No. 1 engine boost pump elbow (SEI 671 and TM 1-2840-248-23).
- (14) Repeat step s. (1) through step s. (13) to test No. 2 fuel selector valve.
- t. Disconnect air source and remove hoses and tees from fuel cell cap pressure test adapters, manometer, and oil/water separator.
- u. Remove fuel cell cap pressure test adapters.
- v. Install gravity fill caps as follows:
 - (1) Place cap in housing and align forward-point arrow.
 - (2) Turn thumb lock clockwise to lock cap.
 - (3) Snap thumb lock closed to seal cap.
- w. Close gravity fuel access covers.
- x. Remove 2 1/4-inch radiator plug from fuel vent at STA 443.0, left lower side of helicopter.
- y. Refuel main fuel cells (PARA 1-3-17, PARA 1-3-18, or PARA 1-3-19).

10-3-2 MAIN FUEL CELLS PRESSURE TEST (ALTERNATE METHOD).

PARAGRAPH BREAKDOWN

	PAGE
10-3-2.1 Main Fuel Cells Pressure Test (Alternate Method)	10-3-2-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Fuel Cell Pressure Test Plate, Locally-Made, Figure H-206, Appendix H (2)
- Manometer, A338
- Oil/Water Separator
- Protection, Eye and Face
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Protective Caps and Plugs
- Tee, AN824-6 (2)

References

- Appendix D
- Appendix H
- PARA 1-3-17
- PARA 1-3-18
- PARA 1-3-19
- PARA 1-3-20
- PARA 1-3-21
- PARA 10-4-17
- PARA 10-4-39

10-3-2.1 Main Fuel Cells Pressure Test (Alternate Method).

- a. Defuel main fuel cells (PARA 1-3-20 or PARA 1-3-21).
- b. Drain fuel cell sumps (PARA 1-3-21).
- c. Remove main fuel system vent valves from both main fuel cells (PARA 10-4-17).

WARNING

Injury to personnel will result if proper eye and face protection is not used while disconnecting lines. Fuel lines may be under pressure. Use proper eye and face protection while disconnecting lines.

- d. On No. 1 main fuel cell, disconnect fuel hose from prime/boost pump fuel breakaway valve (Figure 10-3-2-1, Detail A). Install protective caps and plugs on valves and hoses.
- e. Disconnect No. 1 and No. 2 main fuel hoses from main fuel breakaway valves. Remove and discard packings. Install protective caps and plugs on valves and hoses.
- f. Locally make two fuel cell pressure test plates (Figure H-206, Appendix H).
- g. Install fuel cell pressure test plates and seals on self-sealing breakaway vent valves with bolts, washers, and nuts used to install system vent valves (Figure 10-3-2-1, Detail B).
- h. Close inlet and regulator valves on oil/water separator.
- i. Connect air source to oil/water separator inlet valve.
- j. Connect hoses and tees to oil/water separator outlet valve, fuel cell pressure test plates, and manometer.
- k. Open inlet and outlet valves on oil/water separator.
- l. Slowly open regulator valve on oil/water separator until gage on valve reads 0.5 psig. When manometer scale reads 13.8-inches H₂O, close outlet valve.
- m. After 15-minute period, check manometer scale to see if dimension has decreased, indicating pressure leak. **GREATEST ALLOWABLE PRESSURE DECREASE IS 1/8-INCH ON MANOMETER SCALE FOR 15-MINUTE PERIOD.**
- n. Open outlet valve on oil/water separator.
- o. Slowly open regulator valve until gage on valve reads 1 psig. When manometer scale reads 27.67-inches H₂O, close outlet valve.
- p. After 15-minute period, check manometer scale to see if dimension has decreased, indicating pressure leak. **GREATEST ALLOWABLE PRESSURE DECREASE IS 1/8-INCH ON MANOMETER SCALE FOR 15-MINUTE PERIOD.**
- q. If leakage is beyond limits, check caps on gravity refueling ports for proper seating, and all line and hose connections. Retest main fuel cells for allowable pressure.
- r. Disconnect air source and remove hoses and tees from fuel cell pressure test plates, manometer, and oil/water separator.
- s. Remove bolts, washers, and nuts securing fuel cell pressure test plates and seals to self-sealing breakaway vent valves. Remove plates and seals.
- t. Install main fuel system vent valves (PARA 10-4-17).
- u. Test fuel feed system (PARA 10-4-39).
- v. Lubricate packings with lubricating oil, Item 205, Appendix D. Remove caps and install packings on main fuel breakaway valves (Figure 10-3-2-1, Detail A).
- w. Coat threads of main fuel breakaway valves with antiseize compound, Item 74, Appendix D.
- x. Remove plugs and connect No. 1 and No. 2 main fuel hoses to valves. **TIGHTEN FITTINGS HANDTIGHT.**
- y. On No. 1 main fuel cell connect fuel hose as follows:
 - (1) Remove cap from prime/boost pump fuel breakaway valve.
 - (2) Coat threads of valve with antiseize compound, Item 74, Appendix D.

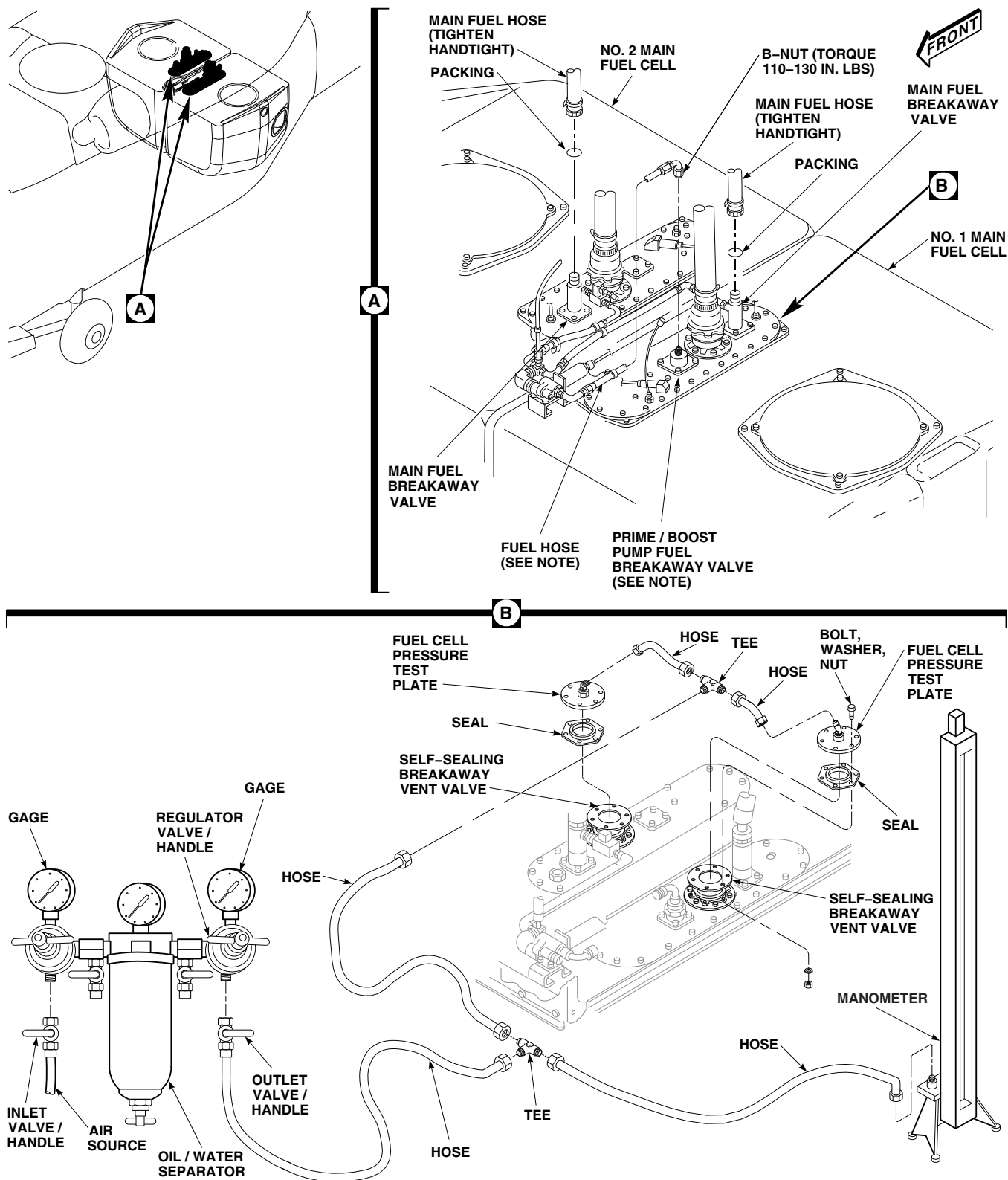


FIGURE 10-3-2-1. MAIN FUEL CELLS PRESSURE TEST (ALTERNATE METHOD)

- (3) Remove protective plug and connect fuel hose to prime/boost pump fuel breakaway valve.
- (4) **TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**
- z. Refuel main fuel cells (PARA 1-3-17, PARA 1-3-18, or PARA 1-3-19).

10-3-3 MAIN FUEL SYSTEM INSPECTIONS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
10-3-3.1	Crossfeed Breakaway Valve	10-3-3-1	10-3-3.4	Main Fuel Cell Ballistic Ring	10-3-3-2
10-3-3.2	Breakaway Tee Valve	10-3-3-1			
10-3-3.3	Engine Compartment Breakaway Valve	10-3-3-1			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 10-4-2
- PARA 10-4-24

- PARA 10-4-26
- PARA 10-4-27
- PARA 10-5-1

10-3-3.1 Crossfeed Breakaway Valve.

- Open engine cowling.
- Inspect crossfeed breakaway valve for visible yellow line around center of valve. If yellow line is visible, replace crossfeed breakaway valve (PARA 10-4-26).
- Close engine cowling.

10-3-3.2 Breakaway Tee Valve.

- Open engine cowling.
- Inspect breakaway tee valve for visible yellow line around center of valve. If yellow line is

visible, replace breakaway tee valve (PARA 10-4-27).

- Close engine cowling.

10-3-3.3 Engine Compartment Breakaway Valve.

- Open engine cowling.
- Inspect engine compartment breakaway valve for visible yellow line around center of valve. If yellow line is visible, replace engine compartment breakaway valve (PARA 10-4-24).
- Close engine cowling.

10-3-3.4 Main Fuel Cell Ballistic Ring.**WARNING**

Injury to personnel will result if care is not taken when opening fuel sump drain door. Fuel may be trapped above door. Take care when opening door.

- a. Unlock latches and turnlock fasteners, and open applicable fuel sump drain door.

NOTE

Activation can be identified by bulging appearance on the inside diameter,

outside diameter, contact surfaces, or by a purple jelly-like substance extruding from the main fuel cell ballistic ring.

- b. Inspect left and right fuel main fuel cell ballistic rings located in fuel cell sump for activation. If activation is suspected, replace main fuel cell (PARA 10-4-2) or ballistic ring (PARA 10-5-1).
- c. Close fuel sump drain door, and lock latches and turnlock fasteners.

SECTION 10-4

MAINTENANCE PROCEDURES

SECTION OVERVIEW

PARAGRAPH	TITLE
10-4-1	Fuel System Priming
10-4-2	Main Fuel Cell
10-4-3	Main Fuel Cell Plate and Attached Components
10-4-4	High Level Shutoff Valves
10-4-5	Pressure Refuel/Defuel Valve
10-4-6	Sump Drain Valve
10-4-7	Prime/Boost Pump Check Valve
10-4-8	Main Fuel Line Check Valve
10-4-9	Low Level Shutoff Valve
10-4-10	Pressure Refueling Interconnect Tube
10-4-11	Extended Range Interconnect Hose
10-4-12	Fuel Boost Pump
10-4-13	Main Fuel Cell Check Valve
10-4-14	Fuel Pressure Switch
10-4-15	Main Engine Prime Fuel Shutoff Valve
10-4-16	Self-Sealing Breakaway Vent Valve
10-4-17	Main Fuel System Vent Valve
10-4-18	Main Fuel Breakaway Valve
10-4-19	Extended Range Breakaway Valve
10-4-20	Prime/Boost Pump

PARAGRAPH	TITLE
10-4-21	Prime/Boost Pump Fuel Outlet Valve
10-4-22	Prime/Boost Pump Fuel Breakaway Valve
10-4-23	Main Fuel System Check Valve
10-4-24	Engine Compartment Breakaway Valve
10-4-25	Fuel Selector Valve
10-4-26	Crossfeed Breakaway Valve
10-4-27	Breakaway Tee Valve
10-4-28	Bulkhead Breakaway Valve
10-4-29	Fuel System Drain Hose and Tubes
10-4-30	Fuel System Vent Lines
10-4-31	Pressure Refueling Receptacle
10-4-32	Refuel Adapter and Nipple
10-4-33	Fuel Quantity Probe
10-4-34	Fuel Cell Wiring Harness
10-4-35	Low Level Warning Conditioner
10-4-36	Fuel Quantity Signal Conditioner
10-4-37	Fixed Bleed-Air Lines and Disconnects
10-4-38	Fixed Fuel Lines and Disconnects
10-4-39	Fuel Feed System
10-4-40	Purge Main Fuel Cell
10-4-41	Upper Deck Main Fuel Hoses
10-4-42	Lower APU Fuel Line
10-4-43	Prime Fuel Lines
10-4-44	Lower Main Fuel Lines

10-4-1 FUEL SYSTEM PRIMING.

PARAGRAPH BREAKDOWN

	PAGE
10-4-1.1 Fuel System Priming	10-4-1-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Gallon

References

- PARA 1-6-16

10-4-1.1 Fuel System Priming.

WARNING

Injury to personnel or damage to equipment will result if normal fire precautions are not followed when priming system. Follow normal fire precautions when priming system.

- Turn on all electrical power (PARA 1-6-16).
- On engine control quadrant in cockpit, move ENG POWER CONT levers in LOCKOUT.
- On engine control quadrant in cockpit, place NO. 1 ENG FUEL SYS and NO. 2 ENG FUEL SYS selector levers in DIR detent.
- Position 1-gallon container under helicopter fuselage engine overboard drain at STA 391.0 and STA 393.0.

CAUTION

Damage to prime pump will result if operated without fuel available at prime

pickup line. Make sure fuel is available at prime pickup line prior to operating pump.

- While holding FUEL PUMP switch on upper console at FUEL PRIME, have assistant watch overboard drain until fuel flows into 1-gallon container.
- On engine control quadrant in cockpit, place NO. 1 ENG FUEL SYS and NO. 2 ENG FUEL SYS selector levers in XFD detent.
- Have assistant watch overboard drain until fuel flows into 1-gallon container.
- On upper console, place FUEL PUMP switch to OFF.
- Turn off all electrical power (PARA 1-6-16).
- On engine control quadrant in cockpit, place NO. 1 ENG FUEL SYS and NO. 2 ENG FUEL SYS selector levers OFF.
- On engine control quadrant in cockpit, place both ENG POWER CONT levers OFF.

10-4-2 MAIN FUEL CELL.

PARAGRAPH BREAKDOWN

	PAGE
10-4-2.1 Replace Main Fuel Cell	10-4-2-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Cargo Tiedown Straps
- Nonmetallic Scraper, Locally-Made
- Rubber Pads
- Small-Pointed Tool
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 11, Appendix D
- Adhesion Promoter, Item 25A, Appendix D
- Adhesive, Item 48, Appendix D
- Antiseize Compound, Item 74, Appendix D
- Insulation Tape, Item 182, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Pressure Sensitive Adhesive Tape, Item 250E, Appendix D
- Protective Caps and Plugs

- Sealing Compound, Item 284, Appendix D
- Sealing Compound, Item 292, Appendix D
- Shim, 70307-03015-101 (Thick)
- Shim, 70307-03015-102 (Thin)

References

- Appendix D
- PARA 1-3-23
- PARA 1-6-16
- PARA 1-7-17
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-2-5
- PARA 10-2-6
- PARA 10-2-7
- PARA 10-3-1
- PARA 10-3-2
- PARA 10-4-1
- PARA 10-4-3
- PARA 10-4-6
- PARA 10-4-30
- PARA 10-4-39

• PARA 13-5-3

• PARA 13-5-6

10-4-2.1 Replace Main Fuel Cell.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedures.

CAUTION

Damage to main fuel cell will result if work area is below 70°F (21°C). Make sure work area is above 70°F (21°C) when removing and/or installing fuel cell.

NOTE

Replacement of No. 1 and No. 2 main fuel cells is the same, except as noted.

10-4-2.1.1 Remove.

- a. Purge fuel feed system (PARA 10-4-39).
- b. Gain access to main fuel cell compartment as follows:
 - (1) Remove rear passenger's seats as required (PARA 2-4-45).
 - (2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).
 - (3) Remove cargo netting from stowage compartments.
 - (4) If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

- c. Unlock fasteners securing left and right panel enclosure on top of main fuel cells. Remove panels.
- d. Remove screws and washers securing front and top panels to airframe. Remove panels from helicopter.
- e. Remove No. 1 and No. 2 vent tubes, vent tube Y, and rear vent tube (PARA 10-4-30).
- f. Remove screws and washers securing rear panel to structure. Remove rear panel.
- g. Remove bolts, washers, and fuel cell access covers from main fuel cells. Remove packings from fuel cell access covers (Figure 10-4-2-1, Sheet 1, Detail C). Discard packings.
- h. Remove main fuel cell plate and attached components from applicable main fuel cell (PARA 10-4-3).

NOTE

Note location of dummy receptacle.

- i. Remove screws and washers securing dummy receptacle to cover panel (Figure 10-4-2-1, Sheet 2, Detail D).
- j. Remove screws and washers securing cover panel to fuel cell compartment structure.
- k. Remove sump drain valve (PARA 10-4-6).
- l. Open gravity refuel door (Figure 10-4-2-1, Sheet 1, Detail B).
- m. Remove bolts, washers, bonding jumper, and gravity fill adapter unit. Discard packing.
- n. Remove bolts and washers securing main fuel cell to gravity refueling panel.
- o. On No. 1 main fuel cell, perform the following (Figure 10-4-2-1, Sheet 1, Detail A):

10-4-2-2 Change 11

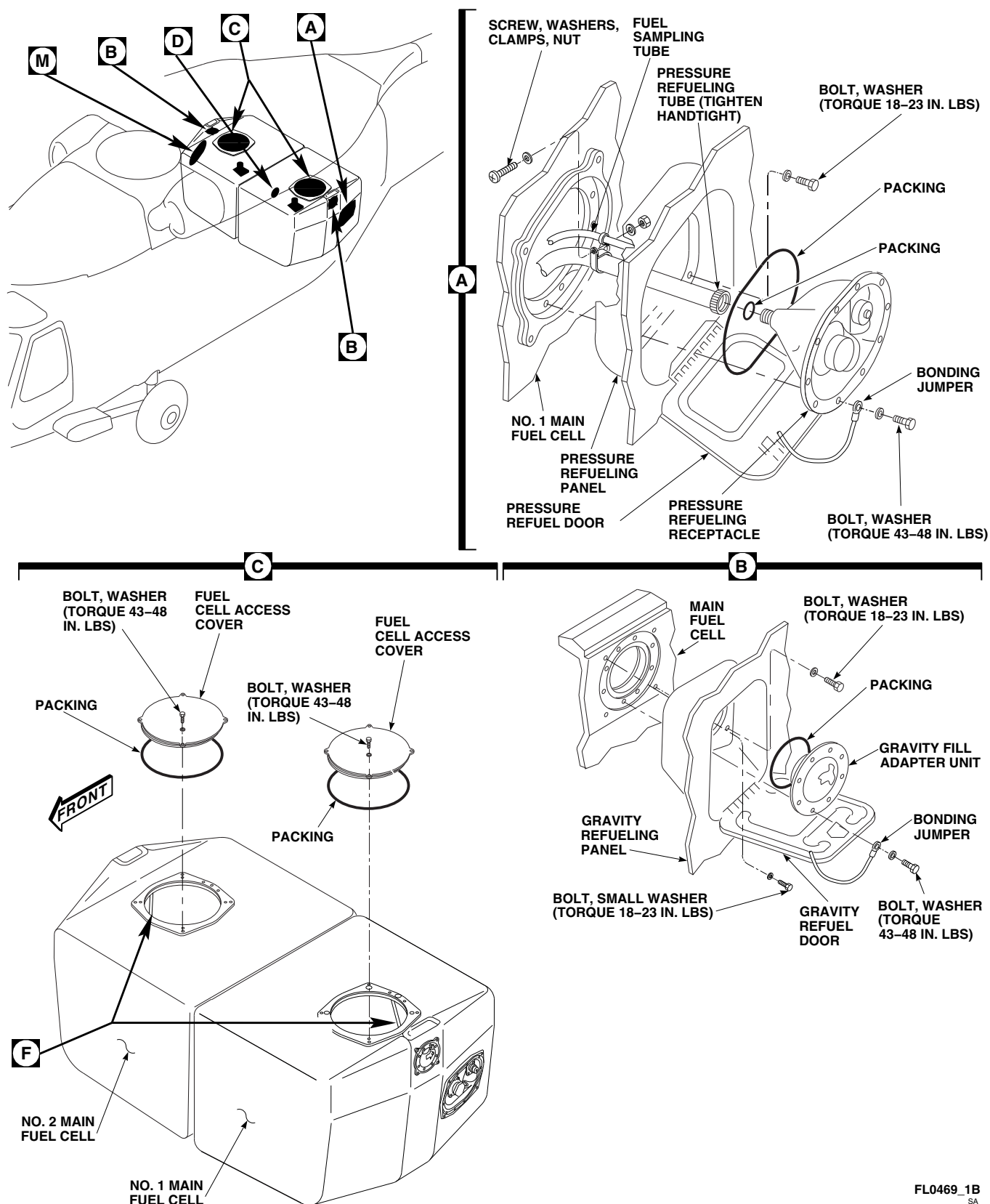


FIGURE 10-4-2-1. REPLACE MAIN FUEL CELL (SHEET 1 OF 5)

FL0469_1B
SA

- (1) Open pressure refuel door.
 - (2) Remove screw, washers, clamps, and nut securing fuel sampling tube to pressure refueling tube.
 - (3) Disconnect pressure refueling tube from pressure refueling receptacle. Remove tube. Discard packing.
 - (4) Remove bolts, washers, packing, bonding jumper, and pressure refueling receptacle from fuel cell. Discard packing.
 - (5) Remove bolts and washers securing fuel cell to pressure refueling panel.
 - (6) Using nonmetallic scraper, remove sealing compound from pressure refueling receptacle and pressure refueling panel.
- p. Remove screws securing brackets and fuel sampling tube to inside top of main fuel cell (Figure 10-4-2-1, Sheet 2, Detail F). Remove tube and brackets from fuel cell.
- q. Remove bolts and washers securing cover panel to main fuel cell at access cover and main fuel cell plate opening (Figure 10-4-2-1, Sheet 2, Detail D).
- r. From inside main fuel cell not being removed, disconnect pressure refueling interconnect tube from fuel cell interconnect breakaway valve. Install protective plug in tube.
- s. Remove fuel cell interconnect breakaway valve as follows:
- (1) From inside each main fuel cell, remove nut from breakaway valve.
 - (2) Remove bolts and washers securing shims and breakaway valve to fuel cell.
 - (3) Remove breakaway valve from both fuel cell interconnect ports. Discard packings.
- sure work area is above 70°F (21°C) when removing fuel cell.
- t. With aid of assistant, remove cover panel from fuel cell compartment.
- u. Loop two cargo tiedown straps through fuel cell access cover and main fuel cell plate openings, one to front of main fuel cell, one to rear (Figure 10-4-2-1, Sheet 3, Detail G).
- v. Connect fixed and movable hooks on front strap to front ring in transition ceiling.
- w. Connect fixed and movable hooks on rear strap to rear ring in transition ceiling.
- x. Using ratchets, remove slack from cargo tiedown straps. Hoist main fuel cell out of cell compartment.
- y. Install two additional cargo tiedown straps around main fuel cell (Figure 10-4-2-1, Sheet 3, Detail H). Position rubber pad against fuel cell under each ratchet to protect fuel cell from damage.
- z. Lower main fuel cell back in cell compartment.
- aa. Remove straps used to hoist main fuel cell.
- ab. With one person standing on top of main fuel cell, alternately tighten each cargo tiedown strap using ratchets and slowly compress fuel cell.
- ac. Compress main fuel cell until gravity fill port is centered and level on top of fuel cell and fuel cell interconnect port is about 6-inches away from compartment bulkhead (Figure 10-4-2-1, Sheet 4, Detail J).
- ad. Using four assistants (one on each corner of main fuel cell), lift compressed fuel cell out of cell compartment and remove from helicopter.
- ae. Remove cargo tiedown straps.
- af. On No. 2 main fuel cell, remove insulation tape, screws, and pressure refueling port cover from fuel cell for installation on replacement fuel cell. Discard packing (Figure 10-4-2-1, Sheet 5, Detail M).

CAUTION

Damage to main fuel cell will result if work area is below 70°F (21°C). Make

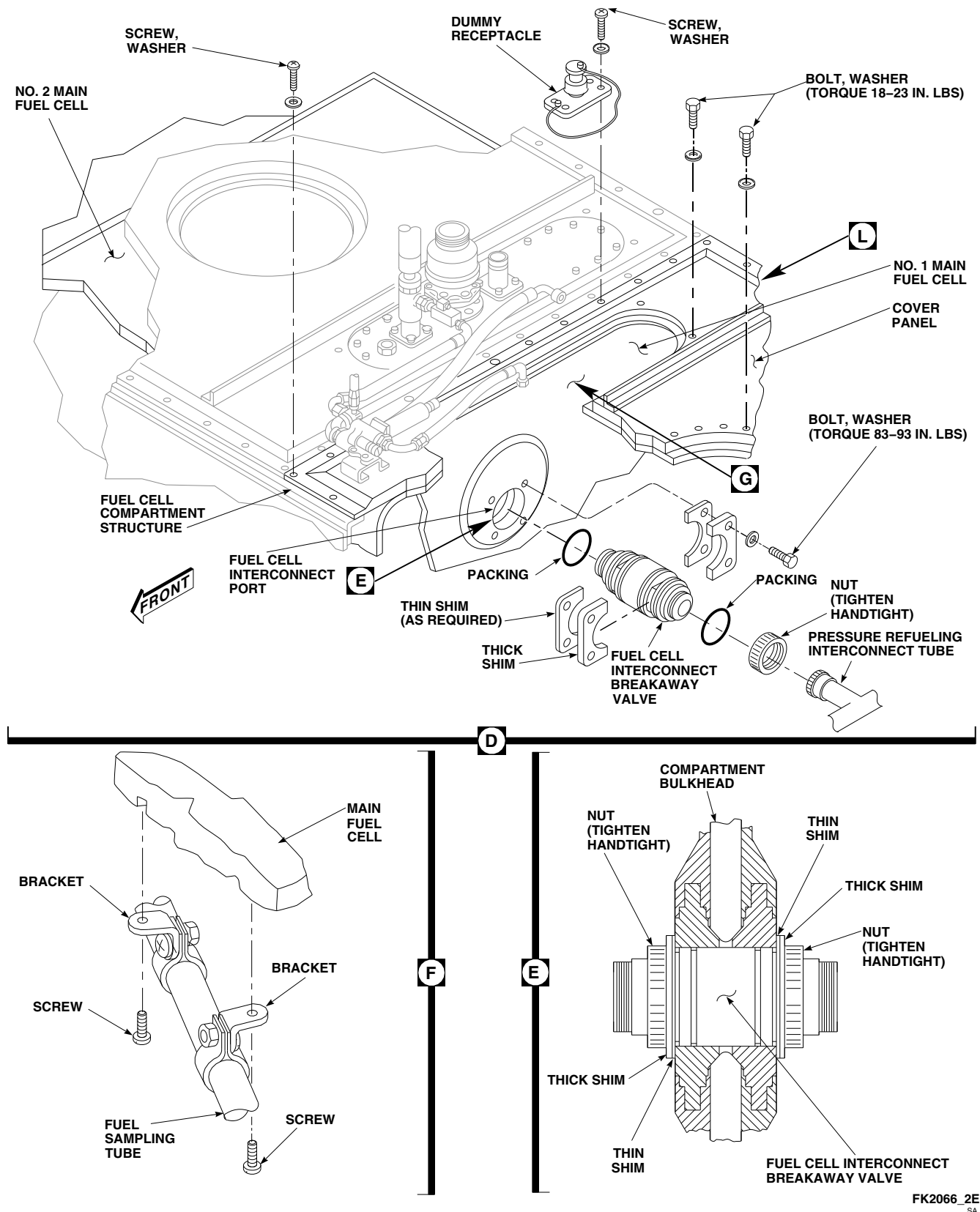


FIGURE 10-4-2-1. REPLACE MAIN FUEL CELL (SHEET 2 OF 5)

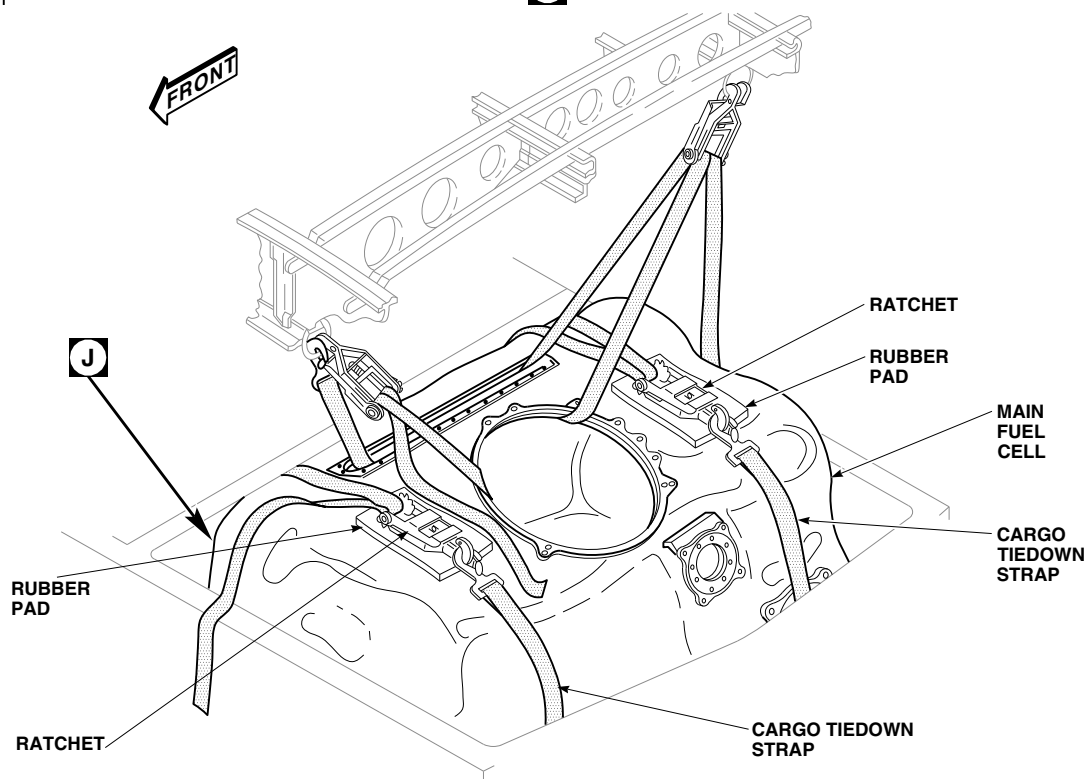
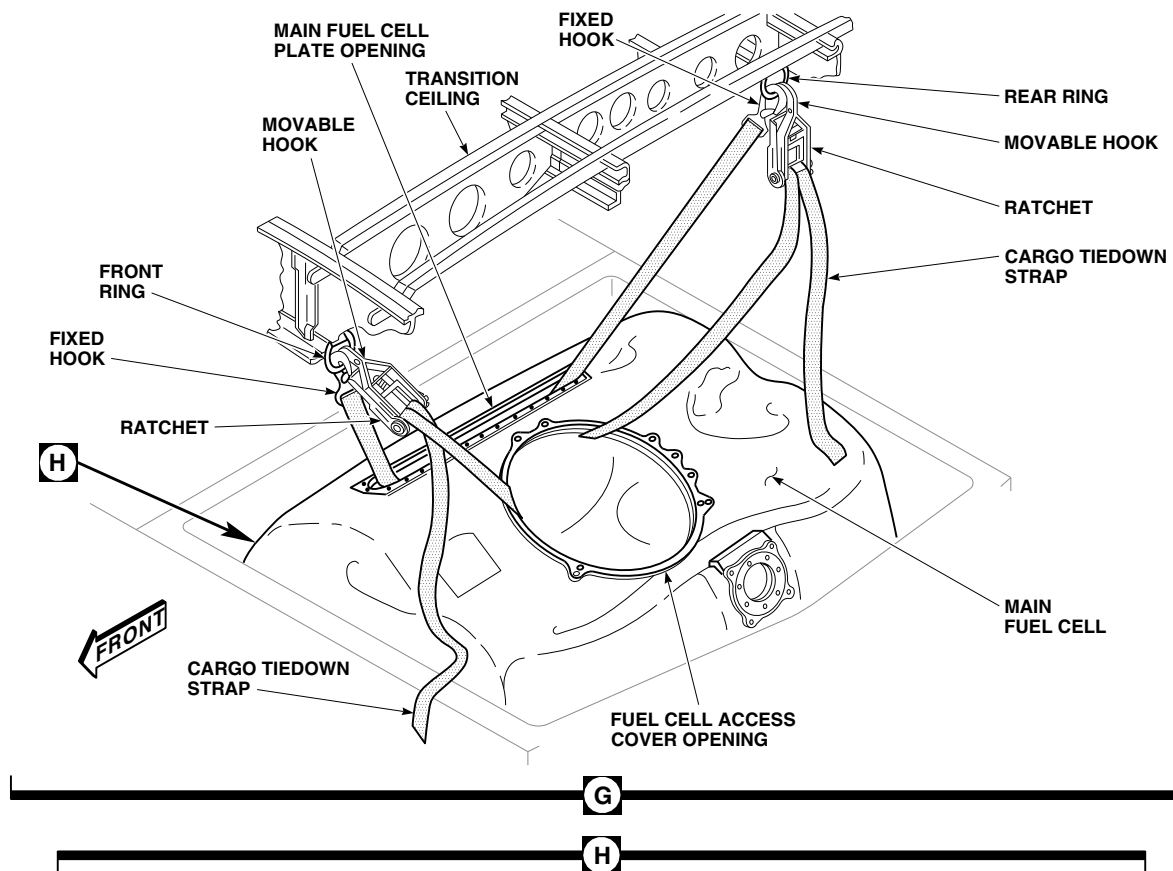
FL0469_3A
SA

FIGURE 10-4-2-1. REPLACE MAIN FUEL CELL (SHEET 3 OF 5)

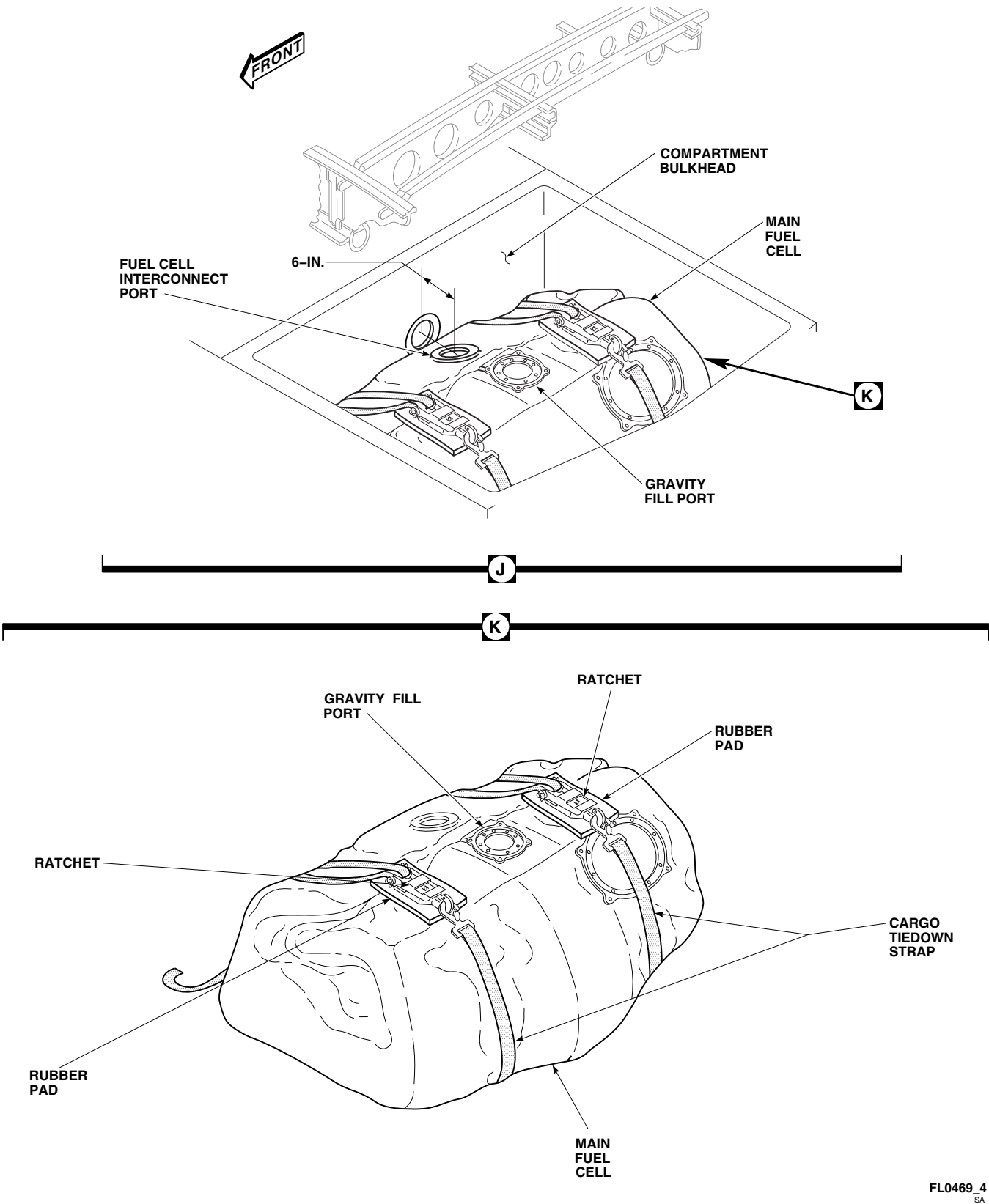


FIGURE 10-4-2-1. REPLACE MAIN FUEL CELL (SHEET 4 OF 5)

FL0469_4
SA

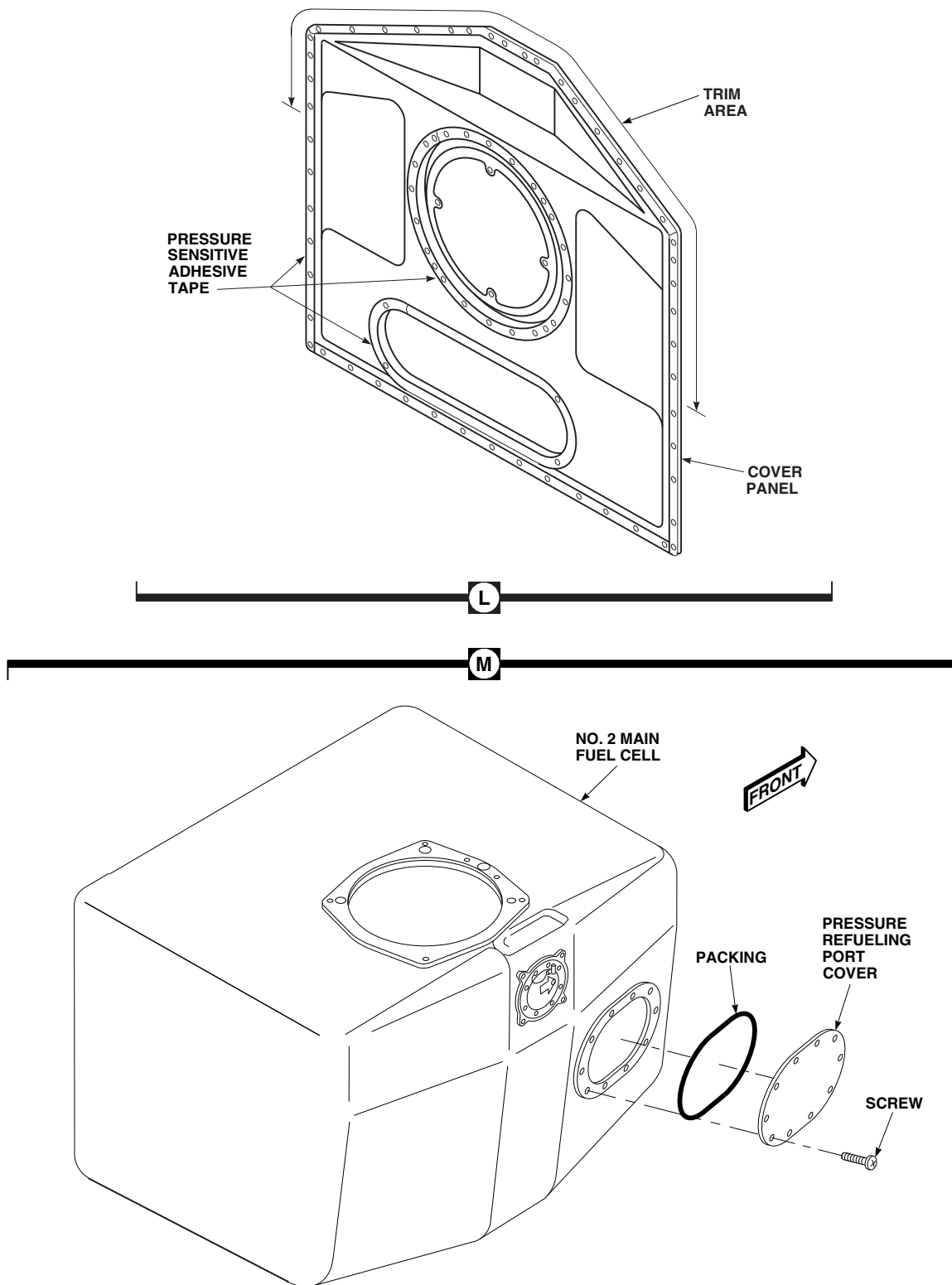
FL0469_5
SA

FIGURE 10-4-2-1. REPLACE MAIN FUEL CELL (SHEET 5 OF 5)

ag. Preserve main fuel cell (PARA 1-3-23) if:

- (1) Fuel cell is being sent to approved maintenance facility, manufacturer, or being placed in storage.
- (2) Fuel cell is to remain defueled for longer than 30 days.

10-4-2.1.2 Inspect.

Inspect transition section tub (PARA 1-7-17).

10-4-2.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Check gravity and pressure refueling panels inside fuel cell compartment to see if any filler blocks on bolt attachment holes have broken off or become disbonded. If required, rebond filler blocks to refueling panels as follows:

- (1) Using methyl ethyl ketone, Item 217, Appendix D, clean bonding surface on refueling panel and filler block.
- (2) Using abrasive cloth, Item 11, Appendix D, rough up bonding surface on refueling panel and filler block.
- (3) Using methyl ethyl ketone, Item 217, Appendix D, clean bonding surfaces again.
- (4) Using adhesive, Item 48, Appendix D, lightly coat bonding surface on refueling panel and filler block.
- (5) Line up boltholes in filler block and refueling panel and press block on panel. Allow adhesive to cure at room temperature for 12 hours.

- c. On No. 2 main fuel cell, install pressure refueling port cover as follows (Figure 10-4-2-1, Sheet 5, Detail M):

- (1) Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing in groove on pressure refueling port cover.
- (2) Install pressure refueling port cover to fuel cell with screws.

- (3) Using methyl ethyl ketone, Item 217, Appendix D, clean pressure refueling port cover and surrounding area.
- (4) Apply insulation tape, Item 182, Appendix D, to pressure refueling port cover and surrounding area.
- (5) Using sealing compound, Item 284, Appendix D, coat taped area and 1/2-inch beyond edge of tape.
- (6) Allow area to air cure at least 24 hours at room temperature.

- d. Make sure fuel cell compartment is clean and free of foreign material.

CAUTION

Damage to main fuel cell will result if work area is below 70°F (21°C). Make sure work area is above 70°F (21°C) when installing fuel cell.

- e. Install two cargo tiedown straps around main fuel cell (Figure 10-4-2-1, Sheet 4, Detail K). Position rubber pad against fuel cell under each ratchet to protect fuel cell against damage. Slowly compress fuel cell by stepping and kneeling on fuel cell and alternately tightening each cargo tiedown strap using ratchets.
- f. Compress main fuel cell until gravity fill port is centered and level on top of fuel cell.
- g. Using four assistants (one on each corner of main fuel cell), lift and place fuel cell in cell compartment. Push four corners of fuel cell into cell compartment, and release tension on cargo tiedown straps using ratchets (Figure 10-4-2-1, Sheet 4, Detail J and Detail K).
- h. Remove cargo tiedown straps from main fuel cell. To ease removal of straps, hook movable hook of strap to front or rear ring in transition ceiling and use ratchet to work strap out (Figure 10-4-2-1, Sheet 3, Detail G and Detail H).

- i. With aid of assistant, line up boltholes on gravity refueling panel and main fuel cell (Figure 10-4-2-1, Sheet 1, Detail B).

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- j. Coat threads of bolts with antiseize compound, Item 74, Appendix D.

NOTE

Small washer is required at upper rear position of gravity refueling panel.

- k. Secure main fuel cell to gravity refueling panel with bolts and washers. **TORQUE BOLTS TO 18 - 23 INCH-POUNDS.**
- l. On No. 1 main fuel cell, perform the following:
 - (1) Line up boltholes on pressure refueling panel and fuel cell (Figure 10-4-2-1, Sheet 1, Detail A).

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- (2) Coat threads of bolts with antiseize compound, Item 74, Appendix D.
 - (3) Secure fuel cell to refueling panel with bolts and washers. **TORQUE BOLTS TO 18 - 23 INCH-POUNDS.**
- m. Apply adhesion promoter, Item 25A, Appendix D, around all openings and outside edges on bottom of cover panel.
- n. Apply pressure sensitive adhesive tape, Item 250E, Appendix D, around all openings and

outside edges on bottom of cover panel (Figure 10-4-2-1, Sheet 5, Detail L). Poke open screw-holes using small-pointed tool.

- o. With aid of assistant, trial-fit cover panel on top of fuel cell compartment with bolts and washers (Figure 10-4-2-1, Sheet 2, Detail D). Do not torque bolts at this time. Check panel for binding against airframe. If binding is found, trim as follows:
- p. Check for cover panel binding against airframe. If binding is found, trim as follows:
 - (1) Remove bolts and washers securing cover panel. Remove cover panel.
 - (2) Trim cover panel (Figure 10-4-2-1, Sheet 5, Detail L).
 - (3) Repeat step p. (1) and p. (2) until no binding occurs.
 - (4) Remove bolts and washers securing cover panel. Remove cover panel.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- q. Coat threads of bolts with antiseize compound, Item 74, Appendix D.
- r. With aid of assistant, secure cover panel to top of main fuel cell at access cover and main fuel cell plate openings with bolts and washers (Figure 10-4-2-1, Sheet 2, Detail D). **TORQUE BOLTS TO 18 - 23 INCH-POUNDS.**
- s. Apply bead of sealing compound, Item 292, Appendix D, to seal gap between fuel cell compartment structure and main fuel cell cover at opening for main fuel cell plate. Do not allow sealing compound to interfere with plate installation.
- t. Lubricate packings with lubricating oil, Item 205, Appendix D. Install packings on fuel cell interconnect breakaway valve.

- u. From inside each main fuel cell, install fuel cell interconnect breakaway valve as follows:
 - (1) Insert interconnect breakaway valve into both fuel cell interconnect ports.
 - (2) Install thick shims on flats on interconnect breakaway valve. Install nut on interconnect breakaway valve. **TIGHTEN NUT HANDTIGHT** (Figure 10-4-2-1, Sheet 2, Detail D and Detail E).
 - (3) Push fuel cells against compartment bulkhead between two cells. Check interconnect breakaway valve for movement by pushing and pulling valve. **NONE ALLOWED.** If movement is felt, perform the following:
 - (a) Install thin shims between each thick shim and fuel cell.
 - (b) Repeat step u. (3) until all movement is removed.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- (4) Remove nut from interconnect breakaway valve. Coat threads on attachment bolts with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads on valve.

CAUTION

Damage to fuel cell interconnect breakaway valve will result if nuts are tightened with pliers or other tools. Do not use pliers or other tools to tighten nuts.

- (5) If required, install thin shims between thick shims and fuel cell on valve flats

with bolts and washers. **TORQUE BOLTS TO 83 - 93 INCH-POUNDS.** Wipe off any squeezeout with machinery towel, Item 211, Appendix D. **TIGHTEN NUT HANDTIGHT** (Figure 10-4-2-1, Sheet 2, Detail D).

- v. From main fuel cell not being installed, remove protective plug and connect pressure refueling interconnect tube to fuel cell interconnect breakaway valve.
- w. Install fuel sampling tube, with brackets attached, into main fuel cell (Figure 10-4-2-1, Sheet 2, Detail F). Secure brackets to top of fuel cell with screws.
- x. On No. 1 main fuel cell, perform the following:
 - (1) Lubricate packings with lubricating oil, Item 205, Appendix D.
 - (2) Install packing on end of pressure refueling receptacle (Figure 10-4-2-1, Sheet 1, Detail A).
 - (3) Install packing in groove of pressure refueling receptacle.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- (4) Coat threads of bolts with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads on receptacle.
- (5) Install pressure refueling receptacle with bolts, washers, and bonding jumper. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS.** Wipe off any squeezeout with machinery towel, Item 211, Appendix D.
- (6) Connect pressure refueling tube to pressure refueling receptacle. **TIGHTEN FITTING HANDTIGHT.**

- (7) Apply bead of sealing compound, Item 292, Appendix D, in cavity between pressure refueling receptacle and pressure refueling panel.
- (8) Close pressure refuel door.
- y. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on gravity fill adapter unit (Figure 10-4-2-1, Sheet 1, Detail B).

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- z. Coat threads of gravity fill adapter unit bolts with antiseize compound, Item 74, Appendix D.
- aa. With directional arrow pointing to front of helicopter, install gravity fill adapter unit and bonding jumper to main fuel cell with bolts and washers. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS.**
- ab. Make sure cavity between gravity fill adapter unit and refueling panel is clean and free of foreign material. Apply bead of sealing compound, Item 292, Appendix D, in cavity.
- ac. Close gravity refuel door.
- ad. Install sump drain valve (PARA 10-4-6).

NOTE

Install dummy receptacle prior to installing screws and washers for cover panel.

- ae. Install dummy receptacle to cover panel with screws and washers (Figure 10-4-2-1, Sheet 2, Detail D).
- af. Secure cover panel to fuel cell compartment structure with remaining screws and washers.

- ag. Install main fuel cell plate and attached components (PARA 10-4-3).
- ah. Test fuel feed system (PARA 10-4-39).
- ai. Lubricate packings with lubricating oil, Item 205, Appendix D. Install packings on fuel cell access covers (Figure 10-4-2-1, Sheet 1, Detail C).
- aj. Inspect area inside main fuel cell and make sure area is free of all foreign material.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- ak. Coat threads of fuel cell access cover attachment bolts with antiseize compound, Item 74, Appendix D.
- al. Install fuel cell access covers on main fuel cells with bolts and washers. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS.**
- am. Pressure test main fuel cells (PARA 10-3-1 or PARA 10-3-2).
- an. Perform operational check of fuel quantity system (PARA 10-2-5 or PARA 10-2-6).
- ao. Perform operational check of engine feed fuel lines (PARA 10-2-7).
- ap. Position rear panel on structure and secure with screws and washers.
- aq. Install No. 1 and No. 2 fuel vent tubes, vent tube Y, and rear vent tube (PARA 10-4-30).
- ar. Make sure area is clean and free of foreign material.
- as. Install front and top panels with screws and washers.
- at. Install left and right panel enclosure on top of main fuel cells. Lock fasteners.

- au. Inspect sump drain valve (PARA 10-4-6).
- av. Prime fuel system (PARA 10-4-1).
- aw. Close fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panels (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

10-4-3 MAIN FUEL CELL PLATE AND ATTACHED COMPONENTS.

PARAGRAPH BREAKDOWN

	PAGE
10-4-3.1 Replace Main Fuel Cell Plate and Attached Components	10-4-3-2

INITIAL SETUP

- PARA 1-7-20

Tools

- PARA 2-4-45

- Aircraft Mechanic’s Toolkit
- Digital Low-Resistance Ohmmeter
- Torque Wrench, 30 - 150 in. lbs

- PARA 2-4-69
- PARA 10-2-5
- PARA 10-2-6
- PARA 10-2-7

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Temporary Covering
- Tiedown Strap, Item 331, Appendix D

- PARA 10-3-1
- PARA 10-3-2
- PARA 10-4-1
- PARA 10-4-10
- PARA 10-4-12
- PARA 10-4-30
- PARA 10-4-39

References

- Appendix D
- PARA 1-6-16

- PARA 10-4-40
- PARA 13-5-3
- PARA 13-5-6

10-4-3.1 Replace Main Fuel Cell Plate and Attached Components.

WARNING

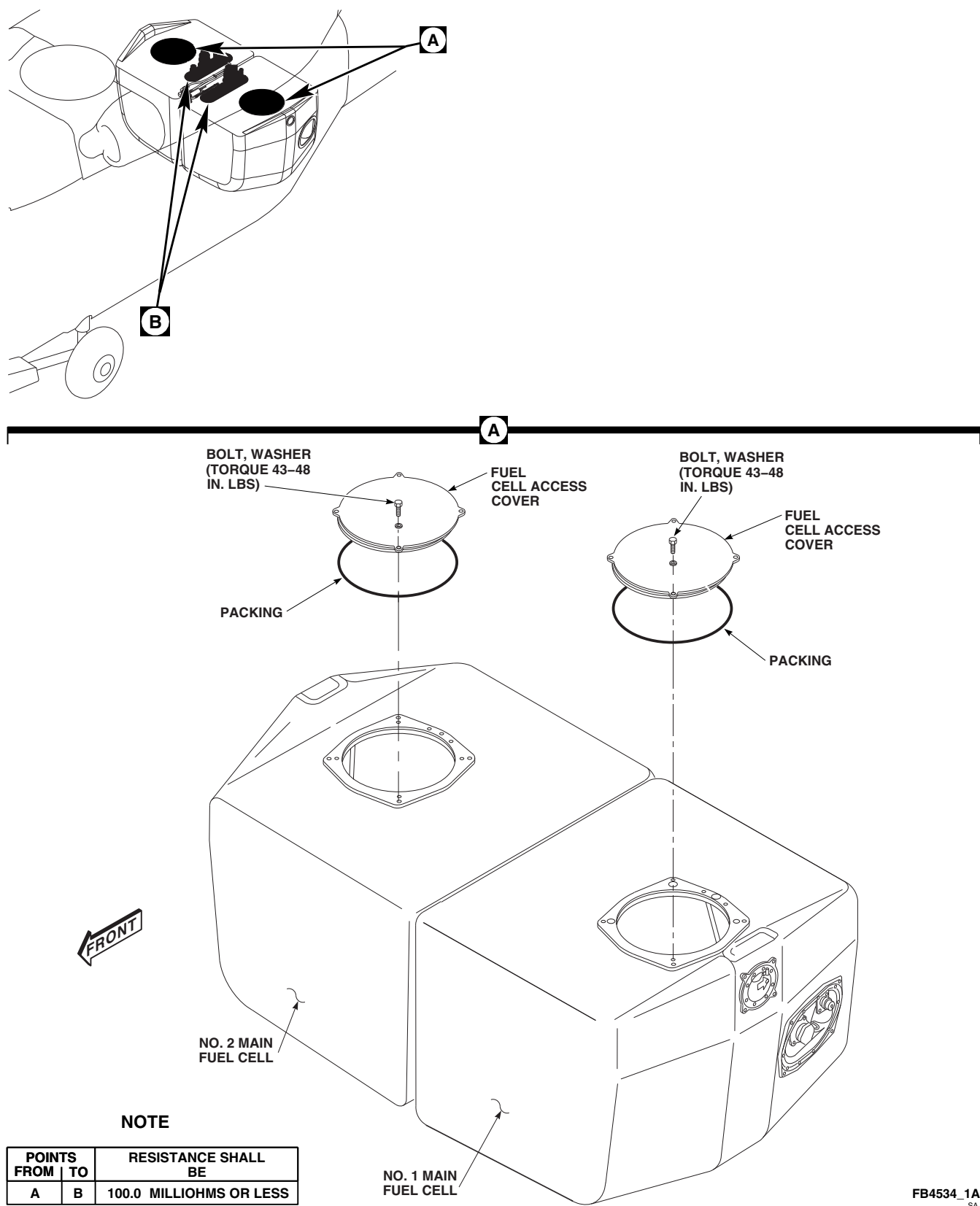
Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

NOTE

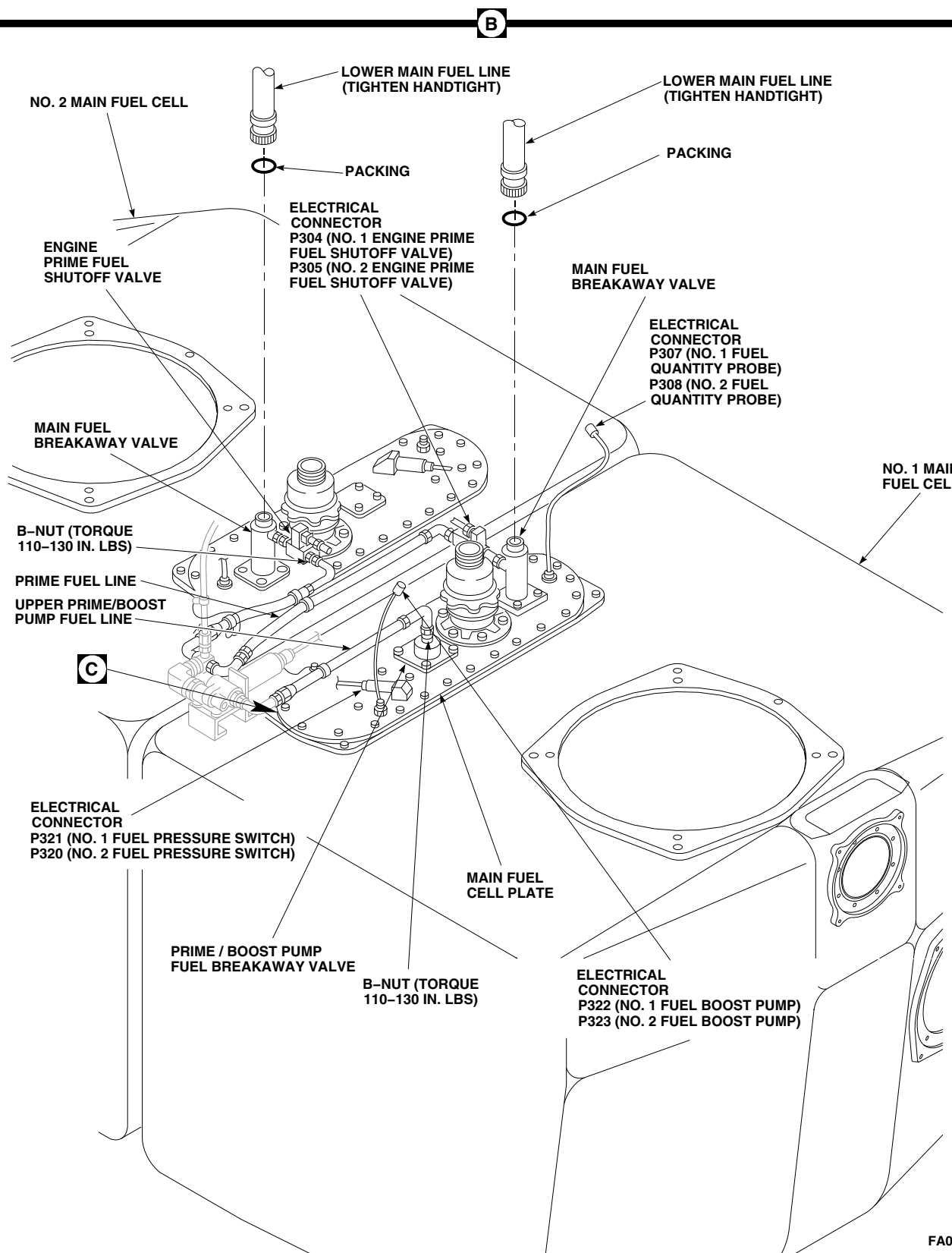
Replacement of No. 1 and No. 2 main fuel cell plates and attached components is the same, except as noted.

10-4-3.1.1 Remove.

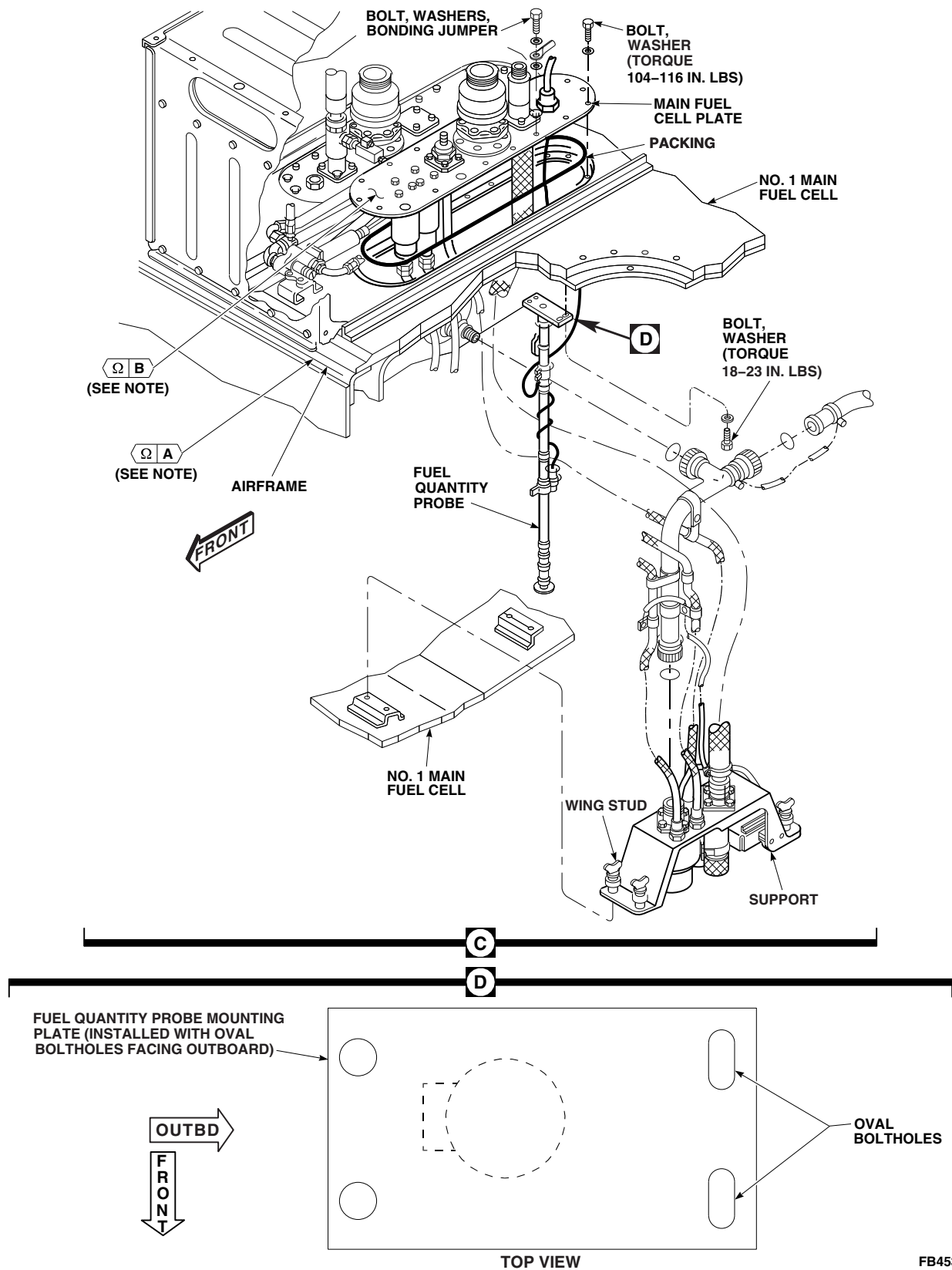
- a. Purge fuel feed system (PARA 10-4-39).
- b. Purge main fuel cell (PARA 10-4-40).
- c. Gain access to main fuel cell compartment as follows:
 - (1) Remove rear passenger's seats as required (PARA 2-4-45).
 - (2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).
 - (3) Remove cargo netting from stowage compartments.
 - (4) If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
- d. Unlock fasteners securing left or right panel enclosure on top of main fuel cells. Remove panel.
- e. Remove screws and washers securing front and top panels to airframe. Remove panels from helicopter.
- f. Remove No. 1 or No. 2 fuel vent tube (PARA 10-4-30).
- g. Remove bolts, washers, and fuel cell access cover from main fuel cell. Remove packing from fuel cell access cover (Figure 10-4-3-1, Sheet 1, Detail A). Discard packing.
- h. Disconnect electrical connector, P304 (No. 1 engine prime fuel shutoff valve), or P305 (No. 2 engine prime fuel shutoff valve) (Figure 10-4-3-1, Sheet 2, Detail B).
- i. Install protective caps and plugs on electrical connectors.
- j. Disconnect electrical connector, P321 (No. 1 fuel pressure switch), or P320 (No. 2 fuel pressure switch).
- k. Install protective caps and plugs on electrical connectors.
- l. Disconnect electrical connector, P322 from J322 (No. 1 fuel boost pump), or P323 from J323 (No. 2 fuel boost pump).
- m. Install protective caps and plugs on electrical connectors.
- n. Disconnect electrical connector, P307 from J307 (No. 1 fuel quantity probe), or P308 from J308 (No. 2 fuel quantity probe). Cut tiedown straps securing wiring harness to lower main fuel line.
- o. Install protective caps and plugs on electrical connectors.
- p. Disconnect prime fuel line from engine prime fuel shutoff valve. Install protective cap and plug on shutoff valve and fuel line.
- q. Disconnect lower main fuel line from main fuel breakaway valve. Discard packing. Install protective cap and plug on breakaway valve and fuel line.
- r. On No. 1 main fuel cell, disconnect upper prime/boost pump fuel line from prime/boost pump fuel breakaway valve. Install protective cap and plug on breakaway valve and fuel line.
- s. Remove bolts and washers securing main fuel cell plate to main fuel cell (Figure 10-4-3-1, Sheet 3, Detail C).



FB4534_1A
 SA

FA0924_2A
SA

**FIGURE 10-4-3-1. REPLACE MAIN FUEL CELL PLATE AND ATTACHED COMPONENTS
(SHEET 2 OF 3)**



FB4534_3B
SA

FIGURE 10-4-3-1. REPLACE MAIN FUEL CELL PLATE AND ATTACHED COMPONENTS (SHEET 3 OF 3)

- t. On No. 1 main fuel cell, remove pressure refueling interconnect tube (PARA 10-4-10).
- u. Remove bolts and washers securing fuel quantity probe to inside top of main fuel cell.
- v. Remove fuel boost pump (PARA 10-4-12).
- w. Unlock wing studs securing support to bottom of main fuel cell.
- x. Carefully remove main fuel cell plate, with attached components and fuel quantity probe, from main fuel cell. Remove and discard packing.
- y. Place temporary covering over main fuel cell plate and fuel cell access cover openings.

10-4-3.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove temporary covering from main fuel cell plate and fuel cell access cover openings.
- c. Lubricate packings with lubricating oil, Item 205, Appendix D. Install packing in groove of main fuel cell plate opening (Figure 10-4-3-1, Sheet 3, Detail C).
- d. Install main fuel cell plate, with attached components and fuel quantity probe, into main fuel cell.
- e. Secure main fuel cell plate to main fuel cell with bolts and washers. **TORQUE BOLTS TO 104 - 116 INCH-POUNDS.**

NOTE

Make sure oval boltholes on mounting plate face outboard when installing fuel quantity probe (Figure 10-4-3-1, Sheet 3, Detail D).

- f. Install fuel quantity probe to inside top of main fuel cell with bolts and washers (Figure 10-4-3-1, Sheet 3, Detail C). **TORQUE BOLTS TO 18 - 23 INCH-POUNDS.**
- g. Lock wing studs securing support to bottom of main fuel cell.

- h. Install fuel boost pump (PARA 10-4-12).
- i. On No. 1 main fuel cell, install pressure refueling interconnect tube (PARA 10-4-10).

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- j. On No. 1 main fuel cell, remove protective cap and coat threads on prime/boost pump fuel breakaway valve with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.

CAUTION

Do not preload prime/boost pump fuel breakaway valve when connecting fuel line. Frangible section of prime/boost fuel breakaway valve is sensitive to bending and tension overload.

- k. Remove protective plug and connect upper prime/boost pump fuel line to prime/boost pump fuel breakaway valve. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS** (Figure 10-4-3-1, Sheet 2, Detail B).
- l. Remove protective cap from main fuel breakaway valve. Install packing on valve.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- m. Coat threads on main fuel breakaway valve with antiseize compound, Item 74, Appendix

D, sparingly ($\frac{1}{4}$ -inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.

CAUTION

Do not preload main fuel breakaway valve when connecting main fuel line. Frangible section of main fuel breakaway valve is sensitive to bending and tension overload.

- n. Remove protective plug and connect lower main fuel line to main fuel breakaway valve. **TIGHTEN FITTING HANDTIGHT.**

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- o. Remove protective cap from engine prime fuel shutoff valve. Coat threads on prime fuel shutoff valve with antiseize compound, Item 74, Appendix D, sparingly ($\frac{1}{4}$ -inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.

CAUTION

Make sure main fuel breakaway valve is not preloaded when connecting fuel line to engine prime fuel shutoff valve. Frangible section of main fuel breakaway valve is sensitive to bending and tension overload.

- p. Remove protective plug and connect prime fuel line to engine prime fuel shutoff valve.

TORQUE B-NUT TO 110 - 130 INCH-POUNDS.

NOTE

Resistance measurements must be taken at least three times.

- q. Using digital low-resistance ohmmeter, measure resistance between airframe and main fuel cell plate (Figure 10-4-3-1, Sheet 3, Detail C). **RESISTANCE SHALL BE 100.0 MILLI-OHMS OR LESS.**
- r. Remove protective caps and plugs from electrical connectors.
- s. Inspect and treat electrical connectors (PARA 1-7-20).
- t. Connect electrical connector, P322 to J322 (No. 1 fuel boost pump), or P323 to J323 (No. 2 fuel boost pump).
- u. Connect electrical connector, P307 to J307 (No. 1 fuel quantity probe), or P308 to J308 (No. 2 fuel quantity probe). Secure wiring harness to lower main fuel line using tiedown straps, Item 331, Appendix D.
- v. Connect electrical connector, P321 (No. 1 fuel pressure switch), or P320 (No. 2 fuel pressure switch).
- w. Connect electrical connector, P304 (No. 1 prime fuel shutoff valve), or P305 (No. 2 prime fuel shutoff valve).
- x. Install No. 1 or No. 2 fuel vent tube (PARA 10-4-30).
- y. Test fuel feed system (PARA 10-4-39).
- z. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on fuel cell access cover.
- aa. Inspect area inside main fuel cell and make sure area is free of all foreign material.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- ab. Coat threads of fuel cell access cover attachment bolts with antiseize compound, Item 74, Appendix D.
- ac. Install fuel cell access cover on main fuel cell with bolts and washers. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS** (Figure 10-4-3-1, Sheet 1, Detail A).
- ad. Pressure test main fuel cells (PARA 10-3-1 or PARA 10-3-2).
- ae. Install front and top panels with screws and washers.

- af. Install left or right panel enclosure on top of main fuel cells. Lock fasteners.
- ag. Perform operational check of fuel quantity system (PARA 10-2-5 or PARA 10-2-6).
- ah. Perform operational check of engine feed fuel lines (PARA 10-2-7).
- ai. Prime fuel system (PARA 10-4-1).
- aj. Close main fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panels (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

10-4-4 HIGH LEVEL SHUTOFF VALVES.

PARAGRAPH BREAKDOWN

	PAGE
10-4-4.1 Replace High Level Shutoff Valves	10-4-4-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-17
- PARA 1-6-16
- PARA 10-4-3

10-4-4.1 Replace High Level Shutoff Valves.

NOTE

- Each main fuel cell contains two high level shutoff valves. If high level shutoff system has malfunctioned, both valves in malfunctioning cell should be replaced.
- Replacement of No. 1 and No. 2 high level shutoff valves is the same.

10-4-4.1.1 Remove.

- Remove main fuel cell plate and attached components from applicable main fuel cell (PARA 10-4-3).
- Disconnect hose from union (Figure 10-4-4-1, Detail A). Install protective plug in hose.
- Unthread union from high level shutoff valve. Remove and discard packing. Install protective caps on union.

- Remove bolts, washers, seal washers, spacers, and high level shutoff valve from main fuel cell plate.

10-4-4.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- Using antiseize compound, coat threads of bolts with antiseize compound, Item 74, Appendix D.
- Install bolts, washers, and seal washers in main fuel cell plate (Figure 10-4-4-1, Detail A).

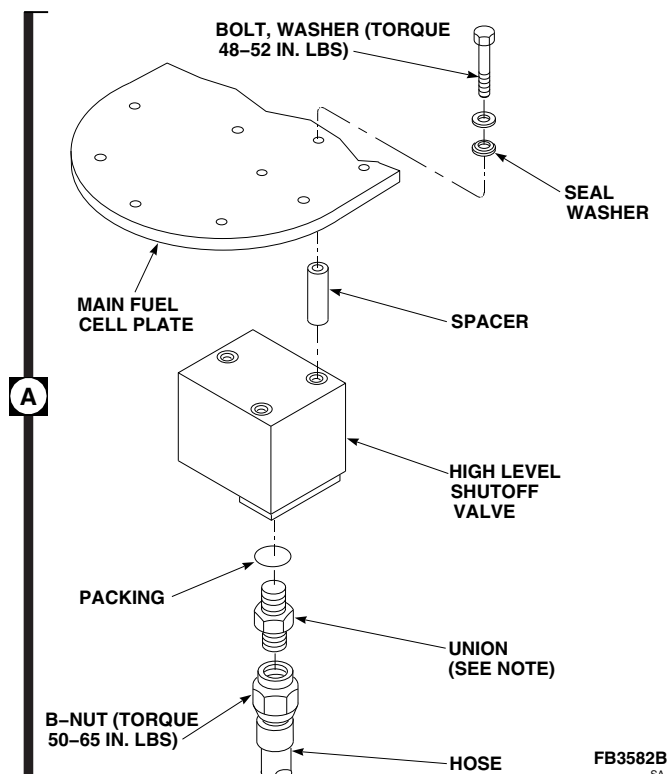
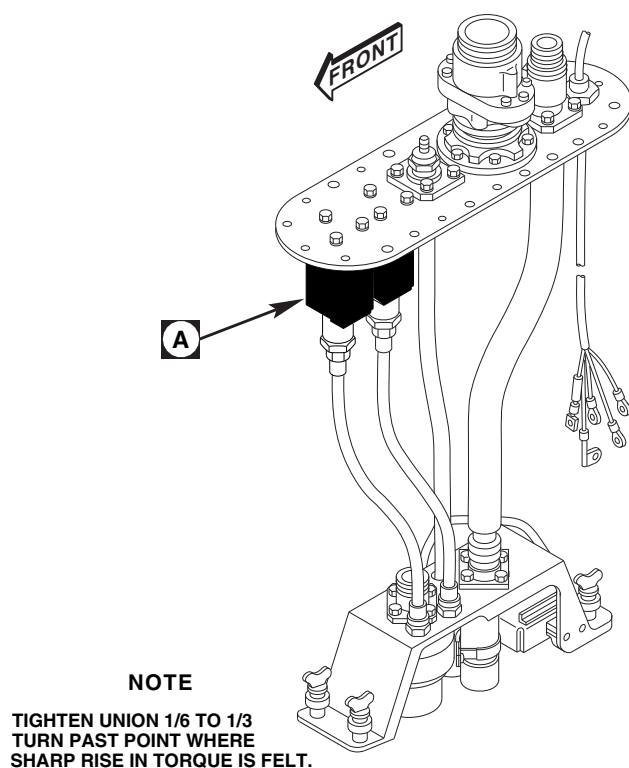


FIGURE 10-4-4-1. REPLACE HIGH LEVEL SHUTOFF VALVES

- d. Place spacers on bolts from beneath main fuel cell plate at each attachment hole.
- e. Position shutoff valve on bolts and secure. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**
- f. Lubricate packing with lubricating oil, Item 205, Appendix D. Remove protective caps from union. Install packing on union end that threads into high level shutoff valve.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- g. Coat threads on union with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.

- h. Thread union into high level shutoff valve. **TIGHTEN UNION 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- i. Remove protective plug from hose. Connect hose to union. **TORQUE B-NUT TO 50 - 65 INCH-POUNDS.**

NOTE

- When installing main fuel cell plate and attached components, make sure fuel cells are pressure refueled.
- Inspection of shutoff valves will occur during pressure refueling of fuel cells.

- j. Install main fuel cell plate and attached components (PARA 10-4-3), and inspect shutoff valves during pressure refueling as follows:

- (1) Make sure main fuel cells are being pressure refueled (PARA 1-3-17).

- (2) Check high level shutoff valves for proper operation by noting that shutoff occurs

before fuel is dumped from vent beneath helicopter.

10-4-5 PRESSURE REFUEL/DEFUEL VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-5.1 Replace Pressure Refuel/Defuel Valve	10-4-5-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lockwire, Item 197, Appendix D

- Lubricating Oil, Item 205, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 10-4-3

10-4-5.1 Replace Pressure Refuel/Defuel Valve.

NOTE

Replacement of No. 1 and No. 2 pressure refuel/defuel valves is the same.

10-4-5.1.1 Remove.

- Remove main fuel cell plate and attached components from applicable main fuel cell (PARA 10-4-3).
- Disconnect hoses from unions on pressure refuel/defuel valve (Figure 10-4-5-1, Detail A). Plug hoses.
- Disconnect tube from union on pressure refuel/defuel valve. Plug tube.
- Remove bolts, washers, union, upper and lower seals, and pressure refuel/defuel valve from support. Discard seals.

- Remove unions and packings from pressure refuel/defuel valve. Discard packings.

10-4-5.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Lubricate packings with lubricating oil, Item 205, Appendix D.
- Install packings on unions (Figure 10-4-5-1, Detail A).



To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into the fuel system.

- Coat male threads on union with antiseize compound, Item 74, Appendix D, sparingly

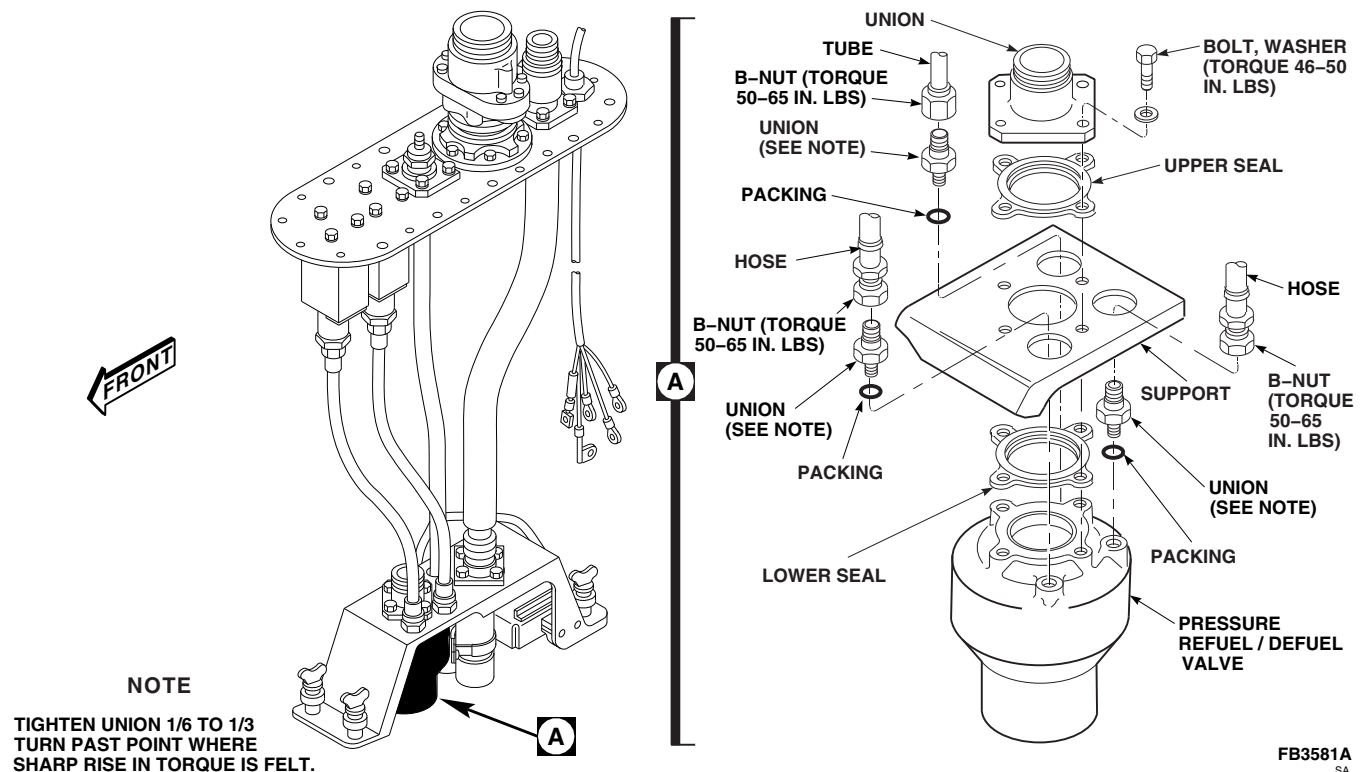


FIGURE 10-4-5-1. REPLACE PRESSURE REFUEL/DEFUEL VALVE

(1/4-inch strip). Wipe off any squeeze out after parts are assembled with machinery towel, Item 211, Appendix D.

- e. Install unions in ports on pressure refuel/defuel valve. **TIGHTEN UNIONS 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into the fuel system.

- f. Coat threads on bolts with antiseize compound, Item 74, Appendix D.
- g. Position lower seal on top of pressure refuel/defuel valve. Position valve beneath support.

- h. Position upper seal on support. Position union on upper seal.
- i. Secure union and valve to support with bolts and washers. **TORQUE BOLTS TO 46 - 50 INCH-POUNDS.**
- j. Lockwire, Item 197, Appendix D, bolts together.
- k. Unplug and connect tube to union on pressure refuel/defuel valve. **TORQUE B-NUT TO 50 - 65 INCH-POUNDS.**
- l. Unplug and connect hoses to unions on pressure refuel/defuel shutoff valve. **TORQUE B-NUTS TO 50 - 65 INCH-POUNDS.**
- m. Install main fuel cell plate and attached components (PARA 10-4-3).

10-4-6 SUMP DRAIN VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-6.1 Replace Sump Drain Valve	10-4-6-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 1-3-21
- PARA 1-6-16
- PARA 10-3-1
- PARA 10-3-2
- PARA 10-4-1
- PARA 10-4-40

10-4-6.1 Replace Sump Drain Valve.

NOTE

Replacement of No. 1 and No. 2 sump drain valves is the same.

10-4-6.1.1 Inspect.

WARNING

Injury to personnel will result if care is not taken when opening fuel sump drain door. Fuel may be trapped above door. Take care when opening door.

- Unlock latches and turnlock fasteners, and open applicable fuel sump drain door (Figure 10-4-6-1, Detail A).

- Check self-sealing area on top of door and around sump for fuel. If no fuel is found, close fuel sump drain door, and lock latches and turnlock fasteners. If fuel is found, perform the following:
 - Wipe area dry with machinery towel, Item 211, Appendix D.
 - Close fuel sump drain door, and lock latches and turnlock fasteners.

NOTE

Helicopter may be operated during this 24-hour period.

- Recheck self-sealing area after 24 hours. If after 24 hours, fuel is found in self-sealing area, perform the following:

- (a) Operate sump drain valve six times using drain valve (PARA 1-3-21).
 - (b) Wipe area dry with machinery towel, Item 211, Appendix D.
 - (c) Close fuel sump drain door, and lock latches and turnlock fasteners.
- (4) Recheck self-sealing area after 24 hours. If after 24 hours, fuel drips from cover, perform the following:
- (a) Remove sump drain valve (PARA 10-4-6.1.2).
 - (b) Remove packings in poppet of valve body (Figure 10-4-6-1, Detail B). Discard packings.
 - (c) Lubricate packings with lubricating oil, Item 205, Appendix D. Install packings in poppet of valve body.
 - (d) Install sump drain valve (PARA 10-4-6.1.3).
 - (e) Check sump drain valve for leakage. **NONE ALLOWED.** If leakage is found, replace sump drain valve.

10-4-6.1.2 Remove.

- a. Purge main fuel cell (PARA 10-4-40).

WARNING

Injury to personnel will result if care is not taken when opening fuel sump drain door. Fuel may be trapped above door. Take care when opening door.

- b. Unlock latches and turnlock fasteners, and open applicable fuel sump drain door (Figure 10-4-6-1, Detail A).
- c. Remove bolts, washers, sump drain valve, and packing from main fuel cell. Discard packing.

10-4-6.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing in groove on sump drain valve (Figure 10-4-6-1, Detail A).

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- c. Coat threads on bolts with antiseize compound, Item 74, Appendix D.
- d. Install sump drain valve in main fuel cell with bolts and washers. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS.**
- e. Pressure test main fuel cells (PARA 10-3-1 or PARA 10-3-2).
- f. Inspect sump drain valve (PARA 10-4-6.1.1).
- g. Make sure area is clean and free of foreign material.
- h. Close fuel sump drain door, and lock latches and turnlock fasteners.
- i. Prime fuel system (PARA 10-4-1).

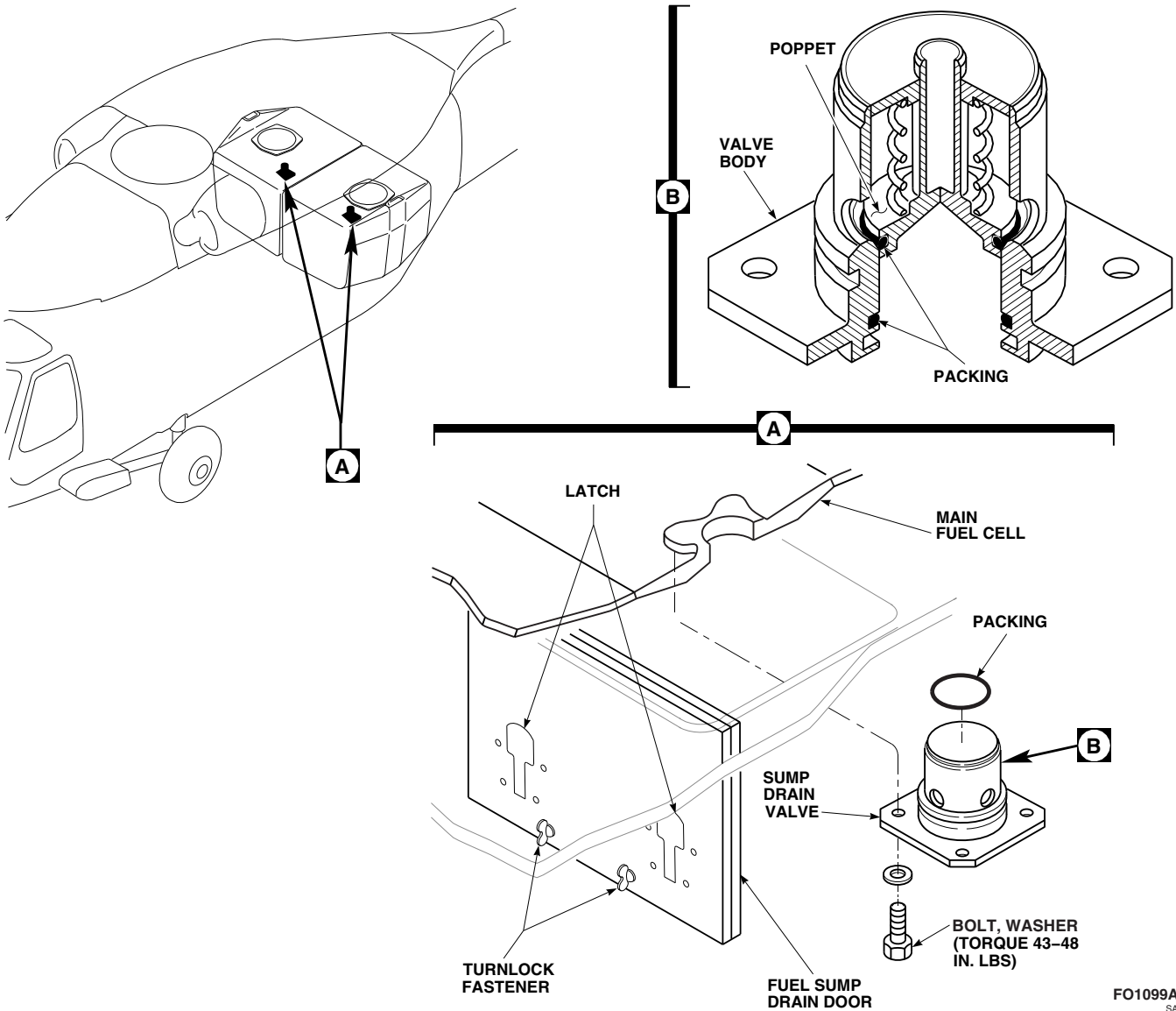


FIGURE 10-4-6-1. REPLACE SUMP DRAIN VALVE

10-4-7 PRIME/BOOST PUMP CHECK VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-7.1 Replace Prime/Boost Pump Check Valve	10-4-7-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Machinery Towel, Item 211, Appendix D

- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 10-4-3

10-4-7.1 Replace Prime/Boost Pump Check Valve.

NOTE

Prime/boost pump check valve is installed on No. 1 main fuel cell only.

- Install clamp on prime/boost pump check valve (Figure 10-4-7-1, Detail A).
- Position clamp, with valve, and spacer to clamp on main fuel line check valve and secure with screw, washers, and nut.

10-4-7.1.1 Remove.

- Remove main fuel cell plate and attached components (PARA 10-4-3).
- Disconnect hose from prime/boost pump check valve (Figure 10-4-7-1, Detail A). Plug hose and cap valve.
- Remove screw, washers, spacer, and nut securing clamps. Remove clamp from prime/boost pump check valve.

10-4-7.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).



To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into the fuel system.

- Uncap valve and coat threads on prime/boost pump check valve with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeeze out after parts are assembled with machinery towel, Item 211, Appendix D.

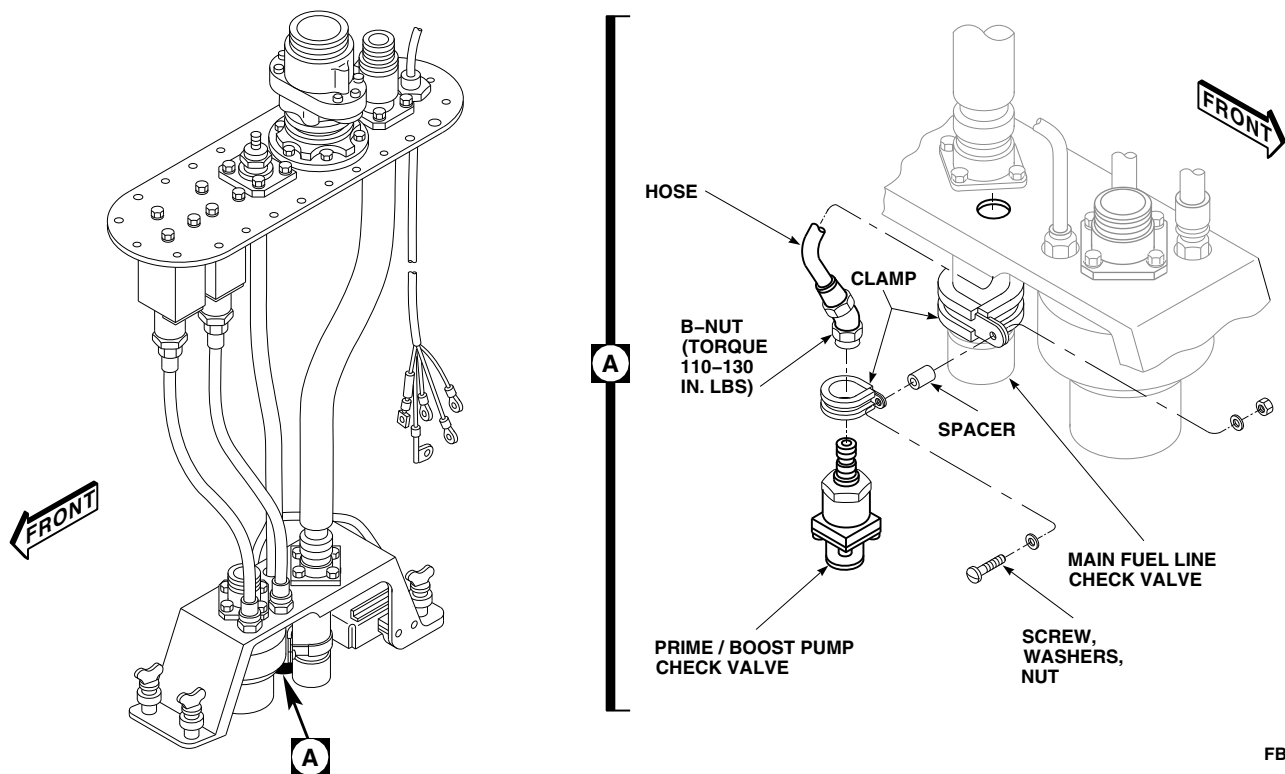
FB3580
SA

FIGURE 10-4-7-1. REPLACE PRIME/BOOST PUMP CHECK VALVE

- e. Unplug and connect hose to prime/boost check valve. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**
- f. Install main fuel cell plate and attached components (PARA 10-4-3).

10-4-8 MAIN FUEL LINE CHECK VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-8.1 Replace Main Fuel Line Check Valve	10-4-8-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Machinery Towel, Item 211, Appendix D
- Protective Cap and Plug

References

- Appendix D
- PARA 1-6-16
- PARA 10-4-3

10-4-8.1 Replace Main Fuel Line Check Valve.

NOTE

Replacement of No. 1 and No. 2 main fuel line check valves is the same, except as noted.

- On No. 1 main fuel line check valve, install clamp on main fuel line check valve (Figure 10-4-8-1, Detail A).
- Lubricate packings with lubricating oil, Item 205, Appendix D. Install packing on main fuel line check valve.

10-4-8.1.1 Remove.

- Remove main fuel cell plate and attached components from applicable main fuel cell (PARA 10-4-3).
- On No. 1 main fuel line check valve, remove screw, washers, spacer, and nut securing clamps (Figure 10-4-8-1, Detail A). Remove clamp from main fuel line check valve.
- Unthread main fuel line check valve from adapter. Discard packing. Plug adapter and cap valve.

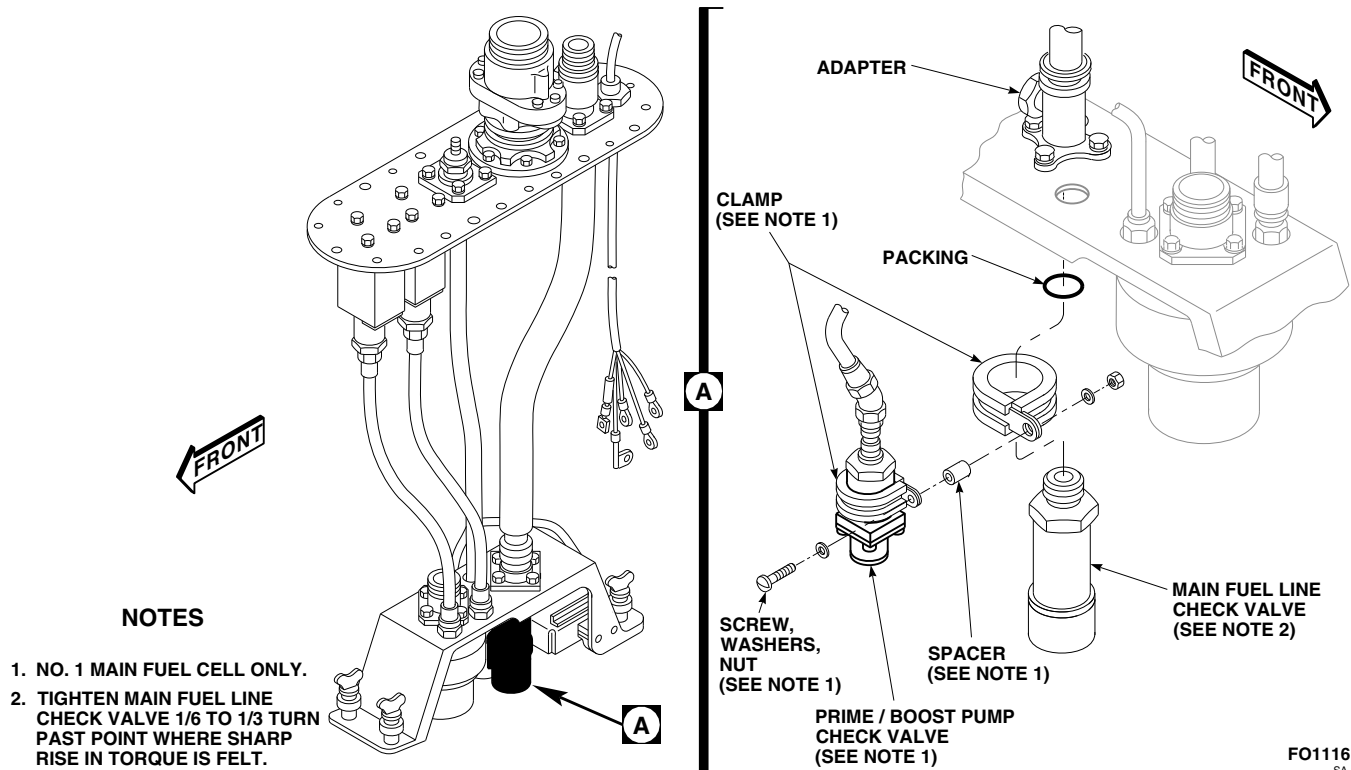
10-4-8.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).



To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into the fuel system.

- Uncap valve and coat threads on main fuel line check valve with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeeze out after parts are assembled with machinery towel, Item 211, Appendix D.

FO1116
SA

e. Remove plug from adapter. Thread main fuel line check valve into adapter. **TIGHTEN MAIN FUEL LINE CHECK VALVE $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

f. On No. 1 main fuel line check valve, position and secure clamp on prime/boost pump check

valve to clamp on main fuel line check valve with screw, washers, spacer, and nut.

g. Install main fuel cell plate and attached components (PARA 10-4-3).

10-4-9 LOW LEVEL SHUTOFF VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-9.1 Replace Low Level Shut-off Valve	10-4-9-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 10-4-3

10-4-9.1 Replace Low Level Shutoff Valve.

NOTE

Replacement of No. 1 and No. 2 low level shutoff valves is the same.

10-4-9.1.1 Remove.

- Remove main fuel cell plate and attached components from applicable main fuel cell (PARA 10-4-3).
- Disconnect tube from union on low level shut-off valve (Figure 10-4-9-1, Detail A). Install protective plug in tube.
- Remove bolts, washers, and nuts securing low level shutoff valve to support. Remove valve.
- Remove union and packing from low level shutoff valve. Remove and discard packing. Install protective caps on union.

10-4-9.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Lubricate packing with lubricating oil, Item 205, Appendix D.
- Install packing on union (Figure 10-4-9-1, Detail A).



To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into the fuel system.

- Apply antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads union. Wipe off any squeeze out after parts are assembled with machinery towel, Item 211, Appendix D.

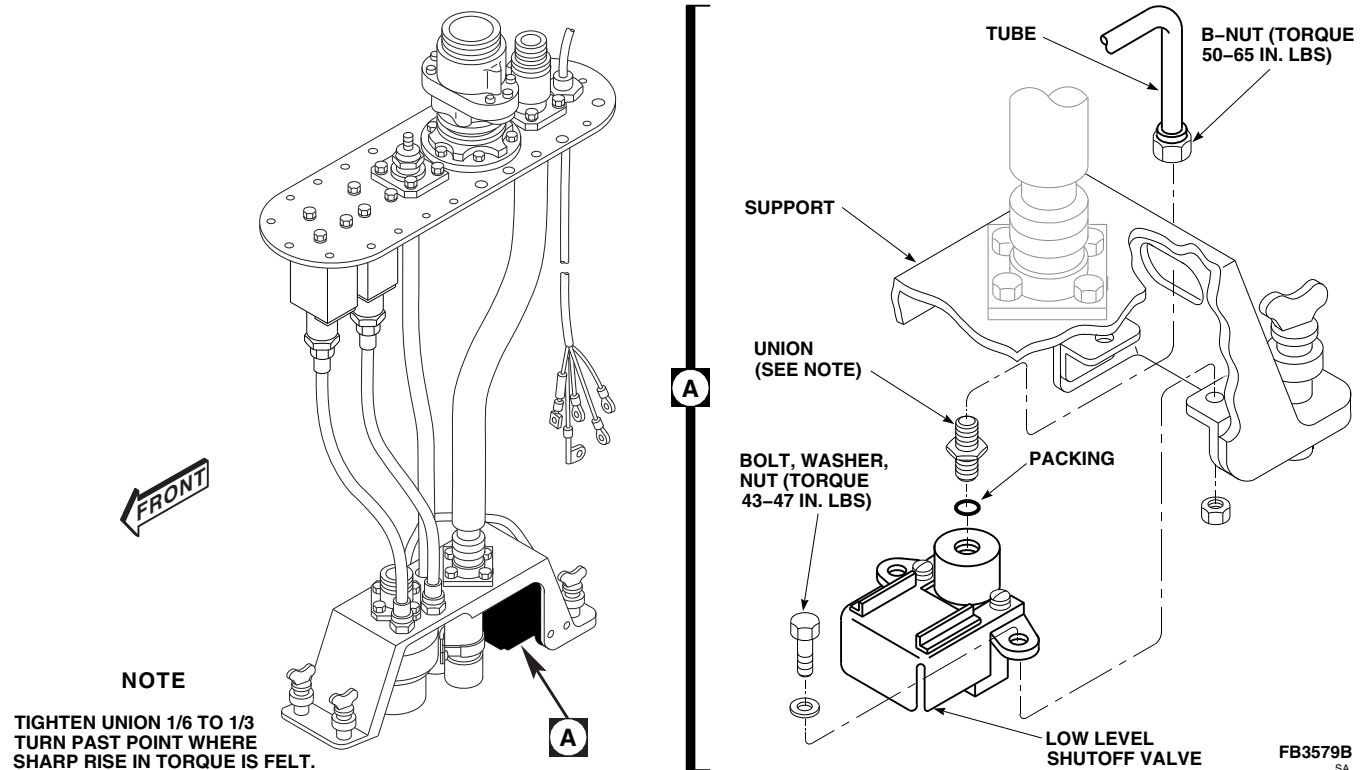


FIGURE 10-4-9-1. REPLACE LOW LEVEL SHUTOFF VALVE

- e. Remove protective caps from union. Thread union into low level shutoff valve. **TIGHTEN UNION $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into the fuel system.

- f. Coat threads on bolts with antiseize compound, Item 74, Appendix D.
- g. Install low level shutoff valve to support with bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
- h. Remove protective plug from tube. Connect tube to union on low level shutoff valve. **TORQUE B-NUT TO 50 - 65 INCH-POUNDS.**
- i. Install main fuel cell plate and attached components (PARA 10-4-3).

10-4-10 PRESSURE REFUELING INTERCONNECT TUBE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-10.1 Replace Pressure Refueling Interconnect Tube	10-4-10-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-17

- PARA 1-3-18
- PARA 1-3-19
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-4-1
- PARA 10-4-40
- PARA 13-5-3
- PARA 13-5-6

10-4-10.1 Replace Pressure Refueling Interconnect Tube.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

NOTE

Replacement of No. 1 and No. 2 pressure refueling interconnect tubes is the same, except as noted.

10-4-10.1.1 Remove.

- Purge main fuel cell (PARA 10-4-40).
- Gain access to main fuel cell compartment as follows:

- (1) Remove rear passenger's seats as required (PARA 2-4-45).
- (2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).
- (3) Remove cargo netting from stowage compartments.
- (4) If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

- c. Remove bolts, washers, and fuel cell access cover from main fuel cell (Figure 10-4-10-1, Detail A). Remove packing from fuel cell access cover. Discard packing.
- d. For No. 1 main fuel cell, remove screws, lockwashers, washers, nuts, and clamps securing bonding jumper to pressure refueling tube and pressure refueling interconnect tube (Figure 10-4-10-1, Detail B and Detail C).
- e. Remove screws, lockwashers, washers, and nuts from clamps securing hose assemblies and bonding jumpers to pressure refueling interconnect tube (Figure 10-4-10-1, Detail B).

NOTE

Install caps and plugs on hoses and fittings as parts are removed and disconnected.

- f. For No. 1 main fuel cell, disconnect pressure refueling interconnect tube from union on pressure refuel/defuel valve, fuel cell interconnect breakaway valve, and pressure refueling tube (Figure 10-4-10-1, Detail B and Detail C). Remove and discard packings.
- g. For No. 2 main fuel cell, disconnect pressure refueling interconnect tube from union on pressure refuel/defuel valve, fuel cell interconnect breakaway valve, and extended range interconnect hose (Figure 10-4-10-1, Detail B and Detail D). Remove and discard packings.

- h. Remove clamps from pressure refueling interconnect tube and retain for installation (Figure 10-4-10-1, Detail B).
- i. Remove pressure refueling interconnect tube from main fuel cell.
- j. Place temporary cover over access cover opening to keep inside of main fuel cell clean.

10-4-10.1.2 Inspect.

Inspect pressure refueling interconnect tube for dents and distortion. **DENTS OR LOCAL DISTORTION SHALL NOT EXCEED 20% OF TUBE DIAMETER.** If limit is exceeded, replace tube.

10-4-10.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Lubricate packings with lubricating oil, Item 205, Appendix D. Install packings on pressure refueling interconnect tube (Figure 10-4-10-1, Detail B, Detail C, and Detail D).
- c. Remove temporary cover from access cover opening.

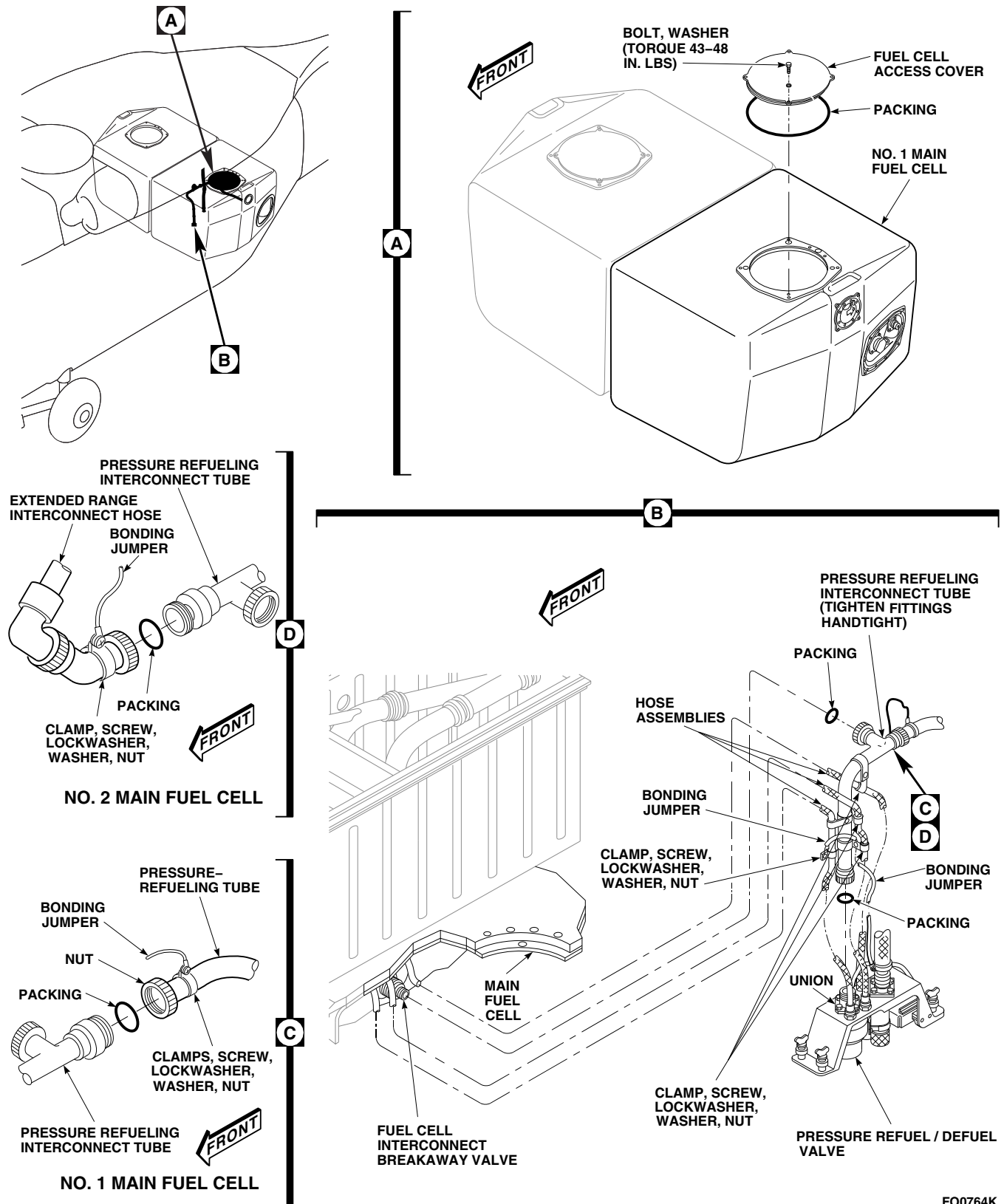
CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

NOTE

Remove caps and plugs from hoses and fittings as parts are installed and connected.

- d. Coat threads on fuel cell interconnect breakaway valve with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.



FO0764K
SA

FIGURE 10-4-10-1. REPLACE PRESSURE REFUELING INTERCONNECT TUBE

- e. Coat threads on pressure refueling interconnect tube with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.
- f. Place pressure refueling interconnect tube in main fuel cell.
- g. Connect pressure refueling interconnect tube as follows:
 - (1) Connect pressure refueling interconnect tube to union on pressure refuel/defuel valve, and to fuel cell interconnect breakaway valve (Figure 10-4-10-1, Detail B).
 - (2) For No. 1 main fuel cell, connect pressure refueling interconnect tube to pressure refueling tube (Figure 10-4-10-1, Detail B and Detail C).
 - (3) For No. 2 main fuel cell, install packing on extended range interconnect hose and connect hose to pressure refueling interconnect tube (Figure 10-4-10-1, Detail B and Detail D).
 - (4) **TIGHTEN FITTINGS HANDTIGHT.**
- h. For No. 1 main fuel cell, perform the following (Figure 10-4-10-1, Detail B and Detail C):
 - (1) Place clamps loosely on pressure refueling tube and pressure refueling interconnect tube.
 - (2) Connect bonding jumper to clamps on tubes with screws, lockwashers, washers, and nuts.
- i. Connect bonding jumpers to clamp on pressure refueling interconnect tube using screw, lockwasher, washers, and nut (Figure 10-4-10-1, Detail B).
- j. Connect clamps securing hose assemblies and bonding jumpers to pressure refueling intercon-

nect tube with screws, lockwashers, washers, and nuts.

- k. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on fuel cell access cover (Figure 10-4-10-1, Detail A).
- l. Inspect area inside main fuel cell and make sure area is free of all foreign material.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- m. Coat threads of fuel cell access cover attachment bolts with antiseize compound, Item 74, Appendix D.
- n. Install fuel cell access cover on main fuel cell with bolts and washers. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS.**
- o. Make sure area is clean and free of foreign material.
- p. Refuel main fuel cells (PARA 1-3-17, PARA 1-3-18, or PARA 1-3-19).
- q. Prime fuel system (PARA 10-4-1).
- r. Close main fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panels (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

10-4-11 EXTENDED RANGE INTERCONNECT HOSE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-11.1 Replace Extended Range Interconnect Hose	10-4-11-1
INITIAL SETUP	• PARA 1-3-18
Tools	• PARA 1-3-19
• Aircraft Mechanic’s Toolkit • Torque Wrench, 30 - 150 in. lbs	• PARA 1-6-16 • PARA 2-4-45
Materials/Parts	• PARA 2-4-69
• Antiseize Compound, Item 74, Appendix D • Lubricating Oil, Item 205, Appendix D	• PARA 10-4-1 • PARA 10-4-40
References	• PARA 13-5-3
• Appendix D • PARA 1-3-17	• PARA 13-5-6

10-4-11.1 Replace Extended Range Interconnect Hose.

NOTE

Extended range interconnect hose is installed on No. 2 main fuel cell only.

10-4-11.1.1 Remove.

- a. Purge main fuel cell (PARA 10-4-40).
- b. Gain access to main fuel cell compartment as follows:
 - (1) Remove right side rear passenger’s seats (PARA 2-4-45).

- (2) Remove soundproofing panel from rear bulkhead covering right stowage compartment (PARA 2-4-69).
- (3) Remove cargo netting from stowage compartment.
- (4) If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
- c. Remove bolts, washers, and fuel cell access cover from main fuel cell. Remove packing from fuel cell access cover (Figure 10-4-11-1, Sheet 1, Detail A). Discard packing.

- d. Remove screw, washers, nut, clamp, and bonding jumper from extended range interconnect hose (Figure 10-4-11-1, Sheet 2, Detail B).
- e. Remove screw, washer, nut, and clamps securing extended range interconnect hose to main fuel hose.
- f. Loosen fitting and disconnect upper end of extended range interconnect hose from extended range breakaway valve. Remove and discard packing.
- g. Loosen nut and disconnect lower end of extended range interconnect hose from pressure refueling interconnect tube. Remove hose from helicopter.
- h. Remove reducer, packing, and nut from extended range interconnect hose. Discard packing.

10-4-11.1.2 Install.

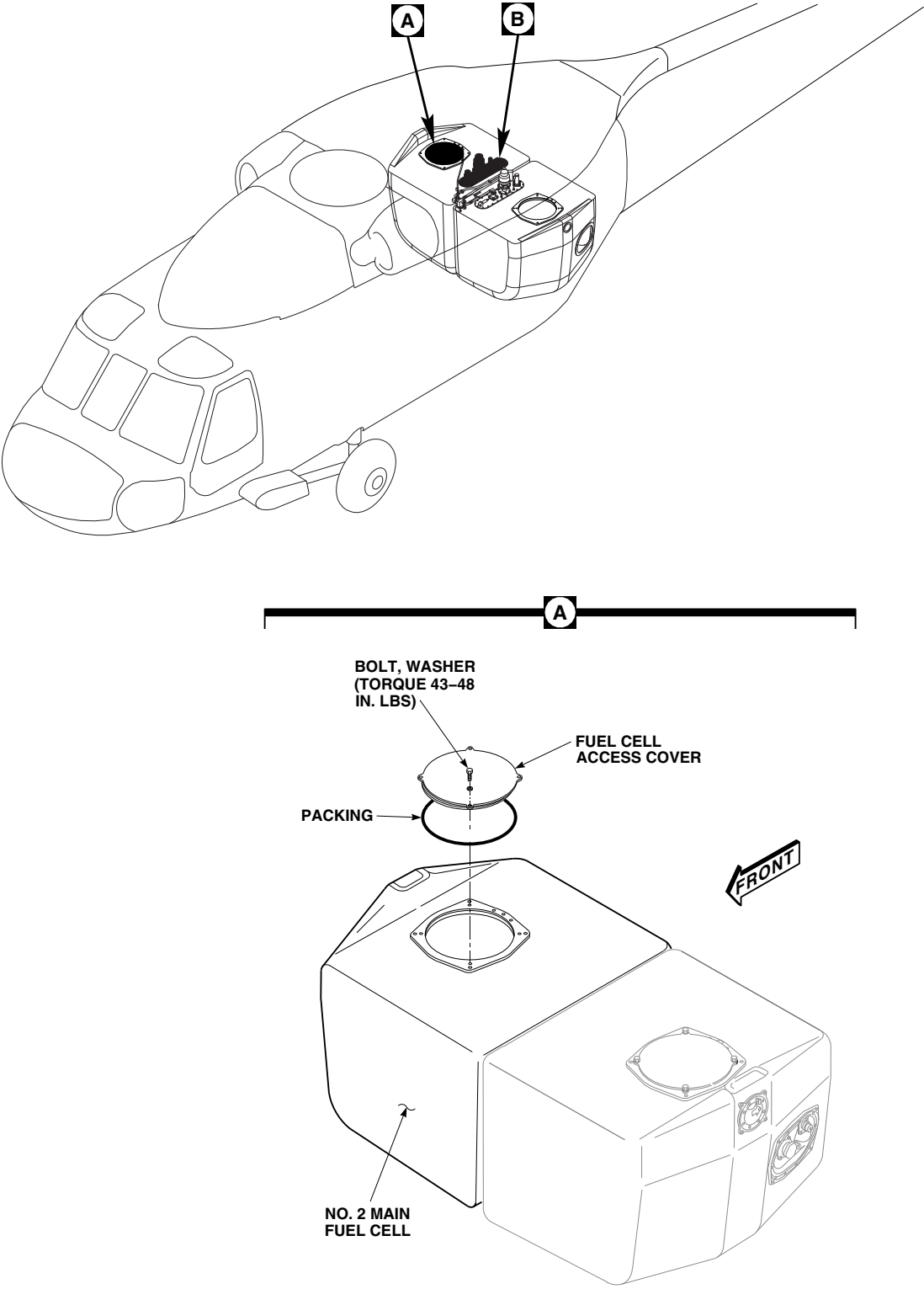
- a. Turn off all electrical power (PARA 1-6-16).
- b. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on reducer (Figure 10-4-11-1, Sheet 2, Detail B).
- c. Install reducer, packing, and nut in extended range interconnect hose.
- d. Install packing on extended range breakaway valve.
- e. Connect fitting end of extended range interconnect hose on extended range breakaway valve. **TIGHTEN FITTING HANDTIGHT.**
- f. Connect reducer end of extended range interconnect hose on pressure refueling interconnect tube. **TIGHTEN NUT HANDTIGHT.**
- g. Secure hose to main fuel hose with clamps, screw, washer, and nut.
- h. Install bonding jumper on extended range interconnect hose with clamps, screws, washers, and nuts.

- i. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on fuel cell access cover (Figure 10-4-11-1, Sheet 1, Detail A).
- j. Inspect area inside main fuel cell and make sure area is free of all foreign material.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- k. Coat threads of fuel cell access cover attachment bolts with antiseize compound, Item 74, Appendix D.
- l. Install fuel cell access cover on main fuel cell with bolts and washers. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS.**
- m. Make sure area is clean and free of foreign material.
- n. Refuel main fuel cells (PARA 1-3-17, PARA 1-3-18, or PARA 1-3-19).
- o. Prime fuel system (PARA 10-4-1).
- p. Close main fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panel (PARA 2-4-69).
 - (4) Install right side rear passenger's seats (PARA 2-4-45).



FB3308_1B
SA

FIGURE 10-4-11-1. REPLACE EXTENDED RANGE INTERCONNECT HOSE (SHEET 1 OF 2)

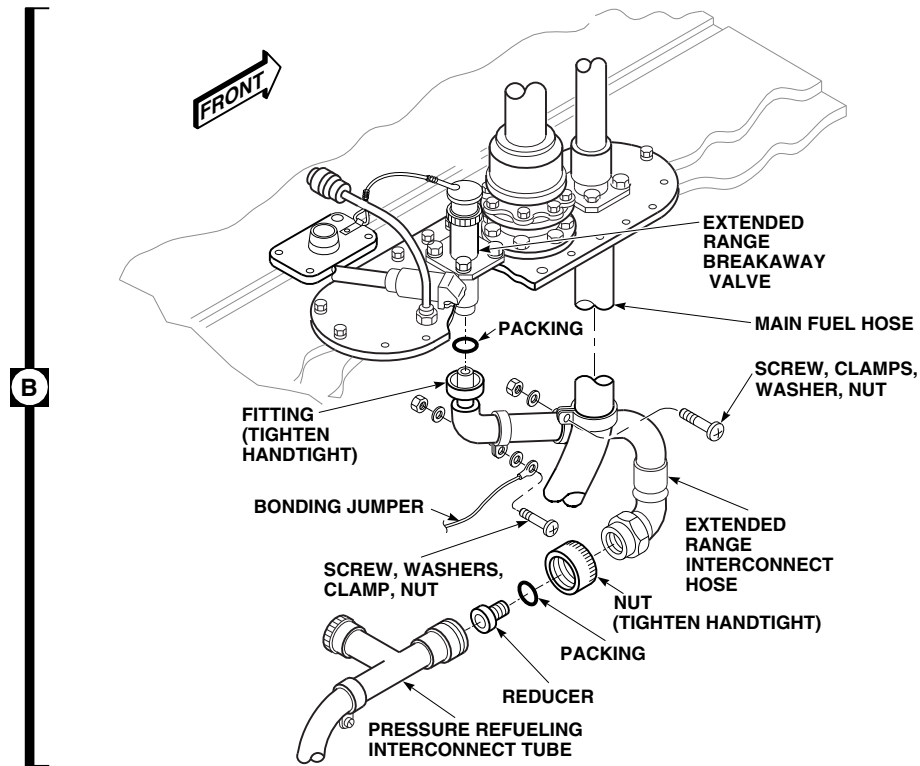
FB3308_2A
SA

FIGURE 10-4-11-1. REPLACE EXTENDED RANGE INTERCONNECT HOSE (SHEET 2 OF 2)

10-4-12 FUEL BOOST PUMP.

PARAGRAPH BREAKDOWN

PAGE

10-4-12.1 Replace Fuel Boost Pump 10-4-12-2

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Wire Snake, Locally-Made

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Insulation Sleeving, Heat-Shrinkable, Item 169, Appendix D
- Lockwire, Item 197, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Tags

- Tape, Item 317B, Appendix D
- Thread, Break-A-Way, Item 327, Appendix D
- Tiedown Strap, Item 331, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-2-5
- PARA 10-2-6
- PARA 10-2-7
- PARA 10-3-1
- PARA 10-3-2
- PARA 10-4-40
- PARA 13-5-3
- PARA 13-5-6

10-4-12.1 Replace Fuel Boost Pump.**NOTE**

Replacement of No. 1 and No. 2 fuel boost pumps is the same, except as noted.

10-4-12.1.1 Remove.

- a. Purge main fuel cell (PARA 10-4-40).
- b. Gain access to main fuel cell compartment as follows:
 - (1) Remove rear passenger's seats as required (PARA 2-4-45).
 - (2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).
 - (3) Remove cargo netting from stowage compartments.
 - (4) If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
- c. Unlock fasteners securing left or right panel enclosure on top of main fuel cells. Remove panel.
- d. Remove screws and front panel from structure.
- e. Disconnect electrical connector, P322 from J322 (No. 1 fuel boost pump), or P323 from J323 (No. 2 fuel boost pump) (Figure 10-4-12-1, Sheet 2, Detail D).
- f. Install protective caps and plugs on electrical connectors.
- g. Using tape, Item 317B, Appendix D, tape wires and remove electrical connector from fuel boost pump electrical harness.
- h. Remove insulation tubing from fuel boost pump wiring harness.
- i. Using nonmetallic scraper, remove sealing compound from scupper and wire leads.
- j. Clean reworked area with machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D.
- k. Tag locations of tiedown straps, mounting plates, and break-a-way thread before removing fuel boost pump electrical harness (Figure 10-4-12-1, Sheet 3, Detail E and Detail F).
- l. Remove tiedown straps, mounting plates, and cut break-a-way thread securing wiring harness.
- m. Remove bolts, washers, and fuel cell access cover from main fuel cell. Remove packing from fuel cell access cover (Figure 10-4-12-1, Sheet 1, Detail A). Discard packing.
- n. Remove screws, washers, and nuts securing bonding jumpers and clamps to pressure sensing hose, electrical wiring hose, and fuel hoses (Figure 10-4-12-1, Sheet 1, Detail B and Sheet 2, Detail C). Remove clamps from hoses.
- o. Inspect bonding jumpers, screws, and clamps for corrosion. **NONE ALLOWED.** If corrosion is found, replace affected hardware.
- p. Disconnect fuel hose from union in fuel boost pump (Figure 10-4-12-1, Sheet 1, Detail B).
- q. Disconnect pressure sensing hose from restrictor in fuel boost pump.
- r. Disconnect electrical wiring hose from fitting on fuel boost pump.
- s. Using nonmetallic scraper, remove sealing compound from edges of long bolt and washers.
- t. Clean area with dry-cleaning solvent, Item 124, Appendix D.
- u. Remove bolts, washers, and bonding jumper securing fuel boost pump to bracket and carefully remove pump and electrical wires from main fuel cell.
- v. Remove restrictor and packing from fuel boost pump. Discard packing.

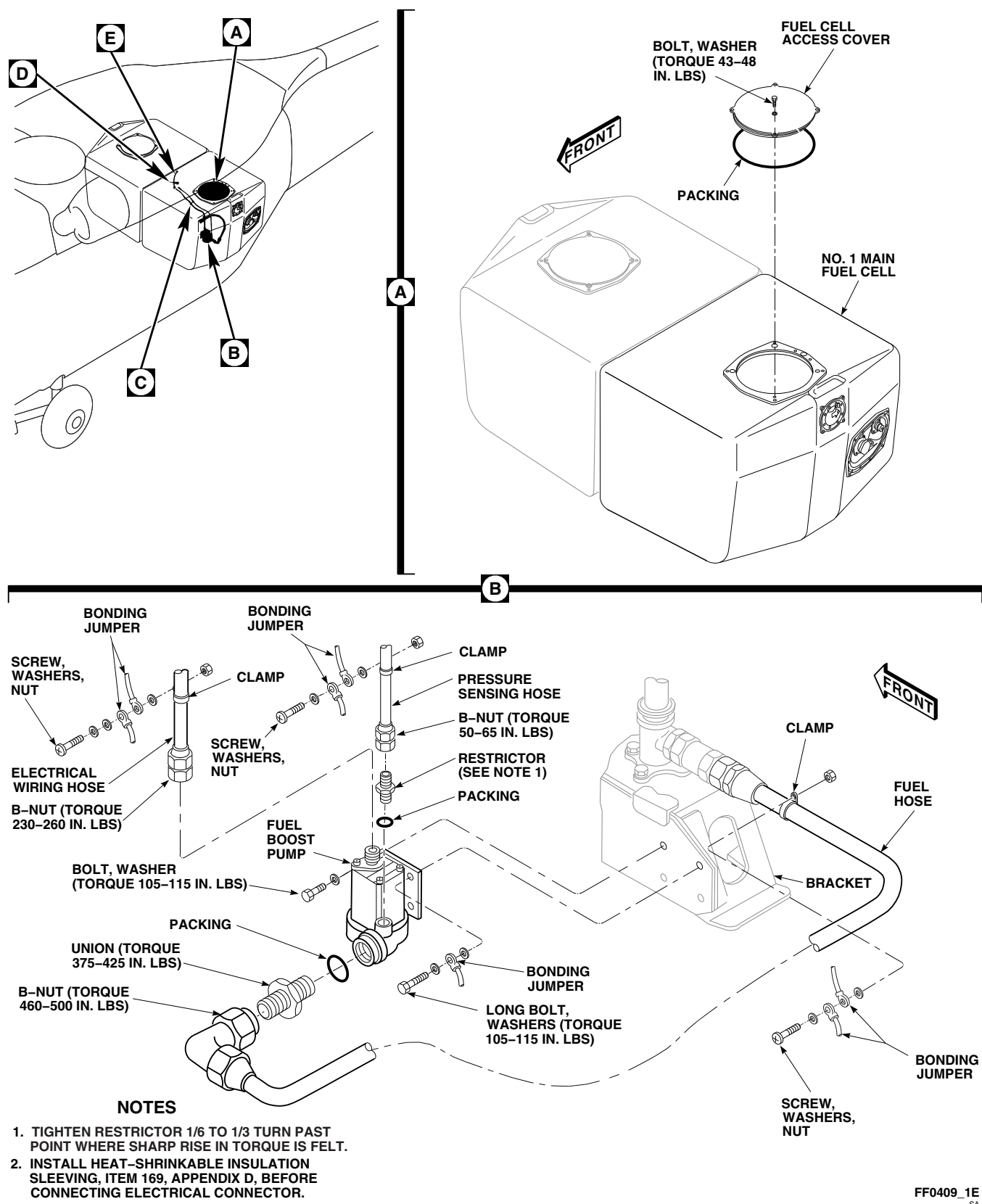
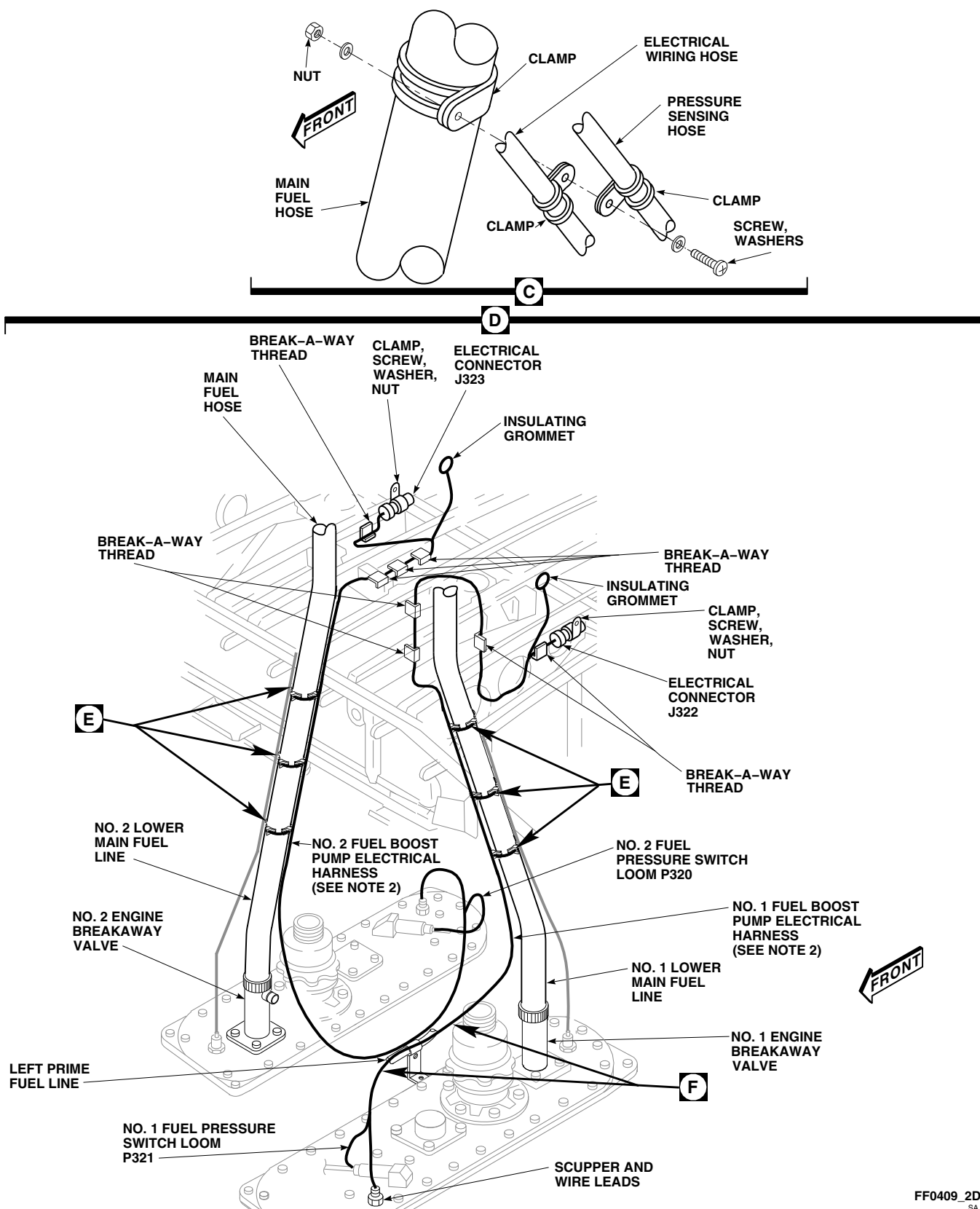
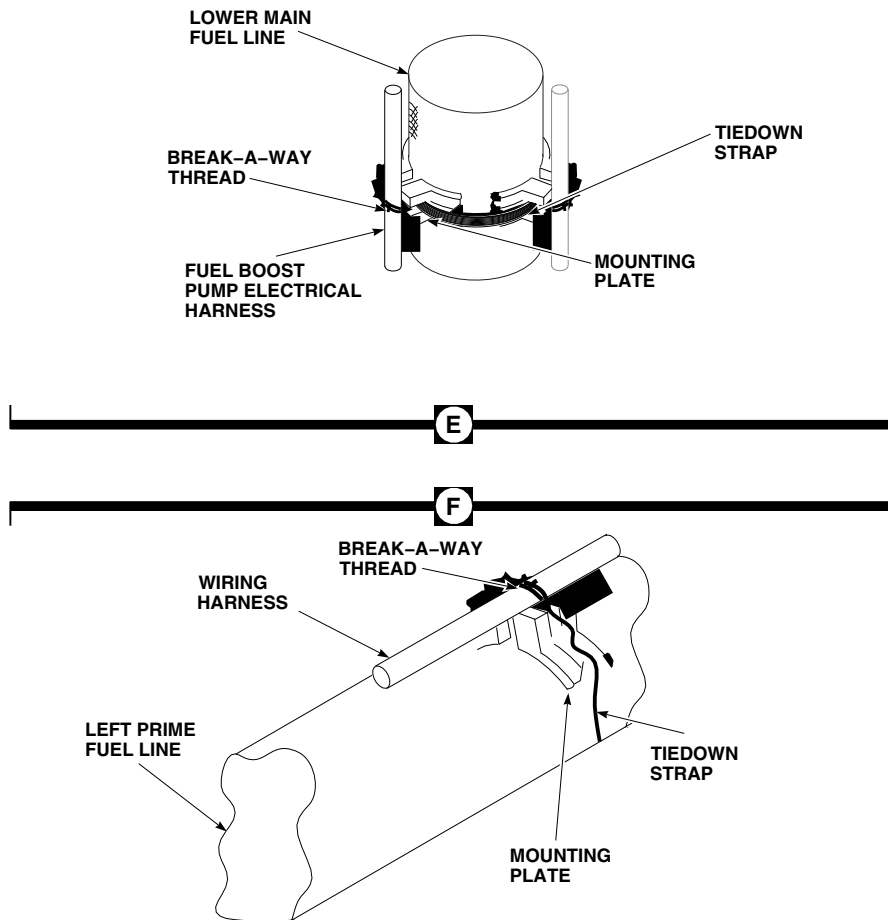


FIGURE 10-4-12-1. REPLACE FUEL BOOST PUMP (SHEET 1 OF 3)



FF0409_2D
SA

FIGURE 10-4-12-1. REPLACE FUEL BOOST PUMP (SHEET 2 OF 3)



FF0409_3
SA

FIGURE 10-4-12-1. REPLACE FUEL BOOST PUMP (SHEET 3 OF 3)

- w. Remove union and packing from fuel boost pump. Discard packing.
- x. Install protective plugs in ports in fuel boost pump.
- d. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on union (Figure 10-4-12-1, Sheet 1, Detail B).

CAUTION

10-4-12.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
 - b. Measure electrical wires on new fuel boost pump. Wires must be at least 95 - 97-inches long.
 - c. If protective plugs were installed, remove from ports in new fuel boost pump.
 - e. Coat threads of union with antiseize compound, Item 74, Appendix D, sparingly ($\frac{1}{4}$ -
- To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

Change 11

10-4-12-5

inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.

- f. Install union in port of fuel boost pump. **TORQUE UNION TO 375 - 425 INCH-POUNDS.**
- g. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on restrictor.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- h. Coat threads of restrictor with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled using machinery towel, Item 211, Appendix D.
- i. Install restrictor in port of fuel boost pump. **TIGHTEN RESTRICTOR 1/6 TO 1/3 PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

NOTE

When installing long bolt in top right location, do not install bonding jumper and additional washer.

- j. Install fuel boost pump on bracket with bolts and washers. Do not torque bolts at this time.
- k. Locally make wire snake from lockwire, Item 197, Appendix D, about 6 1/2-feet long. Make eye on one end of lockwire. Tie ends of electrical wire to eye of lockwire.
- l. Carefully thread electrical wires through electrical wiring hose. Remove lockwire from electrical wires.

- m. Remove long bolt and washer from top right corner of fuel boost pump. Do not remove washer from bolt.
- n. Install bonding jumper and additional washer on long bolt. Install bolt and stack up in fuel boost pump and bracket. **TORQUE ALL BOLTS TO 105 - 115 INCH-POUNDS.**
- o. Seal edges of long bolt and washers with sealing compound, Item 292, Appendix D, to diameter slightly larger than cleaned area.
- p. Connect electrical wiring hose to fitting on fuel boost pump. **TORQUE B-NUT ON ELECTRICAL WIRING HOSE TO 230 - 260 INCH-POUNDS.**
- q. Connect pressure sensing hose to restrictor on fuel boost pump. **TORQUE B-NUT ON PRESSURE SENSING HOSE TO 50 - 65 INCH-POUNDS.**
- r. Connect fuel hose to union on fuel boost pump. **TORQUE B-NUT ON HOSE TO 460 - 500 INCH-POUNDS.**
- s. Position clamps on pressure sensing hose, electrical wiring hose, and fuel hoses (Figure 10-4-12-1, Sheet 1, Detail B, and Sheet 2, Detail C).
- t. Install bonding jumpers to clamps using screws, washers, and nuts (Figure 10-4-12-1, Sheet 1, Detail B).
- u. Install clamps on pressure sensing hose and electrical wiring hose. Secure clamps to main fuel hose clamp with screw, washers, and nut (Figure 10-4-12-1, Sheet 1, Detail B and Sheet 2, Detail C).
- v. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on fuel cell access cover (Figure 10-4-12-1, Sheet 1, Detail A).
- w. Inspect area inside main fuel cell and make sure area is free of all foreign material.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- x. Coat threads of fuel cell access cover attachment bolts with antiseize compound, Item 74, Appendix D.
- y. Install fuel cell access cover on main fuel cell with bolts and washers. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS.**
- z. Install tiedown straps, Item 331, Appendix D, and mounting plates, at tagged locations and remove tags (Figure 10-4-12-1, Sheet 3, Detail E and Detail F).
- aa. Cut heat-shrinkable insulation sleeving, Item 169, Appendix D, to length of wires and install over fuel boost pump electrical wires.

NOTE

Assemble a "single loop" of break-a-way thread for harness diameter of $\frac{3}{16}$ -inch or less. For harnesses with diameters over $\frac{3}{16}$ -inch, assemble "two loops" maximum.

- ab. At tagged locations, install break-a-way thread, Item 327, Appendix D, looping around harness and mounting plate. Tie with square knot and remove tag.
- ac. Install fuel boost pump electrical wires to electrical connector, P322 (No. 1 fuel boost pump), or P323 (No. 2 fuel boost pump) (Figure 10-4-12-1, Sheet 2, Detail D).

- ad. Seal edges of scupper and fill wire leads with sealing compound, Item 292, Appendix D.
- ae. Remove protective caps and plugs from electrical connectors.
- af. Inspect and treat electrical connectors (PARA 1-7-20).
- ag. Connect electrical connector, P322 to J322 (No. 1 fuel boost pump), or P323 to J323 (No. 2 fuel boost pump).
- ah. Pressure test main fuel cells (PARA 10-3-1 or PARA 10-3-2).
- ai. Install front panel and secure with screws.
- aj. Install left or right panel enclosure on top of main fuel cells. Lock fasteners.
- ak. Perform operational check of fuel quantity system (PARA 10-2-5 or PARA 10-2-6).
- al. Perform operational check of engine feed fuel lines (PARA 10-2-7).
- am. Close main fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panels (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

10-4-13 MAIN FUEL CELL CHECK VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-13.1 Replace Main Fuel Cell Check Valve	10-4-13-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-2-5
- PARA 10-2-6
- PARA 10-2-7
- PARA 10-4-40
- PARA 13-5-3
- PARA 13-5-6

10-4-13.1 Replace Main Fuel Cell Check Valve.

NOTE

- The following procedures apply to helicopters with fuel system installation, 70307-03050-014 (main fuel cell plate and attached components).
- Replacement of No. 1 and No. 2 main fuel cell check valves is the same.

10-4-13.1.1 Remove.

- Purge main fuel cell (PARA 10-4-40).

- Gain access to main fuel cell compartment as follows:
 - Remove rear passenger’s seats as required (PARA 2-4-45).
 - Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).
 - Remove cargo netting from stowage compartments.
 - If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

- c. Unlock fasteners securing left or right panel enclosure on top of main fuel cells. Remove panel.
- d. Remove screws and front panel from structure.
- e. Remove bolts, washers, and fuel cell access cover from main fuel cell. Remove packing from fuel cell access cover (Figure 10-4-13-1, Detail A). Discard packing.
- f. Remove screw, washers, lockwasher, and nut securing bonding jumpers and clamp to main fuel cell check valve (Figure 10-4-13-1, Detail B). Remove clamp from check valve.
- g. Disconnect fuel hose from main fuel cell check valve.
- h. Remove main fuel cell check valve and packing from fitting. Discard packing.
- e. Install check valve in fitting with arrow pointing to fitting. **TORQUE CHECK VALVE TO 460 - 500 INCH-POUNDS.**
- f. Install clamp around center area of check valve.
- g. Install bonding jumpers on clamp, using screw, washers, lockwasher, and nut.
- h. Apply sealing compound, Item 292, Appendix D, over hardware at bonding jumper location. Make sure sealing compound completely covers terminal lug of bonding jumper and attaching hardware.
- i. Connect fuel hose to check valve. **TORQUE B-NUT TO 460 - 500 INCH-POUNDS.**
- j. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on fuel cell access cover (Figure 10-4-13-1, Detail A).

10-4-13.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Clean area where clamp is to be installed, using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D.
- c. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on end of check valve nearest head of arrow (Figure 10-4-13-1, Detail B).

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- d. Coat threads of check valve with antiseize compound, Item 74, Appendix D, sparingly ($\frac{1}{4}$ -inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- l. Coat threads of fuel cell access cover attachment bolts with antiseize compound, Item 74, Appendix D.
- m. Install fuel cell access cover on main fuel cell with bolts and washers. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS.**
- n. Perform operational check of fuel quantity system (PARA 10-2-5 or PARA 10-2-6).
- o. Perform operational check of engine feed fuel lines (PARA 10-2-7).
- p. Install front panel and secure with screws.
- q. Install left or right panel enclosure on top of main fuel cells. Lock fasteners.

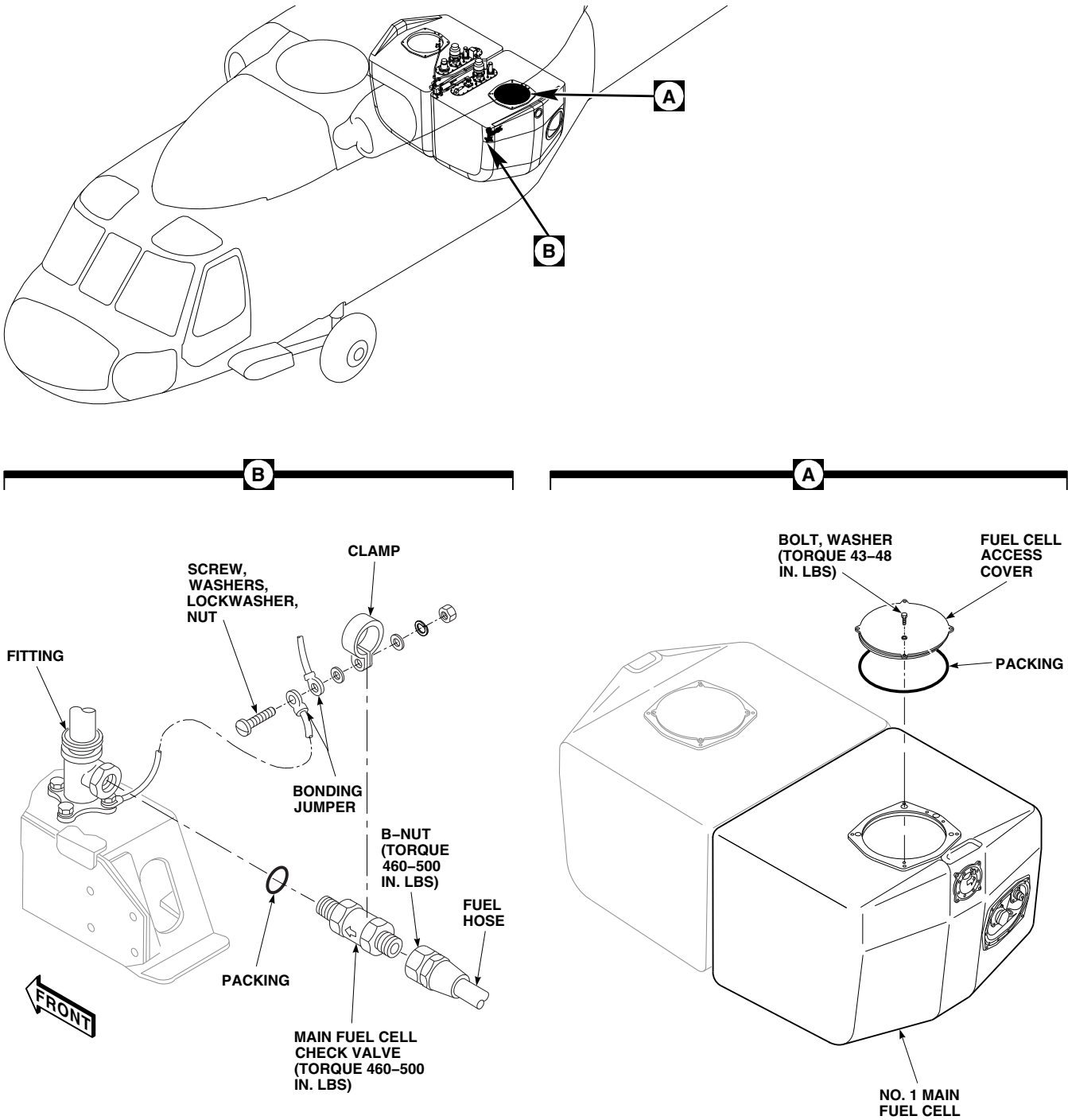


FIGURE 10-4-13-1. REPLACE MAIN FUEL CELL CHECK VALVE

FB3578C
SA

r. Close main fuel cell compartment as follows:

- (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

- (2) Install cargo netting.

- (3) Install soundproofing panels (PARA 2-4-69).

- (4) Install rear passenger's seats (PARA 2-4-45).

10-4-14 FUEL PRESSURE SWITCH.

PARAGRAPH BREAKDOWN

	PAGE
10-4-14.1 Replace Fuel Pressure Switch	10-4-14-1

INITIAL SETUP

- PARA 1-7-20

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

- PARA 2-4-45

- PARA 2-4-69

Materials/Parts

- Lubricating Oil, Item 205, Appendix D
- Protective Caps and Plugs

- PARA 10-2-2

- PARA 13-5-3

- PARA 13-5-6

References

- Appendix D
- PARA 1-6-16

10-4-14.1 Replace Fuel Pressure Switch.

NOTE

Replacement of No. 1 and No. 2 fuel pressure switches is the same, except as noted.

(2) Remove soundproofing panel from rear bulkhead covering left or right stowage compartment (PARA 2-4-69).

(3) Remove cargo netting from stowage compartment.

(4) If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

10-4-14.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Gain access to main fuel cell compartment as follows:
 - Remove rear passenger’s seats as required (PARA 2-4-45).

- Unlock fasteners securing left or right panel enclosure on top of main fuel cells. Remove panel.
- Remove screws and washers securing front and top panels to airframe. Remove panels from helicopter.

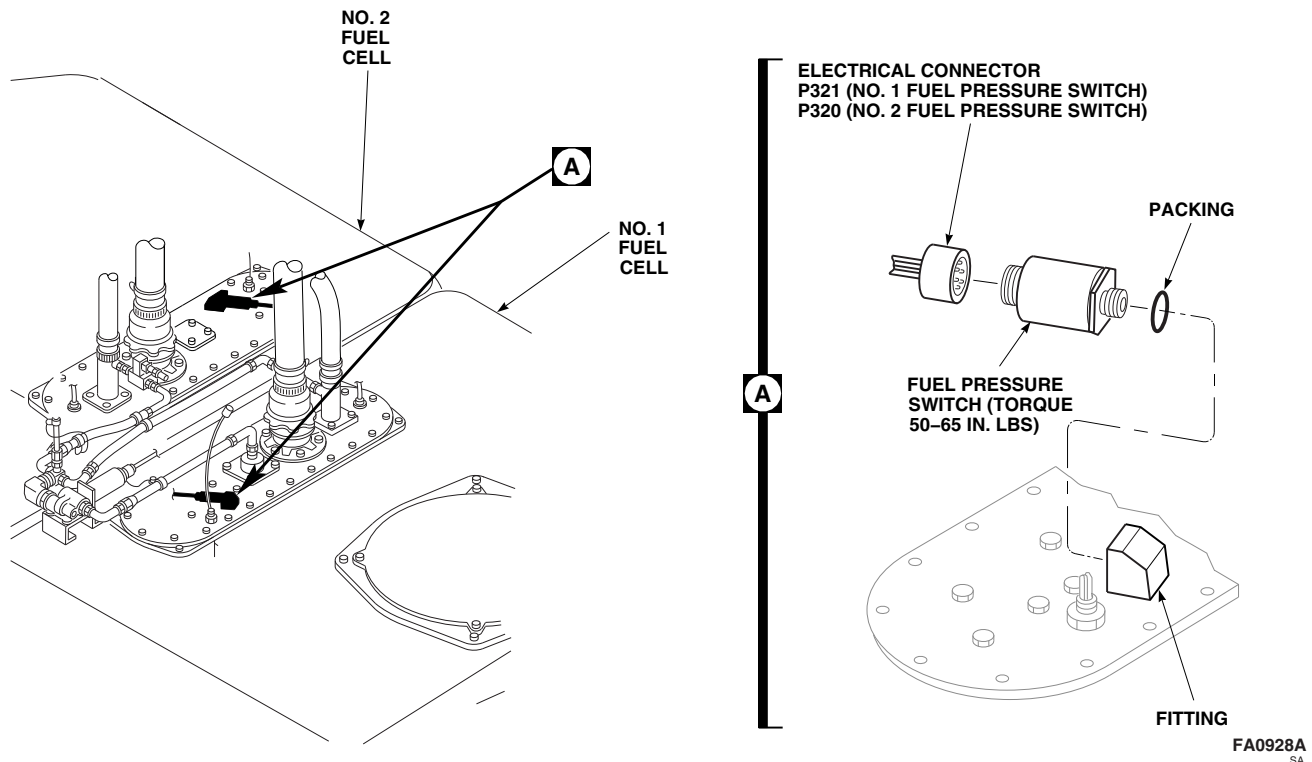


FIGURE 10-4-14-1. REPLACE FUEL PRESSURE SWITCH

- e. Disconnect electrical connector, P321 (No. 1 fuel pressure switch), or P320 (No. 2 fuel pressure switch) (Figure 10-4-14-1, Detail A).
- f. Install protective caps and plugs on electrical connectors.
- g. Remove fuel pressure switch and packing from fitting. Discard packing.
- f. Connect electrical connector, P321 (No. 1 fuel pressure switch), or P320 (No. 2 fuel pressure switch).
- g. Perform operational check of fuel pressure switch (PARA 10-2-2).
- h. Make sure area is clean and free of foreign material.

10-4-14.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on fuel pressure switch (Figure 10-4-14-1, Detail A).
- c. Install fuel pressure switch in fitting. **TORQUE FUEL PRESSURE SWITCH TO 50 - 65 INCH-POUNDS.**
- d. Remove protective caps and plugs from electrical connectors.
- e. Inspect and treat electrical connectors (PARA 1-7-20).
- i. Install front and top panels with screws and washers.
- j. Close main fuel cell compartment as follows:
 - (1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panels (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

10-4-15 MAIN ENGINE PRIME FUEL SHUTOFF VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-15.1 Replace Main Engine Prime Fuel Shutoff Valve	10-4-15-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Gallon
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lockwire, Item 197, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Machinery Towel, Item 211, Appendix D

- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-4-1
- PARA 10-4-39
- PARA 13-5-3
- PARA 13-5-6

10-4-15.1 Replace Main Engine Prime Fuel Shutoff Valve.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

NOTE

Replacement of No. 1 and No. 2 main engine prime fuel shutoff valves is the same, except as noted.

10-4-15.1.1 Remove.

- a. Purge fuel feed system (PARA 10-4-39).
- b. Gain access to main fuel cell compartment as follows:
 - (1) Remove rear passenger's seats as required (PARA 2-4-45).
 - (2) Remove soundproofing panel from rear bulkhead covering left or right stowage compartment (PARA 2-4-69).
 - (3) Remove cargo netting from stowage compartment.
 - (4) If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
- c. Unlock fasteners securing left or right panel enclosure on top of main fuel cells. Remove panel.
- d. Disconnect electrical connector, P304 (No. 1 shutoff valve) or P305 (No. 2 shutoff valve), from engine prime fuel shutoff valve (Figure 10-4-15-1, Detail A).
- e. Install protective caps and plugs on electrical connectors.
- f. Position 1-gallon container to catch fuel. Disconnect fuel hose from shutoff valve and plug hose.
- g. Disassemble shutoff valve as follows (Figure 10-4-15-1, Detail B):
 - (1) Remove screws securing solenoid to valve body, and remove solenoid.
 - (2) Remove and discard packing.
 - (3) Remove setscrew from disc insert and separate sealing disc from disc insert.
- h. Remove valve body from main fuel breakaway valve (Figure 10-4-15-1, Detail A).
- i. Remove packing and jamnut from valve body. Discard packing.
- j. Plug main fuel breakaway valve.

10-4-15.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Assemble engine prime fuel shutoff valve as follows (Figure 10-4-15-1, Detail B):
 - (1) Install sealing disc into disc insert and secure with setscrew.
 - (2) Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing in valve body.
 - (3) Install solenoid on valve body and secure with screws.
 - (4) Lockwire, Item 197, Appendix D, screws.
- c. Install jamnut on shutoff valve (Figure 10-4-15-1, Detail A).
- d. Lubricate packing with lubricating oil, Item 205, Appendix D, and install on shutoff valve.

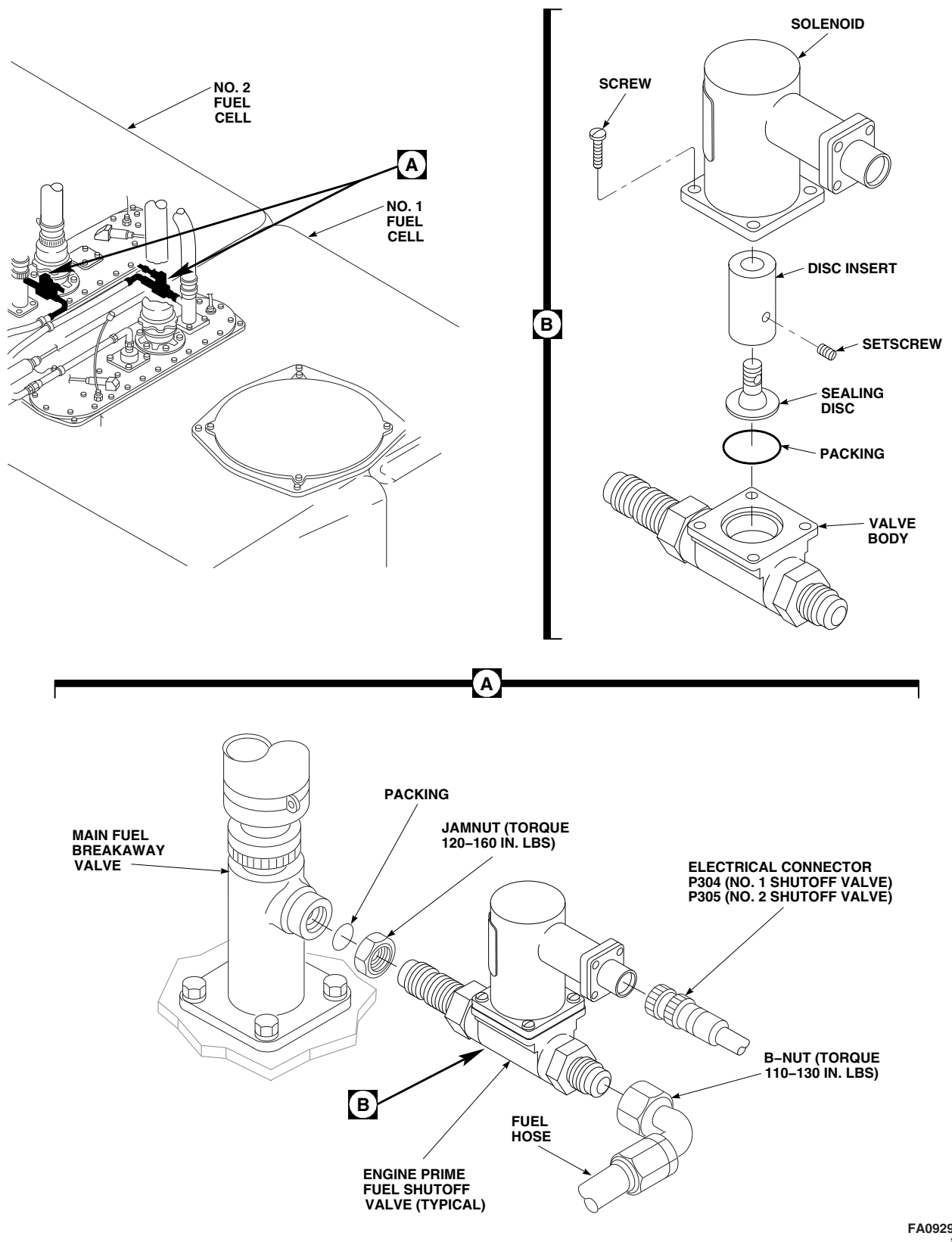


FIGURE 10-4-15-1. REPLACE MAIN ENGINE PRIME FUEL SHUTOFF VALVE

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into the fuel system.

- e. Coat threads on shutoff valve with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeeze out after parts are assembled with machinery towel, Item 211, Appendix D.
- f. Unplug main fuel breakaway valve.

CAUTION

Do not preload main fuel breakaway valve when installing shutoff valve. Frangible section of main fuel breakaway valve is sensitive to bending and tension overload.

- g. Install shutoff valve on main fuel breakaway valve with directional arrow pointing toward main fuel breakaway valve. **TORQUE JAM-NUT TO 120 - 160 INCH-POUNDS.**

CAUTION

Make sure main fuel breakaway valve is not preloaded when installing fuel hose to engine prime fuel shutoff valve.

Frangible section of main fuel breakaway valve is sensitive to bending and tension overload.

- h. Unplug and connect fuel hose to engine prime fuel shutoff valve. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**
- i. Remove protective caps and plugs from electrical connectors.
- j. Inspect and treat electrical connectors (PARA 1-7-20).
- k. Connect electrical connector, P304 (No. 1 shut-off valve) or P305 (No. 2 shutoff valve), to engine prime fuel shutoff valve.
- l. Test fuel feed system (PARA 10-4-39).
- m. Prime fuel system (PARA 10-4-1). Check for leaks.
- n. Make sure area is clean and free of foreign material.
- o. Install left or right panel enclosure on top of main fuel cells. Lock fasteners.
- p. Close main fuel cell compartment as follows:
 - (1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panels (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

10-4-16 SELF-SEALING BREAKAWAY VENT VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-16.1 Replace Self-Sealing Breakaway Vent Valve	10-4-16-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 10-4-17

10-4-16.1 Replace Self-Sealing Breakaway Vent Valve.



Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

NOTE

Replacement of No. 1 and No. 2 self-sealing breakaway vent valves is the same.

10-4-16.1.1 Remove.

- Remove main fuel system vent valve (PARA 10-4-17).



Valve contains frangible sections that damage easily. To prevent damage to self-sealing breakaway vent valve, use care when removing or installing valve or disconnecting/connecting vent tubes and fuel lines.

- Remove bolts, washers, and self-sealing break-away vent valve from main fuel cell plate. Remove packing from groove on vent valve. Discard packing (Figure 10-4-16-1, Detail A).

10-4-16.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing in groove on self-sealing breakaway vent valve mounting flange (Figure 10-4-16-1, Detail A).

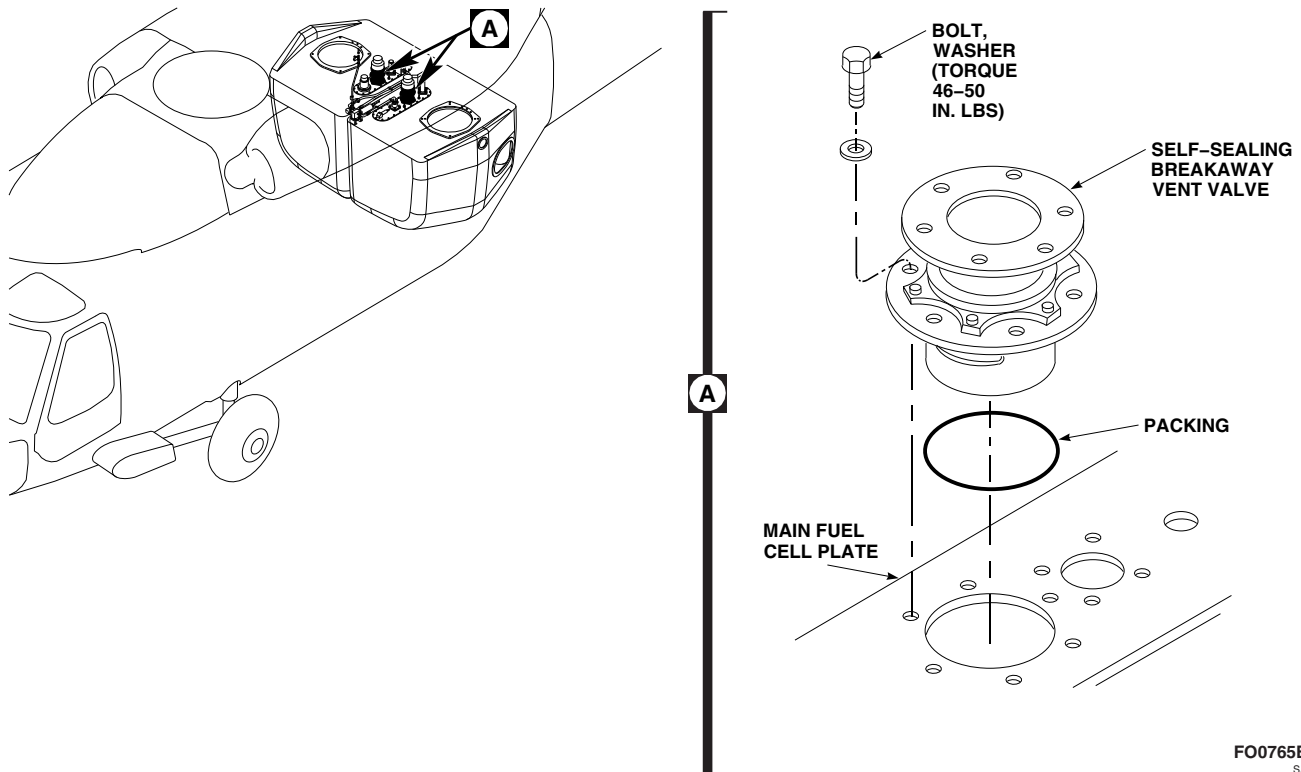
FO0765B
SA

FIGURE 10-4-16-1. REPLACE SELF-SEALING BREAKAWAY VENT VALVE

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into the fuel system.

- c. Coat threads of bolts with antiseize compound, Item 74, Appendix D.

CAUTION

- Valve contains frangible sections that damage easily. To prevent damage to

self-sealing breakaway vent valve, use care when removing or installing valve or disconnecting/connecting vent tubes and fuel lines.

- Do not preload self-sealing breakaway vent valve during installation. Frangible section of self-sealing breakaway valve is sensitive to bending and tension overload.
- d. Install self-sealing breakaway vent valve to main fuel cell plate and secure with bolts and washers. **TORQUE BOLTS TO 46 - 50 INCH-POUNDS.**
- e. Install main fuel system vent valve (PARA 10-4-17).

10-4-17 MAIN FUEL SYSTEM VENT VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-17.1 Replace Main Fuel System Vent Valve	10-4-17-1

INITIAL SETUP	• PARA 2-4-45
Tools	• PARA 2-4-69
• Aircraft Mechanic’s Toolkit • Torque Wrench, 30 - 150 in. lbs	• PARA 10-2-5
Materials/Parts	• PARA 10-2-6
• Antiseize Compound, Item 74, Appendix D • Lubricating Oil, Item 205, Appendix D	• PARA 10-2-7
References	• PARA 13-5-3
• Appendix D • PARA 1-6-16	• PARA 13-5-6

10-4-17.1 Replace Main Fuel System Vent Valve.



Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

NOTE

Replacement of No. 1 and No. 2 main fuel system vent valves is the same.

10-4-17.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Gain access to main fuel cell compartment as follows:
 - (1) Remove rear passenger’s seats as required (PARA 2-4-45).
 - (2) Remove soundproofing panel from rear bulkhead covering left or right stowage compartment (PARA 2-4-69).
 - (3) Remove cargo netting from stowage compartment.
 - (4) If air conditioning is installed, remove environmental control system (ECS)

condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

- c. Unlock fasteners securing left or right panel enclosure on top of main fuel cells. Remove panel.
- d. Remove screws from front panel of fuel system components enclosure. Remove panel.

CAUTION

Valve contains frangible sections that damage easily. To prevent damage to self-sealing breakaway vent valve, use care when removing or installing valve or disconnecting/connecting vent tubes and fuel lines.

- e. Remove screw, washers, bonding jumper, and nut from clamp located on vent tube (Figure 10-4-17-1, Detail A).
- f. Disconnect vent tube from system vent valve. Discard packing.
- g. Remove bolts, washers, nuts, seal, and system vent valve from self-sealing breakaway vent valve. Discard seal.

10-4-17.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install seal between system vent valve and self-sealing breakaway vent valve.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- c. Coat threads of bolts with antiseize compound, Item 74, Appendix D.

CAUTION

- Valve contains frangible sections that damage easily. To prevent damage to self-sealing breakaway vent valve, use care when removing or installing valve or disconnecting/connecting vent tubes and fuel lines.
- Do not preload self-sealing breakaway vent valve when installing system vent valve. Frangible section of self-sealing breakaway valve is sensitive to bending and tension overload.
- d. Install system vent valve onto self-sealing breakaway vent valve and secure with bolts, washers, and nuts (Figure 10-4-17-1, Detail A). **TORQUE NUTS TO 79 - 87 INCH-POUNDS.**
- e. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing in groove on vent tube end.

CAUTION

Make sure self-sealing breakaway vent valve is not preloaded when installing vent tube to system vent valve. Frangible section of self-sealing breakaway valve is sensitive to bending and tension overload.

- f. Connect vent tube to system vent valve. **TIGHTEN FITTING HANDTIGHT.**
- g. Secure bonding jumper to clamp on vent tube with screw, washers, and nut.
- h. Make sure area is clean and free of foreign material.
- i. Install front panel and secure with screws.
- j. Install left or right panel enclosure on top of main fuel cells. Lock fasteners.
- k. Perform operational check of fuel quantity system (PARA 10-2-5 or PARA 10-2-6).

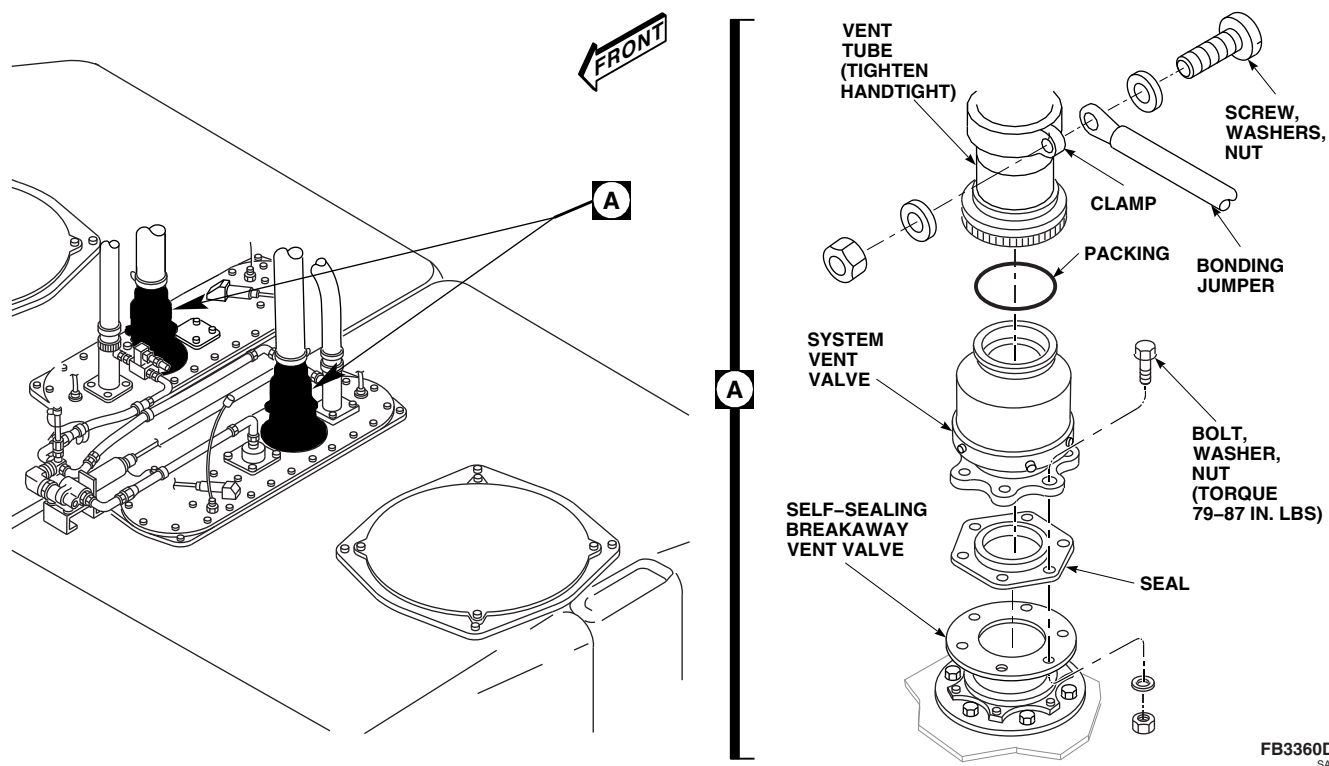


FIGURE 10-4-17-1. REPLACE MAIN FUEL SYSTEM VENT VALVE

1. Perform operational check of engine feed fuel lines (PARA 10-2-7).
- m. Close main fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panel (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

10-4-18 MAIN FUEL BREAKAWAY VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-18.1 Replace Main Fuel Breakaway Valve	10-4-18-1
INITIAL SETUP	• PARA 1-3-18
Tools	• PARA 1-3-19
• Aircraft Mechanic’s Toolkit • Container, 1-Gallon • Torque Wrench, 30 - 150 in. lbs	• PARA 1-6-16 • PARA 2-4-45 • PARA 2-4-69
Materials/Parts	• PARA 10-2-5
• Antiseize Compound, Item 74, Appendix D • Lubricating Oil, Item 205, Appendix D • Machinery Towel, Item 211, Appendix D • Protective Caps and Plugs	• PARA 10-2-6 • PARA 10-2-7 • PARA 10-4-15
References	• PARA 10-4-39
• Appendix D • PARA 1-3-17	• PARA 13-5-3 • PARA 13-5-6

10-4-18.1 Replace Main Fuel Breakaway Valve.



Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

NOTE

Replacement of No. 1 and No. 2 main fuel breakaway valves is the same.

10-4-18.1.1 Remove.

- Purge fuel feed system (PARA 10-4-39).
- Gain access to main fuel cell compartment as follows:

(1) Remove rear passenger's seats as required (PARA 2-4-45).

(2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).

(3) Remove cargo netting from stowage compartments.

(4) Remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

c. Unlock fasteners securing left or right panel enclosure on top of main fuel cells. Remove panel.

d. Remove bolts, washers, and fuel cell access cover from main fuel cell. Remove packing from fuel cell access cover (Figure 10-4-18-1, Detail A). Discard packing.

e. Reach inside main fuel cell and disconnect main fuel hose secured to bottom of main fuel breakaway valve (Figure 10-4-18-1, Detail B). Remove and discard packing. Install protective plug in hose.

f. Remove main engine prime fuel shutoff valve from main fuel breakaway valve (PARA 10-4-15). Install protective cap on shutoff valve.

g. Position 1-gallon container to catch fuel and disconnect lower main fuel line from top of main fuel breakaway valve. Discard packing. Install protective plug in hose.

h. Remove bolts, washers, seal, and main fuel breakaway valve from main fuel cell plate. Discard seal.

10-4-18.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Lubricate packings with lubricating oil, Item 205, Appendix D. Install packings on main fuel breakaway valve (Figure 10-4-18-1, Detail B).

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

c. Coat threads of bolts with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads on valve. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.

d. Install seal between main fuel breakaway valve and main fuel cell plate.

CAUTION

Do not preload main fuel breakaway valve during installation. Frangible section of main fuel breakaway valve is sensitive to bending and tension overload.

e. Install main fuel breakaway valve with bolts and washers. **TORQUE BOLTS TO 89 - 97 INCH-POUNDS.**

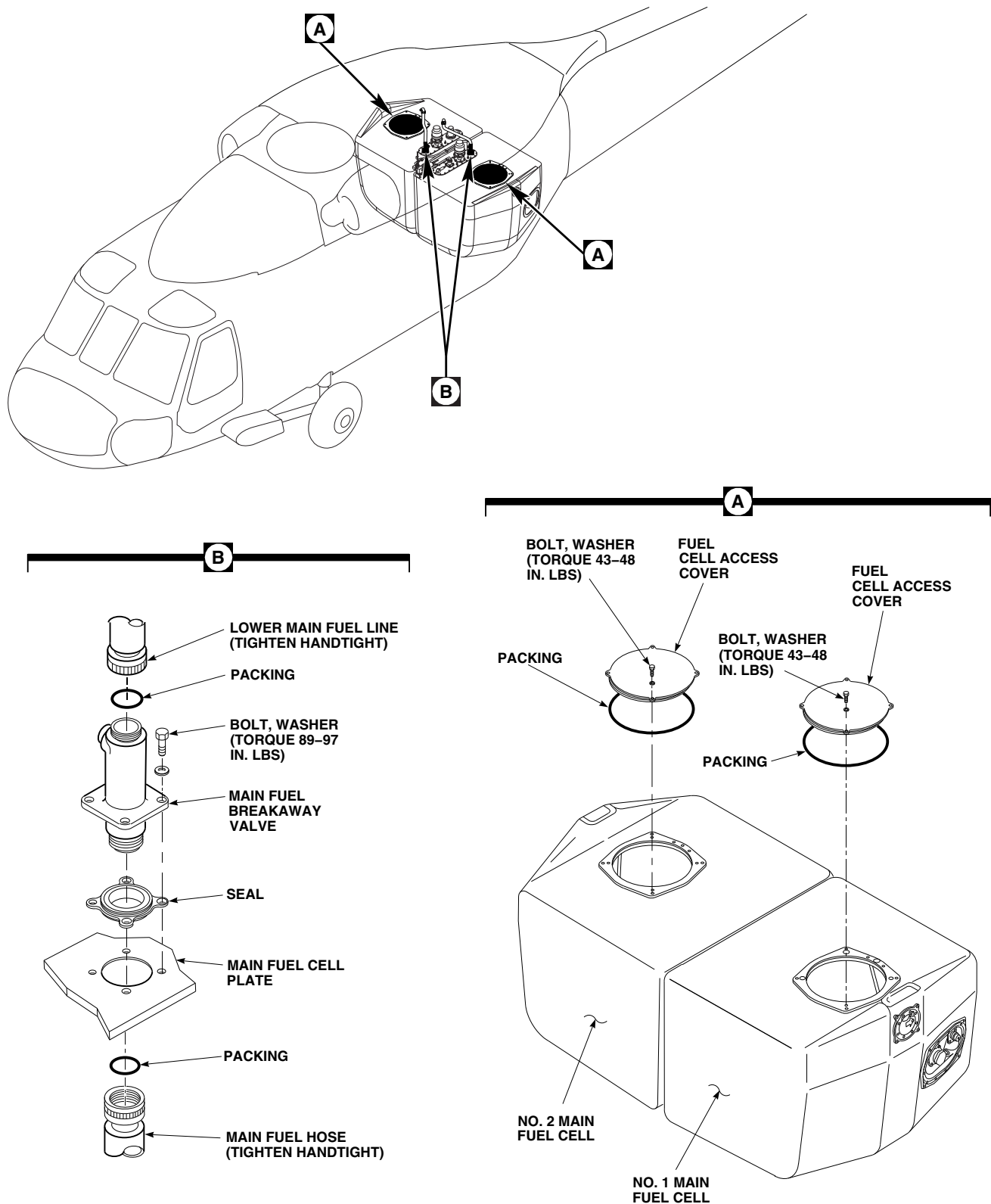
CAUTION

Do not preload main fuel breakaway valve when connecting fuel line. Frangible section of main fuel breakaway valve is sensitive to bending and tension overload.

f. Remove protective plug from lower main fuel line. Connect fuel line to top of main fuel breakaway valve. **TIGHTEN LOWER MAIN FUEL LINE HANDTIGHT.**

g. Remove protective cap from engine prime fuel shutoff valve. Install shutoff valve on main fuel breakaway valve (PARA 10-4-15).

h. Remove protective plug from main fuel hose. Connect fuel hose to bottom of main fuel



FN2104C
SA

FIGURE 10-4-18-1. REPLACE MAIN FUEL BREAKAWAY VALVE

breakaway valve. **TIGHTEN MAIN FUEL HOSE HANDTIGHT.**

- i. Test fuel feed system (PARA 10-4-39).
- j. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on fuel cell access cover (Figure 10-4-18-1, Detail A).
- k. Inspect area inside main fuel cell and make sure area is free of all foreign material.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- l. Coat threads of fuel cell access cover attachment bolts with antiseize compound, Item 74, Appendix D.
- m. Install fuel cell access cover on main fuel cell with bolts and washers. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS.**

- n. Make sure area is clean and free of foreign material.
- o. Install left or right panel enclosure on top of main fuel cells. Lock fasteners.
- p. Refuel main fuel cells (PARA 1-3-17, PARA 1-3-18, or PARA 1-3-19).
- q. Perform operational check of fuel quantity system (PARA 10-2-5 or PARA 10-2-6).
- r. Perform operational check of engine feed fuel lines (PARA 10-2-7).
- s. Close main fuel cell compartment as follows:
 - (1) Install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panels (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

10-4-19 EXTENDED RANGE BREAKAWAY VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-19.1 Replace Extended Range Breakaway Valve	10-4-19-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Assembly Fluid, Item 79, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-2-5
- PARA 10-2-6
- PARA 10-2-7
- PARA 10-4-40
- PARA 13-5-3
- PARA 13-5-6

10-4-19.1 Replace Extended Range Breakaway Valve.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

NOTE

Extended range breakaway valve is installed on No. 2 main fuel cell only.

10-4-19.1.1 Remove.

- Purge main fuel cell (PARA 10-4-40).
- Gain access to main fuel cell compartment as follows:
 - Remove right side rear passenger’s seats (PARA 2-4-45).
 - Remove soundproofing panel from rear bulkhead covering right stowage compartment (PARA 2-4-69).
 - Remove cargo netting from stowage compartment.
 - If air conditioning is installed, remove environmental control system (ECS)

condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

- c. Unlock fasteners securing left or right panel enclosure on top of main fuel cells. Remove panel.
- d. Remove bolts, washers, and fuel cell access cover from No. 2 main fuel cell. Remove packing from fuel cell access cover (Figure 10-4-19-1, Sheet 1, Detail A). Discard packing.
- e. Reach inside main fuel cell and disconnect extended range interconnect hose from bottom of extended range breakaway valve (Figure 10-4-19-1, Sheet 1, Detail B). Discard packing. Install protective plug in extended range interconnect hose.
- f. **ESSS** Disconnect flowmeter fitting, on main fuel hose, from extended range breakaway valve. Install protective plug in main fuel hose.■
- g. **W/O ESSS** Remove cap from extended range breakaway valve and store on dummy receptacle. Remove and discard packing.■
- h. Remove bolts, washers, extended range breakaway valve, and seal from main fuel cell plate. Discard seal.
- i. Install protective plug in main fuel cell plate.
- j. **W/O ESSS** Remove cap from dummy receptacle. Inspect cap, cable, and dummy receptacle for damage. **NONE ALLOWED.** If no damage is found, install cap on dummy receptacle. If damage is found, remove damaged part(s) as follows (Figure 10-4-19-1, Sheet 2, Detail C):■
 - (1) To remove dummy receptacle, remove screws and washers securing dummy receptacle to cover panels. Spread hook securing cable to receptacle and remove receptacle.
 - (2) To remove cable, spread hooks securing cable to dummy receptacle and cap, and remove cable.

- (3) To remove cap, spread hook securing cable to cap, and remove cap.

10-4-19.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. **W/O ESSS** If required, install cap, cable, and/or dummy receptacle as follows (Figure 10-4-19-1, Sheet 2, Detail C):■
 - (1) To install dummy receptacle, position dummy receptacle on cover panels and secure with screws and washers. Connect cable to receptacle and bend hook closed.
 - (2) To install cable, connect cable to receptacle and cap, and bend hooks closed.
 - (3) To install cap, connect cable to cap, and bend hook closed.
- c. Remove protective plug from main fuel cell plate.
- d. Lubricate packings with lubricating oil, Item 205, Appendix D. Install packing on lower half of extended range breakaway valve (Figure 10-4-19-1, Sheet 1, Detail B).
- e. Lubricate seal with assembly fluid, Item 79, Appendix D. Install seal and extended range breakaway valve on main fuel cell plate.

CAUTION

- To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.
- Do not preload extended range breakaway valve during installation. Frangible section of extended range breakaway valve is sensitive to bending and tension overload.
- f. Coat threads of bolts with antiseize compound, Item 74, Appendix D. Install bolts and washers

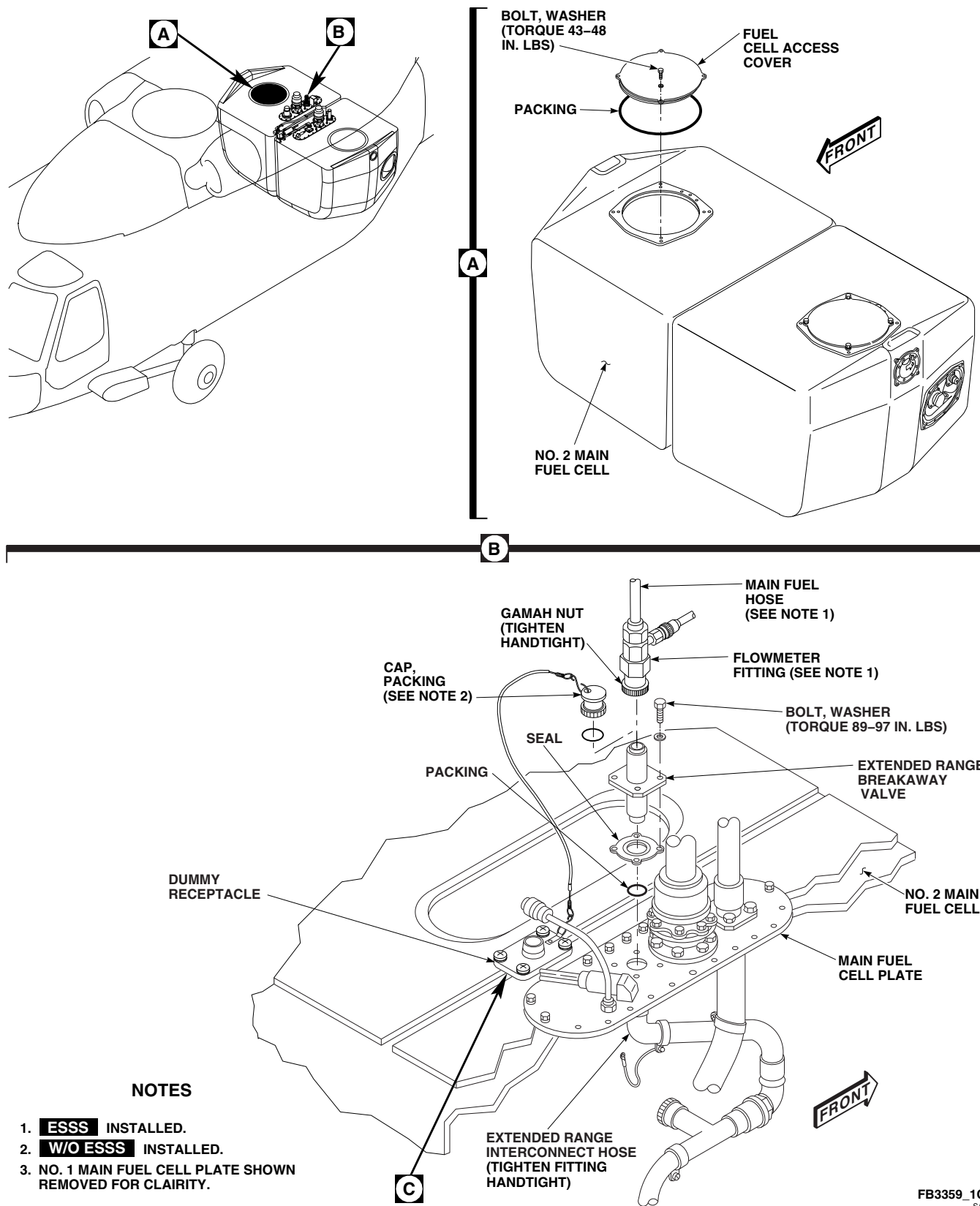
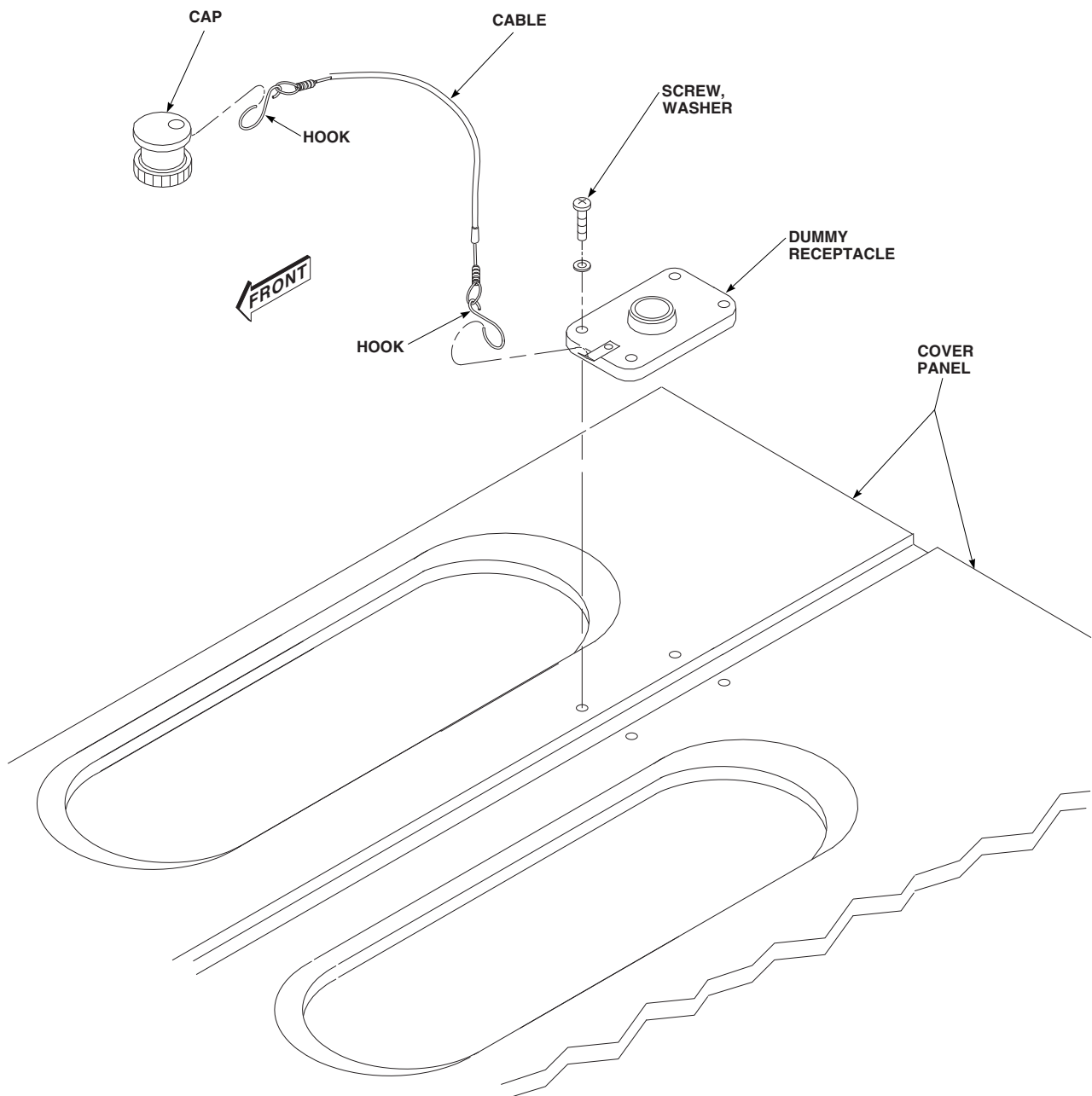


FIGURE 10-4-19-1. REPLACE EXTENDED RANGE BREAKAWAY VALVE (SHEET 1 OF 2)

C

FB3359_2A
SA**FIGURE 10-4-19-1. REPLACE EXTENDED RANGE BREAKAWAY VALVE (SHEET 2 OF 2)**

securing extended range breakaway valve to main fuel cell plate. **TORQUE BOLTS TO 89 - 97 INCH-POUNDS.**

- g. Remove protective plug from extended range interconnect hose and connect hose to bottom of extended range breakaway valve. **TIGHTEN FITTING HANDTIGHT.**



ESSS Do not preload extended range breakaway valve when installing flow-meter fitting. Frangible section of extended range breakaway valve is sensitive to bending and tension overload.■

- h. **ESSS** Remove protective plug from flow-meter fitting and connect flowmeter fitting on main fuel hose to extended range breakaway valve. **TIGHTEN GAMAH NUT ON FLOW-METER FITTING HANDTIGHT.**■



W/O ESSS Do not preload extended range breakaway valve when installing cap. Frangible section of extended range breakaway valve is sensitive to bending and tension overload.■

- i. **W/O ESSS** Install packing on top of extended range breakaway valve and install cap.■
- j. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on fuel cell access cover (Figure 10-4-19-1, Sheet 1, Detail A).

- k. Inspect area inside main fuel cell and make sure area is free of all foreign material.



To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- l. Coat threads of fuel cell access cover attachment bolts with antiseize compound, Item 74, Appendix D.
- m. Install fuel cell access cover on main fuel cell with bolts and washers. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS.**
- n. Install left or right panel enclosure on top of main fuel cells. Lock fasteners.
- o. Perform operational check of fuel quantity system (PARA 10-2-5 or PARA 10-2-6).
- p. Perform operational check of engine feed fuel lines (PARA 10-2-7).
- q. Close main fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panel (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

10-4-20 PRIME/BOOST PUMP.

PARAGRAPH BREAKDOWN

	PAGE
10-4-20.1 Replace Prime/Boost Pump	10-4-20-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-4-1
- PARA 13-5-3
- PARA 13-5-6
- PARA 15-4-9

10-4-20.1 Replace Prime/Boost Pump.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

10-4-20.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Gain access to main fuel cell compartment as follows:

- Remove rear passenger’s seats as required (PARA 2-4-45).
- Remove soundproofing panel from rear bulkhead covering left or right stowage compartment (PARA 2-4-69).
- Remove cargo netting from stowage compartment.
- If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

- Unlock fasteners securing left or right panel enclosure on top of main fuel cells. Remove panel.

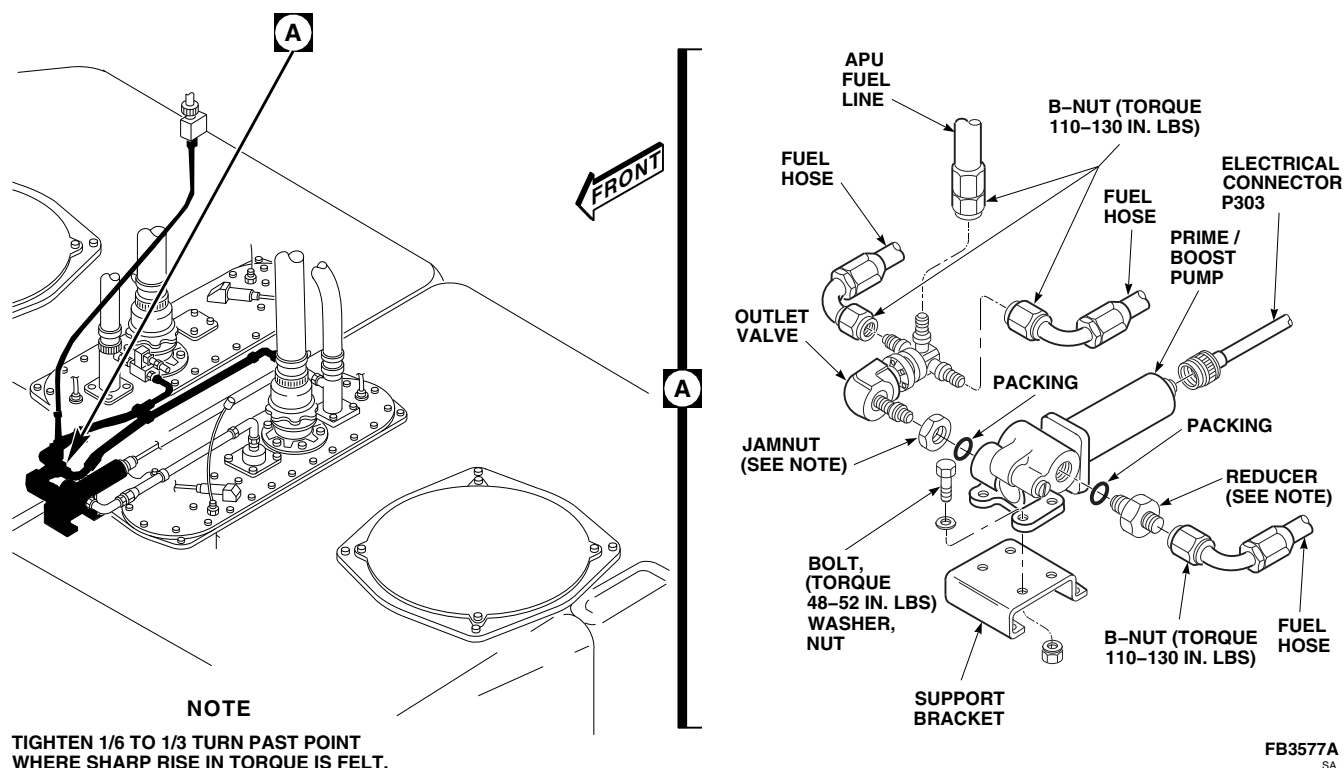


FIGURE 10-4-20-1. REPLACE PRIME/BOOST PUMP

- d. Remove screws from front panel of fuel system components enclosure. Remove panel.
- e. Disconnect electrical connector, P303, from prime/boost pump (Figure 10-4-20-1, Detail A).
- f. Install protective caps and plugs on electrical connectors.
- g. Position 1-quart container to catch fuel and disconnect fuel hose from reducer in prime/boost pump. Plug hose.
- h. Position 1-quart container to catch fuel and disconnect auxiliary power unit (APU) fuel line and fuel hoses from outlet valve. Plug APU fuel line and fuel hoses.
- i. Remove bolts, washers, nuts, and prime/boost pump from support bracket.
- j. Loosen jamnut securing outlet valve to prime/boost pump. Remove valve from pump. Discard packing.

- k. Remove reducer from prime/boost pump. Discard packing.

10-4-20.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Lubricate packing with lubricating oil, Item 205, Appendix D.
- c. Install packing on reducer (Figure 10-4-20-1, Detail A).

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- d. Coat threads on reducer with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeezeout

after parts are assembled with machinery towel, Item 211, Appendix D.

- e. Install reducer in prime/boost pump. **TIGHTEN REDUCER $\frac{1}{8}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- f. Install packing on outlet valve port.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- g. Coat threads on all outlet valve ports with antiseize compound, Item 74, Appendix D, sparingly ($\frac{1}{4}$ -inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.
- h. Install outlet valve in prime/boost pump. **TIGHTEN JAMNUT $\frac{1}{8}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- i. Install prime/boost pump on support bracket with bolts, washers, and nuts. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**
- j. Unplug and connect auxiliary power unit (APU) fuel line and fuel hoses to outlet valve. **TORQUE B-NUTS TO 110 - 130 INCH-POUNDS.**

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- k. Coat threads on reducer in prime/boost pump with antiseize compound, Item 74, Appendix D, sparingly ($\frac{1}{4}$ -inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.
- l. Unplug and connect fuel hose to reducer. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**
- m. Remove protective caps and plugs from electrical connectors.
- n. Inspect and treat electrical connectors (PARA 1-7-20).
- o. Connect electrical connector, P303, to prime/boost pump.
- p. Make sure area is clean and free of foreign material.
- q. Install front panel and secure with screws.
- r. Install left or right panel enclosure on top of main fuel cells. Lock fasteners.
- s. Prime fuel system (PARA 10-4-1) and APU fuel system (PARA 15-4-9). Check for leaks.
- t. Close main fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panel (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

10-4-21 PRIME/BOOST PUMP FUEL OUTLET VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-21.1 Replace Prime/Boost Pump Fuel Outlet Valve	10-4-21-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-4-1
- PARA 13-5-3
- PARA 13-5-6
- PARA 15-4-9

10-4-21.1 Replace Prime/Boost Pump Fuel Outlet Valve.

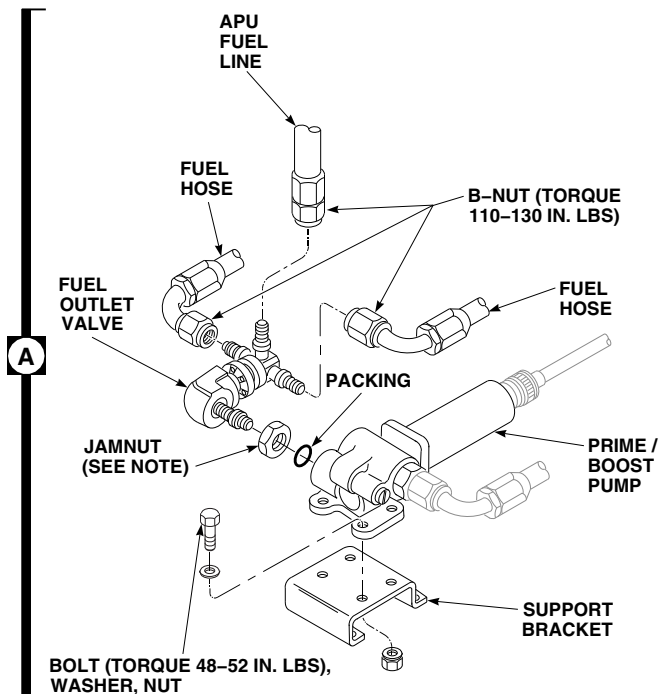
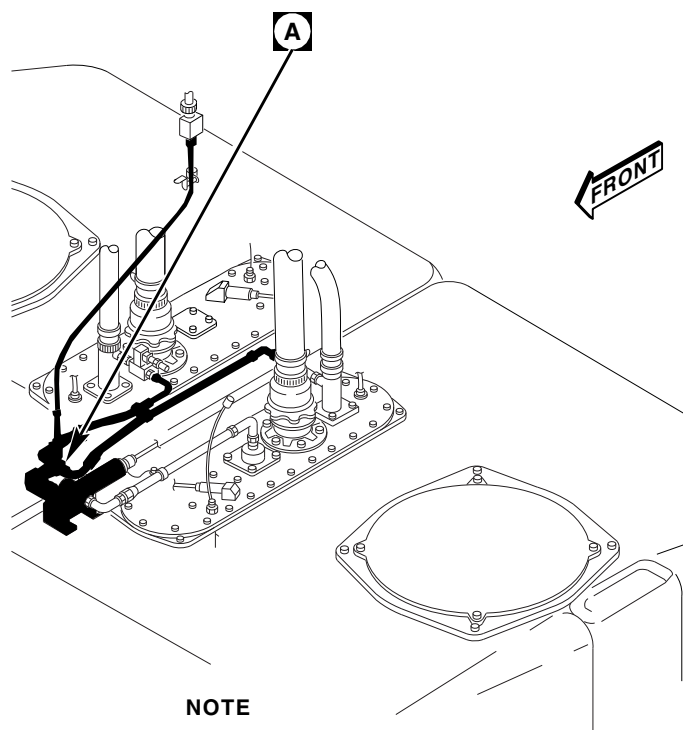
WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

10-4-21.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).

- Gain access to main fuel cell compartment as follows:
 - Remove rear passenger’s seats as required (PARA 2-4-45).
 - Remove soundproofing panel from rear bulkhead covering left or right stowage compartment (PARA 2-4-69).
 - Remove cargo netting from stowage compartment.
 - If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

FB3576A
SA**FIGURE 10-4-21-1. REPLACE PRIME/BOOST PUMP FUEL OUTLET VALVE**

- c. Unlock fasteners securing left or right panel enclosure on top of main fuel cells. Remove panel.
- d. Remove screws from front panel of fuel system components enclosure. Remove panel.
- e. Position 1-quart container to catch fuel and disconnect auxiliary power unit (APU) fuel line and fuel hoses from fuel outlet valve (Figure 10-4-21-1, Detail A). Install protective plugs in APU fuel line and fuel hoses.
- f. Remove bolts, washers, nuts, and prime/boost pump from support bracket.
- g. Loosen jamnut securing fuel outlet valve to prime/boost pump.
- h. Lift up prime/boost pump and remove fuel outlet valve from pump. Discard packing.
- i. Remove jamnut from fuel outlet valve.

10-4-21.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

- b. Install jamnut on fuel outlet valve.
- c. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on fuel outlet valve (Figure 10-4-21-1, Detail A).

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- d. Coat threads on fuel outlet valve ports with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.
- e. Lift up prime/boost pump and install fuel outlet valve in pump. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

- f. Install prime/boost pump on support bracket with bolts, washers, and nuts. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**
- g. Remove protective plugs and connect auxiliary power unit (APU) fuel line and fuel hoses to fuel outlet valve. **TORQUE B-NUTS TO 110 - 130 INCH-POUNDS.**
- h. Prime fuel system (PARA 10-4-1) and APU fuel system (PARA 15-4-9). Check for leaks.
- i. Make sure area is clean and free of foreign material.
- j. Install front panel with screws.
- k. Install left or right panel enclosure on top of main fuel cells. Lock fasteners.
- l. Close main fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panel (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

10-4-22 PRIME/BOOST PUMP FUEL BREAKAWAY VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-22.1 Replace Prime/Boost Pump Fuel Breakaway Valve	10-4-22-1

INITIAL SETUP	• PARA 1-3-18
Tools	• PARA 1-3-19
• Aircraft Mechanic’s Toolkit • Torque Wrench, 30 - 150 in. lbs	• PARA 1-6-16
Materials/Parts	• PARA 2-4-45
• Antiseize Compound, Item 74, Appendix D • Lubricating Oil, Item 205, Appendix D • Machinery Towel, Item 211, Appendix D • Protective Caps and Plugs	• PARA 2-4-69 • PARA 10-4-1 • PARA 10-4-40
References	• PARA 13-5-3
• Appendix D • PARA 1-3-17	• PARA 13-5-6

10-4-22.1 Replace Prime/Boost Pump Fuel Breakaway Valve.

NOTE

Prime/boost pump fuel breakaway valve is installed on No. 1 main fuel cell only.



Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

10-4-22.1.1 Remove.

- a. Purge main fuel cell (PARA 10-4-40).
- b. Gain access to main fuel cell compartment as follows:
 - (1) Remove rear passenger’s seats as required (PARA 2-4-45).

- (2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).
- (3) Remove cargo netting from stowage compartments.
- (4) If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

- c. Unlock fasteners securing left panel enclosure on top of main fuel cells. Remove panel from compartment.
- d. Remove bolts, washers, and fuel cell access cover from main fuel cell. Remove packing from fuel cell access cover (Figure 10-4-22-1, Detail A). Discard packing.
- e. Reach inside main fuel cell and disconnect fuel hose secured to bottom of prime/boost pump fuel breakaway valve (Figure 10-4-22-1, Detail B). Install protective plug in fuel hose.
- f. Disconnect upper prime/boost pump fuel line from top of prime/boost pump fuel breakaway valve. Install protective plug in fuel line.
- g. Remove bolts, washers, seal, and prime/boost pump fuel breakaway valve from main fuel cell plate. Discard seal.

10-4-22.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- b. Coat threads of bolts with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip)

along length of male threads on valve. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.

- c. Install seal between prime/boost pump fuel breakaway valve and main fuel cell plate (Figure 10-4-22-1, Detail B).

CAUTION

Do not preload prime/boost pump fuel breakaway valve during installation. Frangible section of prime/boost fuel breakaway valve is sensitive to bending and tension overload.

- d. Install prime/boost pump fuel breakaway valve, short end inside main fuel cell, with bolts and washers. **TORQUE BOLTS TO 89 - 97 INCH-POUNDS.**

CAUTION

Do not preload prime/boost pump fuel breakaway valve when connecting fuel line. Frangible section of prime/boost fuel breakaway valve is sensitive to bending and tension overload.

- e. Remove plug and connect upper prime/boost pump fuel line to top of prime/boost pump fuel breakaway valve. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**
- f. Reach inside main fuel cell, remove plug and connect fuel hose to bottom of prime/boost pump fuel breakaway valve. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**
- g. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on fuel cell access cover (Figure 10-4-22-1, Detail A).
- h. Inspect area inside main fuel cell and make sure area is free of all foreign material.

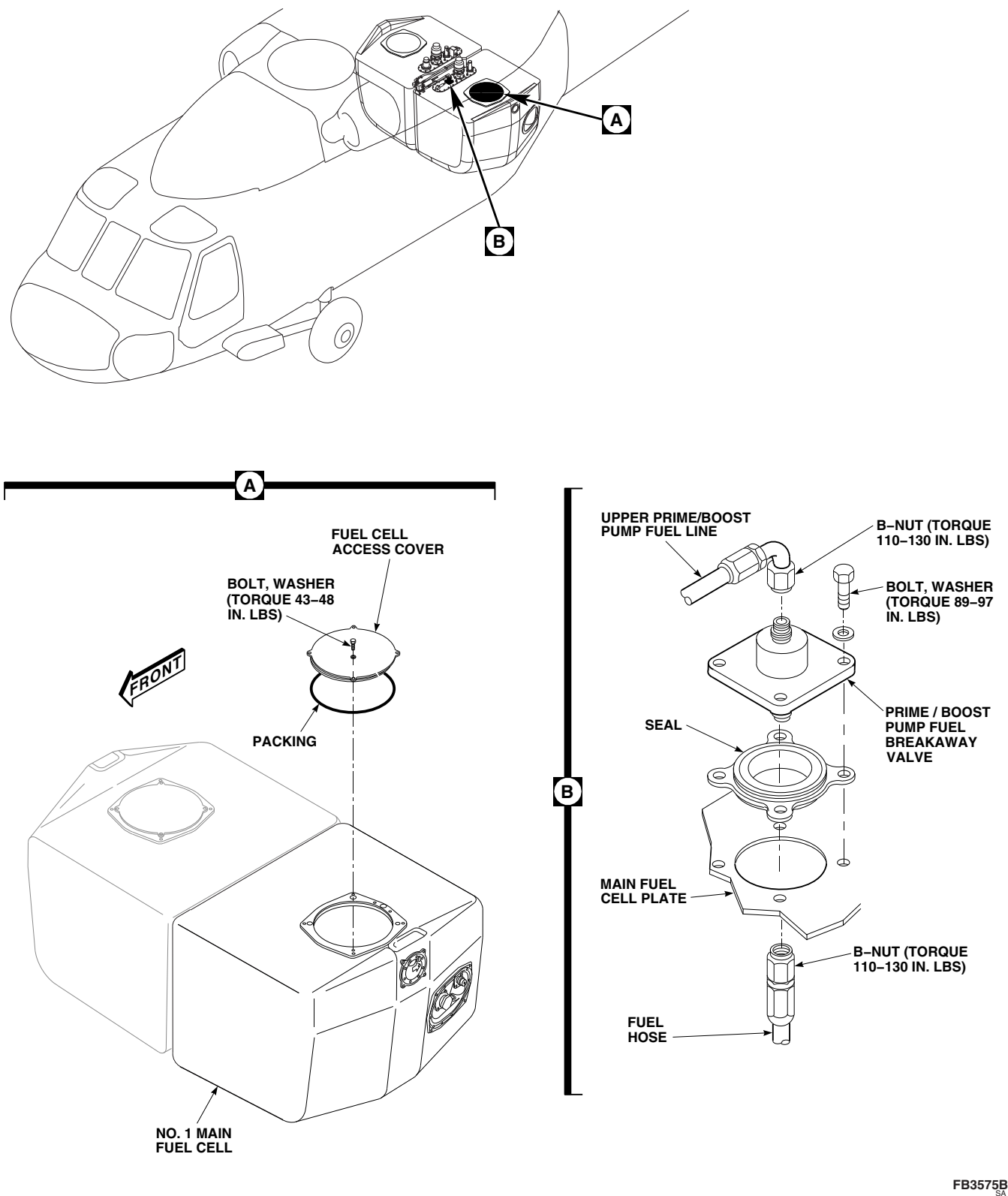


FIGURE 10-4-22-1. REPLACE PRIME/BOOST PUMP FUEL BREAKAWAY VALVE

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- i. Coat threads of fuel cell access cover attachment bolts with antiseize compound, Item 74, Appendix D.
- j. Install fuel cell access cover on main fuel cell with bolts and washers. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS.**
- k. Refuel main fuel cells (PARA 1-3-17, PARA 1-3-18, or PARA 1-3-19).
- l. Prime fuel system (PARA 10-4-1). Check for leaks.

- m. Make sure area is clean and free of foreign material.
- n. Install left panel enclosure on top of main fuel cell. Lock fasteners.
- o. Close main fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panels (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

10-4-23 MAIN FUEL SYSTEM CHECK VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-23.1 Replace Main Fuel System Check Valve	10-4-23-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Gallon
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lubricating Oil, Item 205, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-113
- PARA 10-4-1
- PARA 10-4-39

10-4-23.1 Replace Main Fuel System Check Valve.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

NOTE

Replacement of left and right side main fuel system check valves is the same.

10-4-23.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove left or right oil cooler compartment access door (PARA 2-4-113).

- Purge fuel feed system (PARA 10-4-39).
- Position 1-gallon container to catch fuel and disconnect fuel hose from check valve (Figure 10-4-23-1, Detail A). Plug hose.
- Remove check valve and packing from break-away tee valve. Discard packing. Plug tee valve.

10-4-23.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Lubricate packing with lubricating oil, Item 205, Appendix D.
- Install packing on end of check valve that flow arrow points away from (Figure 10-4-23-1, Detail A).
- Remove plug from breakaway tee valve.

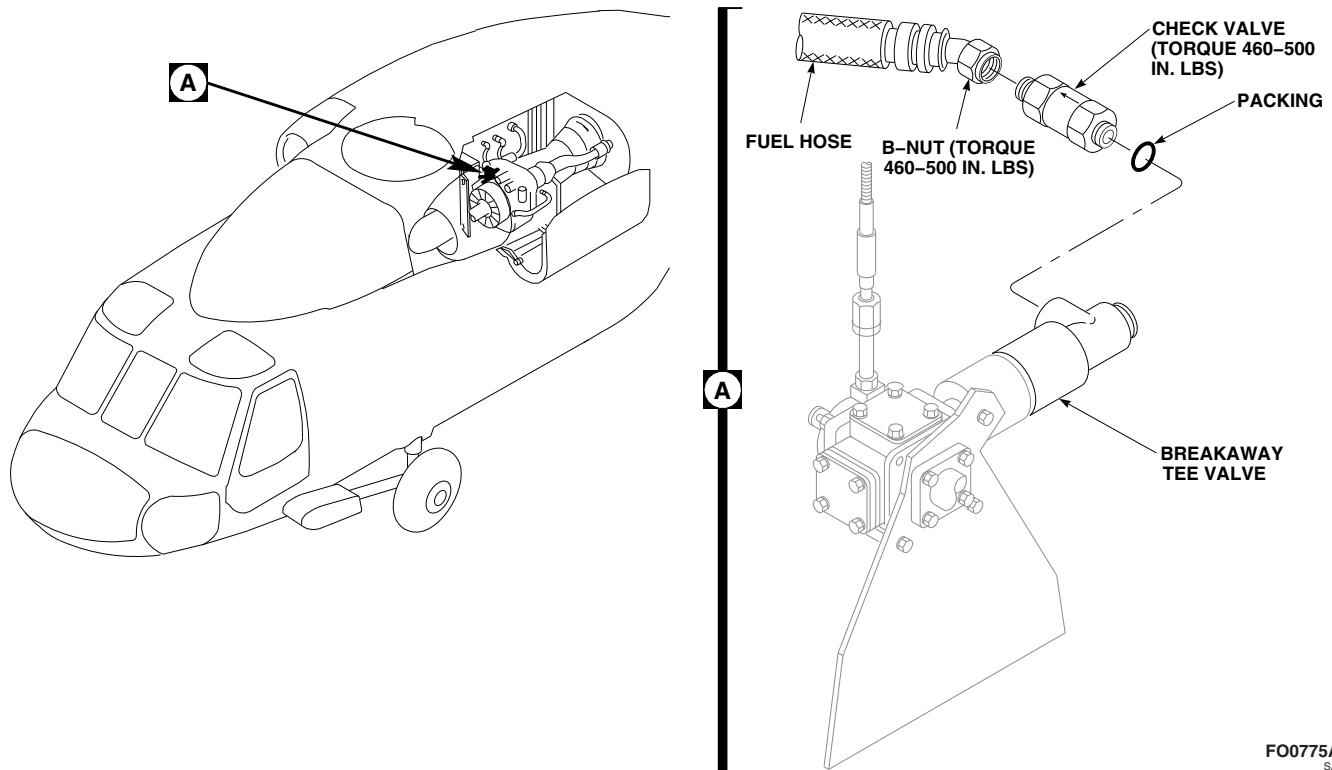
FO0775A
SA

FIGURE 10-4-23-1. REPLACE MAIN FUEL SYSTEM CHECK VALVE

CAUTION

Do not preload breakaway tee valve when installing check valve. Frangible section of breakaway tee valve is sensitive to bending and tension overload.

- e. Install check valve into breakaway tee valve with flow arrow pointing away from breakaway tee valve. **TORQUE CHECK VALVE TO 460 - 500 INCH-POUNDS.**

CAUTION

Make sure breakaway tee valve is not preloaded when installing fuel hose to

check valve. Frangible section of breakaway tee is sensitive to bending and tension overload.

- f. Remove plug and connect fuel hose to check valve. Hold check valve with wrench at end closest to hose. **TORQUE B-NUT TO 460 - 500 INCH-POUNDS.**
- g. Test fuel feed system (PARA 10-4-39).
- h. Prime fuel system (PARA 10-4-1). Check for leaks.
- i. Make sure area is clean and free of foreign material.
- j. Install oil cooler compartment access door (PARA 2-4-113).

10-4-24 ENGINE COMPARTMENT BREAKAWAY VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-24.1 Replace Engine Compartment Breakaway Valve	10-4-24-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Gallon
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lubricating Oil, Item 205, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-113
- PARA 4-4-1
- PARA 10-4-1
- PARA 10-4-39

10-4-24.1 Replace Engine Compartment Breakaway Valve.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

NOTE

Replacement of left and right side engine compartment breakaway valves is the same, except as noted.

10-4-24.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).

- Open engine cowl.
- On right side only, disconnect coupling and remove inlet particle separator (IPS) duct from IPS blower (PARA 4-4-1).
- Remove left or right oil cooler compartment access door (PARA 2-4-113).
- Purge fuel feed system (PARA 10-4-39).
- Position 1-gallon container to catch fuel and disconnect fuel hose from engine compartment breakaway valve (Figure 10-4-24-1, Detail A). Discard packing. Install protective plug in hose.
- Remove bolts, washers, and engine compartment breakaway valve from fuel selector valve on firewall. Discard gasket.
- Cover fuel selector valve opening.

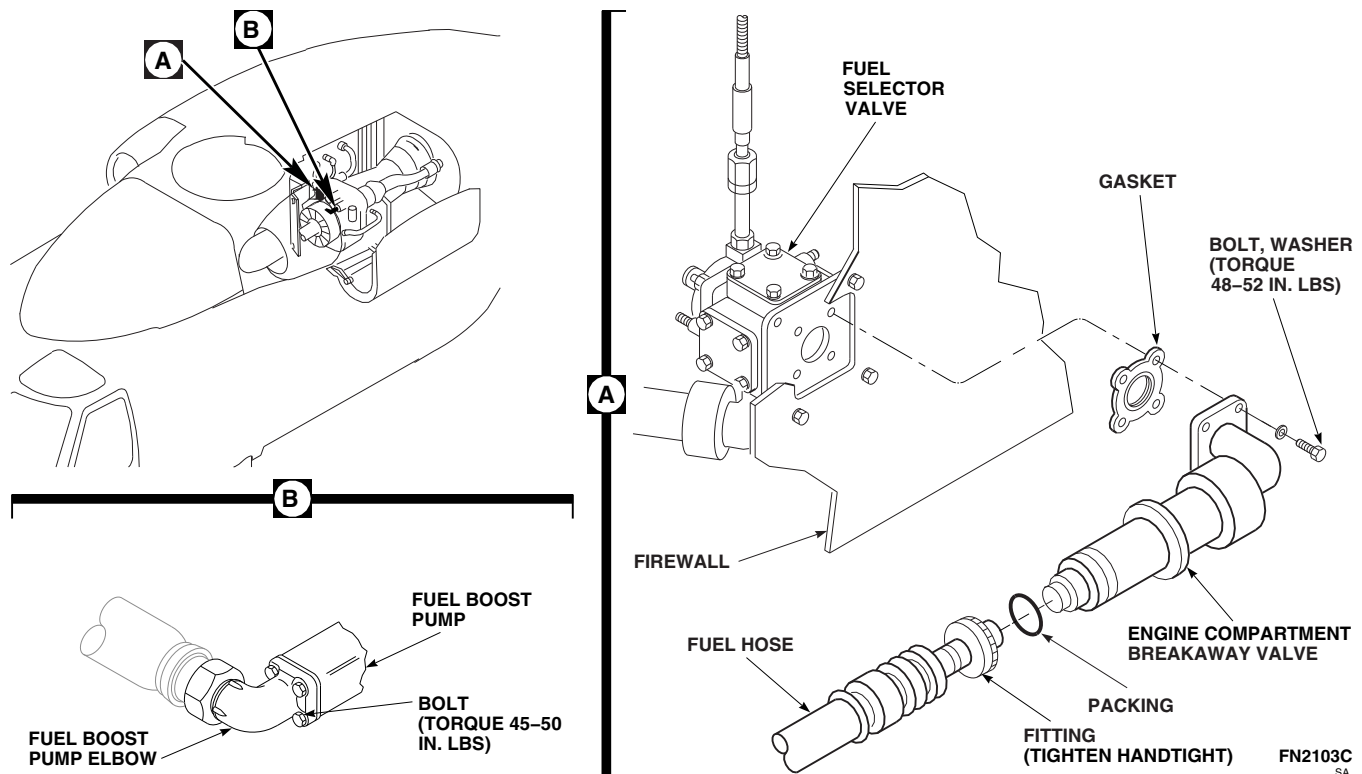


FIGURE 10-4-24-1. REPLACE ENGINE COMPARTMENT BREAKAWAY VALVE

10-4-24.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove cover from fuel selector valve opening.
- c. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on engine compartment breakaway valve end (Figure 10-4-24-1, Detail A).
- e. Loosen bolts securing fuel boost pump elbow to fuel boost pump (Figure 10-4-24-1, Detail B).
- f. Install gasket on engine compartment breakaway valve (Figure 10-4-24-1, Detail A).
- g. Position engine compartment breakaway valve on fuel selector valve on firewall and loosely secure with bolts and washers.

CAUTION

Do not preload engine compartment breakaway valve when installing fuel hose. Frangible section of engine compartment breakaway valve is sensitive to bending and tension overload.

- d. Remove protective plug from fuel hose and connect hose to engine compartment breakaway valve. **TIGHTEN FITTING HANDTIGHT.**

NOTE

Fuel hose for No. 1 engine should be positioned to avoid engine oil filler port and right front angled section of firewall. Additional contact between fuel hose assembly and firewall is acceptable for both engines.

- h. Allow engine compartment breakaway valve and fuel hose assembly to seek neutral position.



Do not preload engine compartment breakaway valve during installation. Frangible section of engine compartment breakaway valve is sensitive to bending and tension overload.

- i. **TORQUE BOLTS SECURING ENGINE COMPARTMENT BREAKAWAY VALVE TO FUEL SELECTOR VALVE TO 48 - 52 INCH-POUNDS.**
- j. **TORQUE BOLTS SECURING FUEL BOOST PUMP ELBOW TO FUEL BOOST PUMP TO 45 - 50 INCH-POUNDS** (Figure 10-4-24-1, Detail B).
- k. Inspect for clearance between No. 2 engine fuel hose and fire extinguisher nozzle. **CLEARANCE TO BE 0.65-INCH MINI-**

MUM. If clearance is less than 0.65-inch minimum, loosen bolts securing engine compartment breakaway valve to fuel selector valve and fuel boost elbow to fuel boost pump. Repeat step h. through step j.

- l. Test fuel feed system (PARA 10-4-39). Check for leaks.
- m. Prime fuel system (PARA 10-4-1). Check for leaks.
- n. Make sure area is clean and free of foreign material.
- o. Install oil cooler compartment access door (PARA 2-4-113).
- p. On right side only, install inlet particle separator (IPS) duct to IPS blower with coupling (PARA 4-4-1).
- q. Close engine cowling.

10-4-25 FUEL SELECTOR VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-25.1 Replace Fuel Selector Valve	10-4-25-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Alcohol, Isopropyl, Item 72, Appendix D
- Cheesecloth, Item 90, Appendix D

- Grease, Item 143, Appendix D
- Lockwire, Item 196, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 10-4-24
- PARA 10-4-26
- PARA 10-4-27

10-4-25.1 Replace Fuel Selector Valve.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

NOTE

Replacement of left and right side fuel selector valves is the same.

10-4-25.1.1 Remove.

- Remove engine compartment breakaway valve (PARA 10-4-24).
- Remove crossfeed breakaway valve (PARA 10-4-26).
- Remove breakaway tee valve (PARA 10-4-27).
- Remove snaping and remove control box and fuel selector valve push/pull cable, as an assembly, cable from fuel selector valve (Figure 10-4-25-1, Detail A).
- Remove bolts, washers, and nuts securing fuel selector valve to firewall. Remove fuel selector valve.

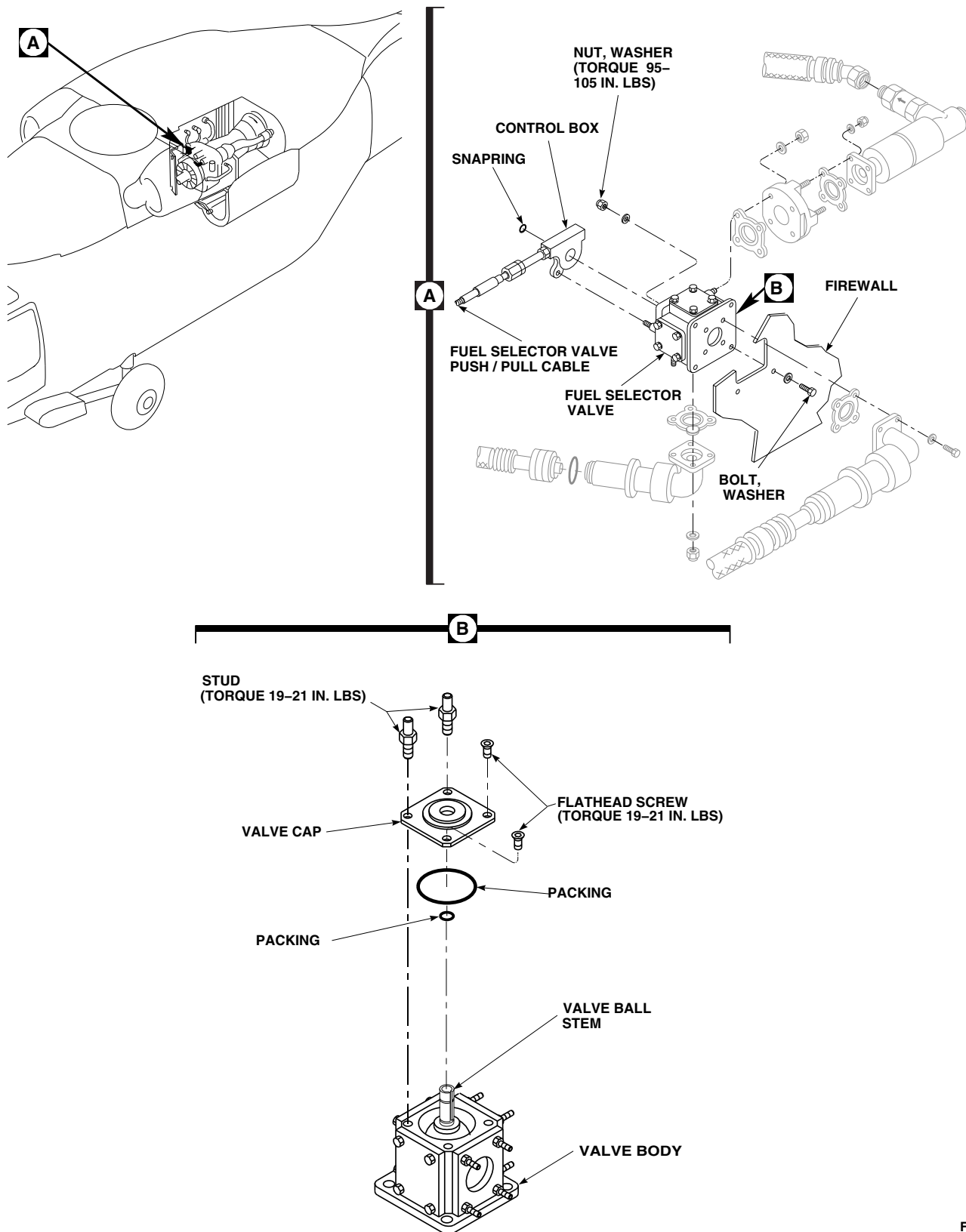
FB4093H
SA

FIGURE 10-4-25-1. REPLACE FUEL SELECTOR VALVE

10-4-25.1.2 Repair.

NOTE

Note location of studs and direction of valve ball stem keyway.

- a. Remove studs and flathead screws securing valve cap to valve body (Figure 10-4-25-1, Detail B).
- b. Remove valve cap from valve body. Remove and discard packing.
- c. Remove and discard packing from valve ball stem.
- d. Clean valve cap using cheesecloth, Item 90, Appendix D, dampened with isopropyl alcohol, Item 72, Appendix D.
- e. Coat packings with grease, Item 143, Appendix D.
- f. Install packing on valve ball stem and valve cap.
- g. Install valve cap on valve body with flathead screws and studs. **TORQUE SCREWS AND**

STUDS IN DIAGONAL SEQUENCE TO 19 - 21 INCH-POUNDS.

- h. Lockwire, Item 196, Appendix D, studs.

10-4-25.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install fuel selector valve on firewall with bolts, washers, and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS** (Figure 10-4-25-1, Detail A).
- c. Line up keyway on shaft of fuel selector valve with key in control box and install control box and fuel selector valve push/pull cable, as an assembly, on fuel selector valve. Install snapping (Figure 10-4-25-1, Detail A and Detail B).
- d. Install breakaway tee valve (PARA 10-4-27).
- e. Install crossfeed breakaway valve (PARA 10-4-26).
- f. Install engine compartment breakaway valve (PARA 10-4-24).

10-4-26 CROSSFEED BREAKAWAY VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-26.1 Replace Crossfeed Breakaway Valve	10-4-26-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Strap Wrench
- Torque Wrench, 30 - 150 in. lbs
- Wrench with Rubber Fittings

Materials/Parts

- Lubricating Oil, Item 205, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-113
- PARA 10-4-1
- PARA 10-4-39

10-4-26.1 Replace Crossfeed Breakaway Valve.

WARNING

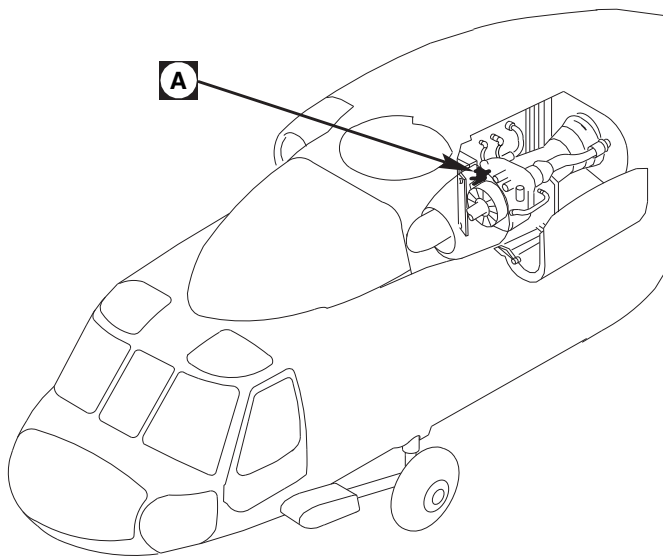
Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedures.

NOTE

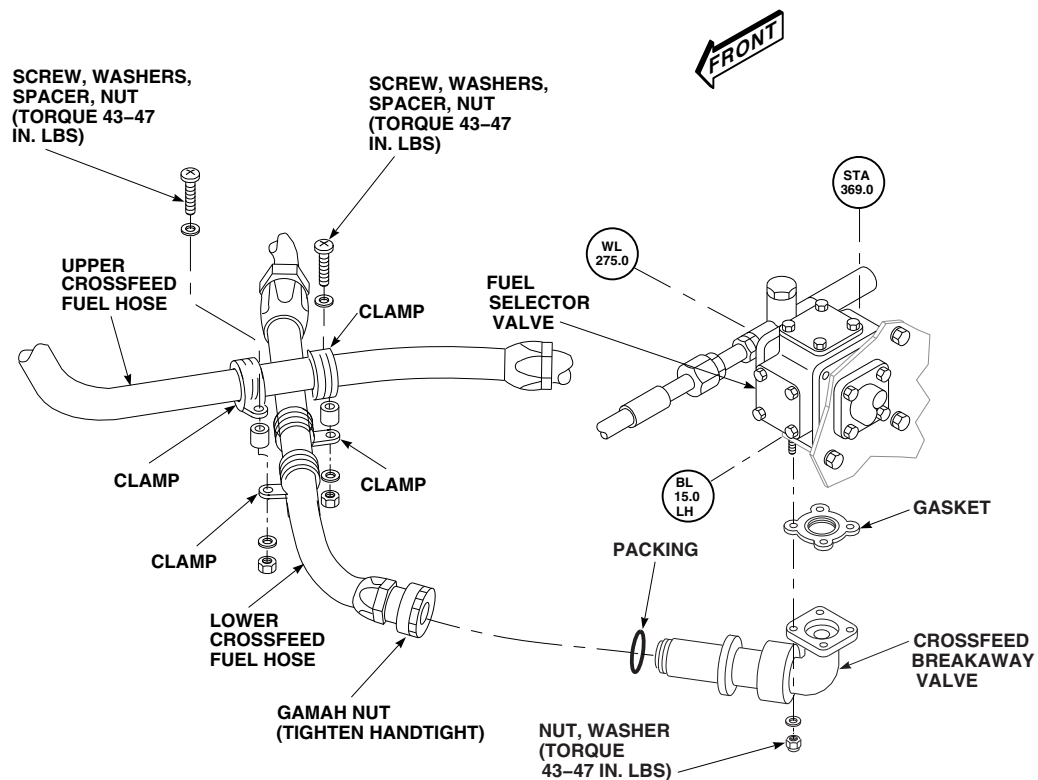
Replacement of left and right side crossfeed breakaway valves is the same.

10-4-26.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open engine cowling.
- Remove left or right oil cooler compartment access door (PARA 2-4-113).
- Purge fuel feed system (PARA 10-4-39).
- Remove screws, washers, spacers, and nuts securing clamps to fuel hoses (Figure 10-4-26-1, Detail A).
- Position 1-quart container to catch fuel and disconnect lower crossfeed fuel hose from crossfeed breakaway valve. Discard packing. Plug hose.



A



FB4482D
SA

FIGURE 10-4-26-1. REPLACE CROSSFEED BREAKAWAY VALVE

10-4-26-2

Change 5

- g. Remove nuts and washers securing crossfeed breakaway valve to fuel selector valve. Remove crossfeed breakaway valve and gasket. Discard gasket.

10-4-26.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

CAUTION

Under no circumstances is the No. 2 (right side) crossfeed breakaway valve (installed on the No. 2 (right side) crossfeed fuel hose) of a helicopter to be removed and installed on a different helicopter, or installed on the No. 1 (left side) crossfeed fuel hose. Any valve installed on the No. 2 (right side) crossfeed fuel hose must remain on the same helicopter until reaching 1500 hours. Valve then must be removed and discarded.

- b. Install gasket on crossfeed breakaway valve (Figure 10-4-26-1, Detail A).

NOTE

Crossfeed breakaway valve should rotate freely on the fuel selector valve.

- c. Install crossfeed breakaway valve on fuel selector valve with washers and nuts. Do not tighten or torque nuts at this time.
- d. Lubricate packing with lubricating oil, Item 205, Appendix D.
- e. Install packing on crossfeed breakaway valve.

CAUTION

To prevent damage to connecting components, use strap wrench or wrench

with rubber fittings to tighten Gamah nut. Do not use pliers or other tools to tighten Gamah nut.

NOTE

Make sure lower crossfeed fuel hose from left crossfeed breakaway valve is routed under upper crossfeed fuel hose from right crossfeed breakaway valve.

- f. Connect lower crossfeed fuel hose to crossfeed breakaway valve. **TIGHTEN GAMAH NUT HANDTIGHT.**

CAUTION

- Damage to crossfeed breakaway valve may occur if valve is preloaded. Do not preload valve.
- Do not preload crossfeed breakaway valve when installing fuel hose. Frangible section of crossfeed breakaway valve is sensitive to bending and tension overload.

NOTE

Allow crossfeed breakaway valve and fuel hose to rotate freely on fuel selector valve. Valve and hose should assume a neutral position to aid in the elimination/reduction of a preload condition.

- g. Allow fuel hose and crossfeed breakaway valve to rotate freely on fuel selector valve. Crossfeed breakaway valve and fuel hose shall seek a neutral position.
- h. Secure fuel hoses together with clamps, screws, spacers, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

CAUTION

- Damage to crossfeed breakaway valve may occur if crossfeed breakaway valve is preloaded. Do not preload crossfeed breakaway valve.
- Do not preload crossfeed breakaway valve when torquing nuts. Frangible section of crossfeed breakaway valve is sensitive to bending and tension overload.

NOTE

Avoid changing crossfeed breakaway valve position/orientation.

- i. **TORQUE NUTS SECURING CROSSFEED BREAKAWAY VALVE TO FUEL SELECTOR VALVE TO 43 - 47 INCH-POUNDS.**
- j. Test fuel feed system (PARA 10-4-39).
- k. Prime fuel system (PARA 10-4-1). Check for leaks.
 - l. Make sure area is clean and free of foreign material.
- m. Install oil cooler compartment access door (PARA 2-4-113).
- n. Close engine cowling.

10-4-27 BREAKAWAY TEE VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-27.1 Replace Breakaway Tee Valve	10-4-27-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lubricating Oil, Item 205, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-113
- PARA 10-4-1
- PARA 10-4-39

10-4-27.1 Replace Breakaway Tee Valve.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

NOTE

Replacement of left and right side breakaway tee valves is the same.

10-4-27.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open engine cowling.

- Remove left or right oil cooler compartment access door (PARA 2-4-113).
- Purge fuel feed system (PARA 10-4-39).
- Position 1-quart container to catch fuel and disconnect fuel hoses from check valve and breakaway tee valve (Figure 10-4-27-1, Detail A). Plug hoses. Discard packing.
- Remove nuts, washers, breakaway tee valve, and gasket from fuel selector valve. Discard gasket.
- Remove check valve and packing from break-away tee valve. Discard packing.

10-4-27.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Lubricate packing with lubricating oil, Item 205, Appendix D.

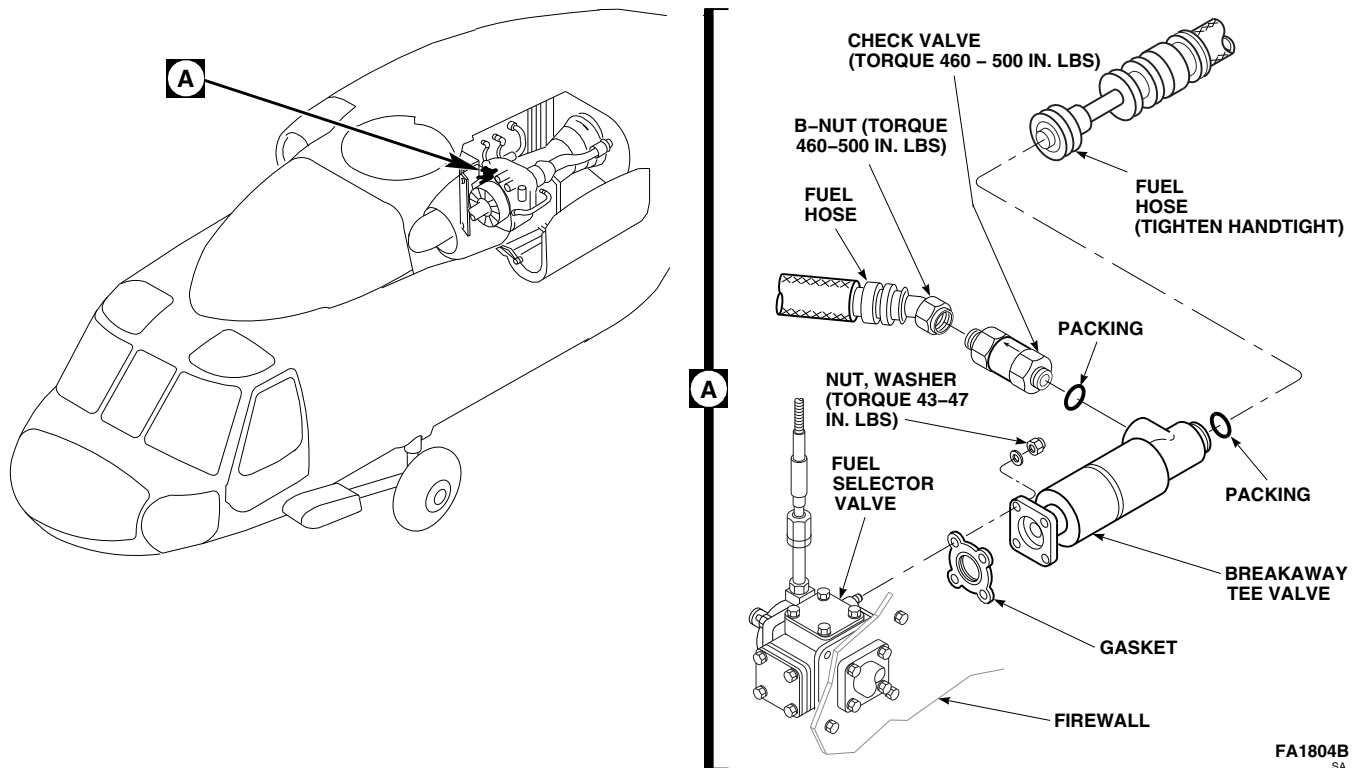


FIGURE 10-4-27-1. REPLACE BREAKAWAY TEE VALVE

- c. Install packing on end of check valve that flow arrow points away from (Figure 10-4-27-1, Detail A).

CAUTION

Do not preload breakaway tee valve when installing check valve. Frangible section of breakaway tee valve is sensitive to bending and tension overload.

- d. Install check valve into breakaway tee valve with flow arrow pointing away from breakaway tee valve. **TORQUE CHECK VALVE TO 460 - 500 INCH-POUNDS.**
- e. Install gasket on breakaway tee valve.

CAUTION

Do not preload breakaway tee valve when installing. Frangible section of

breakaway tee valve is sensitive to bending and tension overload.

- f. Install breakaway tee valve to fuel selector valve with washers and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
- g. Lubricate packing with lubricating oil, Item 205, Appendix D.
- h. Install packing on breakaway tee valve.

CAUTION

Make sure breakaway tee valve is not preloaded when installing fuel hose. Frangible section of breakaway tee valve is sensitive to bending and tension overload.

- i. Remove plug and connect fuel hose to breakaway tee valve. **TIGHTEN FITTING HANDTIGHT.**

- j. Remove plug and connect fuel hose to check valve. Hold check valve with wrench at end closest to hose. **TORQUE B-NUT TO 460 - 500 INCH-POUNDS.**
- k. Test fuel feed system (PARA 10-4-39).
- l. Prime fuel system (PARA 10-4-1). Check for leaks.
- m. Make sure area is clean and free of foreign material.
- n. Install oil cooler compartment access door (PARA 2-4-113).
- o. Close engine cowling.

10-4-28 BULKHEAD BREAKAWAY VALVE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-28.1 Replace Bulkhead Breakaway Valve	10-4-28-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-4-1
- PARA 10-4-39
- PARA 13-5-3
- PARA 13-5-6

10-4-28.1 Replace Bulkhead Breakaway Valve.



Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

NOTE

Replacement of left and right side bulkhead breakaway valves is the same.

10-4-28.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open engine cowling.
- Open auxiliary power unit (APU) access door.
- Purge fuel feed system (PARA 10-4-39).

- e. Gain access to main fuel cell compartment as follows:
 - (1) Remove rear passenger's seats as required (PARA 2-4-45).
 - (2) Remove soundproofing panel from rear bulkhead covering left or right stowage compartment (PARA 2-4-69).
 - (3) Remove cargo netting from stowage compartment.
 - (4) If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
- f. Remove screws and front panel from structure.
- g. Position 1-quart container to catch fuel when fuel hose and line are disconnected.
- h. Disconnect upper deck main fuel hose and lower main fuel line from bulkhead breakaway valve (Figure 10-4-28-1, Sheet 1, Detail B). Remove and discard packings. Install protective plugs in fuel hoses.
- i. Remove bolts securing bulkhead breakaway valve to APU compartment deck.
- j. Using nonmetallic scraper, remove sealing compound from bulkhead breakaway valve flange, swivel joint, and APU compartment deck (Figure 10-4-28-1, Sheet 2, Detail C).
- k. Clean area with dry-cleaning solvent, Item 124, Appendix D.

10-4-28.1.2 Inspect/Repair.

- a. Inspect coated surface of bulkhead breakaway valve for breaks or scratches. **NONE ALLOWED.** If coated surface has breaks or scratches, apply corrosion resistant coating, Item 118, Appendix D.
- b. Inspect activation surface (yellow band area) of bulkhead breakaway valve to verify valve has not been activated after removing sealing compound. If yellow band is visible, replace bulkhead breakaway valve.

10-4-28.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Lubricate packings with lubricating oil, Item 205, Appendix D. Install packings on bulkhead breakaway valve (Figure 10-4-28-1, Sheet 1, Detail B).

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- c. Coat threads of bolts with antiseize compound, Item 74, Appendix D.

CAUTION

Do not preload bulkhead breakaway valve when installing. Frangible section of bulkhead breakaway valve is sensitive to bending and tension overload.

- d. Position bulkhead breakaway valve on auxiliary power unit (APU) compartment deck. Secure valve with bolts wet with sealing compound, Item 292, Appendix D. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- e. Coat threads on bulkhead breakaway valve with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.

CAUTION

Make sure bulkhead breakaway valve is not preloaded when installing fuel hoses. Frangible section of bulkhead breakaway valve is sensitive to bending and tension overload.

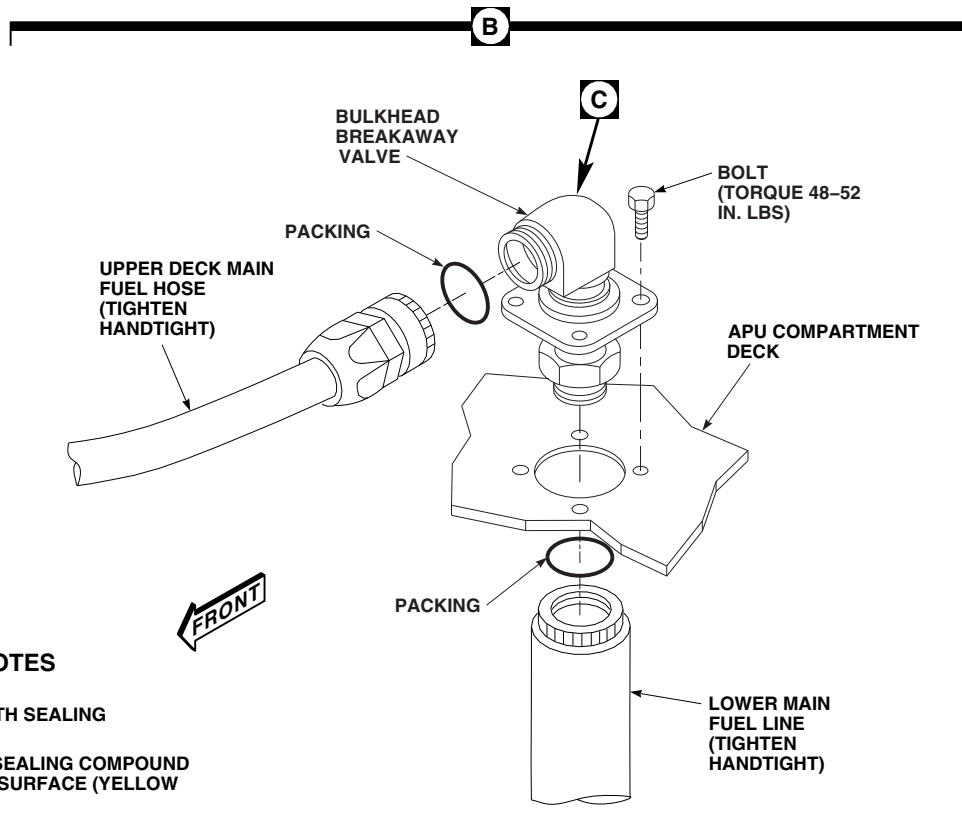
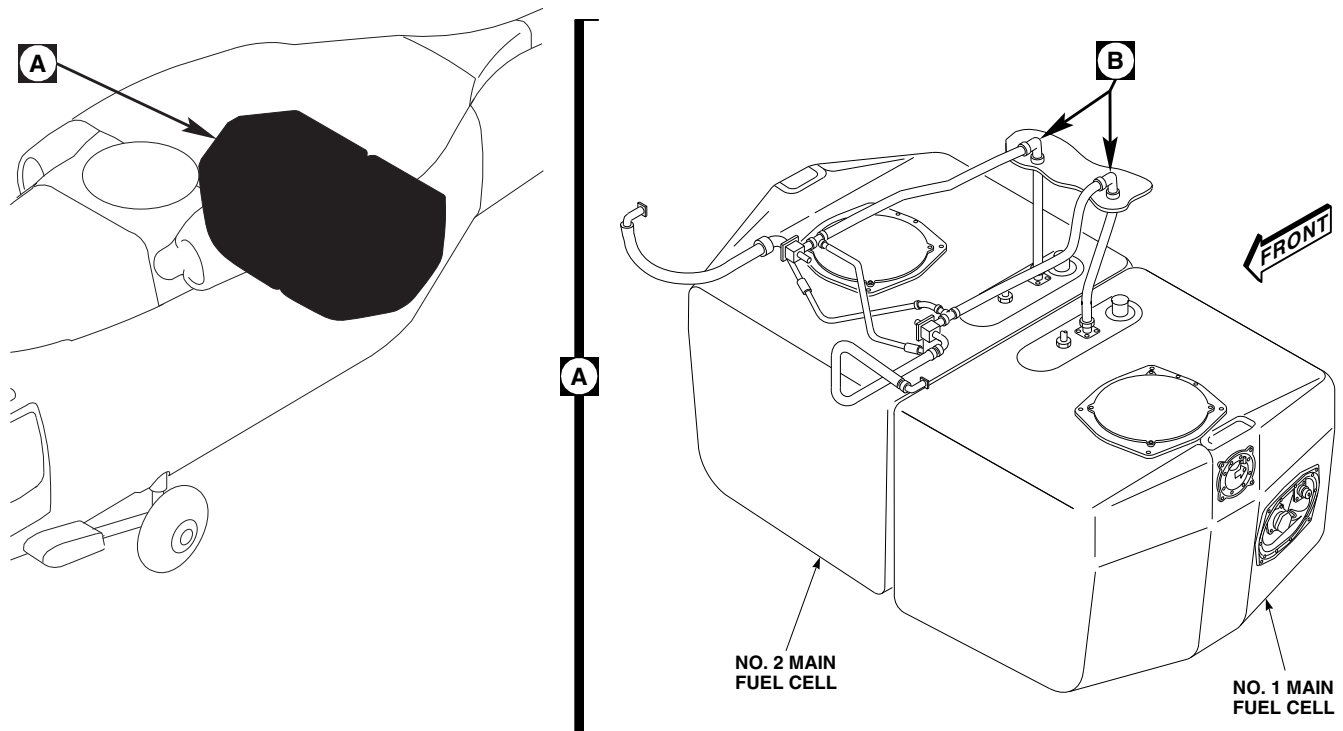
- f. Remove protective plugs from fuel hoses. Connect upper deck main fuel hose and lower main fuel line to bulkhead breakaway valve. **TIGHTEN FITTINGS HANDTIGHT.**

NOTE

Do not apply sealing compound to activation surface (yellow band area) of bulkhead breakaway valve.

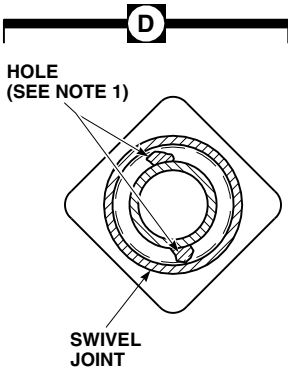
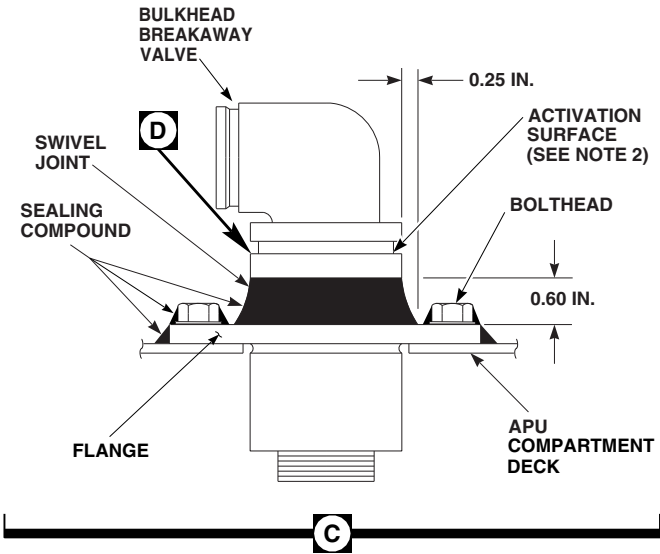
- g. Seal flange, boltheads, and swivel joint area of bulkhead breakaway valve by applying a fillet of sealing compound, Item 292, Appendix D (Figure 10-4-28-1, Sheet 2, Detail C).
- h. Seal two holes in swivel joint of bulkhead breakaway valve with sealing compound, Item 292, Appendix D (Figure 10-4-28-1, Sheet 2, Detail D).

- i. Test fuel feed system (PARA 10-4-39).
- j. Prime fuel system (PARA 10-4-1). Check for leaks.
- k. Make sure area is clean and free of foreign material.
- l. Close main fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panel (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).
- m. Make sure area is clean and free of foreign material.
- n. Close APU access door.
- o. Close engine cowling.



FB3813_1A
SA

FIGURE 10-4-28-1. REPLACE BULKHEAD BREAKAWAY VALVE (SHEET 1 OF 2)



FB3813_2A
 SA

FIGURE 10-4-28-1. REPLACE BULKHEAD BREAKAWAY VALVE (SHEET 2 OF 2)

10-4-29 FUEL SYSTEM DRAIN HOSE AND TUBES.

PARAGRAPH BREAKDOWN

	PAGE
10-4-29.1 Replace Fuel System Drain Hose and Tubes	10-4-29-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Protection, Eye
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Machinery Towel, Item 211, Appendix D

- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 13-5-3
- PARA 13-5-6

10-4-29.1 Replace Fuel System Drain Hose and Tubes.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

NOTE

Replacement of left and right fuel system drain tubes is the same.

10-4-29.1.1 Remove.

CAUTION

Damage to drain tubes will result if tubes are used as supports. Do not use tubes as hand or foot supports.

- Turn off all electrical power (PARA 1-6-16).
- Open engine cowling.
- Gain access to main fuel cell compartment as follows:
 - Remove rear passenger’s seats as required (PARA 2-4-45).

- (2) Remove soundproofing panel from rear bulkhead covering left or right stowage compartment (PARA 2-4-69).
- (3) Remove cargo netting from stowage compartment.
- (4) If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

CAUTION

Damage to drain tubes will result if care is not taken during removal. Do not use force or pressure upon tube or cause tube to bend or distort.

- d. Remove screws, washers, and nuts securing clamps to tubes and brackets from tube(s) to be removed (Figure 10-4-29-1, Sheet 1, Detail B and Sheet 2, Detail C). Remove clamps from tube(s).
- e. To remove engine manifold drain hose, perform the following:
 - (1) Disconnect engine manifold drain hose from elbow on top side of engine compartment rear deck (Figure 10-4-29-1, Sheet 1, Detail A).
 - (2) Disconnect engine manifold drain hose from elbows on intercostal and underside of engine compartment rear deck (Figure 10-4-29-1, Sheet 1, Detail B). Remove engine manifold drain hose.
 - (3) If necessary, remove elbows as follows:
 - (a) Remove nut and washer from underside of engine compartment rear deck, securing elbow, sealing washer, and washer.
 - (b) Remove elbow, sealing washer, and washer from top side of engine compartment rear deck. Discard sealing washer.

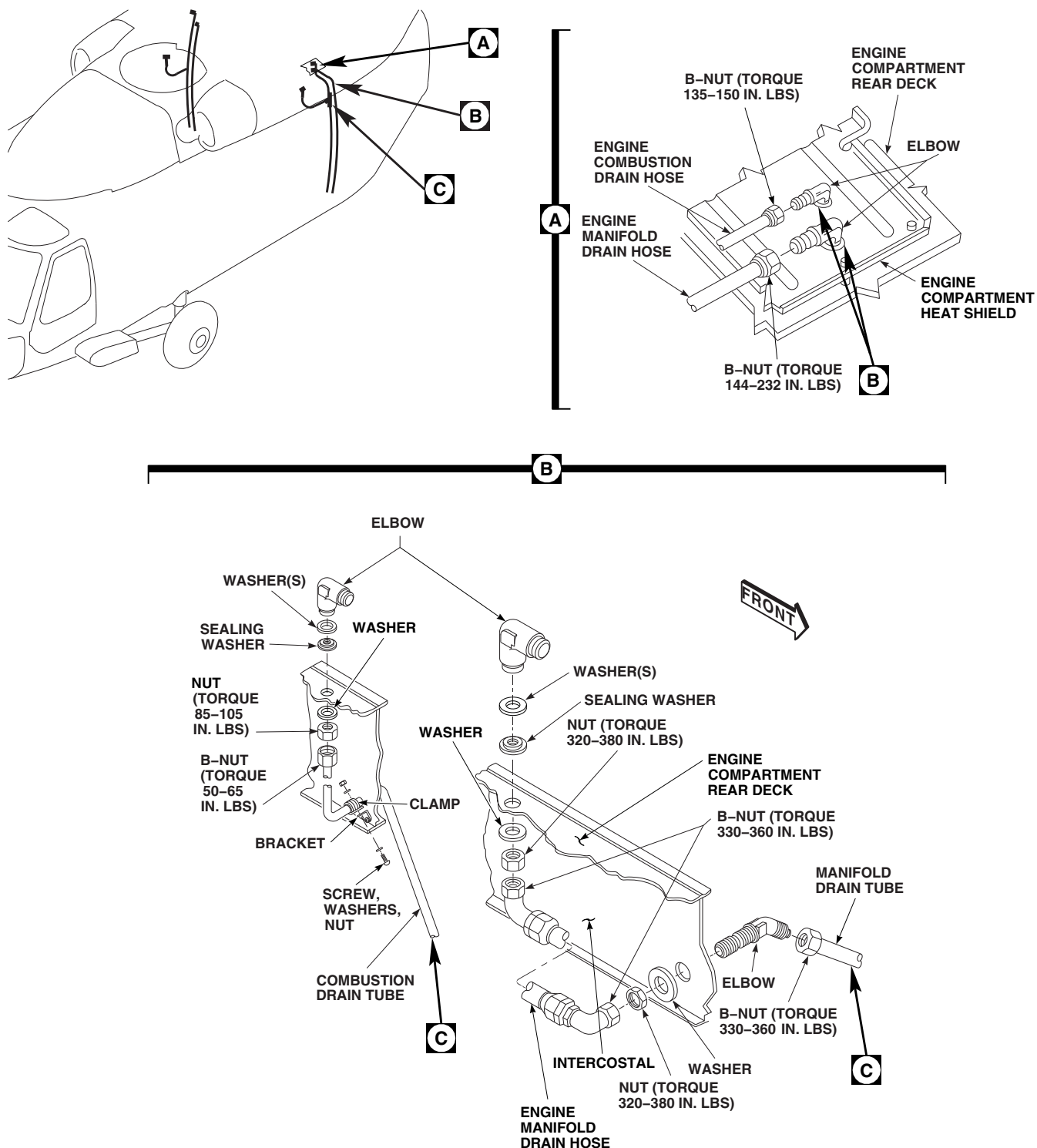
- (c) Disconnect manifold drain tube from elbow on intercostal.
- (d) Remove nut and washer securing elbow to intercostal. Remove elbow.

- f. To remove manifold drain tube, perform the following:

NOTE

Nuts at manifold drain hose and tube ends may be unfastened after clamps are loosened from all tube(s) to be removed.

- (1) Disconnect manifold drain tube from tee (Figure 10-4-29-1, Sheet 2, Detail C).
- (2) Disconnect manifold drain tube from elbow on intercostal (Figure 10-4-29-1, Sheet 1, Detail B). Remove manifold drain tube.
- (3) If necessary, remove elbow as follows:
 - (a) Disconnect engine manifold drain hose from elbow on intercostal.
 - (b) Remove nut and washer securing elbow to intercostal. Remove elbow.
- g. To remove combustion drain tube, perform the following:
 - (1) Disconnect engine combustion drain hose from elbow on top side of engine compartment rear deck (Figure 10-4-29-1, Sheet 1, Detail A).
 - (2) Disconnect engine combustion drain hose from elbow on underside of engine compartment rear deck (Figure 10-4-29-1, Sheet 1, Detail B).
 - (3) Slide combustion drain tube free from grommet located in mounting plate (Figure 10-4-29-1, Sheet 2, Detail C). Remove combustion drain tube.
 - (4) If necessary, remove elbow as follows (Figure 10-4-29-1, Sheet 1, Detail B):



NOTES

1. 70583-70687
2. 70688-SUBQ

FB3305_1E
SA

FIGURE 10-4-29-1. REPLACE FUEL SYSTEM DRAIN HOSE AND TUBES (SHEET 1 OF 2)

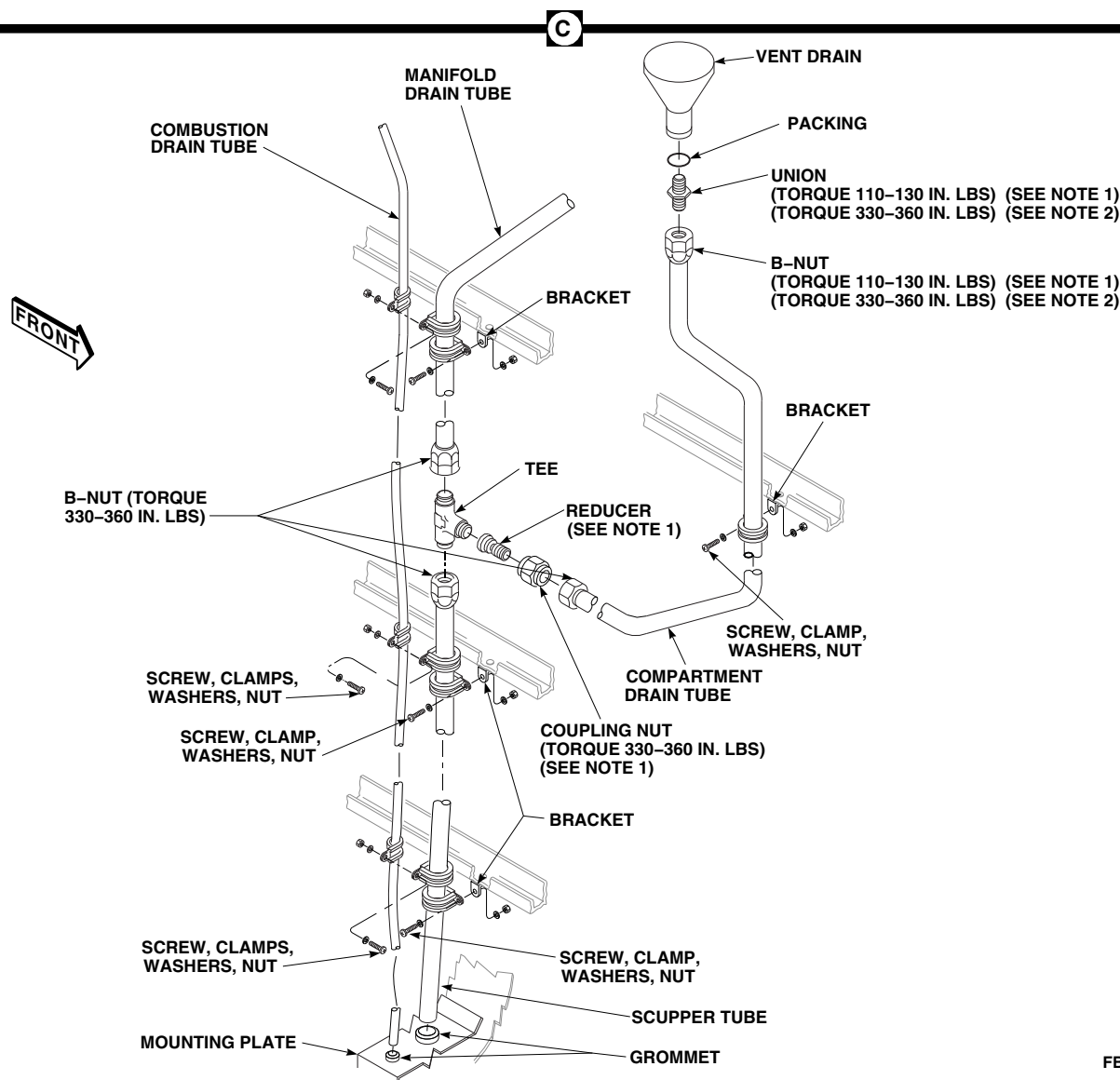


FIGURE 10-4-29-1. REPLACE FUEL SYSTEM DRAIN HOSE AND TUBES (SHEET 2 OF 2)

- (a) Remove nut and washer from underside of engine compartment rear deck, securing elbow, sealing washer, and washer.
 - (b) Remove elbow, sealing washer, and washer from top side of engine compartment rear deck. Discard sealing washer.
- h. **70583-70687** To remove compartment drain tube, perform the following (Figure 10-4-29-1, Sheet 2, Detail C):
 - (1) Disconnect compartment drain tube from union in vent drain and from reducer/coupling nut in tee. Remove compartment drain tube.
 - (2) Remove union and packing from vent drain. Discard packing.
 - (3) Remove coupling nut and reducer from tee.
- i. **70688-SUBQ** To remove compartment drain tube, perform the following (Figure 10-4-29-1, Sheet 2, Detail C):
 - (1) Disconnect compartment drain tube from union in vent drain and from reducer/coupling nut in tee. Remove compartment drain tube.
 - (2) Remove union and packing from vent drain. Discard packing.
 - (3) Remove coupling nut and reducer from tee.

10-4-29-4 Change 10

- (1) Disconnect compartment drain tube from union in vent drain and tee. Remove compartment drain tube.
 - (2) Remove union and packing from vent drain. Discard packing.
- j. To remove scupper tube, perform the following:
- (1) Disconnect scupper tube from tee.
 - (2) Slide scupper tube free from grommet in mounting plate. Remove scupper tube.
- k. Install protective caps and plugs on elbows, hoses, tubes, and connection fittings.

10-4-29.1.2 Inspect.

- a. Check threads of nuts and connection fittings for damage. **NONE ALLOWED.** If damage is found, replace nut or connection fitting.
- b. Inspect hose or tube for dents and distortion. **DENTS OR DISTORTION SHALL NOT EXCEED 10% OF HOSE/TUBE DIAMETER.** If limit is exceeded, replace hose/tube.
- c. Inspect grommets for cracks and wear. **NONE ALLOWED.** If cracked or worn, replace grommet.

10-4-29.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

NOTE

Remove caps and plugs from elbows, hoses, tubes, and connection fittings as parts are installed and connected.

- b. Before installation, blow all tubes clear with dry compressed air.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- c. Coat male threads with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeezeout, after parts are assembled with machinery towel, Item 211, Appendix D.
- d. To install engine manifold drain hose, perform the following:
 - (1) If removed, install elbows as follows (Figure 10-4-29-1, Sheet 1, Detail B):
 - (a) Lubricate sealing washer with petrolatum, Item 234, Appendix D.

NOTE

Additional washers may be used to prevent engine manifold drain hose from chafing against heat shield.

- (b) Install elbow, washer, additional washers (if necessary), and sealing washer through top side of engine compartment rear deck. Position elbow facing front.
- (c) Secure elbow to underside of engine compartment rear deck with washer and nut. **TORQUE NUT TO 320 - 380 INCH-POUNDS.**
- (d) Install elbow through opening in intercostal. Position elbow downward.

- (e) Secure elbow to intercostal with washer and nut. **TORQUE NUT TO 320 - 380 INCH-POUNDS.**
- (f) Connect manifold drain tube to elbow on intercostal. **TORQUE B-NUT TO 330 - 360 INCH-POUNDS.**
- (2) Connect engine manifold drain hose to elbows on intercostal and underside of engine compartment rear deck. **TORQUE B-NUTS TO 330 - 360 INCH-POUNDS.**
- (3) Connect engine manifold drain hose to elbow on top side of engine compartment rear deck (Figure 10-4-29-1, Sheet 1, Detail A). **TORQUE B-NUT TO 144 - 232 INCH-POUNDS.**

NOTE

Install and support all tubing to prevent chafing. Make sure all fuel lines have a minimum clearance of 1/2-inch from all engine control cables.

- e. To install manifold drain tube, perform the following:

- (1) If removed, install elbow as follows (Figure 10-4-29-1, Sheet 1, Detail B):
 - (a) Install elbow through opening in intercostal. Position elbow downward.
 - (b) Secure elbow to intercostal with washer and nut. **TORQUE NUT TO 320 - 380 INCH-POUNDS.**
 - (c) Connect engine manifold drain hose to elbow on intercostal. **TORQUE B-NUT TO 330 - 360 INCH-POUNDS.**

NOTE

As manifold drain tube is being installed, make sure tube does not chafe against engine main frame, swirl frame,

or scroll case. If chafing occurs, loosen B-nuts on tube hose, and nut securing elbow. Reposition elbow and tube, then retighten nut and B-nuts to proper torque.

- (2) Connect manifold drain tube to elbow on intercostal. **TORQUE B-NUTS TO 330 - 360 INCH-POUNDS.**
- (3) Connect manifold drain tube to tee. Position tee facing front (Figure 10-4-29-1, Sheet 2, Detail C). **TORQUE B-NUT TO 330 - 360 INCH-POUNDS.**
- (4) Install clamps on manifold drain tube, lining up clamps with bracket and clamp on combustion drain tube. Secure clamps with screws, washers, and nuts.

- f. To install combustion drain tube, perform the following:

- (1) If removed, install elbow as follows (Figure 10-4-29-1, Sheet 1, Detail B):
 - (a) Lubricate sealing washer with petrolatum, Item 234, Appendix D.

NOTE

Additional washers may be used to prevent combustion drain hose from chafing against heat shield.

- (b) Install elbow, washer, additional washers (if necessary), and sealing washer through top side of engine compartment rear deck. Position elbow facing front.
- (c) Secure elbow to underside of engine compartment rear deck with washer and nut. **TORQUE NUT TO 85 - 105 INCH-POUNDS.**
- (2) Connect engine combustion drain hose to elbow on top side of engine compartment rear deck (Figure 10-4-29-1, Sheet 1, Detail A). **TORQUE B-NUT TO 135 - 150 INCH-POUNDS.**

- (3) Carefully feed combustion drain tube through hole in intercostal (Figure 10-4-29-1, Sheet 1, Detail B).
 - (4) Insert combustion drain tube in grommet on mounting plate (Figure 10-4-29-1, Sheet 2, Detail C).
 - (5) Connect combustion drain tube to elbow on underside of engine compartment rear deck (Figure 10-4-29-1, Sheet 1, Detail B). **TORQUE B-NUT TO 50 - 65 INCH-POUNDS.**
 - (6) Install clamps on combustion drain tube, lining up clamps with clamps on manifold drain tube and scupper tube. Secure clamps with screws, washers, and nuts (Figure 10-4-29-1, Sheet 2, Detail C).
 - (7) Install clamp on combustion drain tube, lining up clamp with bracket on intercostal. Secure clamp with screw, washer, and nut (Figure 10-4-29-1, Sheet 1, Detail B).
- g. **70583-70687** To install compartment drain tube, perform the following (Figure 10-4-29-1, Sheet 2, Detail C):
- (1) Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing in groove on union.
 - (2) Install union in vent drain. **TORQUE UNION TO 110 - 130 INCH-POUNDS.**
 - (3) Install reducer and coupling nut on tee. **TORQUE COUPLING NUT TO 330 - 360 INCH-POUNDS.**
 - (4) Connect compartment drain tube to union. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**
 - (5) Install compartment drain tube on reducer. **TORQUE B-NUT TO 330 - 360 INCH-POUNDS.**
 - (6) Install clamp on compartment drain tube, lining up clamp with bracket. Secure clamp with screw, washer, and nut.
- h. **70688-SUBQ** To install compartment drain tube, perform the following (Figure 10-4-29-1, Sheet 2, Detail C):
- (1) Lubricate packing with lubricating oil, Item 205, Appendix D.
 - (2) Install packing in groove on union. Install union in vent drain. **TORQUE UNION TO 330 - 360 INCH-POUNDS.**
 - (3) Connect compartment drain tube to union. **TORQUE B-NUT TO 330 - 360 INCH-POUNDS.**
 - (4) Connect compartment drain tube to tee. **TORQUE B-NUT TO 330 - 360 INCH-POUNDS.**
 - (5) Install clamp on compartment drain tube, lining up clamp with bracket. Secure clamp with screw, washer, and nut.
- i. To install scupper tube, perform the following (Figure 10-4-29-1, Sheet 2, Detail C):
- (1) Insert scupper tube into grommet on mounting plate.
 - (2) Connect scupper tube to tee. **TORQUE B-NUT TO 330 - 360 INCH-POUNDS.**
 - (3) Install clamps on scupper tube, lining up clamps with brackets and clamps on combustion drain tube. Secure clamps with screws, washers, and nuts.
- j. Pressure test fuel system drain lines for leaks at 15 psig for 5 minutes. **NONE ALLOWED.** If leakage is found, perform the following:
- (1) Make sure all connections are properly torqued.
 - (2) If leakage persists, replace packing on union or sealing washers on elbows.
- k. Close main fuel cell compartment as follows:
- (1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

- (2) Install cargo netting.
 - (3) Install soundproofing panel (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).
1. Close engine cowling.

10-4-30 FUEL SYSTEM VENT LINES.

PARAGRAPH BREAKDOWN

	PAGE
10-4-30.1 Replace Fuel System Vent Lines	10-4-30-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Protective Covering
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Machinery Towel, Item 211, Appendix D

- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 13-5-3
- PARA 13-5-6

10-4-30.1 Replace Fuel System Vent Lines.

NOTE

Replacement of No. 1 and No. 2 fuel system vent lines is the same.

10-4-30.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Gain access to main fuel cell compartment as follows:
 - Remove rear passenger’s seats as required (PARA 2-4-45).
 - Remove soundproofing panel from rear bulkhead covering left or right stowage compartment (PARA 2-4-69).

- Remove cargo netting from stowage compartment.
- If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

- Open avionics compartment access door.
- Remove screws securing front panel to fuel cell compartment. Remove panel.
- Unlock fasteners securing left or right panel of enclosure on top of main fuel cells. Remove panel.
- Remove screw, lockwasher, washers, nut, and bonding jumper from clamp on vent tube Y (Figure 10-4-30-1, Detail B).

- g. Remove screws, washers, nuts, and bonding jumpers, if installed, from clamps on No. 1 and No. 2 fuel vent tubes (Figure 10-4-30-1, Detail A).
- h. Disconnect No. 1 and No. 2 fuel vent tubes from vent valves and vent tube Y. Remove fuel vent tubes. Discard packings.
- i. Disconnect vent tube Y from fuel vent tube elbow. Remove vent tube Y. Discard packing.
- j. Remove screws, lockwasher, washers, bonding jumpers, and nuts securing fuel vent tube elbow to rear panel.

CAUTION

To avoid damaging UHF antenna in transition section tub, protect with protective covering before entering tail cone area.

- k. Install protective covering over UHF antenna in transition section tub.

NOTE

Install protective plugs in valves and tubes as they are disconnected.

- l. Disconnect rear vent tube from fuel vent tube elbow and transition section bottom. Remove rear vent tube. Discard packings.
- m. Remove screw, lockwasher, washers, and nut securing bonding jumper to clamp on rear vent tube.

10-4-30.1.2 Inspect.

- a. Inspect tubing for dents and distortion. **DENTS OR DISTORTION SHALL NOT EXCEED 10% OF TUBE DIAMETER.** If limit is exceeded, replace tube.
- b. Inspect clamps for distortion and security. **NONE ALLOWED.** If damaged, replace clamp.

10-4-30.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).

NOTE

- Remove protective caps and plugs from valves and tubes as required.
- Install and support all tubing to avoid chafing. Make sure all vent tubes have a minimum clearance of 1/2-inch from all engine control cables.

- b. Make sure all lines are clean and clear.

- c. Lubricate packings with lubricating oil, Item 205, Appendix D.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- d. Coat threads of vent tubes with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.
- e. Install fuel vent tube elbow on rear panel and secure with screws, lockwasher, washers, bonding jumpers, and nuts (Figure 10-4-30-1, Detail A). **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
- f. Install packings on ends of vent tube Y and upper end of fuel vent tube elbow.
- g. Screw vent tube Y to fuel vent tube elbow. **TIGHTEN FITTING HANDTIGHT.**
- h. Install packings on No. 1 and No. 2 fuel vent tube ends.

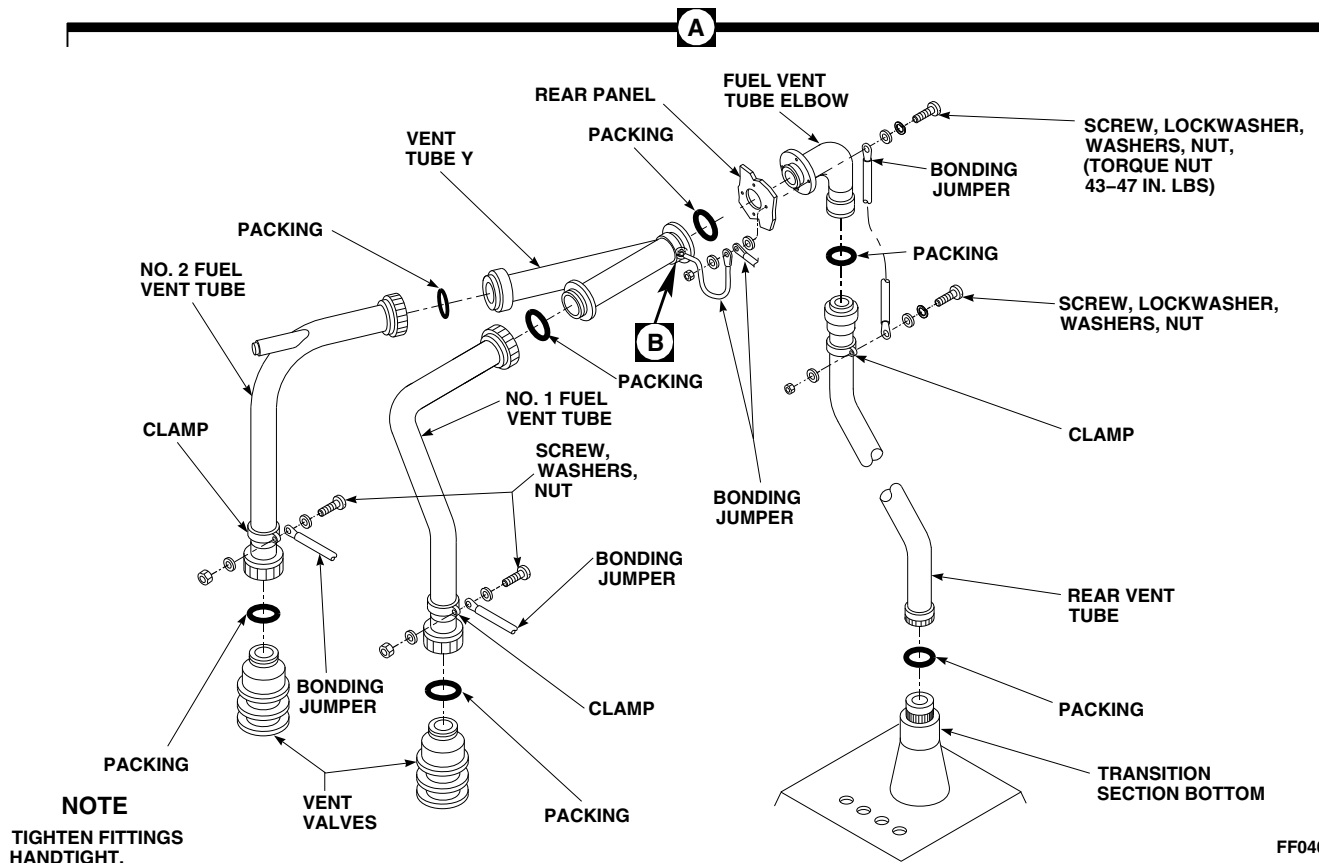
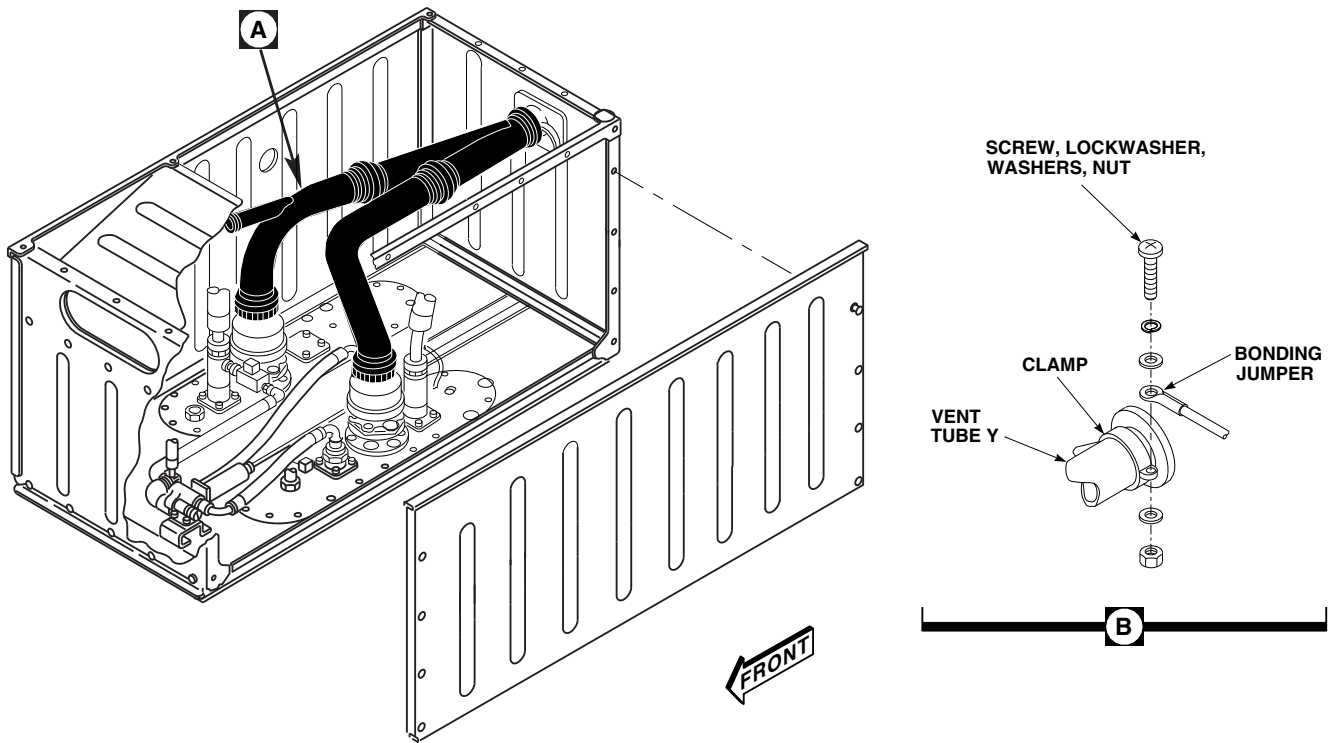


FIGURE 10-4-30-1. REPLACE FUEL SYSTEM VENT LINES

FF0406D
SA

CAUTION

Make sure self-sealing breakaway vent valve is not preloaded when installing vent tube to system vent valve. Frangible section of self-sealing breakaway valve is sensitive to bending and tension overload.

- i. Screw No. 1 and No. 2 fuel vent tubes to vent valves and vent tube Y. **TIGHTEN FITTINGS HANDTIGHT.**
- j. Install bonding jumpers to clamps on No. 1 and No. 2 fuel vent tubes with screws, washers, and nuts.
- k. Install bonding jumper to clamp on vent tube Y with screw, lockwasher, washers, and nut (Figure 10-4-30-1, Detail B).
- l. Install packings on rear vent tube ends (Figure 10-4-30-1, Detail A).
- m. Screw rear vent tube to fuel vent tube elbow and to transition section bottom. **TIGHTEN FITTINGS HANDTIGHT.**
- n. Install bonding jumper from rear panel to rear vent tube with clamp, screw, lockwasher, washers, and nut.
- o. Seal all bonding jumpers using sealing compound, Item 292, Appendix D.

CAUTION

Damage to fuel tanks will occur if tanks are pressurized. Do not pressurize fuel

tanks. Install protective caps on vent tubes at inlet to tanks when pressurizing vent tubes.

- p. Install protective caps and plugs in vent tubes.
- q. Pressure test fuel system drain lines for leaks at 15 psig for 5 minutes. **NONE ALLOWED.** If leakage is found, perform the following:
 - (1) Make sure all connections are properly torqued.
 - (2) If leakage persists, replace packings.
- r. Make sure area is clean and free of foreign material.
- s. Install front panel with screws.
- t. Install left or right panel enclosure on top of main fuel cells. Lock fasteners.
- u. Remove protective covering from UHF antenna.
- v. Close avionics compartment access door.
- w. Close main fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panel (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

10-4-31 PRESSURE REFUELING RECEPTACLE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-31.1 Replace Pressure Refueling Receptacle	10-4-31-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-3-1
- PARA 10-3-2
- PARA 10-4-1
- PARA 10-4-40
- PARA 13-5-3
- PARA 13-5-6

10-4-31.1 Replace Pressure Refueling Receptacle.

NOTE

Pressure refueling receptacle is installed on No. 1 main fuel cell only.

10-4-31.1.1 Remove.

- Purge main fuel cell (PARA 10-4-40).
- Gain access to main fuel cell compartment as follows:
 - Remove rear passenger’s seats as required (PARA 2-4-45).

- Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).
- Remove cargo netting from stowage compartments.
- If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
- Remove bolts, washers, and fuel cell access cover from main fuel cell. Remove packing from fuel cell access cover (Figure 10-4-31-1, Detail A). Discard packing.

- d. Disconnect pressure refueling tube from pressure refueling receptacle (Figure 10-4-31-1, Detail B). Remove and discard packing.

NOTE

Note location of bonding jumper prior to removing.

- e. Remove bolts, washers, bonding jumper, and pressure refueling receptacle from main fuel cell. Remove and discard packing.
- f. Using nonmetallic scraper, remove sealing compound from pressure refueling receptacle and pressure refueling panel.

10-4-31.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Lubricate packings with lubricating oil, Item 205, Appendix D. Install packings on end of pressure refueling receptacle (Figure 10-4-31-1, Detail B).
- c. Install packing in groove on pressure refueling receptacle.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- d. Coat threads of bolts with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads on receptacle. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.
- e. Position bonding jumper on pressure refueling receptacle. Secure pressure refueling receptacle and bonding jumper with bolts and washers. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS.**

- f. Connect pressure refueling tube to pressure refueling receptacle. **TIGHTEN FITTING HANDTIGHT.**

- g. Apply bead of sealing compound, Item 292, Appendix D, in cavity between pressure refueling receptacle and pressure refueling panel.
- h. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on fuel cell access cover (Figure 10-4-31-1, Detail A).
- i. Inspect area inside main fuel cell and make sure area is free of all foreign material.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- j. Coat threads of fuel cell access cover attachment bolts with antiseize compound, Item 74, Appendix D.
- k. Install fuel cell access cover on main fuel cell with bolts and washers. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS.**
- l. Pressure test main fuel cells (PARA 10-3-1 or PARA 10-3-2).
- m. Close main fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panels (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).
- n. Prime fuel system (PARA 10-4-1).

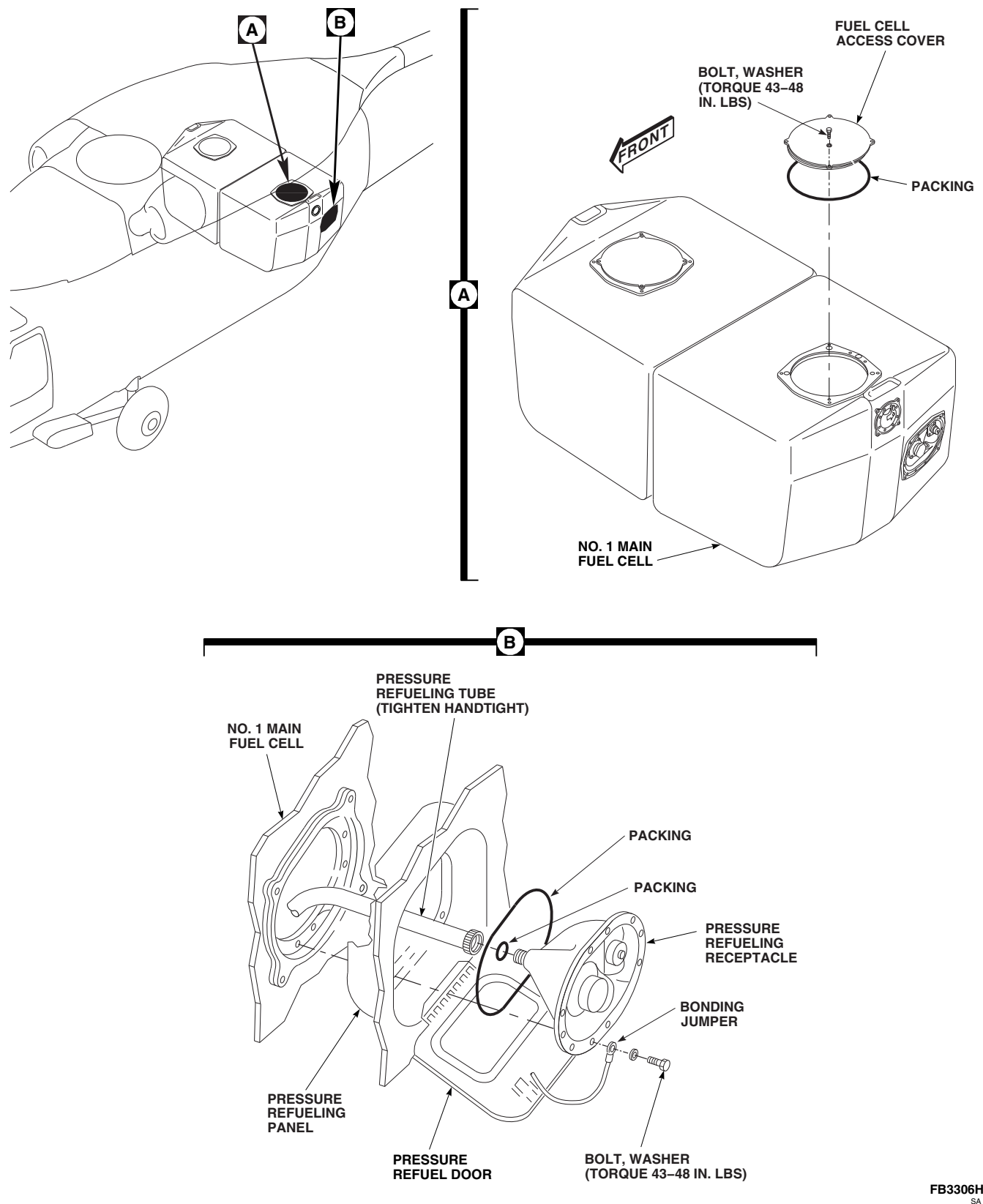


FIGURE 10-4-31-1. REPLACE PRESSURE REFUELING RECEPTACLE

FB3306H
SA

10-4-32 REFUEL ADAPTER AND NIPPLE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-32.1 Replace Refuel Adapter and Nipple	10-4-32-1

INITIAL SETUP

- PARA 1-3-18

Tools

- PARA 1-3-19

- Aircraft Mechanic’s Toolkit

- PARA 1-6-16

Materials/Parts

- Lubricating Oil, Item 205, Appendix D

- PARA 10-4-1

References

- PARA 10-4-40

- Appendix D
- PARA 1-3-17

10-4-32.1 Replace Refuel Adapter and Nipple.

10-4-32.1.2 Install.

NOTE

Refuel adapter and nipple are installed on No. 1 main fuel cell only.

10-4-32.1.1 Remove.

- Purge main fuel cell (PARA 10-4-40).
- Open pressure refuel door.
- Remove cap from refuel adapter (Figure 10-4-32-1, Detail A).
- Remove screws and refuel adapter from pressure refueling receptacle. Discard packing.
- Remove cap from nipple.
- Remove screws and nipple from pressure refueling receptacle. Discard packing.

- Turn off all electrical power (PARA 1-6-16).
- Lubricate packing with lubricating oil, Item 205, Appendix D.
- Install packing in groove on refuel adapter (Figure 10-4-32-1, Detail A).
- Install refuel adapter in pressure refueling receptacle with screws.
- Place cap on refuel adapter.
- Lubricate packing with lubricating oil, Item 205, Appendix D.
- Install packing in groove on nipple.
- Install nipple in pressure refueling receptacle with screws.

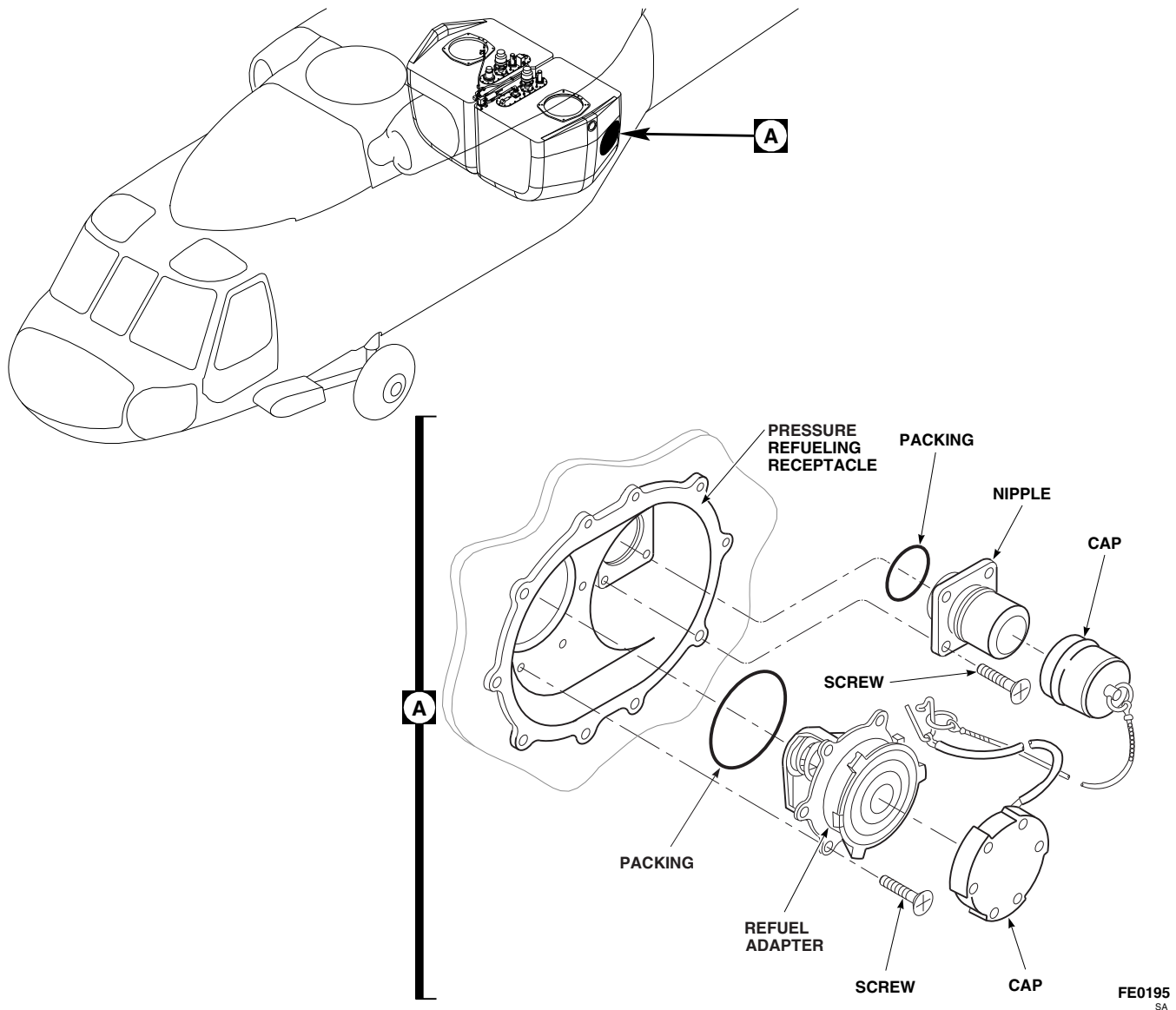


FIGURE 10-4-32-1. REPLACE REFUEL ADAPTER AND NIPPLE

- i. Install cap on nipple.
- j. Refuel main fuel cells (PARA 1-3-17, PARA 1-3-18, or PARA 1-3-19).
- k. Prime fuel system (PARA 10-4-1).

10-4-33 FUEL QUANTITY PROBE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-33.1 Replace Fuel Quantity Probe	10-4-33-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Tags
- Tiedown Strap, Item 331, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-2-5
- PARA 10-2-6
- PARA 10-2-7
- PARA 10-4-40
- PARA 13-5-3
- PARA 13-5-6

10-4-33.1 Replace Fuel Quantity Probe.

NOTE

Replacement of No. 1 and No. 2 fuel quantity probes is the same.

10-4-33.1.1 Remove.

- Purge main fuel cell (PARA 10-4-40).
- Gain access to main fuel cell compartment as follows:
 - Remove rear passenger’s seats as required (PARA 2-4-45).

- Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).
- Remove cargo netting from stowage compartments.
- If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
- Unlock fasteners securing left or right panel enclosure on top of main fuel cells. Remove panel.
- Remove bolts, washers, and fuel cell access cover from main fuel cell. Remove packing

from fuel cell access cover (Figure 10-4-33-1, Detail A). Discard packing.

- e. Disconnect fuel cell wiring harness from fuel quantity probe as follows (Figure 10-4-33-1, Detail C):

- (1) Loosen captive screws and disconnect HIGH Z and LOW Z terminals from probe.
- (2) Loosen screw on clip and remove green wire (HIGH Z wire) from clip on probe.
- (3) Loosen clamps securing green and red wires (HIGH Z and LOW Z wires) and remove wires from wire supports on probe.
- (4) Tag color-coded wiring connected to low-level sensing unit.
- (5) Loosen captive screws and disconnect white, brown, and blue wiring from low-level sensing unit.
- (6) Remove tiedown straps securing harness to probe.

- f. Remove bolts and washers securing fuel quantity probe to inside top of main fuel cell (Figure 10-4-33-1, Detail B).

10-4-33.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

NOTE

Make sure oval boltholes on mounting plate face outboard when installing fuel quantity probe (Figure 10-4-33-1, Detail D).

- b. Install fuel quantity probe to inside top of main fuel cell with bolts and washers (Figure 10-4-33-1, Detail B). **TORQUE BOLTS TO 18 - 23 INCH-POUNDS.**
- c. Connect fuel cell wiring harness to fuel quantity probe as follows (Figure 10-4-33-1, Detail C):

- (1) Position HIGH Z and LOW Z terminals on probe and secure with captive screws.
 - (2) Position ferrule on green wire (HIGH Z wire) in clip on probe and tighten screw.
 - (3) Position green and red wires (HIGH Z and LOW Z wires) in wire supports on probe and secure with clamp.
 - (4) Position white, brown, and blue wiring on low-level sensing unit and secure with captive screws. Remove tags.
 - (5) Install tiedown straps, Item 331, Appendix D, securing harness to probe.
- d. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on fuel cell access cover (Figure 10-4-33-1, Detail A).
- e. Inspect area inside main fuel cell and make sure area is free of all foreign material.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- f. Coat threads of fuel cell access cover attachment bolts with antiseize compound, Item 74, Appendix D.
- g. Install fuel cell access cover on main fuel cell with bolts and washers. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS.**
- h. Install left or right panel enclosure on top of main fuel cells. Lock fasteners.
- i. Close main fuel cell compartment as follows:
- (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.

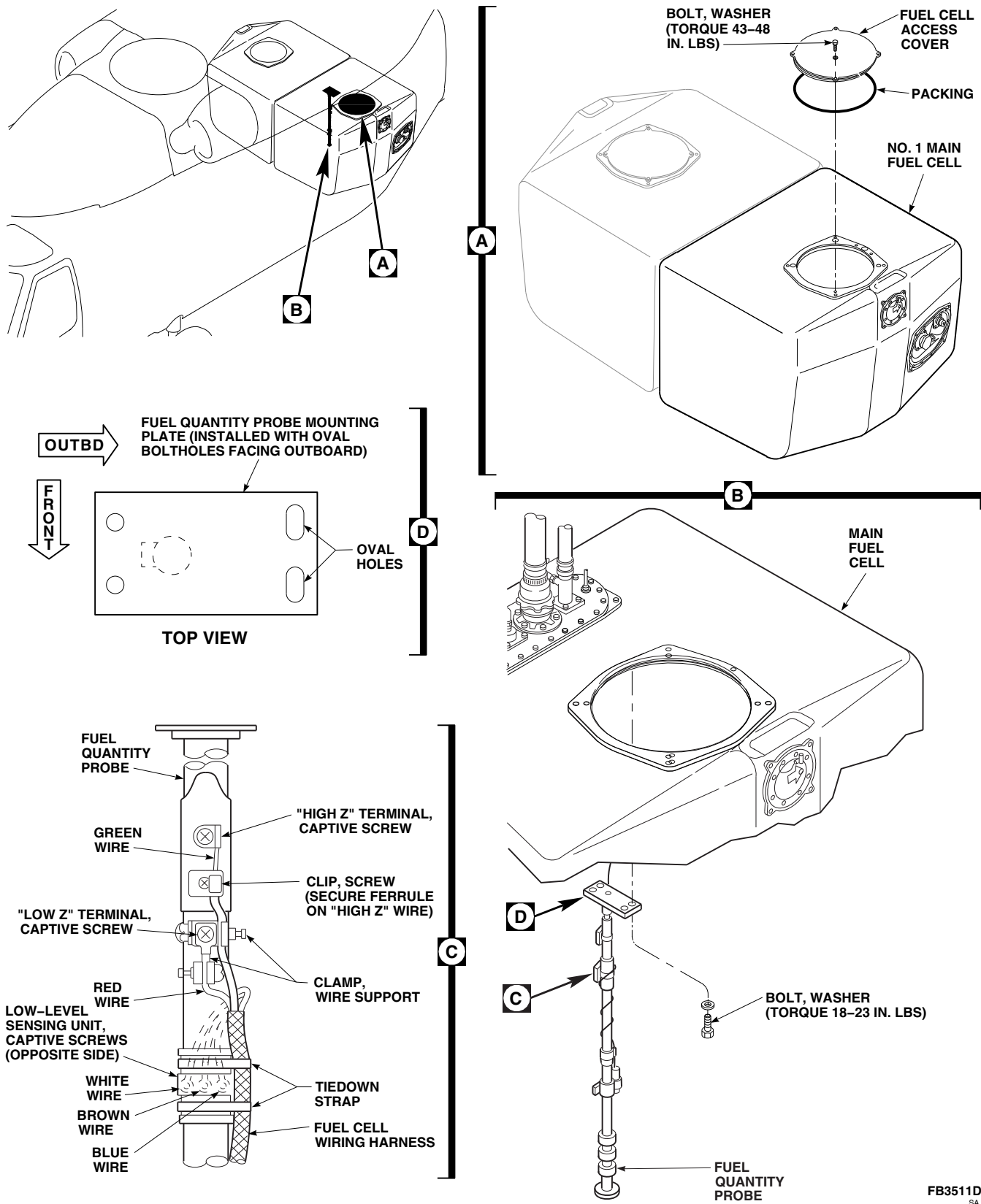


FIGURE 10-4-33-1. REPLACE FUEL QUANTITY PROBE

FB3511D
SA

- (3) Install soundproofing panels (PARA 2-4-69).
- (4) Install rear passenger's seats (PARA 2-4-45).
- j. Perform operational check of fuel quantity system (PARA 10-2-5 or PARA 10-2-6).
- k. Perform operational check of engine feed fuel lines (PARA 10-2-7).

10-4-34 FUEL CELL WIRING HARNESS.

PARAGRAPH BREAKDOWN

	PAGE
10-4-34.1 Replace Fuel Cell Wiring Harness	10-4-34-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Protective Caps and Plugs
- Tags
- Thread, Break-A-Way, Item 327, Appendix D
- Tiedown Strap, Item 331, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-2-5
- PARA 10-2-6
- PARA 10-2-7
- PARA 10-4-40
- PARA 13-5-3
- PARA 13-5-6

10-4-34.1 Replace Fuel Cell Wiring Harness.

NOTE

Replacement of No. 1 and No. 2 main fuel cell wiring harnesses is the same, except as noted.

10-4-34.1.1 Remove.

- Purge main fuel cell (PARA 10-4-40).
- Gain access to main fuel cell compartment as follows:

- Remove rear passenger’s seats as required (PARA 2-4-45).
- Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).
- Remove cargo netting from stowage compartments.
- If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

- c. Unlock fasteners securing left or right panel enclosure on top of main fuel cells. Remove panel.
- d. Disconnect electrical connector, P307 from J307 (No. 1 main fuel cell), or P308 from J308 (No. 2 main fuel cell) (Figure 10-4-34-1, Sheet 1, Detail A).
- e. Install protective caps and plugs on electrical connectors.
- f. Remove bolts, washers, and fuel cell access cover from main fuel cell (Figure 10-4-34-1, Sheet 1, Detail B). Discard packing.
- g. Tag locations of tiedown straps, mounting plates, and break-a-way thread before removing fuel cell wiring harness.
- h. Disconnect fuel cell wiring harness from fuel quantity probe as follows (Figure 10-4-34-1, Sheet 1, Detail C):
 - (1) Loosen captive screws and disconnect HIGH Z and LOW Z terminals from probe.
 - (2) Loosen screw on clip and remove green wire (HIGH Z wire) from clip on probe.
 - (3) Loosen clamps securing green and red wires (HIGH Z and LOW Z wires) and remove wires from wire supports on probe.
 - (4) Tag color-coded wiring connected to low-level sensing unit.
 - (5) Loosen captive screws and disconnect white, brown, and blue wiring from low-level sensing unit.
 - (6) Remove tiedown straps securing harness to probe.
- i. Straighten tab on lockwasher. Remove jamnut, lockwasher, and wiring harness from main fuel cell plate (Figure 10-4-34-1, Sheet 1, Detail D). Remove and discard packing.
- j. Remove tiedown straps, mounting plates, and cut break-a-way thread securing fuel cell wir-

ing harness to lower main fuel line and airframe (Figure 10-4-34-1, Sheet 2, Detail E and Detail F).

- k. Remove wiring harness.

10-4-34.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install packing on fuel cell wiring harness feed-through (Figure 10-4-34-1, Sheet 1, Detail D).
- c. Install fuel cell wiring harness and feed-through in main fuel cell plate. Secure feed-through to plate with lockwasher and jamnut. **TIGHTEN JAMNUT $\frac{1}{4}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Bend over tab of lockwasher against flat on jamnut.
- d. Connect fuel cell wiring harness to fuel quantity probe as follows (Figure 10-4-34-1, Sheet 1, Detail C):
 - (1) Position HIGH Z and LOW Z terminals on probe and secure with captive screws.
 - (2) Position ferrule on green wire (HIGH Z wire) in clip on probe and tighten screw.
 - (3) Position green and red wires (HIGH Z and LOW Z wires) in wire supports on probe and secure with clamp.
 - (4) Position white, brown, and blue wiring on low-level sensing unit and secure with captive screws. Remove tags.
 - (5) Install tiedown straps, Item 331, Appendix D, securing harness to probe.
- e. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on fuel cell access cover (Figure 10-4-34-1, Sheet 1, Detail B).
- f. Inspect area inside main fuel cell and make sure area is free of all foreign material.

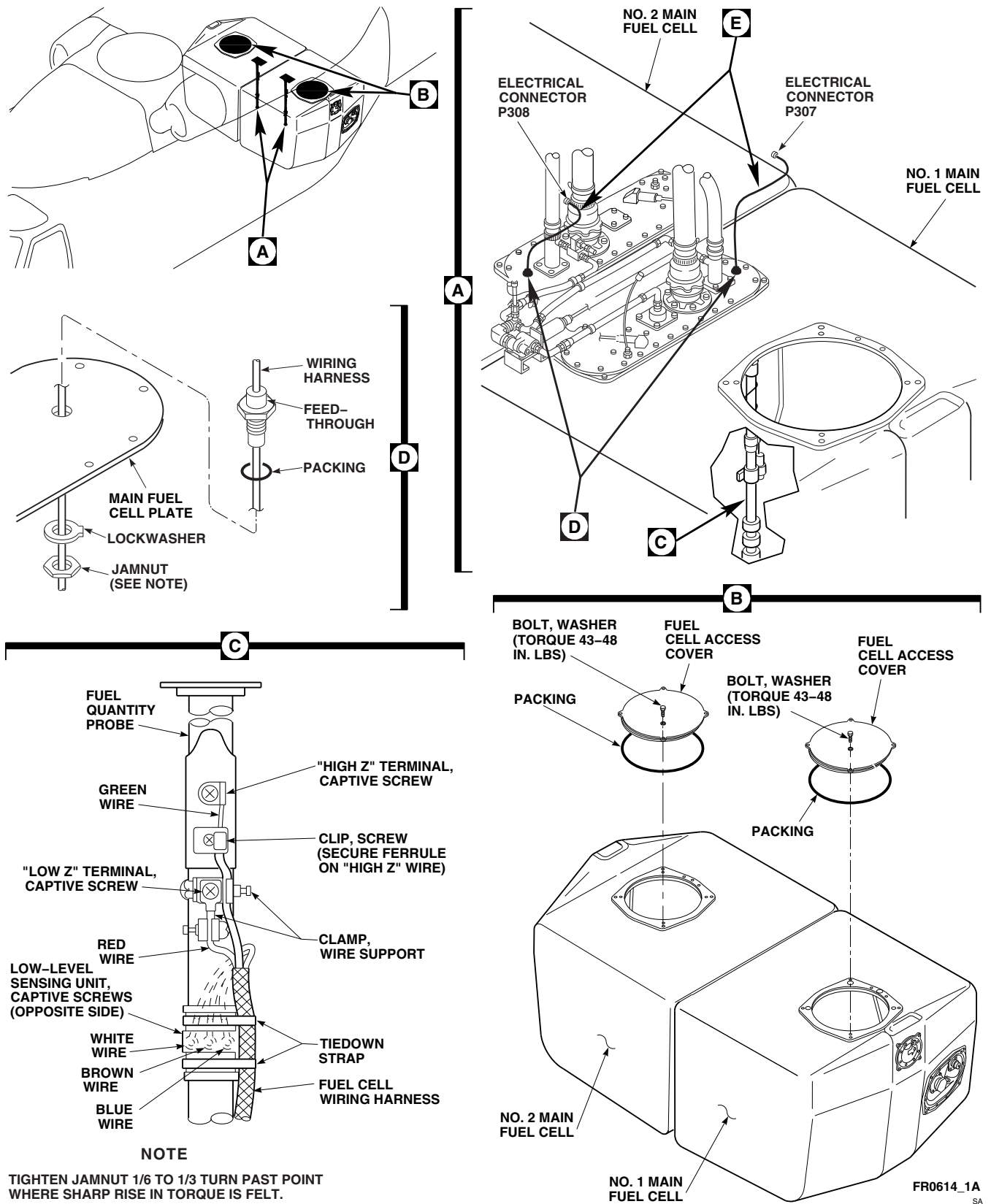
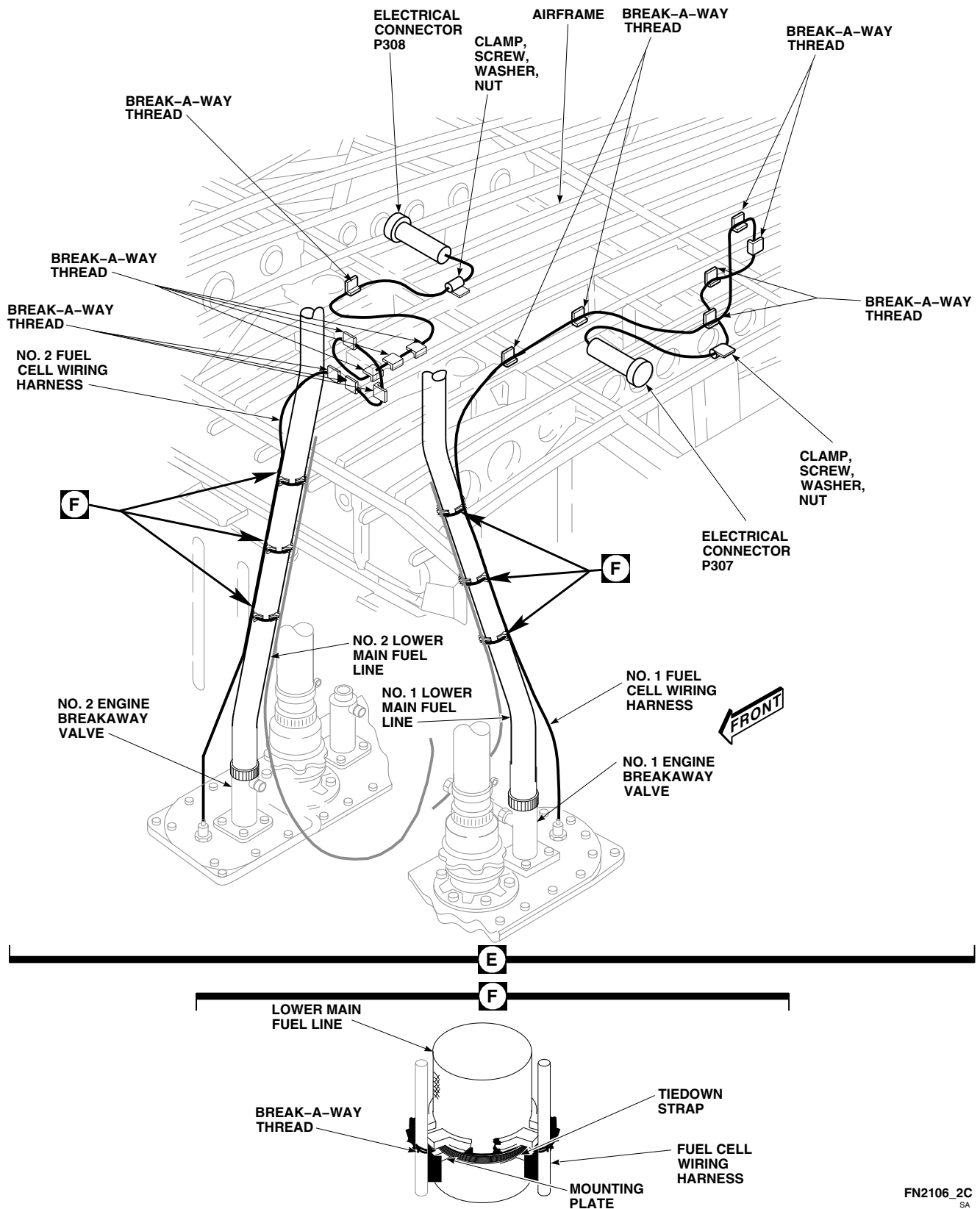


FIGURE 10-4-34-1. REPLACE FUEL CELL WIRING HARNESS (SHEET 1 OF 2)



FN2106_2C
SA

FIGURE 10-4-34-1. REPLACE FUEL CELL WIRING HARNESS (SHEET 2 OF 2)

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- g. Coat threads of fuel cell access cover attachment bolts with antiseize compound, Item 74, Appendix D.
- h. Install fuel cell access cover on main fuel cell with bolts and washers. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS.**
- i. Remove protective caps and plugs from electrical connectors.
- j. Inspect and treat electrical connectors (PARA 1-7-20).
- k. Connect electrical connector, P307 to J307 (No. 1 main fuel cell), or P308 to J308 (No. 2 main fuel cell) (Figure 10-4-34-1, Sheet 1, Detail A).
- l. Install tiedown straps, Item 331, Appendix D, at tagged locations securing wiring harness to lower main fuel line and remove tags (Figure 10-4-34-1, Sheet 2, Detail E and Detail F).
- m. Install mounting plates at tagged locations and remove tags.

NOTE

Assemble a "single loop" of break-a-way thread for harness diameter of $\frac{3}{16}$ -inch or less. For harnesses with diameters over $\frac{3}{16}$ -inch, assemble "two loops" maximum.

- n. Install break-a-way thread, Item 327, Appendix D, at tagged location securing wiring harness to airframe, looping around wiring harness and mounting plate. Tie with knot and remove tags.
- o. Install left or right panel enclosure on top of main fuel cells. Lock fasteners.
- p. Close main fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panels (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).
- q. Perform operational check of fuel quantity system (PARA 10-2-5 or PARA 10-2-6).
- r. Perform operational check of engine feed fuel lines (PARA 10-2-7).

10-4-35 LOW LEVEL WARNING CONDITIONER.

PARAGRAPH BREAKDOWN

	PAGE
10-4-35.1 Replace Low Level Warning Conditioner	10-4-35-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D

- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69

10-4-35.1 Replace Low Level Warning Conditioner.

dampened with dry-cleaning solvent, Item 124, Appendix D.

10-4-35.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove soundproofing panels (PARA 2-4-69).
- Disconnect electrical connector, P255, from low level warning conditioner (Figure 10-4-35-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove screws, lockwashers, washers, and low level warning conditioner from overhead supports.

- Place low level warning conditioner on overhead supports with receptacle facing left side of cabin, and fasten with screws, lock-washers, and washers (Figure 10-4-35-1, Detail A).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P255, to low level warning conditioner.
- Turn on electrical power (PARA 1-6-16).
- On MISC SWITCH panel on lower console, press FUEL GAGE TEST button and observe as follows:
 - #1 FUEL LOW and #2 FUEL LOW capsules on caution/advisory panel light up and flash.

10-4-35.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Clean surface around all mounting holes using machinery towel, Item 211, Appendix D,

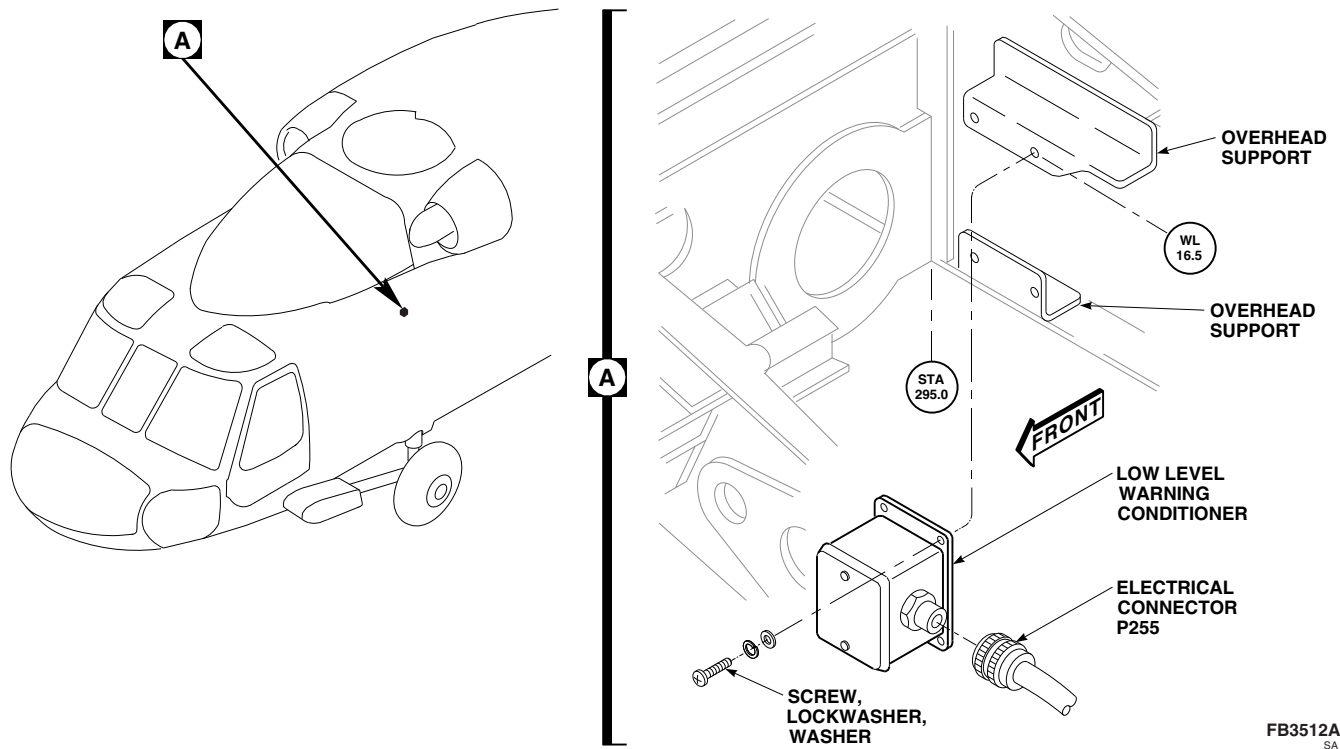


FIGURE 10-4-35-1. REPLACE LOW LEVEL WARNING CONDITIONER

- (2) MASTER CAUTION lights on master warning panels light up.
- i. Turn off all electrical power (PARA 1-6-16).
- j. Make sure area is clean and free of foreign material.
- k. Install soundproofing panels (PARA 2-4-69).

10-4-36 FUEL QUANTITY SIGNAL CONDITIONER.

PARAGRAPH BREAKDOWN

	PAGE
10-4-36.1 Replace Fuel Quantity Signal Conditioner	10-4-36-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Digital Low-Resistance Ohmmeter
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 2-6-5
- PARA 10-2-5
- PARA 10-2-6
- PARA 10-2-7

10-4-36.1 Replace Fuel Quantity Signal Conditioner.

10-4-36.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove soundproofing panels (PARA 2-4-69).
- Disconnect electrical connector, P905, from fuel quantity signal conditioner (Figure 10-4-36-1, Detail A).
- Install protective caps and plugs on electrical connectors.

NOTE

EMEP Do not place fuel quantity signal conditioner on its chemical conversion-coated surfaces.

- Remove screws, lockwashers, and washers securing fuel quantity signal conditioner **EMEP** and bonding straps to overhead supports.

10-4-36.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Clean around all mounting holes with cheesecloth, Item 90, Appendix D, dampened

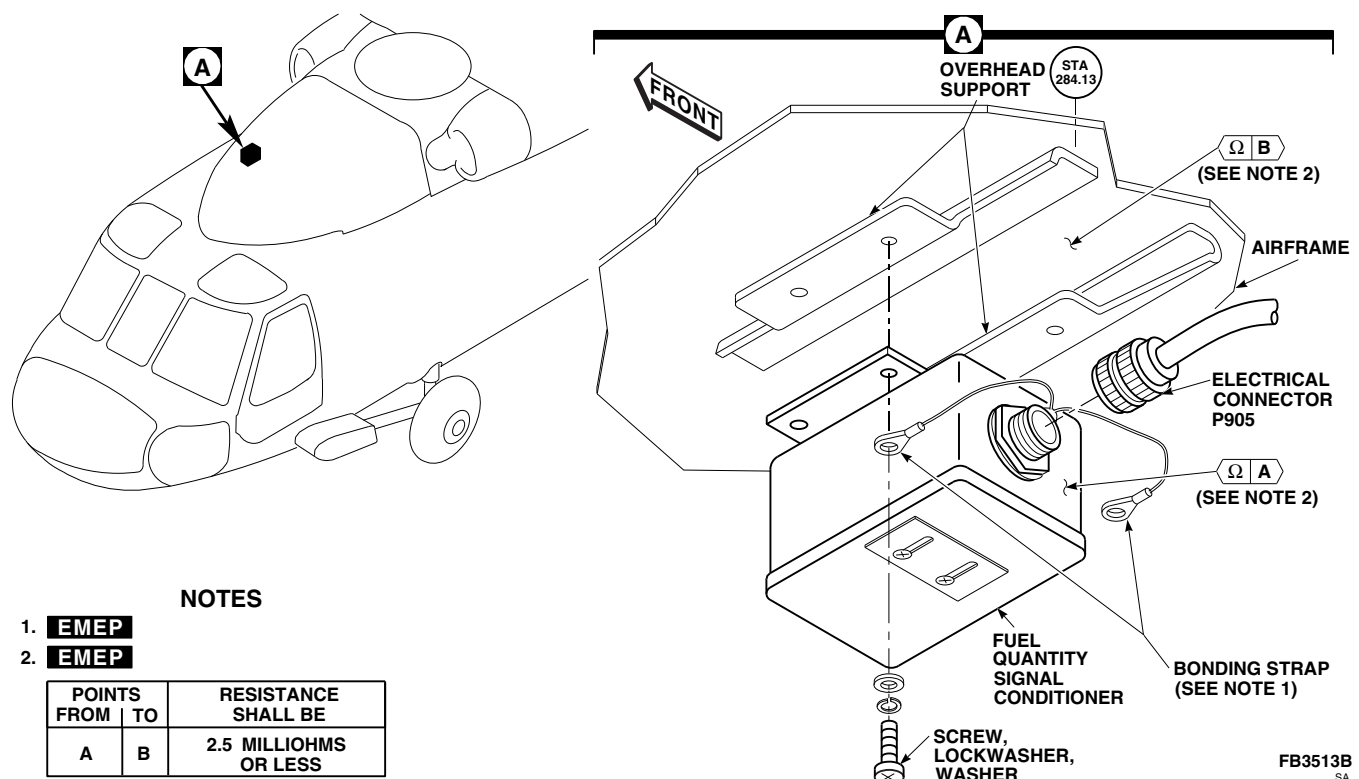


FIGURE 10-4-36-1. REPLACE FUEL QUANTITY SIGNAL CONDITIONER

with dry-cleaning solvent, Item 124, Appendix D. Dry surfaces with clean, dry cheesecloth, Item 90, Appendix D.

- c. **EMEP** Inspect chemical conversion-coated surfaces for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5). 
- d. Position fuel quantity signal conditioner **EMEP** and bonding straps  on overhead supports, with receptacle facing left side of cabin, and secure with screws, lockwashers, and washers (Figure 10-4-36-1, Detail A).
- e. **EMEP** Perform EME resistance check of fuel quantity signal conditioner as follows: 

NOTE

Resistance measurement must be taken at least three times.

- (1) Using digital low-resistance ohmmeter, measure resistance from enclosure of fuel

quantity signal conditioner and airframe.
**RESISTANCE SHALL BE 2.5 MILLI-
OHMS OR LESS.**

- (2) If resistance is greater than 2.5 milliohms, remove conditioner and replace chemical conversion coating (PARA 2-6-5).
- f. Remove protective caps and plugs from electrical connectors.
- g. Inspect and treat electrical connectors (PARA 1-7-20).
- h. Connect electrical connector, P905, to fuel quantity signal conditioner.
- i. Perform operational check of fuel quantity system (PARA 10-2-5 or PARA 10-2-6).
- j. Perform operational check of engine feed fuel lines (PARA 10-2-7).
- k. Make sure area is clean and free of foreign material.
- l. Install soundproofing panels (PARA 2-4-69).

10-4-37 FIXED BLEED-AIR LINES AND DISCONNECTS.

PARAGRAPH BREAKDOWN

	PAGE
10-4-37.1 Replace Fixed Bleed-Air Lines and Disconnects	10-4-37-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16

10-4-37.1 Replace Fixed Bleed-Air Lines and Disconnects.

NOTE

Replacement of left and right bleed-air lines and disconnects is the same.

10-4-37.1.1 Remove.

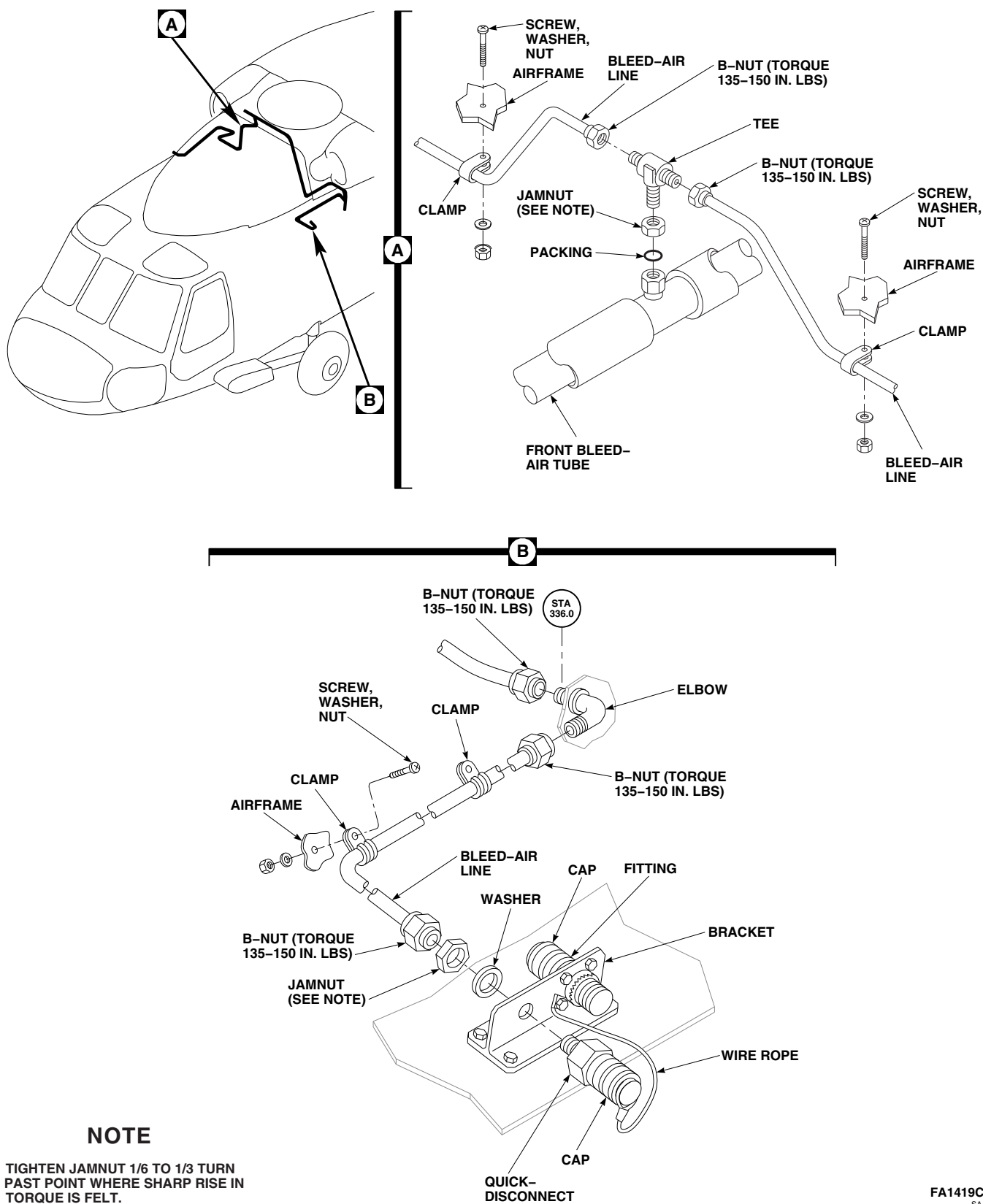
- Turn off all electrical power (PARA 1-6-16).
- Open main rotor pylon sliding cover.
- Remove screws, washers, and nuts securing clamps and bleed-air line to airframe (Figure 10-4-37-1, Detail A). Remove clamps from bleed-air line.

NOTE

Keep attaching hardware with tubing for future use.

- Disconnect bleed-air line from tee in front bleed-air tube.

- Remove tee from front bleed-air tube. Remove packing and jamnut from tee. Discard packing. Install protective cap or plug on tee.
- Disconnect bleed-air line from elbow in airframe at STA 336.0, and remove line (Figure 10-4-37-1, Detail B). Install protective caps or plugs on lines.
- Open rear fairing on upper external stores support system fitting fairing and platform.
- Remove screws, washers, nuts, and clamps securing bleed-air line to airframe.
- Disconnect bleed-air line from elbow in airframe at STA 336.0.
- Disconnect bleed-air line from quick-disconnect and remove line. Install protective cap or plug on line.
- Remove protective cap from quick-disconnect. Remove jamnut and washer from quick-disconnect and remove quick-disconnect from bracket.



10-4-37.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install quick-disconnect in bracket using jam-nut and washer (Figure 10-4-37-1, Detail B). **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- c. Install cap on quick-disconnect with wire rope ends properly secured. **MAXIMUM GAP ALLOWED IS 0.30-INCH BETWEEN INTERFACE OF CAP AND PLATE.** If limit is exceeded, inspect threads on cap and quick-disconnect for damage. **NONE ALLOWED.** If damage is found, replace cap or quick-disconnect.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- d. Coat threads of quick-disconnect, installed in bracket, with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D. Connect bleed-air line to quick-disconnect. **TORQUE B-NUT ON LINE TO 135 - 150 INCH-POUNDS.**
- e. Secure bleed-air line to elbow in airframe at STA 336.0. **TORQUE B-NUT ON LINE TO 135 - 150 INCH-POUNDS.**
- f. Install clamps on bleed-air line and secure line and clamps to airframe with screws, washers, and nuts.

- g. Close rear fairing on upper external stores support system fitting fairing and platform.
- h. Lubricate packing with lubricating oil, Item 205, Appendix D. Install jamnut and packing on tee (Figure 10-4-37-1, Detail A).

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- i. Coat threads of tee with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.
- j. Connect bleed-air line to elbow in airframe (Figure 10-4-37-1, Detail B).
- k. Line up and connect bleed-air line to tee (Figure 10-4-37-1, Detail A). **TIGHTEN JAMNUT ON TEE 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT. TORQUE B-NUTS ON LINES AT ELBOWS AND TEE TO 135 - 150 INCH-POUNDS** (Figure 10-4-37-1, Detail A and Detail B).
- l. Install clamps on bleed-air line and secure line and clamps to airframe with screws, washers, and nuts (Figure 10-4-37-1, Detail A).
- m. Make sure area is clean and free of foreign material.
- n. Close main rotor pylon sliding cover.

10-4-38 FIXED FUEL LINES AND DISCONNECTS.

PARAGRAPH BREAKDOWN

	PAGE		PAGE
10-4-38.1 Replace Fixed Fuel Lines and Tension Clips	10-4-38-1	10-4-38.2 Replace Fitting Assembly and Bulkhead Connectors	10-4-38-5

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Gallon
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 2, Appendix D
- Acetone, Item 25, Appendix D
- Adhesive, Item 49, Appendix D
- Adhesive, Item 53, Appendix D

- Antiseize Compound, Item 74, Appendix D
- Cheesecloth, Item 90, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Machinery Towel, Item 211, Appendix D
- Spiral Wrap, Item 315A, Appendix D
- Tiedown Strap, Item 329, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69
- PARA 16-4-24

10-4-38.1 Replace Fixed Fuel Lines and Tension Clips.

NOTE

Replacement of left and right side fixed fuel lines and tension clips is the same, except as noted.

10-4-38.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove soundproofing panels (PARA 2-4-69).

- If required, disconnect ESSS horizontal stores support (HSS) fuel line from elbow quick-disconnect valve (PARA 16-4-24).



Use 1-gallon container to catch remaining fuel as lines are disconnected.

- Disconnect fuel line from breakaway valve located at STA 398.0 (Figure 10-4-38-1, Sheet 2, Detail G).

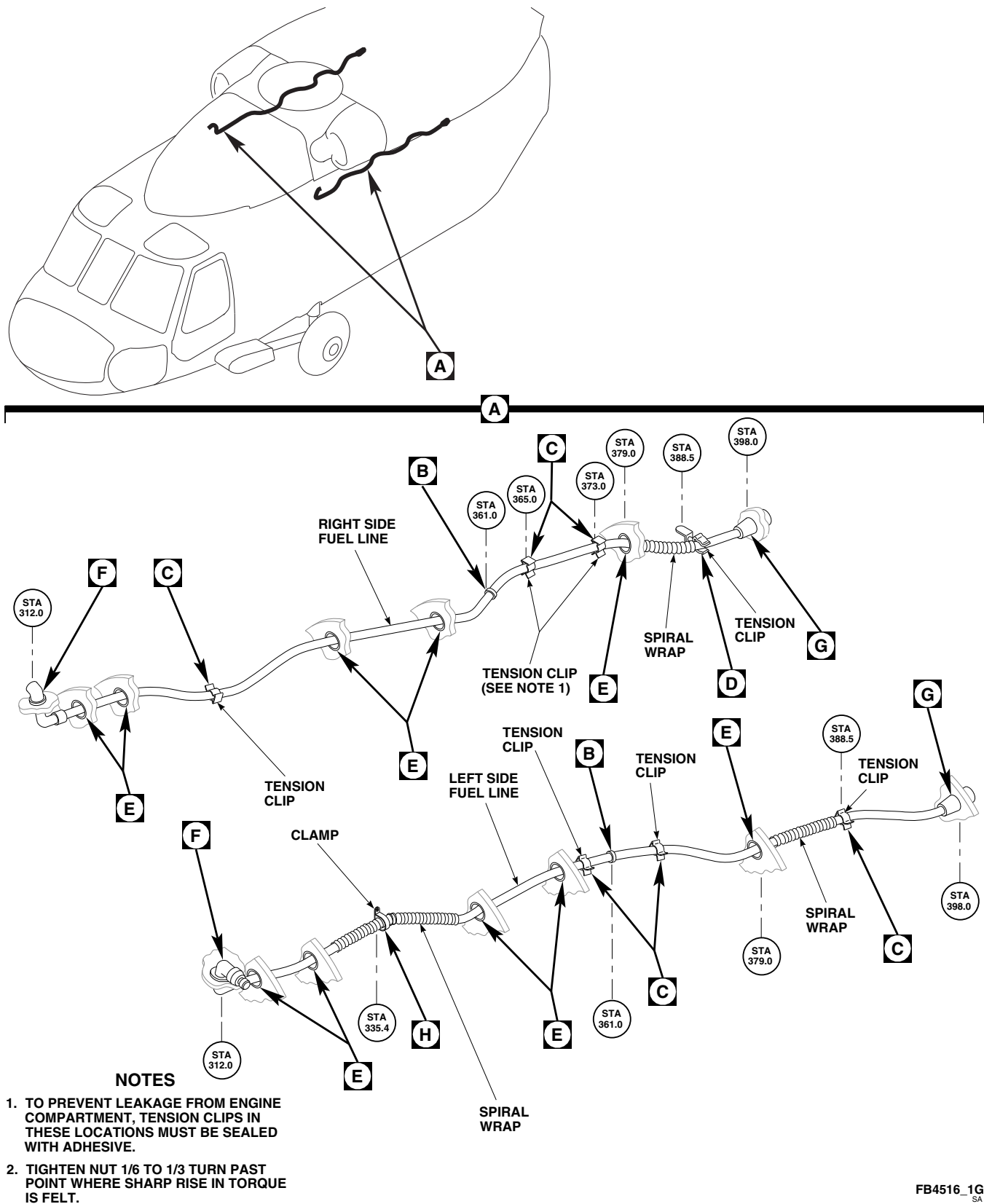
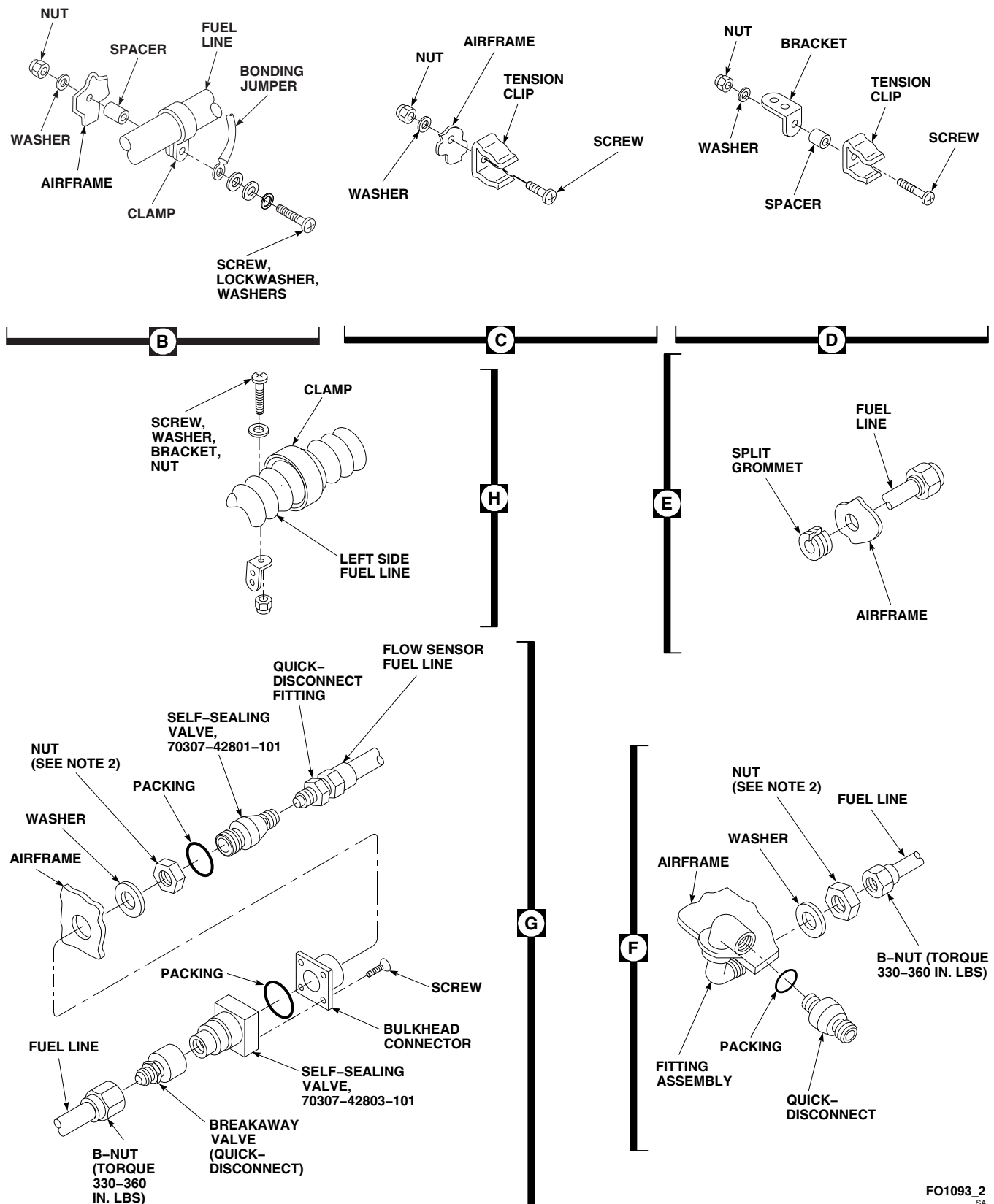


FIGURE 10-4-38-1. REPLACE FIXED FUEL LINES AND DISCONNECTS (SHEET 1 OF 2)



FO1093_2
SA

FIGURE 10-4-38-1. REPLACE FIXED FUEL LINES AND DISCONNECTS (SHEET 2 OF 2)

- e. Disconnect fuel line from fitting assembly located at STA 312.0 (Figure 10-4-38-1, Sheet 2, Detail F).
 - f. Remove split grommets from airframe and fuel line (Figure 10-4-38-1, Sheet 2, Detail E). Discard grommets.
 - g. Remove screw, lockwasher, washers, spacer, and nut securing fuel line, clamps, and bonding jumper to airframe (Figure 10-4-38-1, Sheet 2, Detail B). Remove clamp.
 - h. On left side fuel line, remove screw, washer, and nut securing clamp and fuel line to bracket at STA 335.4. Remove clamp from fuel line (Figure 10-4-38-1, Sheet 2, Detail H).
 - i. Cut tiedown straps securing spiral wrap to fuel lines. Remove spiral wrap from fuel lines and discard (Figure 10-4-38-1, Sheet 1, Detail A).
 - j. Remove fuel line from tension clips, and pull fuel line from airframe.
- (a) Using nonmetallic scraper and cheesecloth, Item 90, Appendix D, dampened with acetone, Item 25, Appendix D, remove any adhesive around mounting holes. Do not allow solvent to dry on airframe. Keep turning cloth after each wipe.
 - (b) Coat threads of screw with adhesive, Item 53, Appendix D.
 - (c) Install tension clip to airframe with screw, washer, and nut.
 - (d) Apply a bead of adhesive, Item 53, Appendix D, where nut contacts airframe.
- (3) Install all other tension clips to airframe, using screw, washer, and nut (Figure 10-4-38-1, Sheet 2, Detail C).
- c. Using nonmetallic scraper and cheesecloth, Item 90, Appendix D, dampened with acetone, Item 25, Appendix D, clean edge of holes where split grommets are installed. Do not allow solvent to dry on airframe. Keep turning cloth after each wipe.

NOTE

Removal of tension clips are identical, except for tension clip located at STA 388.5 right side.

- k. If required, remove screw, washer, nut, and, if present, spacer securing tension clip to airframe and/or bracket. Remove tension clip from aircraft (Figure 10-4-38-1, Sheet 2, Detail C and Detail D).

10-4-38.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. If required, install tension clips as follows:
 - (1) Install tension clip at STA 388.5 right side to bracket using screw, spacer, washer, and nut (Figure 10-4-38-1, Sheet 2, Detail D).
 - (2) Install tension clips at STA 365.0 right side and STA 373.0 right side as follows (Figure 10-4-38-1, Sheet 2, Detail C):
 - g. Install grommets in hole in airframe (Figure 10-4-38-1, Sheet 2, Detail E). For best results, allow adhesive to cure for 24 hours.
 - h. Connect fuel line to fitting assembly located at STA 312.0 (Figure 10-4-38-1, Sheet 2, Detail F). **TORQUE B-NUT ON FUEL LINE TO 330 - 360 INCH-POUNDS.**
 - i. Install breakaway valve to fuel line (Figure 10-4-38-1, Sheet 2, Detail G). **TORQUE B-NUT ON FUEL LINE TO 330 - 360 INCH-**

POUNDS. Connect breakaway valve quick-disconnect fitting to self-sealing valve located at STA 398.0.

- j. Install clamp on fuel line. Secure fuel line, clamp, and bonding jumper to airframe with clamp, screw, lockwasher, washers, spacer, and nut (Figure 10-4-38-1, Sheet 2, Detail B).

NOTE

Make sure fuel lines are depressurized following pressure test.

- k. Pressure test fuel lines for leaks with regulated air, set at 60 psi, for five minutes. **NONE ALLOWED.** If leakage is found, remove and reinstall fuel line.
- l. Install spiral wrap, Item 315A, Appendix D, on fuel line. Wrap line starting 1-inch from STA 379.0 and wrap to within 1-inch of tension clip located at STA 388.5. Secure both ends of spiral wrap with tiedown straps, Item 329, Appendix D (Figure 10-4-38-1, Sheet 1, Detail A).
- m. On left side fuel line, locate bracket at STA 335.4. Measure 7.0-inches forward from bracket and mark fuel line. Measure 5.0-inches from bracket toward rear and mark fuel line. Starting at forward mark, wrap fuel line between measurements using spiral wrap, Item 315A, Appendix D. Secure both ends of spiral wrap with tiedown straps, Item 329, Appendix D.
- n. On left fuel line, install clamp on fuel line and secure to bracket at STA 335.4 with screw, washer, and nut (Figure 10-4-38-1, Sheet 2, Detail H).
- o. Install soundproofing panels (PARA 2-4-69).
- p. If required, connect ESSS horizontal stores support (HSS) fuel line to elbow quick-disconnect valve (PARA 16-4-24).

10-4-38.2 Replace Fitting Assembly and Bulkhead Connectors.

NOTE

Replacement of left and right side fitting assembly and bulkhead connectors is the same.

10-4-38.2.1. Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove soundproofing panels (PARA 2-4-69).
- c. If required, disconnect ESSS horizontal stores support (HSS) fuel line from fitting assembly quick-disconnect valve (PARA 16-4-24).



Use 1-gallon container to catch fuel as lines are disconnected.

- d. Remove fitting assembly as follows:
 - (1) Unscrew cap from fitting assembly and place on dummy receptacle.
 - (2) Disconnect fuel line from fitting assembly (Figure 10-4-38-1, Sheet 2, Detail F).
 - (3) Remove washer and nut securing fitting assembly to airframe. Remove fitting assembly from airframe.
 - (4) If required, remove quick-disconnect and packing from fitting assembly. Discard packing.



Use 1-gallon container to catch fuel as lines are disconnected.

- e. Remove bulkhead connectors as follows:

- (1) Disconnect breakaway valve quick-disconnect fitting, 70307-42803-101, from self-sealing valve (Figure 10-4-38-1, Sheet 2, Detail G).
- (2) Disconnect flow sensor fuel line quick-disconnect fitting from self-sealing valve, 70307-42801-101.
- (3) Disconnect self-sealing valve, 70307-42801-101, from bulkhead connector. Remove and discard packing.
- (4) Remove nut and washer securing bulkhead connector to airframe. Remove bulkhead connector and attached components from aircraft.
- (5) Remove screws securing bulkhead connector to self-sealing valve, 70307-42803-101. Remove and discard packing.

10-4-38.2.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Lubricate packings with lubricating oil, Item 205, Appendix D, prior to installation.
- c. Install fitting assembly as follows:
 - (1) If required, install quick-disconnect to fitting assembly as follows:
 - (a) Install packing on quick-disconnect (Figure 10-4-38-1, Sheet 2, Detail F).

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into the fuel system.

- (b) Coat threads of quick-disconnect with antiseize compound, Item 74, Appendix D, sparingly ($\frac{1}{4}$ -inch strip) along length of male threads at

connection points. Wipe off any squeeze out after parts are assembled with machinery towel, Item 211, Appendix D. Install quick-disconnect on fitting assembly.

- (2) Install fitting assembly into airframe and secure, using washer and nut. **TIGHTEN NUT $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- (3) Connect fuel line to fitting assembly. **TORQUE B-NUT ON FUEL LINE TO 330 - 360 INCH-POUNDS.**
- (4) Remove cap from dummy receptacle and install on quick-disconnect.

NOTE

Installation of left and right side bulkhead connectors is identical.

- d. Install bulkhead connectors as follows:

- (1) Install packing in groove located on bulkhead connector (Figure 10-4-38-1, Sheet 2, Detail G).
- (2) Secure bulkhead connector to self-sealing valve, 70307-42803-101, using screws.
- (3) Install bulkhead connector to airframe and secure using washer and nut. **TIGHTEN NUT $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- (4) Install packing on self-sealing valve, 70307-42801-101.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into the fuel system.

- (5) Coat threads of self-sealing valve, 70307-42801-101, with antiseize compound,

Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeeze out after parts are assembled with machinery towel, Item 211, Appendix D. Install on bulkhead connector.

- (6) Connect flow sensor fuel line quick-disconnect fitting to self-sealing valve, 70307-42801-101.
- (7) Connect breakaway valve quick-disconnect fitting to self-sealing valve, 70307-42803-101.

NOTE

Make sure fuel lines are depressurized following pressure test.

- e. Pressure test fuel lines for leaks with regulated air, set at 60 psi, for five minutes. **NONE ALLOWED.** If leakage is found, remove and reinstall fitting assembly and/or bulkhead connector.
- f. Install soundproofing panels (PARA 2-4-69).
- g. If required, connect ESSS horizontal stores support (HSS) fuel line to elbow quick-disconnect valve (PARA 16-4-24).

10-4-39 FUEL FEED SYSTEM.

PARAGRAPH BREAKDOWN

	PAGE		PAGE
10-4-39.1	Purge Fuel Feed System	10-4-39-2	10-4-39.2
		Test Fuel Feed System	10-4-39-6

INITIAL SETUP

Tools

- Air Source, Pressure-Regulated
- Aircraft Mechanic’s Toolkit
- Cap, AN929-6
- Cap, AN929-16
- Container, 1-Gallon
- Fuel Feed System Test Hose Assembly, Locally-Made, Figure H-229, Appendix H
- Metering Valve, MV200B
- Plug, AN806-6
- Plug, AN806-12
- Reducer, AN919-16
- Shutoff Valve, MS29530-4
- Tee, AN824-4
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Union, AN814-4
- Union, AN816-4
- Vacuum/Pressure Gage, TE-302T60
- Vacuum/Pump, 70700-77325-046

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Leak Detection Fluid, Item 189, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- Appendix H
- Appropriate Flight Manual
- PARA 1-3-17
- PARA 1-3-18
- PARA 1-3-19
- PARA 1-3-20
- PARA 1-3-21
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 2-4-115
- PARA 8-2-4
- PARA 10-2-2
- PARA 10-2-5

- PARA 10-2-6
- PARA 10-2-7
- PARA 10-4-1
- PARA 10-4-8
- PARA 10-4-13
- PARA 10-4-40
- PARA 13-5-3
- PARA 13-5-6

10-4-39.1 Purge Fuel Feed System.

NOTE

- The following procedure purges the entire fuel feed system to reduce the amount of spillage when lines are disconnected.
- The following procedure shall be performed any time maintenance or testing is performed on fuel feed system.
- The following procedure can be performed independently on either No. 1 or No. 2 fuel feed system.
- Purging of No. 1 and No. 2 fuel feed systems is the same, except as noted.

10-4-39.1.1 Purge.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

- a. Turn off all electrical power (PARA 1-6-16).

NOTE

If performing purge of fuel feed system components outside main fuel cells, partial defueling is required and gravity fill caps must be removed to prevent internal fuel pressure buildup.

- b. If performing purge of fuel feed system components outside main fuel cells, partially defuel helicopter until No. 1 and No. 2 main fuel cells are below 900 pounds (PARA 1-3-20 or PARA 1-3-21).
- c. If performing purge of entire fuel feed system, perform the following:
 - (1) Defuel main fuel cells (PARA 1-3-20 or PARA 1-3-21).
 - (2) Drain fuel cell sump (PARA 1-3-21).
 - (3) Purge main fuel cell (PARA 10-4-40).
- d. Gain access to main fuel cell compartment as follows:
 - (1) Remove rear passenger's seats as required (PARA 2-4-45).
 - (2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).
 - (3) Remove cargo netting from stowage compartments.
 - (4) If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
- e. Unlock fasteners securing left or right panel enclosure on top of main fuel cells. Remove panel.
- f. If performing purge of entire fuel feed system, remove bolts, washers, and fuel cell access cover from main fuel cell (Figure 10-4-39-1, Sheet 1, Detail A). Discard packing.

- g. Open left and/or right engine cowling(s).
- h. Remove IR jammer panel (PARA 2-4-115).
- i. Disconnect front fuel hose as follows:
 - (1) If purging No. 1 fuel feed system, position 1-gallon container to catch fuel and disconnect No. 1 front fuel hose from No. 1 engine pump elbow (Figure 10-4-39-1, Sheet 1, Detail B).
 - (2) If purging No. 2 fuel feed system, position 1-gallon container to catch fuel and disconnect No. 2 front fuel hose from No. 2 engine pump elbow (Figure 10-4-39-1, Sheet 1, Detail C).
- j. Connect pressure-regulated air source, with shutoff valve and pressure gage, to front fuel hose.
- k. Set engine control quadrant controls as follows (Figure 10-4-39-1, Sheet 2, Detail D):
 - (1) If purging No. 1 fuel feed system, on engine control quadrant in cockpit, place NO. 1 ENG FUEL SYS selector lever to DIR, and place NO. 2 ENG FUEL SYS selector lever to OFF.
 - (2) If purging No. 2 fuel feed system, on engine control quadrant in cockpit, place NO. 2 ENG FUEL SYS selector lever to DIR, and place NO. 1 ENG FUEL SYS selector lever to OFF.
- l. Remove No. 1 and/or No. 2 gravity fill cap(s) as follows (Figure 10-4-39-1, Sheet 2, Detail E):
 - (1) Release thumb lock on top of cap by pulling up on thumb lock.



Injury to personnel will result if main fuel cell pressure is not removed prior to removing filler cap. Before turning

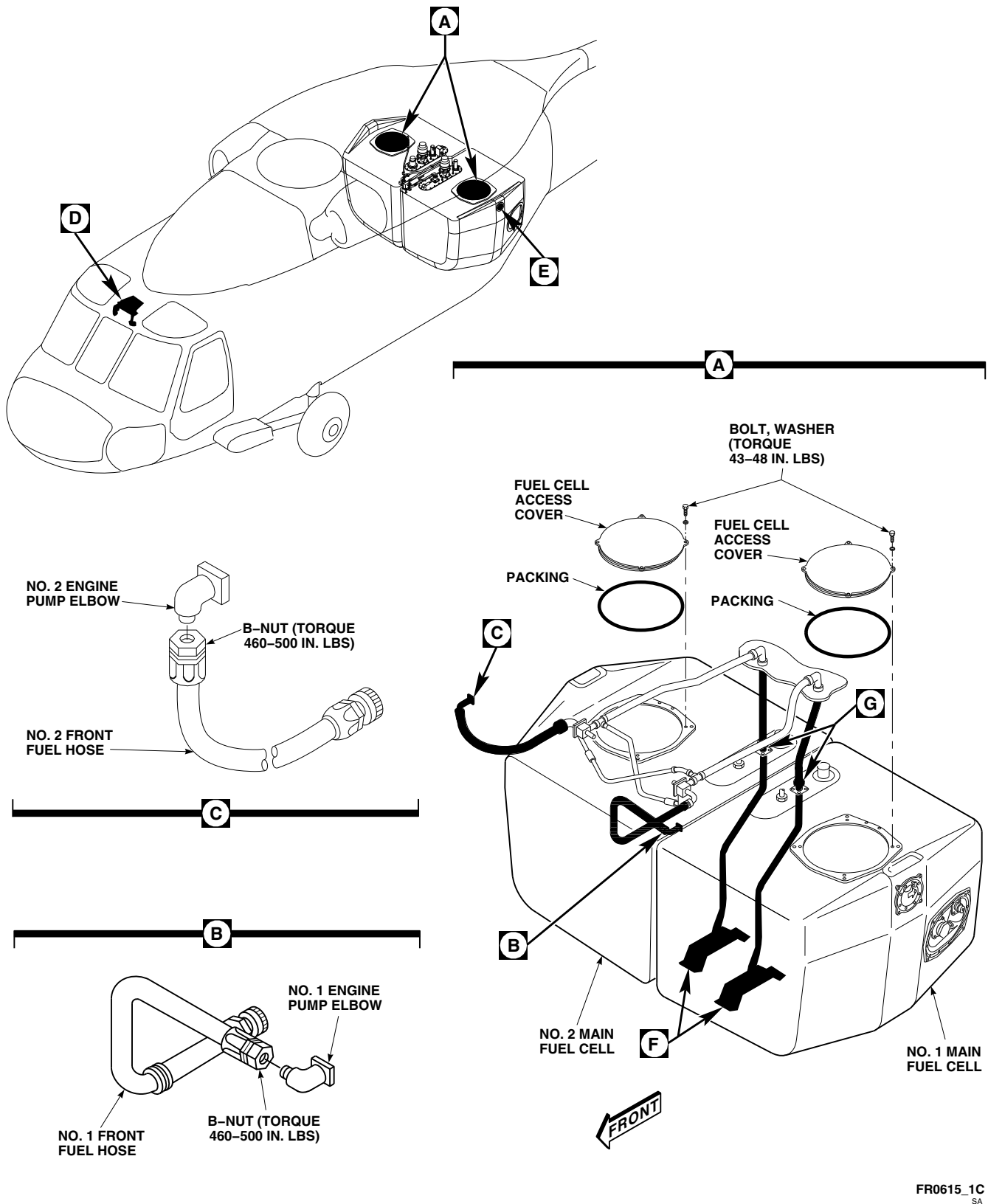
thumb lock, listen for escaping air and make sure no pressure remains in fuel cell.

- (2) Turn thumb lock counterclockwise until it stops.
- (3) Remove cap and let it hang by lanyard.



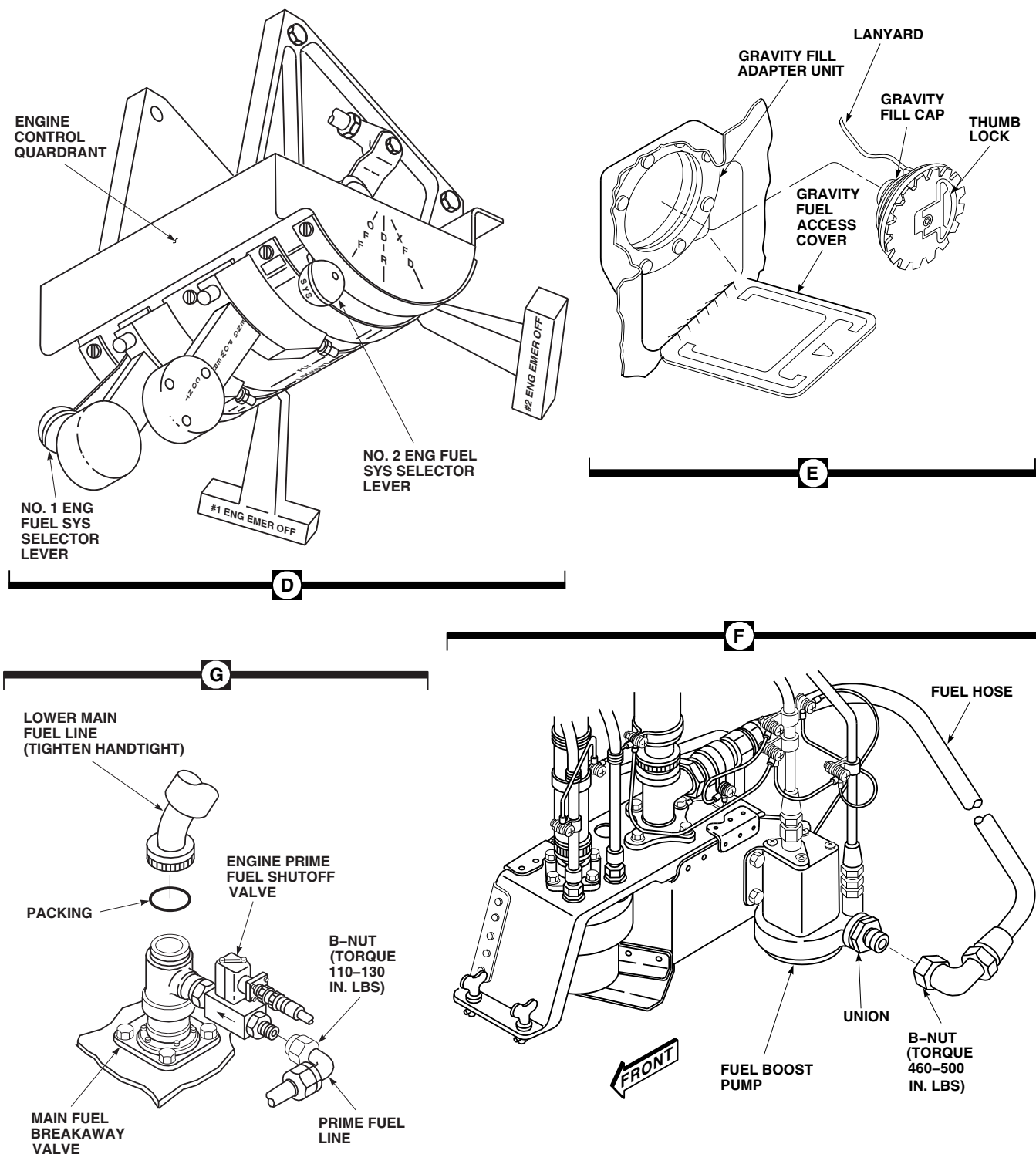
Damage to equipment will result if air pressure is not regulated. Make sure regulated air pressure does not exceed 60 psig.

- m. Apply 50 psig filtered air to closed shutoff valve.
- n. Slowly open shutoff valve to pressurize fuel feed system.
- o. Maintain 50 psig in fuel feed system for 15 minutes.
- p. After purging fuel feed system for 15 minutes, close shutoff valve. Bleed system of any trapped air pressure.
- q. Install gravity fill cap as follows:
 - (1) Place cap in housing and align front-point arrow.
 - (2) Turn thumb lock clockwise to lock cap.
 - (3) Snap thumb lock closed to seal cap.
- r. Close No. 1 and or No. 2 gravity fuel access cover(s).
- s. Install reducer in No. 1 or No. 2 front fuel hose.
- t. Connect 8-inch hose to reducer. Place hose in 1-gallon container to catch fuel from crossfeed system.
- u. Set engine control quadrant controls as follows (Figure 10-4-39-1, Sheet 2, Detail D):



FR0615_1C
SA

FIGURE 10-4-39-1. TEST FUEL FEED SYSTEM (SHEET 1 OF 2)



FN2105_2
SA

FIGURE 10-4-39-1. TEST FUEL FEED SYSTEM (SHEET 2 OF 2)

- (1) If purging No. 1 fuel feed system, on engine control quadrant in cockpit, place NO. 1 ENG FUEL SYS selector lever to DIR, and place NO. 2 ENG FUEL SYS selector lever to XFD.
- (2) If purging No. 2 fuel feed system, on engine control quadrant in cockpit, place NO. 2 ENG FUEL SYS selector lever to DIR, and place NO. 1 ENG FUEL SYS selector lever to XFD.

CAUTION

Damage to equipment will result if air pressure is not regulated. Make sure regulated air pressure does not exceed 5 psig.

- v. Apply 3 psig of filtered air to closed shutoff valve.
- w. Slowly open shutoff valve to pressurize fuel crossfeed system.
- x. Maintain 3 psig in crossfeed system until only air is discharging from front fuel hose.
- y. After purging crossfeed system, close shutoff valve.
- z. Disconnect air source from front fuel hose.
- aa. Remove reducer and hose from front fuel hose.
- ab. Vacuum test fuel feed system (PARA 10-4-39.2.1).

10-4-39.2 Test Fuel Feed System.

NOTE

- The following test covers the entire fuel feed system and should be performed to the extent required to make sure portion of system in question is acceptable. For example, if components in No. 1 fuel feed system are removed, replaced, or are in question, only No. 1 fuel feed

system should be tested. It would not be necessary to test No. 2 fuel feed system.

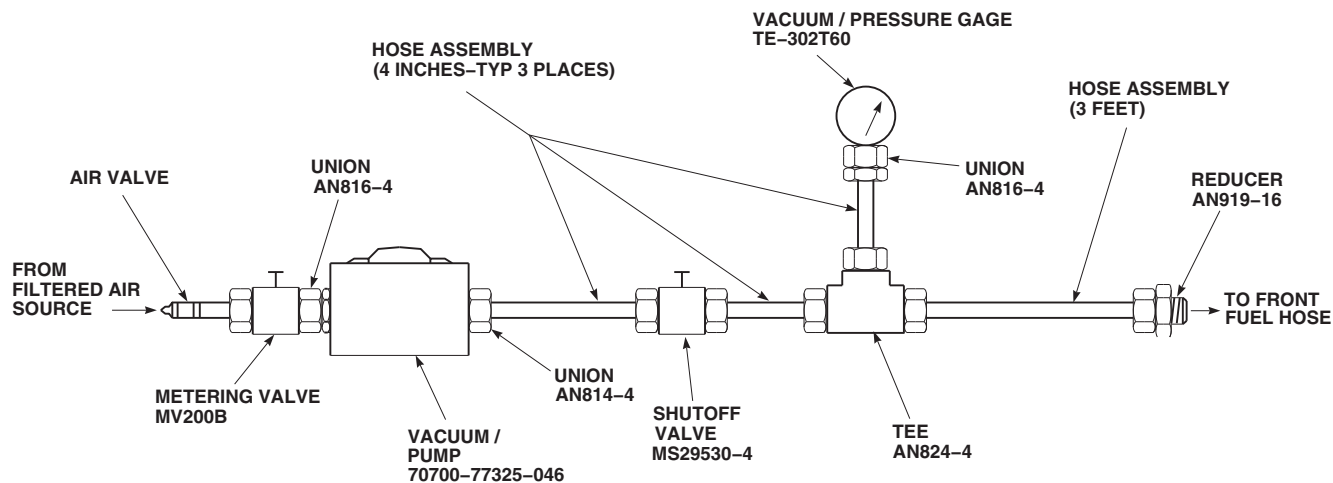
- The following procedure does not require defueling helicopter if fuel feed system components outside of main fuel cells are removed, replaced, or are in question.
- Testing of No. 1 and No. 2 fuel feed system is the same, except as noted.

10-4-39.2.1 Vacuum Test.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

- a. Turn off all electrical power (PARA 1-6-16).
- b. If not previously performed, purge fuel feed system (PARA 10-4-39.1).
- c. Assemble test setup of fuel system integrity test kit as follows (Figure 10-4-39-2):
 - (1) Locally make fuel feed system test hose assemblies (Figure H-229, Appendix H).
 - (2) Unscrew air valve from vacuum/pump, 70700-77325-046.
 - (3) Install air valve in inlet of metering valve, MV200B.
 - (4) Install inlet of vacuum/pump to outlet of metering valve using union, AN816-4.
 - (5) Install union, AN814-4, to outlet of vacuum/pump.
 - (6) Connect 4-inch hose assembly to union.
 - (7) Install inlet of shutoff valve, MS29530-4, to 4-inch hose assembly.



FB3307_3C
SA

FIGURE 10-4-39-2. FUEL SYSTEM INTEGRITY TEST - TEST SETUP

- (8) Connect another 4-inch hose assembly to outlet of shutoff valve.
- (9) Install tee, AN824-4, to 4-inch hose assembly.
- (10) Connect 4-inch hose assembly and 3-foot hose assembly to both ports of tee.
- (11) Install union, AN816-4, and vacuum/pressure gage, TE-302T60, on 4-inch hose assembly.
- (12) Install reducer, AN919-16, on 3-foot hose assembly.
- (13) Perform pressure test of test setup as follows:
 - (a) Install cap, AN929-16, on reducer.
 - (b) Pressure test fuel system integrity test kit for leaks.
 - (c) Remove cap after pressure test.
- d. Position 1-gallon container to catch fuel and disconnect left and/or right front fuel hoses from engine pump elbows (Figure 10-4-39-1, Sheet 1, Detail B and Detail C).

NOTE

Either components and lines outside main fuel cell or entire fuel feed system may be vacuum tested.

- e. Vacuum test fuel feed system as follows:
 - (1) If testing No. 1 fuel feed system, connect test setup to left front fuel hose. Install plug, AN806-12, in right front fuel hose.
 - (2) If testing No. 2 fuel feed system, connect test setup to right front fuel hose. Install plug, AN806-12, in left front fuel hose.
 - (3) If integrity is to be checked for fuel feed system components outside main fuel cell, perform the following:
 - (a) Disconnect lower main fuel lines from main fuel breakaway valves

- (Figure 10-4-39-1, Sheet 2, Detail G). Discard packings.
- (b) Install fuel feed line test plugs and packing on fuel lines.
 - (c) Install protective caps on main fuel breakaway valves.
- (4) If integrity is to be checked for entire fuel feed system, perform the following:
- (a) Remove main fuel line check valves (PARA 10-4-8).
 - (b) Disconnect fuel hose from union on fuel boost pump (Figure 10-4-39-1, Sheet 2, Detail F). Install plug, AN806-12, and packing in fuel hose.
 - (c) Install protective caps and plugs.
 - (d) Disconnect prime fuel line from main engine prime fuel shutoff valve (Figure 10-4-39-1, Sheet 2, Detail G). Install cap, AN929-6, on engine prime fuel shutoff valve. Install packings and plug, AN806-6, on fuel line.
- (5) On engine control quadrant in cockpit, place NO. 1 and/or NO. 2 ENG FUEL SYS selector lever(s) to DIR (Figure 10-4-39-1, Sheet 2, Detail D).
- (6) Close metering valve and place vacuum/pump to ON position. Open shutoff valve.

CAUTION

Damage to equipment will result if air pressure is not regulated. Make sure regulated air pressure does not exceed 80 psig.

- (7) Apply filtered air to metering valve. Slowly open metering valve and regulate to negative 6-inches of mercury on vacuum/pressure gage. Hold vacuum for minimum of five minutes. Close shutoff valve.

- (8) Monitor vacuum/pressure gage for vacuum loss for minimum of five minutes. **NONE ALLOWED.** If vacuum loss is noted, perform the following:
- (a) Check test setup equipment for leaks.
 - (b) Repeat vacuum test. If vacuum loss is still noted, perform pressure test to isolate leak (PARA 10-4-39.2.2).

f. Vacuum test crossfeed system as follows:

- (1) If testing No. 1 crossfeed system, on engine control quadrant in cockpit, place NO. 1 ENG FUEL SYS selector lever to XFD (Figure 10-4-39-1, Sheet 2, Detail D).
- (2) If testing No. 2 crossfeed system, on engine control quadrant in cockpit, place NO. 2 ENG FUEL SYS selector lever to XFD.
- (3) Close metering valve. Place vacuum/pump to ON position. Open shutoff valve.

CAUTION

Damage to equipment will result if air pressure is not regulated. Make sure regulated air pressure does not exceed 80 psig.

- (4) Apply filtered air to metering valve. Slowly open metering valve and regulate to negative 6-inches of mercury on vacuum/pressure gage. Hold vacuum for minimum of five minutes. Close shutoff valve.
- (5) Monitor vacuum/pressure gage for vacuum loss for minimum of five minutes. **NONE ALLOWED.** If vacuum loss is noted, perform the following:
 - (a) Check test setup equipment for leaks.

- (b) Repeat vacuum test. If vacuum loss is still noted, perform pressure test to isolate leak (PARA 10-4-39.2.2).
- g. On No. 1 fuel feed system, remove test setup from No. 1 engine fuel feed hose. Remove plug from No. 2 engine fuel feed hose.
- h. On No. 2 fuel feed system, remove test setup from No. 2 engine fuel feed hose. Remove plug from No. 1 fuel feed hose.
- i. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on lower main fuel line. Connect fuel line to main fuel breakaway valve. **TIGHTEN HANDTIGHT** (Figure 10-4-39-1, Sheet 2, Detail G).
- j. Perform pressure test (PARA 10-4-39.2.2).

10-4-39.2.2 Pressure Test.

NOTE

Vacuum test will be performed prior to pressure test.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove vacuum/pump and inlet and discharge unions of vacuum/pump from test setup. Install metering valve to inlet tube of shutoff valve.
- c. If vacuum test was performed on entire fuel feed system, perform the following:

NOTE

There is no need to disconnect fuel hose from engine prime fuel shutoff valve. The hose was disconnected during the previously performed vacuum test.

- (1) Remove plugs and packings from main fuel line check valve adaptor.
- (2) Install main fuel line check valves (PARA 10-4-8).
- (3) Remove plug and packing from fuel boost pump union and fuel hose. Discard pack-

ing. Connect fuel hose to union on fuel boost pump (Figure 10-4-39-1, Sheet 2, Detail F). **TORQUE B-NUT TO 460 - 500 INCH-POUNDS.**

NOTE

Either components and lines outside main fuel cell or entire fuel feed system may be pressure tested.

- d. Pressure test fuel feed system as follows:
 - (1) If testing No. 2 fuel feed system, connect test setup to No. 2 front fuel hose. Install plug, AN806-12, in No. 1 engine fuel feed hose.
 - (2) If testing No. 1 fuel feed system, connect test setup to No. 1 front fuel hose. Install plug, AN806-12, in No. 2 engine fuel feed hose.
 - (3) If pressure testing No. 1 fuel feed system, on engine control quadrant in cockpit, place NO. 1 ENG FUEL SYS selector lever to DIR, and position NO. 2 ENG FUEL SYS selector lever to OFF (Figure 10-4-39-1, Sheet 2, Detail D).
 - (4) If pressure testing No. 2 fuel feed system, on engine control quadrant in cockpit, place NO. 2 ENG FUEL SYS selector lever to DIR, and position NO. 1 ENG FUEL SYS selector lever to OFF.
 - (5) Close metering valve and open shutoff valve of test setup.

CAUTION

Damage to equipment will result if air pressure is not regulated. Make sure regulated air pressure does not exceed 40 psig.

- (6) Apply filtered air to metering valve. Slowly open metering valve and regulate to 30 psig on vacuum/pressure gage. Hold pressure for minimum of five minutes. Close shutoff valve.

- (7) Monitor vacuum/pressure gage for pressure loss for minimum of five minutes. **NONE ALLOWED.** If pressure loss is noted, perform the following:

NOTE

Maximum allowable leakage of the main fuel inlet check valve is one drop in six minutes. Acceptability of the check valve can only be verified during a bench test using calibration fluid or fuel.

- (a) Repeat pressure test at 10 psig. If no pressure loss is noted, replace main fuel line check valve (PARA 10-4-13).
 - (b) Repeat pressure test at 30 psig. If pressure loss is still noted, inspect fuel hoses and components for leaks using leak detection fluid, Item 189, Appendix D. If required, repair leaks. Repeat pressure test until no leaks are noted.
- (8) If testing No. 1 fuel feed system, on engine control quadrant in cockpit, place NO. 1 ENG FUEL SYS selector lever to XFD, and position NO. 2 ENG FUEL SYS selector lever to OFF.
- (9) If testing No. 2 fuel feed system, on engine control quadrant in cockpit, place NO. 2 ENG FUEL SYS selector lever to XFD, and position NO. 1 ENG FUEL SYS selector lever to OFF.
- (10) Close metering valve and open shutoff valve of test setup.

CAUTION

Damage to equipment will result if air pressure is not regulated. Make sure regulated air pressure does not exceed 40 psig.

- (11) Apply filtered air to metering valve. Slowly open metering valve and regulate

to 30 psig on vacuum/pressure gage. Hold pressure for minimum of five minutes. Close shutoff valve.

- (12) Monitor vacuum/pressure gage for pressure loss for minimum of five minutes. **NONE ALLOWED.** If pressure loss is noted, perform the following:

- (a) Repeat pressure test at 10 psig. If no pressure loss is noted, replace main fuel line check valve (PARA 10-4-13).
 - (b) Repeat pressure test at 30 psig. If pressure loss is still noted, inspect fuel hoses and components for leaks using leak detection fluid, Item 189, Appendix D. If required, repair leaks. Repeat pressure test until no leaks are noted.
- e. Relieve all pressure from fuel feed system.
- f. Connect No. 1 or No. 2 fuel hose to engine pump elbow (Figure 10-4-39-1, Sheet 1, Detail B or Detail C). **TORQUE B-NUT 460 - 500 INCH-POUNDS.**
- g. Lubricate packing with lubricating oil, Item 205, Appendix D. Install packing on fuel cell access cover (Figure 10-4-39-1, Sheet 1, Detail A).

CAUTION

Make sure main fuel breakaway valve is not preloaded when connecting fuel line to engine prime fuel shutoff valve. Frangible section of main fuel breakaway valve is sensitive to bending and tension overload.

- h. Remove protective caps and plugs from engine prime fuel shutoff valve and prime fuel line. Connect prime fuel line to main engine prime shutoff valve. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS** (Figure 10-4-39-1, Sheet 2, Detail G).

- i. Inspect area inside main fuel cell and make sure area is free of all foreign material.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- j. Coat threads of fuel cell access cover attachment bolts with antiseize compound, Item 74, Appendix D.
- k. If purge/test was performed on entire fuel feed system, install fuel cell access cover on main fuel cell with bolts and washers. **TORQUE BOLTS TO 43 - 48 INCH-POUNDS** (Figure 10-4-39-1, Sheet 1, Detail A).
- l. Install left or right panel enclosure on top of main fuel cells. Lock fasteners.
- m. Refuel main fuel cells (PARA 1-3-17, PARA 1-3-18, or PARA 1-3-19).
- n. Prime fuel system (PARA 10-4-1).
- o. Perform operational check of fuel quantity system (PARA 10-2-5 or PARA 10-2-6).
- p. Perform operational check of engine feed fuel lines (PARA 10-2-7).
- q. Close main fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panels (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

- r. Install IR jammer panel (PARA 2-4-115).
- s. Close left and/or right engine cowling(s).

10-4-39.2.3 Low Fuel Ground Run Test.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedures.

CAUTION

Personnel performing these procedures must be properly trained and familiar with and comply with all Occupational Safety and Health Administration (OSHA) and local safety standing operating procedures.

NOTE

Low fuel ground run procedure is required for all subsequent fuel system actions requiring a fuel feed system check.

- a. Verify low fuel pressure warning lamps (# 1 FUEL PRESS and # 2 FUEL PRESS) are operational on caution/advisory panel (PARA 8-2-4).
- b. Defuel main fuel cells (PARA 1-3-20 or PARA 1-3-21).
- c. Drain fuel cell sump (PARA 1-3-21).
- d. When fuel cells are completely empty, using gravity fill or pressure refuel methods, add 30 gallons (approximately 200 lbs) of jet fuel, JP4, JP5, or JP8, to No. 1 or No. 2 main fuel cell (PARA 1-3-17 or PARA 1-3-18).

WARNING

Refer to all warnings, cautions, and notes listed in each applicable operators manual for engine run-up and shutdown procedures.

- e. Set engine control quadrant controls as follows:
 - (1) If performing low fuel ground run test on No. 1 main fuel cell, with fuel boost pumps off, position NO. 1 ENG FUEL SYS selector lever to DIR, and place NO. 2 ENG FUEL SYS selector lever to XFD.
 - (2) If performing low fuel ground run test on No. 2 main fuel cell, with fuel boost pumps off, position NO. 2 ENG FUEL SYS selector lever to DIR, and place NO. 1 ENG FUEL SYS selector lever to XFD.
- f. Perform engine start for both engines. Position ENGINE POWER CONT levers to FLY. Note time on clock and monitor fuel pressure lights.
- g. Perform the following:
 - (1) If helicopter is equipped with a vertical instrument display system (VIDS), run at

100% Nr until 100 lbs. amber segment extinguishes.

- (2) If helicopter is equipped with multifunction displays (MFD), run at 100% Nr until fuel level indication reaches 70 lbs.
- h. If fuel pressure light illuminates, turn on respective boost pump switch to prevent engine flameout. Perform shutdown for both engines (Appropriate Flight Manual).
- i. If fuel pressure light(s) does not illuminate, retard PCL to IDLE, and continue to run at IDLE for additional two minutes. Perform engine shutdown (Appropriate Flight Manual).
- j. Repeat step b. through step i. for other main fuel cell.
- k. If no indications of low fuel pressure light illumination or no flameout occurs, test is complete. No further maintenance actions are required at this time. Refuel main fuel cells (PARA 1-3-17 or PARA 1-3-18).
- l. If either low fuel pressure light illuminated during the operation at FLY, or subsequent ground run, troubleshoot submerged fuel boost pump system (PARA 10-2-2).

10-4-40 PURGE MAIN FUEL CELL.

PARAGRAPH BREAKDOWN

	PAGE		PAGE
10-4-40.1 Purge Main Fuel Cell (Primary Method)	10-4-40-1	10-4-40.2 Purge Main Fuel Cell (Alternate Method)	10-4-40-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Combustible Gas Indicator, 51-8018
- Reduced Pressure Hose
- Workstands

Materials/Parts

- Carbon Dioxide
- Fire Extinguisher

- Lubricating Oil, Item 207, Appendix D
- Nitrogen, Item 221, Appendix D
- Tags

References

- Appendix D
- PARA 1-3-20
- PARA 1-3-21
- PARA 1-6-16

10-4-40.1 Purge Main Fuel Cell (Primary Method).

WARNING

- Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.
- Injury to personnel and damage to equipment will result if the following precautions are not observed while preparing main fuel cells for purging or storage.

- Helicopter and all equipment used to purge main fuel cells, including defueling equipment, workstands, and any powered or pneumatic devices, must be properly grounded.
- Workstands must be equipped with a personnel static discharge plate of copper or zinc, mounted in such a position that personnel will contact the plate before coming in contact with the helicopter.
- Main fuel cells shall not be drained at the end of the work day and be allowed to stand empty overnight. Residual fuel will drain down and form puddles in fuel cell. Fuel puddles will evaporate

into the air and create an explosive fuel-air ratio inside the fuel cell.

NOTE

- This procedure is used when adequate supply of lubricating oil, Item 207, Appendix D, is available.
- Purging of No. 1 and No. 2 main fuel cells is the same.

10-4-40.1.1 Prepare Helicopter.

- Turn off all electrical power (PARA 1-6-16).
- Defuel main fuel cells (PARA 1-3-20 or PARA 1-3-21).
- Drain fuel cell sump (PARA 1-3-21).

10-4-40.1.2 Purge.

- Make sure main fuel cell is completely drained of fuel.
- Reduce flash point of empty main fuel cells by pouring approximately five gallons of lubricating oil, Item 207, Appendix D, into main fuel cell.
- Drain lubricating oil from main fuel cell. When first five gallons of oil are completely drained, close drain valve.
- Fill main fuel cell with lubricating oil, Item 207, Appendix D. Allow oil to remain in fuel cell for at least eight to ten hours.
- Drain lubricating oil from main fuel cell. Save to purge other fuel cells.
- After two or three hours, test main fuel cell for presence of fuel vapors, using combustible gas indicator. If unsafe condition exists, repeat step b. through step f. until safe reading is obtained.
- Attach tag to fuel filler cap with the following information printed on it: "THIS FUEL CELL

HAS BEEN PRESERVED IN ACCORDANCE WITH TM 1-70-23 AND IS GAS FREE. REMOVE THIS TAG UPON FILLING FUEL CELL WITH FUEL." Sign and date tag.

10-4-40.2 Purge Main Fuel Cell (Alternate Method).

WARNING

- Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.
- Injury to personnel and damage to equipment will result if the following precautions are not observed while preparing main fuel cells for purging or storage.
- Helicopter and all equipment used to purge main fuel cells, including defueling equipment, workstands, and any powered or pneumatic devices, must be properly grounded.
- Workstands must be equipped with a personnel static discharge plate of copper or zinc, mounted in such a position that personnel will contact the plate before coming in contact with the helicopter.
- Main fuel cells shall not be drained at the end of the work day and be allowed to stand empty overnight. Residual fuel will drain down and form puddles in fuel cell. Fuel puddles will evaporate into the air and create an explosive fuel-air ratio inside the fuel cell.

NOTE

- This procedure is used when an adequate supply of lubricating oil, Item 207, Appendix D, is not available.
- Purging of No. 1 and No. 2 main fuel cells is the same.

10-4-40.2.1 Prepare Helicopter.

- Turn off all electrical power (PARA 1-6-16).
- Defuel main fuel cells (PARA 1-3-20 or PARA 1-3-21).
- Drain fuel cell sump (PARA 1-3-21).

10-4-40.2.2 Purge.

- Make sure main fuel cell is completely drained of fuel.
- Reduce flashpoint of empty main fuel cell by pouring approximately five gallons of lubricating oil, Item 207, Appendix D, into main fuel cell.
- Drain lubricating oil from main fuel cell and purge fuel cell with carbon dioxide or nitrogen gas as follows:

CAUTION

Damage to fuel system will result if water is allowed to enter main fuel cell. To prevent water from entering fuel cell, use only dehydrated air.

- Open all vents, drains, and filler cap on main fuel cell. Insert reduced pressure hose into filler neck to supply 50 psi of air into fuel cell.
- Purge main fuel cell with air for one half hour. Close all drains.

WARNING

To avoid generating static electrical charge when using carbon dioxide fire extinguisher, remove horn from hose before inserting into filler neck. Ground fire extinguisher to helicopter.

NOTE

- Use fire extinguisher as source of carbon dioxide.
 - Nitrogen, Item 221, Appendix D, or other inert gas may be used in place of carbon dioxide.
 - Use at least three pounds of carbon dioxide or nitrogen to purge each main fuel cell.
- Insert hose into fuel cell filler neck from carbon dioxide or nitrogen tank. Discharge gas into main fuel cell at rate of not more than one pound of purging gas per minute.
 - After two or three hours, test main fuel cell for presence of fuel vapors using combustible gas indicator. If unsafe condition exists, repeat step b. through step c. (4). with fresh oil until safe reading is obtained.
 - Coat all interior surfaces of main fuel cell with lubricating oil, Item 207, Appendix D.
 - Attach tag to fuel filler cap with the following information printed on it: "THIS FUEL CELL HAS BEEN PRESERVED IN ACCORDANCE WITH TM 1-70-23 AND IS GAS FREE. REMOVE THIS TAG UPON FILLING FUEL CELL WITH FUEL." Sign and date tag.

10-4-41 UPPER DECK MAIN FUEL HOSES.

PARAGRAPH BREAKDOWN

	PAGE
10-4-41.1 Replace Upper Deck Main Fuel Hoses	10-4-41-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Protection, Eye
- Strap Wrench
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Wrench (with Rubber Fittings)

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-113
- PARA 6-4-50
- PARA 10-4-1
- PARA 10-4-39

10-4-41.1 Replace Upper Deck Main Fuel Hoses.

10-4-41.1.1 Remove.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

- Purge fuel feed system (PARA 10-4-39).
- Open applicable left or right engine cowling.
- Remove left or right oil cooler compartment access door (PARA 2-4-113).
- If removing rear fuel hoses, remove oil cooler (PARA 6-4-50).
- Position 1-quart container to catch excess fuel when disconnecting fuel hoses.
- Remove right front fuel hose as follows (Figure 10-4-41-1, Sheet 2, Detail B):

CAUTION

To prevent damage to fuel hose fittings, use strap wrench or wrench with rubber fittings when disconnecting Gamah nuts.

- (1) Disconnect right front fuel hose from elbow.
- (2) Install protective cap and plug on hose.

CAUTION

Make sure not to preload engine compartment breakaway valve when disconnecting front fuel hose. Frangible section of engine compartment breakaway valve is sensitive to bending and tension overload.

- (3) Disconnect right front fuel hose from engine compartment breakaway valve.
 - (4) Remove right front fuel hose from helicopter.
 - (5) Remove and discard packing from right front fuel hose.
- g. Remove right rear fuel hose as follows (Figure 10-4-41-1, Sheet 2, Detail C):

CAUTION

To prevent damage to fuel hose fittings, use strap wrench or wrench with rubber fittings when disconnecting Gamah nuts.

- (1) Remove screws, washers, and nuts securing clamps on right rear fuel hose to bracket and airframe. Remove clamps from right rear fuel hose.

CAUTION

Make sure not to preload breakaway tee valve when disconnecting right rear fuel hose. Frangible section of engine breakaway tee valve is sensitive to bending and tension overload.

- (2) Disconnect right rear fuel hose from breakaway tee valve.

CAUTION

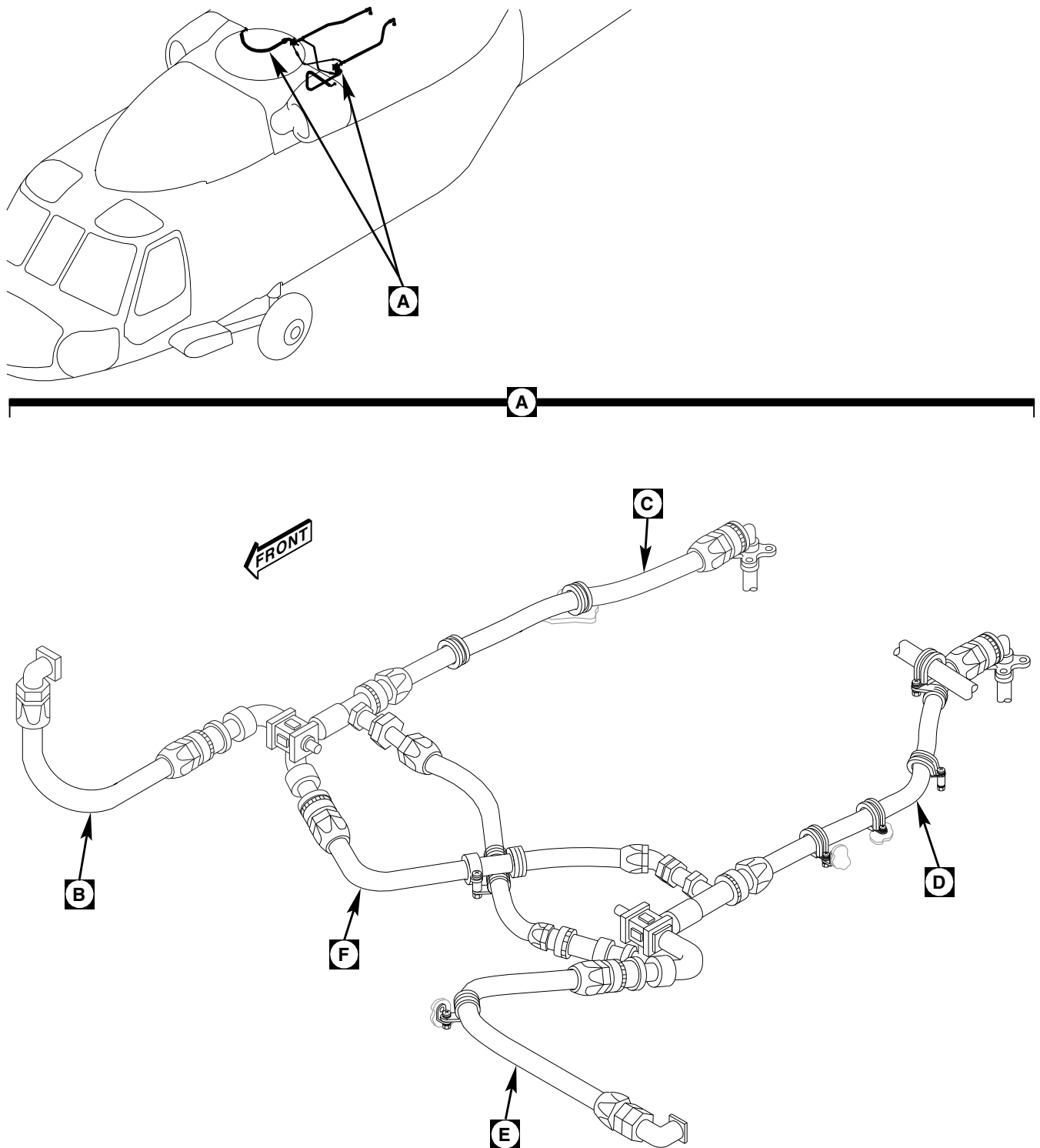
Make sure not to preload bulkhead breakaway valve when disconnecting right rear fuel hose. Frangible section of bulkhead breakaway valve is sensitive to bending and tension overload.

- (3) Disconnect right rear fuel hose from bulkhead breakaway valve.
 - (4) Install protective cap and plug on hose.
 - (5) Remove right rear fuel hose from helicopter.
 - (6) Remove and discard packings from right rear fuel hose.
- h. Remove left rear fuel hose as follows (Figure 10-4-41-1, Sheet 3, Detail D):

CAUTION

To prevent damage to fuel hose fittings, use strap wrench or wrench with rubber fittings when disconnecting Gamah nuts.

- (1) Remove screws, washers, spacers, and nuts securing clamps on left rear fuel hose



NOTE

LOCATE AND CLAMP HOSE TO AVOID
CONTACT WITH FIREWALL ANGLE AND
ENGINE OIL PORT.

FB3571_1A
SA

FIGURE 10-4-41-1. REPLACE UPPER DECK MAIN FUEL HOSES (SHEET 1 OF 4)

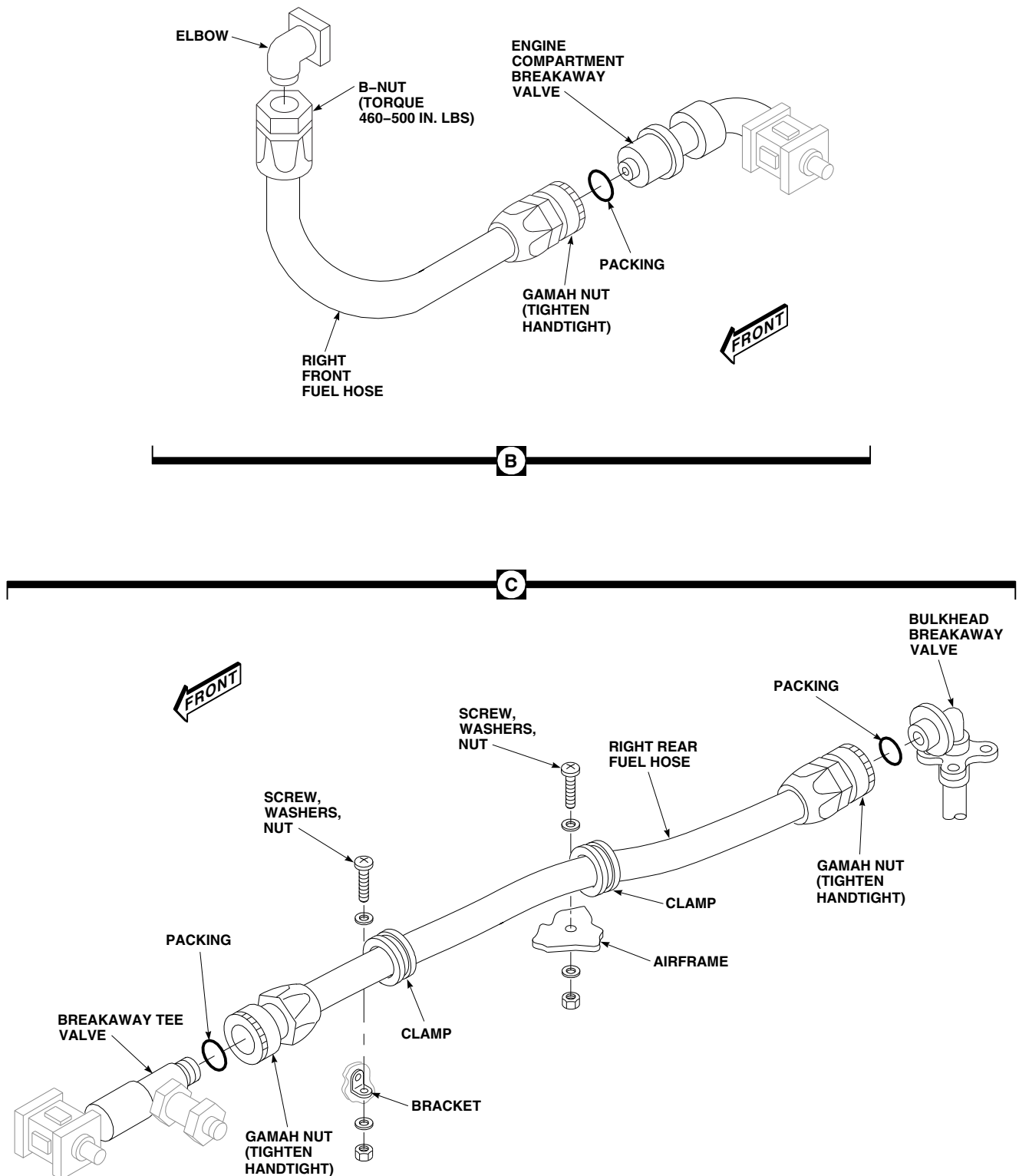
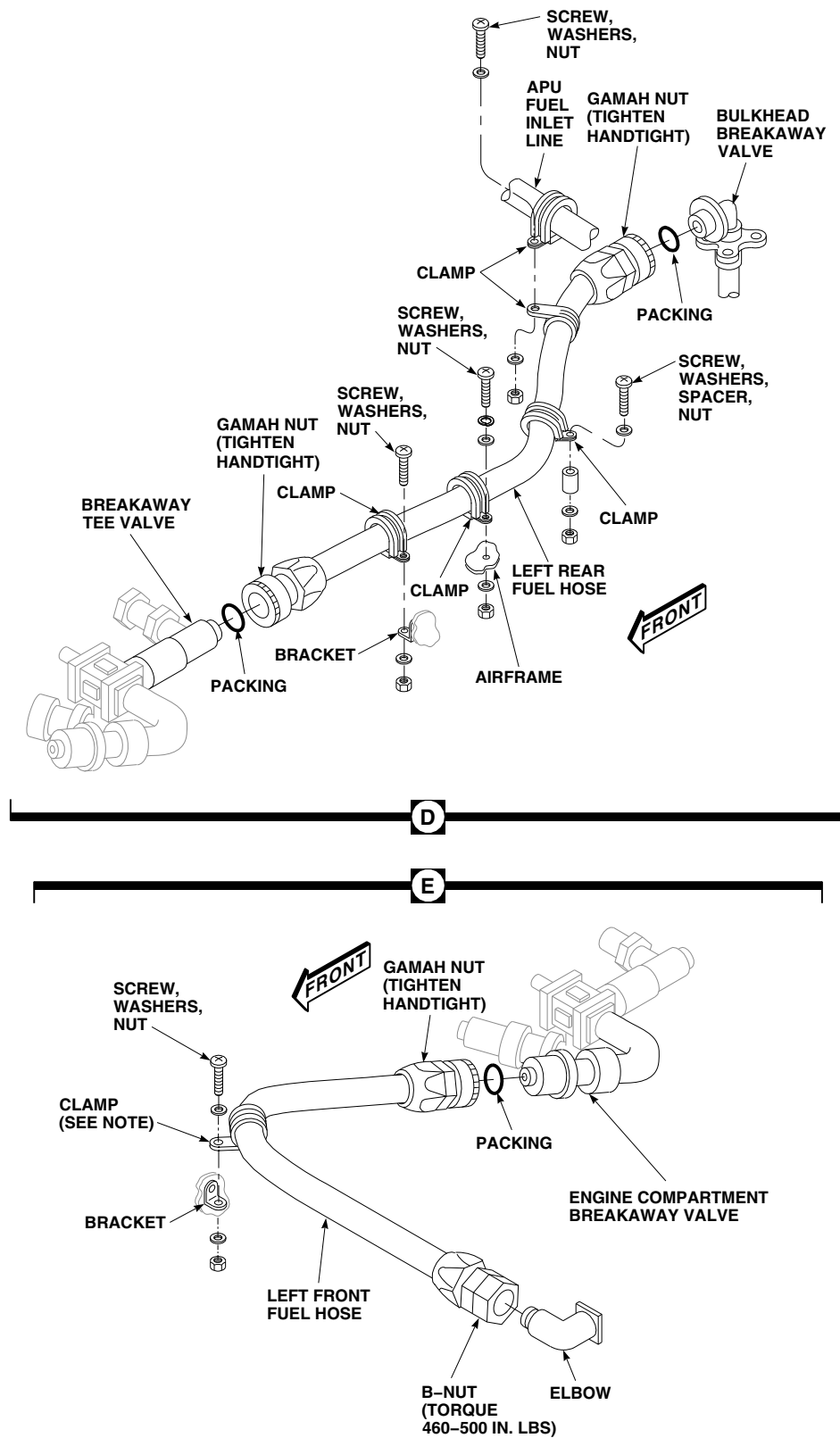
FB3571_2C
SA

FIGURE 10-4-41-1. REPLACE UPPER DECK MAIN FUEL HOSES (SHEET 2 OF 4)



FB3571_3C
SA

FIGURE 10-4-41-1. REPLACE UPPER DECK MAIN FUEL HOSES (SHEET 3 OF 4)

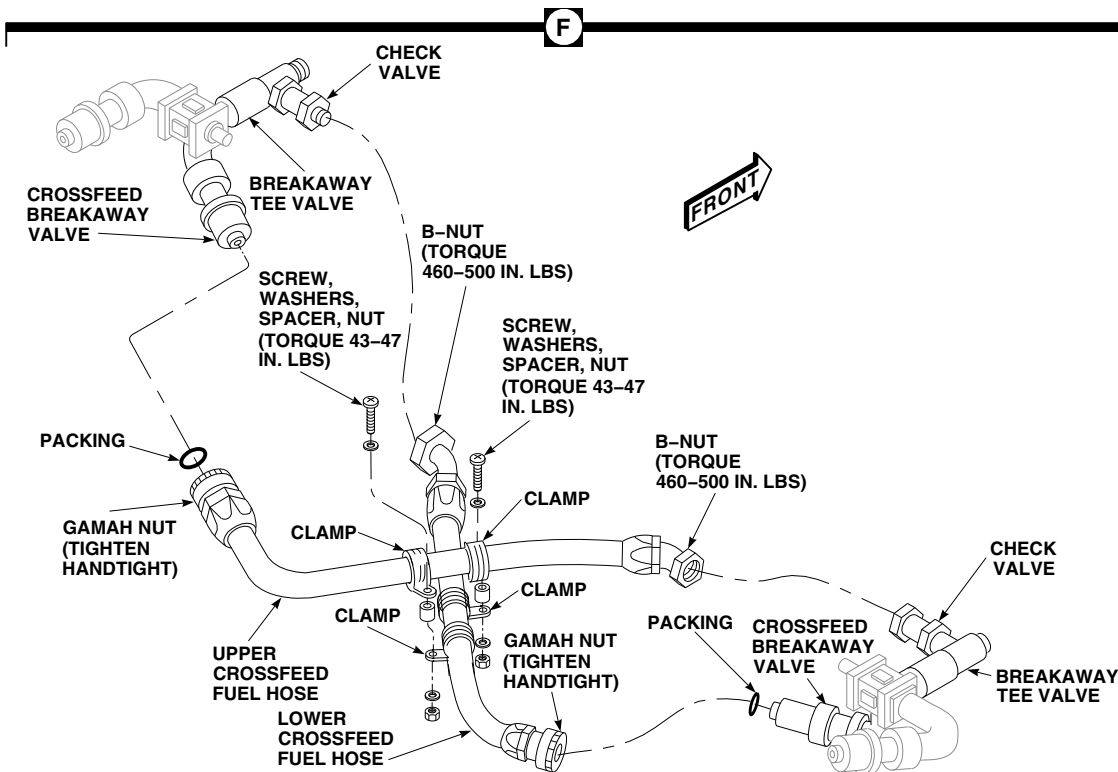


FIGURE 10-4-41-1. REPLACE UPPER DECK MAIN FUEL HOSES (SHEET 4 OF 4)

to bracket, airframe, and auxiliary power unit fuel inlet line. Remove clamps from left rear fuel hose.

CAUTION

Make sure not to preload breakaway tee valve when disconnecting left rear fuel hose. Frangible section of breakaway tee valve is sensitive to bending and tension overload.

- (2) Disconnect left rear fuel hose from breakaway tee valve.

CAUTION

Make sure not to preload bulkhead breakaway valve when disconnecting left rear fuel hose. Frangible section of bulkhead breakaway valve is sensitive to bending and tension overload.

- (3) Disconnect left rear fuel hose from bulkhead breakaway valve.
 - (4) Install protective cap and plug on hose.
 - (5) Remove left rear fuel hose from helicopter.
 - (6) Remove and discard packings from left rear fuel hose.
- i. Remove left front fuel hose as follows (Figure 10-4-41-1, Sheet 3, Detail E):

CAUTION

To prevent damage to fuel hose fittings, use strap wrench or wrench with rubber fittings when disconnecting Gamah nuts.

- (1) Remove screw, washers, and nut securing clamp on left front fuel hose to bracket. Remove clamp from left front fuel hose.

- (2) Disconnect left front fuel hose from elbow.

CAUTION

Make sure not to preload engine compartment breakaway valve when disconnecting left front fuel hose. Frangible section of engine compartment breakaway valve is sensitive to bending and tension overload.

- (3) Disconnect left front fuel hose from engine compartment breakaway valve.
 - (4) Install protective cap and plug on hose.
 - (5) Remove left front fuel hose from helicopter.
 - (6) Remove and discard packing from left front fuel hose.
- j. Remove crossfeed fuel hoses (upper and/or lower) as follows (Figure 10-4-41-1, Sheet 4, Detail F):

CAUTION

- To prevent damage to fuel hose fittings, use strap wrench or wrench with rubber fittings when disconnecting Gamah nuts.
- To prevent damage to oil cooler, if installed, be careful when loosening Gamah nuts.

NOTE

When removing crossfeed fuel hose from crossfeed check valve, make sure to hold valve steady by grasping hex section closest to attaching fitting.

- (1) Remove screws, washers, spacers, and nuts securing clamps on upper and lower crossfeed fuel hoses. Remove clamps from upper and lower crossfeed fuel hoses.

- (2) Disconnect upper crossfeed fuel hose as follows:

CAUTION

Make sure not to preload breakaway tee valve when disconnecting upper crossfeed fuel hose. Frangible section of breakaway tee valve is sensitive to bending and tension overload.

- (a) Disconnect upper crossfeed fuel hose from check valve.

CAUTION

Make sure not to preload breakaway tee valve when disconnecting upper crossfeed fuel hose. Frangible section of breakaway tee valve is sensitive to bending and tension overload.

- (b) Disconnect upper crossfeed fuel hose from crossfeed breakaway valve.
 - (c) Remove upper crossfeed fuel hose from helicopter.
 - (d) Remove and discard packings from upper crossfeed fuel hose.
- (3) Disconnect lower crossfeed fuel hose as follows:

CAUTION

Make sure not to preload breakaway tee valve when disconnecting lower crossfeed fuel hose. Frangible section of breakaway tee valve is sensitive to bending and tension overload.

- (a) Disconnect lower crossfeed fuel hose from check valve.

CAUTION

Make sure not to preload breakaway tee valve when disconnecting lower cross-feed fuel hose. Frangible section of breakaway tee valve is sensitive to bending and tension overload.

- (b) Disconnect lower crossfeed fuel hose from crossfeed breakaway valve.
- (c) Remove lower crossfeed fuel hose from helicopter.
- (d) Remove and discard packings from lower crossfeed fuel hose.

10-4-41.1.2 Install.**WARNING**

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

- a. Turn off all electrical power (PARA 1-6-16).

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- b. Before installation, blow all hoses clear with dry compressed air.

- c. Install right front fuel hose as follows (Figure 10-4-41-1, Sheet 2, Detail B):

CAUTION

To prevent damage to connecting components, use strap wrench or wrench with rubber fittings to tighten Gamah nuts. Do not use pliers or other tools to tighten Gamah nuts.

NOTE

- Make sure right front fuel hose has minimum clearance of 1/2-inch from all control cables.
- Contact between right front fuel hose and firewall is normal and acceptable unless noted.
 - (1) Lubricate packing and mating surface sparingly with lubricating oil, Item 205, Appendix D.
 - (2) Remove protective cap and plug from hose.
 - (3) Insert packing into hose fitting.

CAUTION

Do not preload engine compartment breakaway valve when installing fuel hose. Frangible section of engine compartment breakaway valve is sensitive to bending and tension overload.

- (4) Connect right front fuel hose to engine compartment breakaway valve.
TIGHTEN GAMAH NUT HANDTIGHT.
- (5) Connect right front fuel hose to elbow.
TORQUE B-NUT TO 460 - 500 INCH-POUNDS.
- d. Install right rear fuel hose as follows (Figure 10-4-41-1, Sheet 2, Detail C):

CAUTION

- To prevent damage to connecting components, use strap wrench or wrench with rubber fittings to tighten Gamah nuts. Do not use pliers or other tools to tighten Gamah nuts.
- To prevent damage to oil cooler, if installed, be careful when tightening hex nuts.

NOTE

- Make sure right rear fuel hose has minimum clearance of 1/2-inch from all control cables.
 - Make sure right rear fuel hose maintains constant rise from engine feed breakaway valve to tee breakaway valve when installed.
- (1) Lubricate packings and mating surfaces sparingly with lubricating oil, Item 205, Appendix D.
 - (2) Remove protective cap and plug from hose.
 - (3) Insert packings into hose fittings.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- (4) Coat threads on bulkhead breakaway valve with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection

points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.

CAUTION

Make sure bulkhead breakaway valve is not preloaded when installing right rear fuel hose. Frangible section of bulkhead breakaway valve is sensitive to bending and tension overload.

- (5) Connect right rear fuel hose to bulkhead breakaway valve. **TIGHTEN GAMAH NUT HANDTIGHT.**

CAUTION

Make sure breakaway tee valve is not preloaded when installing right rear fuel hose. Frangible section of breakaway tee valve is sensitive to bending and tension overload.

- (6) Connect right rear fuel hose to breakaway tee valve. **TIGHTEN GAMAH NUT HANDTIGHT.**

CAUTION

To prevent damage to equipment, make sure there is no preload conditions on fuel hoses. Adjust brackets before securing to clamps so that no preload conditions occur.

- (7) Secure right rear fuel hose to bracket and airframe using clamps, screws, washers, and nuts.
- e. Install left rear fuel hose as follows (Figure 10-4-41-1, Sheet 3, Detail D):

CAUTION

- To prevent damage to connecting components, use strap wrench or wrench with rubber fittings to tighten Gamah nuts. Do not use pliers or other tools to tighten Gamah nuts.
- To prevent damage to oil cooler, if installed, be careful when tightening hex nuts.

NOTE

- Make sure left rear fuel hose has minimum clearance of 1/2-inch from all control cables.
 - Make sure left rear fuel hose maintains constant rise from engine feed breakaway valve to breakaway tee valve when installed.
- (1) Lubricate packings and mating surfaces sparingly with lubricating oil, Item 205, Appendix D.
 - (2) Remove protective cap and plug from hose.
 - (3) Insert packings into hose fittings.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- (4) Coat threads on bulkhead breakaway valve with antiseize compound, Item 74, Appendix D, sparingly (1/4-inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.

CAUTION

Make sure bulkhead breakaway valve is not preloaded when installing left rear fuel hose. Frangible section of bulkhead breakaway valve is sensitive to bending and tension overload.

- (5) Connect left rear fuel hose to bulkhead breakaway valve. **TIGHTEN GAMAH NUT HANDTIGHT.**

CAUTION

Make sure breakaway tee valve is not preloaded when installing rear fuel hose. Frangible section of breakaway tee valve is sensitive to bending and tension overload.

- (6) Connect left rear fuel hose to breakaway tee valve. **TIGHTEN GAMAH NUT HANDTIGHT.**

CAUTION

To prevent damage to equipment, make sure there is no preload conditions on fuel hoses. Adjust brackets before attaching to clamps so that no preload conditions occur.

- (7) Secure left rear fuel hose to bracket and airframe using clamps, screws, washers, spacers, and nuts.

CAUTION

To prevent damage to equipment, make sure there is no preload conditions on fuel hoses. Adjust brackets before attaching to clamps so that no preload conditions occur.

NOTE

When securing auxiliary power unit (APU) fuel inlet line to left rear fuel hose, make sure APU fuel inlet line is on top of left rear fuel hose.

- (8) Secure left rear fuel hose to bracket APU fuel inlet line using clamps, screw, washers, and nut.
- f. If removed, install left front fuel hose as follows (Figure 10-4-41-1, Sheet 3, Detail E):

CAUTION

To prevent damage to connecting components, use strap wrench or wrench with rubber fittings to tighten Gamah nuts. Do not use pliers or other tools to tighten Gamah nuts.

NOTE

- Make sure left front fuel hose has minimum clearance of 1/2-inch from all control cables.
- Contact between left front fuel hose and firewall is normal and acceptable, except where noted.
- (1) Lubricate packings and mating surfaces sparingly with lubricating oil, Item 205, Appendix D.
- (2) Insert packing into hose fitting.

CAUTION

Do not preload engine compartment breakaway valve when installing left front fuel hose. Frangible section of engine compartment breakaway valve is sensitive to bending and tension overload.

- (3) Connect left front fuel hose to engine compartment breakaway valve. **TIGHTEN GAMAH NUT HANDTIGHT.**
- (4) Connect left front fuel hose to elbow. **TORQUE B-NUT TO 460 - 500 INCH-POUNDS.**

CAUTION

To prevent damage to equipment, make sure there is no preload conditions on fuel hoses. Adjust brackets before attaching to clamps so that no preload conditions occur.

NOTE

Locate clamp and left front fuel hose to avoid contact with firewall angle and engine oil port.

- (5) Secure left front fuel hose to bracket using clamp, screw, washers, and nut.
- g. Install crossfeed fuel hoses (upper and/or lower) as follows (Figure 10-4-41-1, Sheet 4, Detail F):

CAUTION

- To prevent damage to connecting components, use strap wrench or wrench with rubber fittings to tighten Gamah nuts. Do not use pliers or other tools to tighten Gamah nuts.
- To prevent damage to oil cooler, if installed, be careful when installing Gamah nuts.

NOTE

- When installing crossfeed fuel hoses together, make sure crossfeed fuel hose running from left check valve to right crossfeed breakaway valve is on top of other crossfeed fuel hose.
- When installing crossfeed fuel hoses (upper and lower) to crossfeed check valve, make sure to hold valve steady by grasping hex section closest to attaching fitting.
 - (1) Lubricate packing(s) and mating surface(s) sparingly with lubricating oil, Item 205, Appendix D.
 - (2) Remove protective cap and plug from hose.
 - (3) Insert packing(s), into line hose(s).
 - (4) Connect lower crossfeed fuel hose as follows:

CAUTION

- Damage to crossfeed breakaway valve may occur if valve is preloaded. Do not preload valve.
- Do not preload crossfeed breakaway valve when installing fuel hose.

Frangible section of crossfeed breakaway valve is sensitive to bending and tension overload.

NOTE

Allow crossfeed breakaway valve and fuel hose to rotate freely on fuel selector valve. Valve and hose should assume neutral position to aid in elimination/reduction of preload condition.

- (a) Connect lower crossfeed fuel hose to crossfeed breakaway valve. **TIGHTEN GAMAH NUT HANDTIGHT.**

CAUTION

Make sure breakaway tee valve is not preloaded when installing lower crossfeed fuel hose. Frangible section of breakaway tee valve is sensitive to bending and tension overload.

- (b) Connect lower crossfeed fuel hose to check valve. **TORQUE B-NUT TO 460 - 500 INCH-POUNDS.**
- (5) Connect upper crossfeed fuel hose as follows:

CAUTION

- Damage to crossfeed breakaway valve may occur if crossfeed breakaway valve is preloaded. Do not preload crossfeed breakaway valve.
- Do not preload crossfeed breakaway valve when installing fuel hose. Frangible section of crossfeed breakaway valve is sensitive to bending and tension overload.

NOTE

Allow crossfeed breakaway valve and fuel hose to rotate freely on fuel selector valve. Valve and hose should assume neutral position to aid in elimination/reduction of a preload condition.

- (a) Connect upper crossfeed fuel hose to crossfeed breakaway valve. **TIGHTEN GAMAH NUT HANDTIGHT.**

CAUTION

Make sure breakaway tee valve is not preloaded when installing upper crossfeed fuel hose. Frangible section of breakaway tee valve is sensitive to bending and tension overload.

- (b) Connect upper crossfeed fuel hose to check valve. **TORQUE B-NUT TO 460 - 500 INCH-POUNDS.**

CAUTION

To prevent chafing of crossfeed fuel hoses, make sure lower crossfeed fuel

hose is not touching engine compartment deck. Lower crossfeed fuel hose should be suspended slightly above engine compartment deck.

- (6) Secure upper and lower crossfeed fuel hoses together using clamps, screws, washers, spacers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
- h. If removed, install oil cooler (PARA 6-4-50).
- i. Make sure area is clean and free of foreign material.
- j. Test fuel feed system (PARA 10-4-39).
- k. Prime fuel system (PARA 10-4-1). Check for leaks. If leakage is detected, make sure all connections are properly tightened or install new packings.
- l. Install oil cooler compartment access door (PARA 2-4-113).
- m. Close engine cowling, as required.

10-4-42 LOWER APU FUEL LINE.

PARAGRAPH BREAKDOWN

	PAGE
10-4-42.1 Replace Lower APU Fuel Line	10-4-42-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Protection, Eye
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-3-17
- PARA 1-3-18
- PARA 1-3-19
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-4-1
- PARA 10-4-39
- PARA 13-5-3
- PARA 13-5-6
- PARA 15-4-9

10-4-42.1 Replace Lower APU Fuel Line.

10-4-42.1.1 Remove.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

- Purge fuel feed system (PARA 10-4-39).
- Gain access to main fuel cell compartment as follows:
 - Remove rear passenger’s seats as required (PARA 2-4-45).
 - Remove soundproofing panel from rear bulkhead covering left or right stowage compartment (PARA 2-4-69).
 - Remove cargo netting from stowage compartment.

- (4) If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
- c. Unlock studs securing left and right panels to main fuel cells and airframe. Remove panels from helicopter.
- d. Remove screws and washers securing front and top panels to airframe. Remove panels from helicopter.
- e. Remove screws, washers, lockwasher, and nuts securing clamps, spacer, and bonding jumper to lower auxiliary power unit (APU) fuel line (Figure 10-4-42-1, Detail A).
- f. Position 1-quart container to catch excess fuel when disconnecting lower APU fuel line.
- g. Disconnect lower APU fuel line from APU fuel solenoid valve and main fuel prime/boost pump outlet valve. Remove lower APU fuel line from helicopter.
- h. Install protective caps and plugs on fuel line.

10-4-42.1.2 Install.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove protective caps and plugs from fuel line.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel

debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- c. Before installation, blow lower auxiliary power unit (APU) fuel line clear with dry compressed air.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- d. Apply antiseize compound, Item 74, Appendix D, sparingly ($\frac{1}{4}$ -inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.
- e. Connect lower APU fuel line to APU fuel solenoid valve and main fuel prime/boost pump outlet valve. **TORQUE B-NUTS TO 110 - 130 INCH-POUNDS.**
- f. Secure lower APU fuel line to bracket using clamp, screw, washers, spacer, and nut.
- g. Install bonding jumper to clamp on lower APU fuel line with screw, washers, lockwasher, and nut.
- h. Seal bonding jumper using sealing compound, Item 292, Appendix D.
- i. Make sure area is clean and free of foreign material.
- j. Test fuel feed system (PARA 10-4-39).
- k. Refuel main fuel cells (PARA 1-3-17, PARA 1-3-18, or PARA 1-3-19).



Do not operate main fuel prime/boost pump without fuel at main fuel prime/boost pump outlet valve. Pump failure may occur by running pump dry.

- l. Prime fuel system (PARA 10-4-1). Check for leaks. If leakage is detected, make sure all connections are properly tightened or install new packings.
- m. Prime APU lines and check for leaks (PARA 15-4-9). If leakage is detected, make sure all connections are properly tightened or install new packings.
- n. Install front and top panels with screws and washers.
- o. Install left and right panels enclosing fuel system components. Lock fastener studs.
- p. Close main fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panel (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

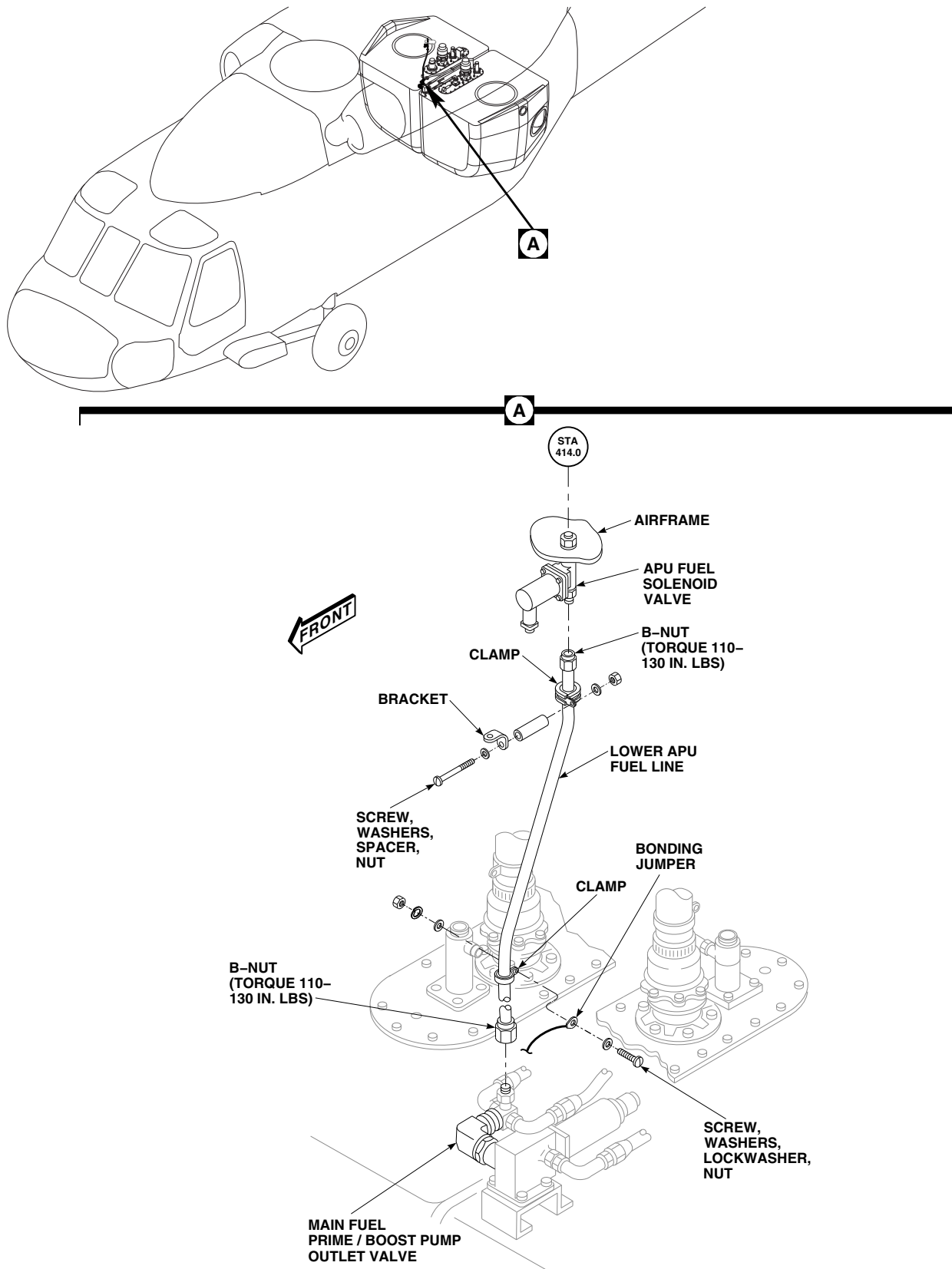
FB3572A
SA

FIGURE 10-4-42-1. REPLACE LOWER APU FUEL LINE

10-4-43 PRIME FUEL LINES.

PARAGRAPH BREAKDOWN

	PAGE
10-4-43.1 Replace Prime Fuel Lines	10-4-43-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Protection, Eye
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-3-17
- PARA 1-3-18
- PARA 1-3-19
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-4-1
- PARA 10-4-39
- PARA 13-5-3
- PARA 13-5-6

10-4-43.1 Replace Prime Fuel Lines.

NOTE

Replacement of left, right, and upper prime/boost pump fuel lines is the same, except as noted.

10-4-43.1.1 Remove.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow

normal fire precautions when performing the following procedure.

- Purge fuel feed system (PARA 10-4-39).
- Gain access to main fuel cell compartment as follows:
 - Remove rear passenger’s seats as required (PARA 2-4-45).
 - Remove soundproofing panel from rear bulkhead covering left or right stowage compartment (PARA 2-4-69).
 - Remove cargo netting from stowage compartment.

(4) If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

- c. Unlock studs fastening left and right panels to main fuel cells and airframe. Remove panels from helicopter.
- d. Remove screws and washers securing front and top panels to airframe. Remove panels from helicopter.
- e. Remove clamps, screws, washers, lockwasher, bonding jumpers, and nuts from prime fuel lines (Figure 10-4-43-1, Detail A).
- f. Use machinery towel, Item 211, Appendix D, to catch excess fuel when disconnecting prime fuel lines.

CAUTION

Make sure not to preload breakaway valves when disconnecting prime fuel lines. Frangible section of breakaway valve is sensitive to bending and tension overload.

- g. Disconnect prime fuel line from valve and/or pump, as required. Remove prime fuel line from helicopter.
- h. Install protective caps and plugs on hoses.

10-4-43.1.2 Install.

WARNING

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove protective caps and plugs before clearing lines.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- c. Before installation, blow all lines clear with dry compressed air.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- d. Apply antiseize compound, Item 74, Appendix D, sparingly ($\frac{1}{4}$ -inch strip) along length of male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.
- e. Connect right prime fuel line as follows (Figure 10-4-43-1, Detail A):

CAUTION

Make sure main fuel breakaway valve is not preloaded when installing right prime fuel line. Frangible section of main fuel breakaway valve is sensitive to bending and tension overload.

- (1) Connect right prime fuel line to engine prime fuel shutoff valve. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**

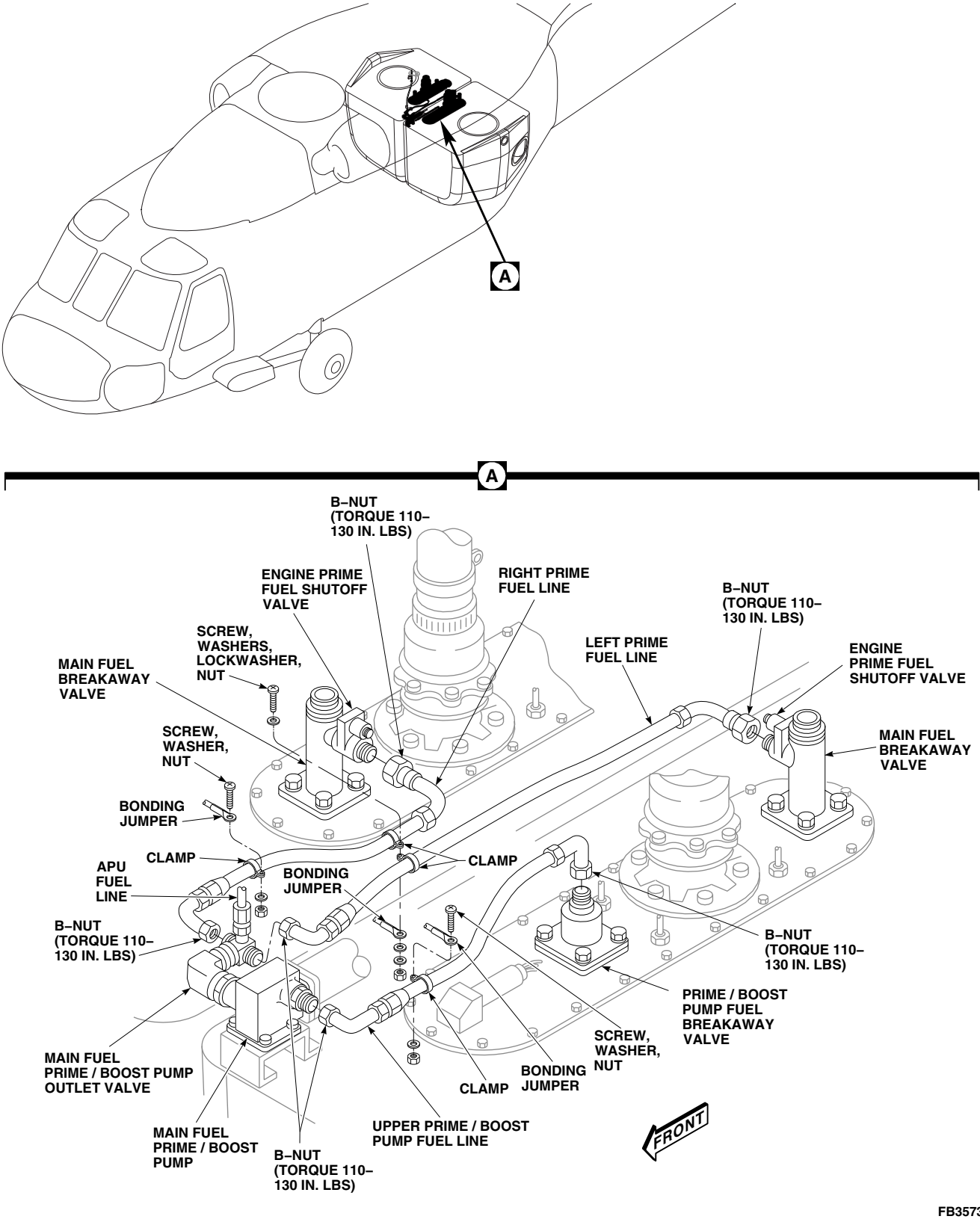


FIGURE 10-4-43-1. REPLACE PRIME FUEL LINES

- (2) Connect right prime fuel line to main fuel prime/boost pump outlet valve. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**

f. Connect left prime fuel line as follows:

CAUTION

Make sure main fuel breakaway valve is not preloaded when installing left prime fuel line. Frangible section of main fuel breakaway valve is sensitive to bending and tension overload.

- (1) Connect left prime fuel line to engine prime fuel shutoff valve. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**
- (2) Connect left prime fuel line to main fuel prime/boost pump outlet valve. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**

g. Connect upper prime/boost pump fuel line as follows:

CAUTION

Make sure prime/boost pump fuel breakaway valve is not preloaded when installing upper prime/boost pump fuel line. Frangible section of prime/boost pump fuel breakaway valve is sensitive to bending and tension overload.

- (1) Connect upper prime/boost pump fuel line to prime/boost pump fuel breakaway valve. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**
- (2) Connect upper prime/boost pump fuel line to main fuel prime/boost pump. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**

- h. Secure prime fuel lines using clamps, screws, washers, lockwasher, bonding jumpers, and nuts.
- i. Seal all bonding jumpers using sealing compound, Item 292, Appendix D.
- j. Make sure area is clean and free of foreign material.
- k. Test fuel feed system (PARA 10-4-39).
- l. Refuel main fuel cells (PARA 1-3-17, PARA 1-3-18, or PARA 1-3-19).

CAUTION

Do not operate main fuel prime/boost pump without fuel at main fuel prime/boost pump outlet valve. Pump failure may occur by running pump dry.

- m. Prime fuel system (PARA 10-4-1). Check for leaks. If leakage is detected, make sure all connections are properly tightened or install new packings.
- n. Install front and top panels with screws and washers.
- o. Install left and right panels enclosing fuel system components. Lock fastener studs.
- p. Close main fuel cell compartment as follows:
 - (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panel (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

10-4-44 LOWER MAIN FUEL LINES.

PARAGRAPH BREAKDOWN

	PAGE
10-4-44.1 Replace Lower Main Fuel Lines	10-4-44-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Protection, Eye
- Strap Wrench
- Wrench (with Rubber Fittings)

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-3-17
- PARA 1-3-18
- PARA 1-3-19
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-4-1
- PARA 10-4-34
- PARA 10-4-39
- PARA 13-5-3
- PARA 13-5-6

10-4-44.1 Replace Lower Main Fuel Lines.**NOTE**

Replacement of left and right lower main fuel lines is the same, except as noted.

10-4-44.1.1 Remove.**WARNING**

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

- a. Purge fuel feed system (PARA 10-4-39).
- b. Gain access to main fuel cell compartment as follows:
 - (1) Remove rear passenger's seats as required (PARA 2-4-45).
 - (2) Remove soundproofing panel from rear bulkhead covering left or right stowage compartment (PARA 2-4-69).
 - (3) Remove cargo netting from stowage compartment.
 - (4) If air conditioning is installed, remove environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
- c. Unlock studs securing left and right panels to main fuel cells and airframe. Remove panels from helicopter.
- d. Remove screws and washers securing front and top panels to airframe. Remove panels from helicopter.
- e. Remove fuel cell wiring harness to lower main fuel lines (PARA 10-4-34).
- f. On left lower main fuel line, disconnect electrical connector, P306, from J306 (Figure 10-4-44-1, Detail A).

- g. On left lower main fuel line, install protective caps and plugs on electrical connectors.
- h. On right lower main fuel line, remove clamp, screw, washers, spacer, and nut from lower main fuel line.
- i. Remove screw, washers, bonding jumpers, and nut from lower main fuel line.
- j. Position 1-quart container to catch excess fuel when disconnecting fuel lines.

CAUTION

- To prevent damage to fuel line fittings, use a strap wrench or wrench with rubber fittings when disconnecting Gamah nuts.
- Make sure not to preload main fuel breakaway valve when disconnecting lower main fuel line. Frangible section of main fuel breakaway valve is sensitive to bending and tension overload.

NOTE

Install protective caps and plugs as parts are removed.

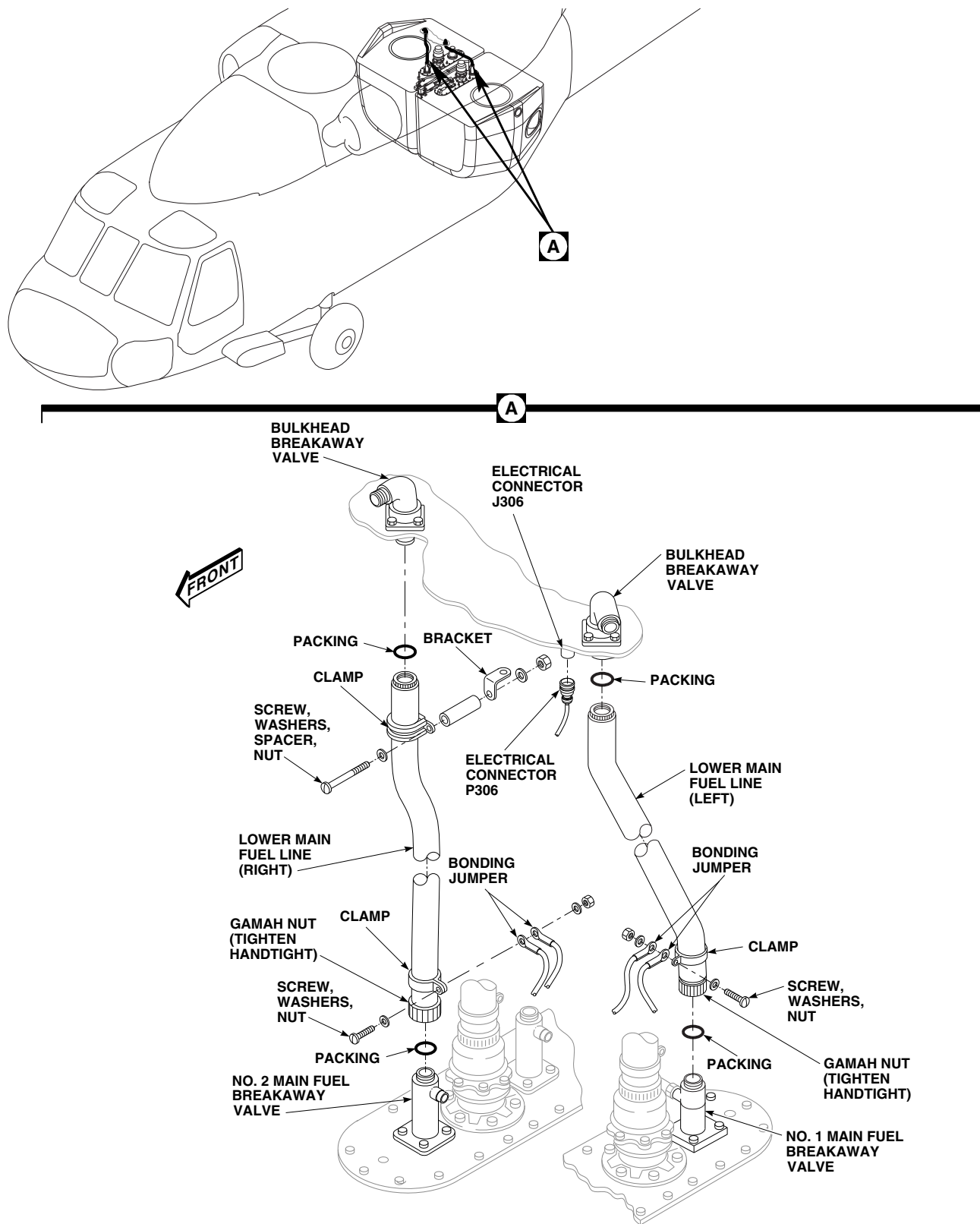
- k. Disconnect lower main fuel line from No. 1 and/or No. 2 main fuel breakaway valve, as required.

CAUTION

Make sure not to preload bulkhead breakaway valve when disconnecting lower main fuel line. Frangible section of main fuel breakaway valve is sensitive to bending and tension overload.

- l. Disconnect lower main fuel line from bulkhead breakaway valve. Remove lower main fuel line from helicopter.
- m. Remove packings from lower main fuel lines. Discard packings.

10-4-44-2 Change 6



FB3654B
SA

FIGURE 10-4-44-1. REPLACE LOWER MAIN FUEL LINES

10-4-44.1.2 Install.**WARNING**

Injury to personnel and damage to equipment will result if normal fire precautions are not followed. Follow normal fire precautions when performing the following procedure.

- a. Turn off all electrical power (PARA 1-6-16).

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- b. Before installation, remove protective caps and plugs and blow all lines clear with dry compressed air.
- c. Lubricate packings and mating surfaces sparingly with lubricating oil, Item 205, Appendix D.
- d. Install packings into line fittings.

CAUTION

To prevent contamination of fuel system and/or damage to equipment, care should be taken not to get antiseize compound into fuel system.

- e. Apply antiseize compound, Item 74, Appendix D, sparingly ($\frac{1}{4}$ -inch strip) along length of

male threads at connection points. Wipe off any squeezeout after parts are assembled with machinery towel, Item 211, Appendix D.

CAUTION

- To prevent damage to connecting components, use a strap wrench or wrench with rubber fittings to tighten Gamah nuts. Do not use pliers or other tools to tighten Gamah nuts.
- To prevent damage to packings, install lower main fuel lines to engine feed breakaway valves first.
- Make sure bulkhead breakaway valve is not preloaded when installing fuel lines. Frangible section of bulkhead breakaway valve is sensitive to bending and tension overload.

NOTE

Remove protective caps and plugs as parts are installed.

- f. Connect lower main fuel lines to bulkhead breakaway valves (Figure 10-4-44-1, Detail A). **TIGHTEN GAMAH NUT HANDTIGHT.**

CAUTION

Do not preload main fuel breakaway valve when installing lower main fuel lines. Frangible section of main fuel breakaway valve is sensitive to bending and tension overload.

- g. Connect lower main fuel line to No. 1 and/or No. 2 main fuel breakaway valve, as required. **TIGHTEN GAMAH NUT HANDTIGHT.**
- h. On right lower main fuel line, secure right lower main fuel line to bracket using clamp, screw, washers, spacer, and nut.
- i. Secure bonding jumpers to lower main fuel line using clamp, screw, washers, and nut.

- j. Seal all bonding jumpers using sealing compound, Item 292, Appendix D.

NOTE

Make sure that fuel cell wiring harness is still connected to mounting plates specified in PARA 10-4-34 with break-away thread. If fuel cell wiring harness is not connected to mounting plates, reattach with breakaway thread as specified in PARA 10-4-34.

- k. Secure fuel cell wiring harness to lower main fuel lines (PARA 10-4-34).

NOTE

This step is only performed if left lower main fuel line was removed.

- l. On left lower main fuel line, connect electrical connector as follows:
 - (1) Remove protective caps and plugs from electrical connectors.
 - (2) Inspect and treat electrical connectors (PARA 1-7-20).
 - (3) Connect electrical connector, P306 to J306.

- m. Make sure area is clean and free of foreign material.

- n. Test fuel feed system (PARA 10-4-39).

- o. Refuel main fuel cells (PARA 1-3-17, PARA 1-3-18, or PARA 1-3-19).

- p. Prime fuel system (PARA 10-4-1). Check for leaks. If leakage is detected, make sure all connections are properly tightened or install new packings.

- q. Install front and top panels with screws and washers.

- r. Install left and right panels enclosing fuel system components. Lock fastener studs.

- s. Close main fuel cell compartment as follows:

- (1) If removed, install environmental control system (ECS) condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
- (2) Install cargo netting.
- (3) Install soundproofing panel (PARA 2-4-69).
- (4) Install rear passenger's seats (PARA 2-4-45).

SECTION 10-5

MAINTENANCE PROCEDURES (BENCH)

SECTION OVERVIEW

PARAGRAPH	TITLE
10-5-1	Main Fuel Cell Ballistic Ring (BENCH)

10-5-1/(10-5-2 Blank)

10-5-1 MAIN FUEL CELL BALLISTIC RING (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
10-5-1.1 Replace Main Fuel Cell Ballistic Ring (BENCH)	10-5-1-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Clamping and Locating Device, 320-4-44816
- Knife

Materials/Parts

- Abrasive Cloth, Item 2, Appendix D
- Adhesive, Item 65, Appendix D

- Aircraft Wax, Item 69, Appendix D
- Ballistic Ring, ACA8002053
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D

10-5-1.1 Replace Main Fuel Cell Ballistic Ring (BENCH).



NOTE

Replacement of the ballistic ring for both main fuel cells is the same.

Damage to fuel cell will result if care is not taken when cutting bondline. Do not cut into molded fitting, flange fabric, or cord plies.

10-5-1.1.1 Remove.

- Cover mating surface of all metal fittings with 1-inch masking tape, Item 212, Appendix D.
- Clean main fuel cell.
- Position main fuel cell with sump drain valve fitting and ballistic ring up.

NOTE

Ballistic ring material at cutline will be glossy purple. Molded fitting flange material at cutline will be dull black. If material on main fuel cell wall at the cutline is dull black, you are cutting too deep.

- d. Using knife, make cut parallel to bondline and between bondline and Line A (Figure 10-5-1-1, Detail A and Detail B). Once cut is made, pry up one end of ballistic ring to permit visibility and continue cutting from outside diameter toward inside diameter, until ballistic ring is removed.
- e. Remove remaining bondline cement or ballistic ring material from main fuel cell, using abrasive cloth, Item 2, Appendix D.

10-5-1.1.2 Install.

CAUTION

Damage to fuel cell will result if care is not taken when buffing bondline. Do not buff into molded fitting flange fabric or cord plies.

NOTE

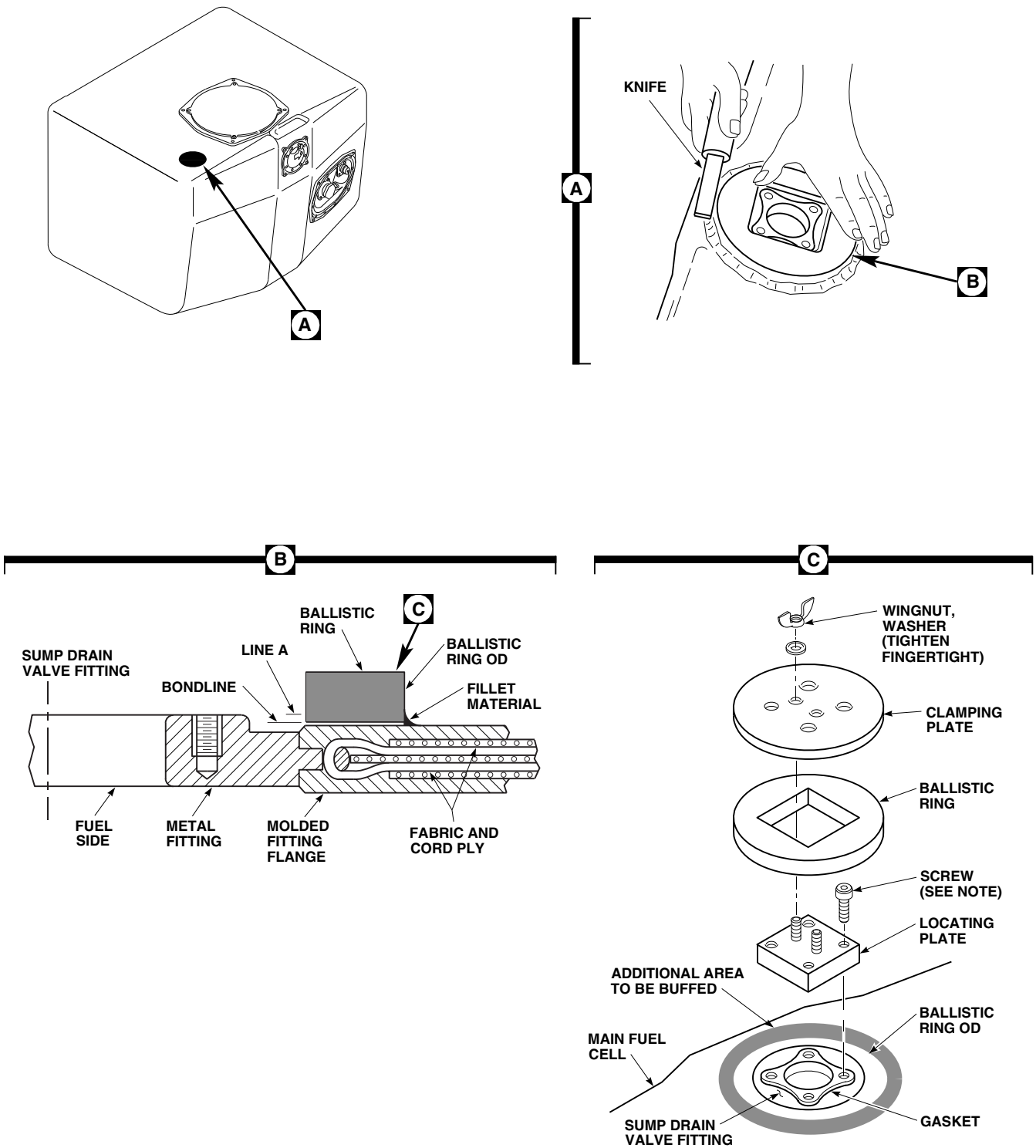
Buffing should be uniform and enough to cover area with fine scratches.

- a. Buff main fuel cell wall $\frac{3}{4}$ -inch larger in every direction than area where ballistic ring is to be bonded, using abrasive cloth, Item 2, Appendix D.
- b. Clean buffed area with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- c. Buff dull side and outside diameter surfaces of new ballistic ring, using abrasive cloth, Item 2, Appendix D.
- d. Clean buffed area with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- e. Disassemble clamping and locating device.

NOTE

Be careful not to touch coated surfaces or allow release agent to contact buffed and cleaned surfaces to be bonded.

- f. Apply light coat of aircraft wax, Item 69, Appendix D, to gasket and side surfaces of locating plate.
- g. Attach locating plate to sump drain valve fitting with screws (Figure 10-5-1-1, Detail C). **TIGHTEN SCREWS UNTIL SHARP RISE IN TORQUE IS FELT.**
- h. Mix adhesive, Item 65, Appendix D, per manufacturer's specification.
- i. Apply an even coat of adhesive to prepared mating surfaces of ballistic ring and main fuel cell.
- j. Position ballistic ring, coated-side down, on main fuel cell.
- k. Apply light coat of aircraft wax, Item 69, Appendix D, to cutout side of clamping plate.
- l. Position clamping plate, coated-side down, over ballistic ring, with studs protruding through clamping plate.
- m. Secure clamping plate to locating plate with washers and wingnuts. **TIGHTEN WINGNUTS FINGERTIGHT.**
- n. Smooth extra adhesive at ballistic ring outside diameter into a fillet (Figure 10-5-1-1, Detail B).
- o. Fill voids at ballistic ring outside diameter with remaining adhesive and smooth into a fillet.
- p. Allow adhesive to cure for at least 6 hours, at not less than 70°F (21°C).
- q. Remove wingnuts, washers, clamping plate, and locating plate from fuel cell (Figure 10-5-1-1, Detail C).
- r. Remove tape from metal fittings.



NOTE

TIGHTEN SCREWS UNTIL A SHARP RISE IN TORQUE IS FELT.

FO0767B
SA

FIGURE 10-5-1-1. REPLACE MAIN FUEL CELL BALLISTIC RING (BENCH)

